Elementary Analysis

Chapter 2: Sequences

All 117 exercises, with every subpart
Detailed worked solutions

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Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Chapter 2. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

Coverage. Complete Chapter 2, Sections 7–16.

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7 Limits of sequences

A sequence assigns a real number s_n to each index n. Unless the exercise states otherwise, the indices here are $n = 1, 2, 3, \ldots$. The order of the terms matters, even when their values repeat.

We use Ross's Definition 7.1: $s_n \to s$ means that, for every $\varepsilon > 0$, there is a number N such that

$$n > N \quad \Longrightarrow \quad |s_n - s| < \varepsilon \qquad (n \in \mathbb{N}).$$

The choice of N may depend on ε but cannot depend on the later index n. A real threshold N is permitted; the Archimedean property ensures that there are integer indices past it. Convergence here means convergence to a finite real number. Divergence means that no such real limit exists.

Although several exercises in this section ask only for answers, we include the algebra and estimates that explain those answers. The proofs use elementary inequalities and the limit definition, rather than the limit laws introduced in Section 9.

7.1 Exercise 7.1: The first five terms of four sequences

What we must show. Compute the terms with indices $n = 1, 2, 3, 4, 5$ for

$$\text{(a) } s_n = \frac{1}{3n+1}, \qquad \text{(b) } b_n = \frac{3n+1}{4n-1}, \qquad \text{(c) } c_n = \frac{n}{3^n}, \qquad \text{(d) } d_n = \sin\left(\frac{n\pi}{4}\right).$$

Plan. Substitute the index into every occurrence of n, evaluate the numerator and denominator separately, and simplify where possible. Indices start at 1. In part (a), the denominator $3n+1$ means 3 times n, followed by addition of 1.

1. Part (a): A reciprocal with a linear denominator

For $n \in \mathbb{N}$, we have $3n + 1 \geq 4 > 0$, so all the indicated fractions are defined. The five substitutions give

$$\begin{aligned}
s_1 &= \frac{1}{3 \cdot 1 + 1} = \frac{1}{3+1} = \frac{1}{4}, \\
s_2 &= \frac{1}{3 \cdot 2 + 1} = \frac{1}{6+1} = \frac{1}{7}, \\
s_3 &= \frac{1}{3 \cdot 3 + 1} = \frac{1}{9+1} = \frac{1}{10}, \\
s_4 &= \frac{1}{3 \cdot 4 + 1} = \frac{1}{12+1} = \frac{1}{13}, \\
s_5 &= \frac{1}{3 \cdot 5 + 1} = \frac{1}{15+1} = \frac{1}{16}.
\end{aligned}$$

2. Part (b): A quotient of two linear expressions

Here $4n - 1 \geq 3 > 0$ for every $n \geq 1$, so there is again no zero denominator. Compute both parts of each fraction:

$$b_1 = \frac{3 \cdot 1 + 1}{4 \cdot 1 - 1} = \frac{3+1}{4-1} = \frac{4}{3},$$

$$b_2 = \frac{3\cdot 2+1}{4\cdot 2-1} = \frac{6+1}{8-1} = \frac{7}{7} = 1,$$
$$b_3 = \frac{3\cdot 3+1}{4\cdot 3-1} = \frac{9+1}{12-1} = \frac{10}{11},$$
$$b_4 = \frac{3\cdot 4+1}{4\cdot 4-1} = \frac{12+1}{16-1} = \frac{13}{15},$$
$$b_5 = \frac{3\cdot 5+1}{4\cdot 5-1} = \frac{15+1}{20-1} = \frac{16}{19}.$$

3. Part (c): A linear numerator and an exponential denominator

The denominator is the positive integer power 3^n. In particular, $3^1 = 3$, $3^2 = 9$, $3^3 = 27$, $3^4 = 81$, and $3^5 = 243$. Substitution yields

$$c_1 = \frac{1}{3^1} = \frac{1}{3},$$
$$c_2 = \frac{2}{3^2} = \frac{2}{9},$$
$$c_3 = \frac{3}{3^3} = \frac{3}{27} = \frac{1}{9},$$
$$c_4 = \frac{4}{3^4} = \frac{4}{81},$$
$$c_5 = \frac{5}{3^5} = \frac{5}{243}.$$

The cancellation in the third term divides numerator and denominator by 3; it does not change the value of that term.

4. Part (d): Sine at multiples of an eighth-turn angle

On the unit circle at angle $\pi/4$, the two coordinates are equal and positive. If each is t, then $t^2+t^2 = 1$, so $t^2 = 1/2$ and $t = \sqrt{2}/2$. Thus $\sin(\pi/4) = \sqrt{2}/2$. The unit-circle values at $\pi/2$ and π are 1 and 0. Using these values and the identities $\sin(\pi-\theta) = \sin\theta$ and $\sin(\pi+\theta) = -\sin\theta$, we obtain

$$d_1 = \sin\left(\frac{1\pi}{4}\right) = \sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2},$$
$$d_2 = \sin\left(\frac{2\pi}{4}\right) = \sin\left(\frac{\pi}{2}\right) = 1,$$
$$d_3 = \sin\left(\frac{3\pi}{4}\right) = \sin\left(\pi - \frac{\pi}{4}\right) = \frac{\sqrt{2}}{2},$$
$$d_4 = \sin\left(\frac{4\pi}{4}\right) = \sin\pi = 0,$$
$$d_5 = \sin\left(\frac{5\pi}{4}\right) = \sin\left(\pi + \frac{\pi}{4}\right) = -\frac{\sqrt{2}}{2}.$$

Conclusion. The requested five-term lists are

Part	1	2	3	4	5
(a)	$\frac{1}{4}$	$\frac{1}{7}$	$\frac{1}{10}$	$\frac{1}{13}$	$\frac{1}{16}$
(b)	$\frac{4}{3}$	1	$\frac{10}{11}$	$\frac{13}{15}$	$\frac{16}{19}$
(c)	$\frac{1}{3}$	$\frac{2}{9}$	$\frac{1}{9}$	$\frac{4}{81}$	$\frac{5}{243}$
(d)	$\frac{\sqrt{2}}{2}$	1	$\frac{\sqrt{2}}{2}$	0	$-\frac{\sqrt{2}}{2}$

7.2 Exercise 7.2: Convergence of the four preceding sequences

What we must show. Decide which sequences in Exercise 7.1 converge and give each finite limit when it exists.

Plan. Use exact error estimates for the first three sequences. For the sine sequence, find values 1 and -1 at arbitrarily large indices, then compare them with any proposed common limit. The arguments use the limit definition directly: $u_n \to L$ means that for every $\varepsilon > 0$ there is a threshold N such that $n > N$ implies $|u_n - L| < \varepsilon$.

1. Part (a): The limit of $1/(3n+1)$

For every positive integer n,

$$\left|\frac{1}{3n+1} - 0\right| = \frac{1}{3n+1} < \frac{1}{3n}.$$

The reciprocal inequality follows because $3n+1 > 3n > 0$. Let $\varepsilon > 0$. By the Archimedean property, choose an integer $N > 1/(3\varepsilon)$. If $n > N$, then $3n > 1/\varepsilon$ and

$$\left|\frac{1}{3n+1} - 0\right| < \frac{1}{3n} < \varepsilon.$$

This verifies the definition and proves $s_n \to 0$.

2. Part (b): The limit of $(3n+1)/(4n-1)$

The ratio of the leading coefficients suggests $3/4$. Compute the error over a common denominator to verify it:

$$\begin{aligned} b_n - \frac{3}{4} &= \frac{3n+1}{4n-1} - \frac{3}{4} \\ &= \frac{4(3n+1) - 3(4n-1)}{4(4n-1)} \\ &= \frac{12n + 4 - 12n + 3}{4(4n-1)} \\ &= \frac{7}{4(4n-1)}. \end{aligned}$$

The denominator is positive for $n \geq 1$, so this is also the absolute error. Moreover, $4n - 1 - 3n = n - 1 \geq 0$, hence $4n - 1 \geq 3n$. Therefore

$$\left|b_n - \frac{3}{4}\right| = \frac{7}{4(4n-1)} \leq \frac{7}{12n}.$$

Given $\varepsilon > 0$, choose an integer $N > \max\{1, 7/(12\varepsilon)\}$. For $n > N$ we get

$$\left|b_n - \frac{3}{4}\right| \leq \frac{7}{12n} < \varepsilon.$$

Thus $b_n \to 3/4$. No quotient limit law is needed for this conclusion.

3. Part (c): Bound the exponential denominator by a square

We first prove $3^n \geq n^2$ for every integer $n \geq 2$. For $n = 2$, this reads $9 \geq 4$. Suppose it holds for an integer $n \geq 2$. Then

$$3^{n+1} = 3 \cdot 3^n \geq 3n^2.$$

Expanding the next square gives

$$3n^2 - (n+1)^2 = 3n^2 - (n^2 + 2n + 1) = 2n(n-1) - 1 \geq 3 > 0,$$

where $n \geq 2$ gives $2n(n-1) \geq 4$. Hence $3n^2 \geq (n+1)^2$, so $3^{n+1} \geq (n+1)^2$. Mathematical induction proves the claimed bound.

Because all denominators are positive, the bound implies, for $n \geq 2$,

$$0 < \frac{n}{3^n} \leq \frac{n}{n^2} = \frac{1}{n}.$$

Let $\varepsilon > 0$ and choose an integer $N > \max\{2, 1/\varepsilon\}$. Then $n > N$ gives

$$\left| \frac{n}{3^n} - 0 \right| = \frac{n}{3^n} \leq \frac{1}{n} < \varepsilon.$$

It follows directly that $c_n \to 0$.

4. Part (d): Values separated by 2 persist arbitrarily far out

For every nonnegative integer k, periodicity of sine gives

$$d_{8k+2} = \sin\left(\frac{(8k+2)\pi}{4}\right) = \sin\left(2k\pi + \frac{\pi}{2}\right) = 1,$$
$$d_{8k+6} = \sin\left(\frac{(8k+6)\pi}{4}\right) = \sin\left(2k\pi + \frac{3\pi}{2}\right) = -1.$$

Both families of indices can be made larger than any given real threshold N: choose an integer $k > \max\{0, N/8\}$. Then $8k + 2 > N$ and $8k + 6 > N$.

Suppose, for a contradiction, that $d_n \to L \in \mathbb{R}$. Use $\varepsilon = 1/2$ in the definition. There is a threshold N such that $n > N$ implies $|d_n - L| < 1/2$. Choose k so that both indices above exceed this N. Then

$$|1 - L| < \frac{1}{2}, \qquad |-1 - L| < \frac{1}{2}.$$

The triangle inequality, together with $|L + 1| = |-1 - L|$, would imply

$$2 = |1 - (-1)| = |(1 - L) + (L + 1)| \leq |1 - L| + |L + 1| < \frac{1}{2} + \frac{1}{2} = 1.$$

This contradicts $2 > 1$. Therefore the sine sequence has no real limit.

Conclusion. Parts (a), (b), and (c) converge to 0, 3/4, and 0, respectively. Part (d) diverges.

7.3 Exercise 7.3: A catalogue of twenty sequences and their limits

What we must show. Determine convergence or divergence for every sequence in parts (a)–(t), and give the real limit whenever one exists. Although the exercise does not require proofs, we include a justification of every answer. The arguments use the definition of convergence directly, rather than the limit laws developed later in Section 9.

Plan. For a convergent sequence, subtract the proposed limit and bound the resulting error by a reciprocal that becomes arbitrarily small. For a divergent sequence, find either arbitrarily late values that are arbitrarily large, or two arbitrarily late values separated by a fixed positive distance.

Elementary facts used in the individual parts

Fact 1: Reciprocal error estimates imply convergence.

The definition says that $u_n \to L$ if, for every $\varepsilon > 0$, there is a threshold N such that

$$n > N \quad \Longrightarrow \quad |u_n - L| < \varepsilon.$$

Suppose we prove $|u_n - L| \le C/n$ for every $n \ge n_0$, where $C > 0$ is fixed. Given $\varepsilon > 0$, choose a positive integer $N > \max\{n_0, C/\varepsilon\}$, using the Archimedean property. Then $n > N$ implies both $n \ge n_0$ and $n > C/\varepsilon$. Multiplying the latter inequality by $\varepsilon > 0$ and dividing by $n > 0$ gives

$$|u_n - L| \le \frac{C}{n} < \varepsilon.$$

This proves convergence from the definition. In particular, $1/n \to 0$. If an error is identically 0, it is less than every $\varepsilon > 0$ without any additional estimate.

We will also use a shifted version. If, for all sufficiently large n, $|u_n - L| \le C/(n - r)$, with fixed real r and $n > r$, choose N larger than the starting index and $r + C/\varepsilon$. Then $n > N$ gives $n - r > C/\varepsilon$, and the same calculation gives $|u_n - L| < \varepsilon$. This proves the shifted version without any new limit theorem.

Only the tail is used in either proof. Thus a finite number of unusual initial values does not affect the conclusion: increase N to exceed all their indices, and every $n > N$ still satisfies the required estimate. This observation will matter when an initial denominator is negative but never zero.

Fact 2: Powers of two dominate a simple linear expression.

For every integer $m \ge 0$,

$$2^m \ge m + 1.$$

At $m = 0$, both sides equal 1. If the inequality holds at m, then multiplication by $2 > 0$ gives

$$2^{m+1} = 2 \cdot 2^m \ge 2(m + 1) = 2m + 2 \ge m + 2,$$

where the last difference is $(2m + 2) - (m + 2) = m \ge 0$. Induction proves the assertion. Taking reciprocals of positive quantities consequently gives

$$0 < 2^{-m} \le \frac{1}{m + 1}.$$

Fact 3: Separated arbitrarily late values rule out a real limit.

Suppose some fixed $D > 0$ has this property: for every threshold N, there are indices $p, q > N$ with $|u_p - u_q| \ge D$. Assume, to obtain a contradiction, that $u_n \to L \in \mathbb{R}$. Apply the definition with $\varepsilon = D/3$. It supplies N such that both late terms obey $|u_p - L| < D/3$ and $|u_q - L| < D/3$. The triangle inequality would then give

$$|u_p - u_q| = |(u_p - L) + (L - u_q)| \le |u_p - L| + |L - u_q| < \frac{D}{3} + \frac{D}{3} = \frac{2D}{3} < D,$$

contradicting the stipulated separation. Thus such a sequence does not converge to a real number. This argument does not assume any theorem about subsequences.

Fact 4: Arbitrarily large late values also rule out a real limit.

Suppose that for every $B \in \mathbb{R}$ and every threshold N, some $n > N$ satisfies $u_n > B$. If $u_n \to L \in \mathbb{R}$, the definition with $\varepsilon = 1$ would supply N for which $|u_n - L| < 1$ whenever $n > N$. Choose a late value with $u_n > L + 1$. It satisfies $u_n - L > 1$, and therefore $|u_n - L| = u_n - L > 1$, a contradiction. This is a direct nonconvergence argument; it does not invoke a later boundedness theorem for convergent sequences.

Solutions of parts (a)–(t)

(a) The sequence $a_n = n/(n+1)$.

For every $n \in \mathbb{N}$, $n + 1 \geq 2 > 0$, so every term is defined. To test the candidate 1, subtract it using the common denominator $n + 1$:

$$a_n - 1 = \frac{n}{n+1} - \frac{n+1}{n+1} = \frac{n - (n+1)}{n+1} = -\frac{1}{n+1}.$$

It follows that

$$|a_n - 1| = \frac{1}{n+1} \leq \frac{1}{n}.$$

The inequality uses $n + 1 \geq n > 0$ and the reversal of order for positive reciprocals. Fact 1 with $C = 1$ now proves the limit.

Conclusion. Part (a) converges, and $\lim_{n\to\infty} \dfrac{n}{n+1} = 1$.

(b) The sequence $b_n = (n^2 + 3)/(n^2 - 3)$.

The denominator is nonzero at every positive integer. At $n = 1$ it is $1 - 3 = -2$, while $n \geq 2$ implies $n^2 \geq 4$ and $n^2 - 3 \geq 1 > 0$. Thus there is no undefined term, including in the initial part of the sequence.

Subtracting the candidate 1 gives

$$b_n - 1 = \frac{n^2 + 3 - (n^2 - 3)}{n^2 - 3} = \frac{n^2 + 3 - n^2 + 3}{n^2 - 3} = \frac{6}{n^2 - 3}.$$

For $n \geq 3$, we have $n^2 \geq 9 \geq 6$, so $3 \leq n^2/2$. Subtracting 3 from n^2 therefore gives

$$n^2 - 3 \geq n^2 - \frac{n^2}{2} = \frac{n^2}{2} > 0.$$

Consequently

$$|b_n - 1| = \frac{6}{n^2 - 3} \leq \frac{6}{n^2/2} = \frac{12}{n^2} \leq \frac{12}{n}.$$

The last inequality follows from $n^2 \geq n > 0$ for $n \geq 1$. Fact 1 with $C = 12$ and starting index 3 proves convergence. The negative first denominator does not enter the tail estimate.

Conclusion. Part (b) converges to 1.

(c) The sequence $c_n = 2^{-n}$.

The expression means the positive reciprocal $1/2^n$. Fact 2 gives

$$|c_n - 0| = \frac{1}{2^n} \leq \frac{1}{n+1} \leq \frac{1}{n} \qquad (n \geq 1).$$

The error is thus bounded by $1/n$. Applying Fact 1 proves the limit directly.

Conclusion. Part (c) converges to 0.

(d) The sequence $t_n = 1 + 2/n$.

Since $n > 0$, every term is defined, and subtracting 1 gives

$$|t_n - 1| = \left|1 + \frac{2}{n} - 1\right| = \left|\frac{2}{n}\right| = \frac{2}{n}.$$

For each $\varepsilon > 0$, choosing $N > 2/\varepsilon$ makes this error smaller than ε whenever $n > N$. This is Fact 1 with $C = 2$.

Conclusion. Part (d) converges to 1.

(e) The sequence $x_n = 73 + (-1)^n$.

If n is even, $(-1)^n = 1$ and $x_n = 73 + 1 = 74$. If n is odd, $(-1)^n = -1$ and $x_n = 73 - 1 = 72$. These values do not merely occur at the start: for any threshold N, choose a positive integer $m > N$. Then $2m > N$ and $2m + 1 > N$, and

$$x_{2m} = 74, \qquad x_{2m+1} = 72, \qquad |x_{2m} - x_{2m+1}| = |74 - 72| = 2.$$

Thus Fact 3 applies with $D = 2$. No real number can be approached by every term, because every tail still contains these two values at distance 2.

Conclusion. Part (e) does not converge.

(f) The sequence $s_n = 2^{1/n}$.

Write $r_n = 2^{1/n}$ for the positive real nth root of 2; thus $r_n > 0$ and $r_n^n = 2$. We prove the concrete estimate

$$0 \le r_n - 1 \le \frac{1}{n}.$$

It will imply convergence by Fact 1 without any theorem about limits of roots or logarithms.

First prove the auxiliary Bernoulli inequality. For every $x \ge 0$ and every positive integer m,

$$(1 + x)^m \ge 1 + mx.$$

Fix $x \ge 0$. At $m = 1$ the two sides are equal. If the inequality holds at m, multiplying by the positive number $1 + x$ gives

$$\begin{aligned}(1 + x)^{m+1} &\ge (1 + mx)(1 + x)\\ &= 1 + x + mx + mx^2\\ &= 1 + (m + 1)x + mx^2\\ &\ge 1 + (m + 1)x,\end{aligned}$$

because $mx^2 \ge 0$. This is the required step at $m + 1$, so induction proves the inequality for the fixed x, and hence for every $x \ge 0$.

Apply it with $x = 1/n$ and $m = n$. We obtain

$$\left(1 + \frac{1}{n}\right)^n \ge 1 + n\frac{1}{n} = 2.$$

For positive real numbers, raising to a positive integer power preserves strict order. For example, if $u > v > 0$, then $u/v > 1$, so the product of n factors u/v is greater than 1; multiplication by $v^n > 0$ gives $u^n > v^n$. If $r_n > 1 + 1/n$, this order property would give

$$2 = r_n^n > \left(1 + \frac{1}{n}\right)^n \geq 2,$$

a contradiction. Hence $r_n \leq 1 + 1/n$. Also $r_n \geq 1$: if $0 < r_n < 1$, multiplying n factors each less than 1 would give $r_n^n < 1$, contradicting $r_n^n = 2$.

Thus $1 \leq r_n \leq 1 + 1/n$, and subtraction of 1 yields

$$|r_n - 1| = r_n - 1 \leq \frac{1}{n}.$$

Fact 1 proves the asserted limit.

Conclusion. Part (f) converges to 1.

(g) The sequence $y_n = n!$.

For $n \geq 1$, the factorial can be written

$$n! = n\,(1 \cdot 2 \cdots (n-1)).$$

At $n = 1$, the parenthesized product is empty and equals 1. For $n \geq 2$, every factor in that product is at least 1, so its product is at least 1. In all cases, therefore,

$$n! \geq n.$$

Now let a real number B and a threshold N be arbitrary. Choose $n \in \mathbb{N}$ with $n > \max\{N, B, 1\}$. Then $n > N$ and $y_n = n! \geq n > B$. Thus arbitrarily late values exceed every prescribed real size. Fact 4 rules out convergence to any real number. This argument establishes the needed nonconvergence without citing a boundedness theorem.

Conclusion. Part (g) does not converge to a real limit; its values grow beyond every real bound.

(h) The sequence $d_n = (-1)^n n$.

At an even index $2m$ the value is $2m$, while at an odd index $2m + 1$ the value is $-(2m + 1)$. Given any threshold N, choose a positive integer $m > N$. Both indices $2m$ and $2m + 1$ are then greater than N, and

$$\begin{aligned} |d_{2m} - d_{2m+1}| &= |2m - (-(2m+1))| \\ &= |4m + 1| \\ &= 4m + 1 \geq 5 > 1. \end{aligned}$$

Here $m \geq 1$, so $4m + 1$ is positive, justifying the removal of the absolute value. Every tail has two values separated by at least 1. Fact 3 with $D = 1$ rules out a real limit.

Conclusion. Part (h) does not converge. The signs alternate while the magnitudes increase.

(i) The sequence $u_n = (-1)^n/n$.

Although its sign alternates, its distance from 0 is

$$|u_n - 0| = \frac{|(-1)^n|}{|n|} = \frac{1}{n},$$

because $|(-1)^n| = 1$ and $n > 0$. Fact 1 applies with $C = 1$. In contrast with (h), the alternating values now have small magnitude for every sufficiently large index.

Conclusion. Part (i) converges to 0.

(j) The sequence $u_n = (7n^3 + 8n)/(2n^3 - 3)$.

At $n = 1$, the denominator is $2 - 3 = -1$, so it is nonzero. For $n \geq 2$, $n^3 \geq 8$, giving $2n^3 - 3 \geq 16 - 3 = 13 > 0$. Thus the sequence is defined at every positive integer.

We test the candidate $7/2$ by exact subtraction:

$$\begin{aligned} u_n - \frac{7}{2} &= \frac{2(7n^3 + 8n) - 7(2n^3 - 3)}{2(2n^3 - 3)} \\ &= \frac{14n^3 + 16n - 14n^3 + 21}{2(2n^3 - 3)} \\ &= \frac{16n + 21}{2(2n^3 - 3)}. \end{aligned}$$

For $n \geq 2$, we have $n^3 \geq 8 > 3$, so

$$2n^3 - 3 \geq n^3 > 0.$$

Also $n \geq 1$ implies $21 \leq 21n$ and hence $16n + 21 \leq 37n$. Using these positive denominator bounds gives

$$\left| u_n - \frac{7}{2} \right| = \frac{16n + 21}{2(2n^3 - 3)} \leq \frac{37n}{2n^3} = \frac{37}{2n^2} \leq \frac{37}{2n}.$$

The last step again uses $n^2 \geq n > 0$. Fact 1, with $C = 37/2$ and starting index 2, proves convergence.

Conclusion. Part (j) converges to $7/2$.

(k) The sequence $u_n = (9n^2 - 18)/(6n + 18)$.

The denominator $6n + 18$ is positive for every $n \geq 1$. We establish a lower estimate that becomes arbitrarily large. For $n \geq 3$, $n^2 \geq 9$, so $3n^2 \geq 27 > 18$ and

$$9n^2 - 18 \geq 9n^2 - 3n^2 = 6n^2 > 0.$$

Also $n \geq 3$ implies $18 \leq 6n$, giving

$$0 < 6n + 18 \leq 12n.$$

First lower the numerator while keeping the positive denominator fixed, then increase that denominator:

$$u_n = \frac{9n^2 - 18}{6n + 18} \geq \frac{6n^2}{6n + 18} \geq \frac{6n^2}{12n} = \frac{n}{2}.$$

The second comparison follows from order reversal for positive reciprocals, followed by multiplication by $6n^2 > 0$.

Let $B \in \mathbb{R}$ and a threshold N be arbitrary. Choose a positive integer $n > \max\{N, 3, 2B\}$. The estimate applies and gives $u_n \geq n/2 > B$, with $n > N$. Fact 4 therefore rules out convergence to every real candidate.

Conclusion. Part (k) does not converge to a real limit; its values grow beyond every real bound.

(l) The sequence $u_n = \sin(n\pi/2)$.

Every positive integer has the form $n = 4q + r$, with $q \geq 0$ an integer and $r \in \{1, 2, 3, 4\}$. Since $n\pi/2 = 2q\pi + r\pi/2$, the period 2π of sine gives the repeating values

r	1	2	3	4
$\sin(r\pi/2)$	1	0	-1	0

The repetition alone suggests nonconvergence; we now check the arbitrarily late separation required by Fact 3. Given a threshold N, choose $m \in \mathbb{N}$ with $m > N$. Then $p = 4m+1 > N$ and $q = 4m+3 > N$, and periodicity gives

$$u_p = \sin(2m\pi + \pi/2) = 1, \qquad u_q = \sin(2m\pi + 3\pi/2) = -1.$$

Their distance is $|1 - (-1)| = 2$. Fact 3 applies with $D = 2$ and proves nonconvergence.

Conclusion. Part (l) does not converge.

(m) The sequence $u_n = \sin(n\pi)$.

For every integer n, the angle $n\pi$ lies at an integer multiple of a half-turn, so its sine is 0. Explicitly, if $n = 2m$ is even, then $\sin(n\pi) = \sin(2m\pi) = 0$ by periodicity. If $n = 2m + 1$ is odd, then

$$\sin(n\pi) = \sin(2m\pi + \pi) = \sin \pi = 0.$$

These two cases cover every positive integer. Thus $u_n = 0$ for every index, and for every $\varepsilon > 0$,

$$|u_n - 0| = 0 < \varepsilon$$

holds at every index. This verifies convergence directly from the definition.

Conclusion. Part (m) converges to 0.

(n) The sequence $u_n = \sin(2n\pi/3)$.

Write $n = 3q + r$, with $q \geq 0$ and $r \in \{1, 2, 3\}$. Then $2n\pi/3 = 2q\pi + 2r\pi/3$, giving the period table

r	1	2	3
$\sin(2r\pi/3)$	$\sqrt{3}/2$	$-\sqrt{3}/2$	0

For any threshold N, choose a positive integer $m > N$. The indices $p = 3m + 1$ and $q = 3m + 2$ both exceed N, and

$$u_p = \frac{\sqrt{3}}{2}, \qquad u_q = -\frac{\sqrt{3}}{2}.$$

Their distance is

$$|u_p - u_q| = \left| \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} \right| = \sqrt{3} > 0.$$

Consequently Fact 3 with $D = \sqrt{3}$ excludes every real limit.

Conclusion. Part (n) does not converge.

(o) The sequence $u_n = (\sin n)/n$.

Here the argument is n radians, with no factor of π. We do not need to classify the behavior of $\sin n$ itself. The elementary trigonometric inequality $|\sin x| \leq 1$ for every real x gives

$$|u_n - 0| = \left| \frac{\sin n}{n} \right| = \frac{|\sin n|}{n} \leq \frac{1}{n},$$

where division by $n > 0$ preserves the inequality. Fact 1 with $C = 1$ proves the limit.

Conclusion. Part (o) converges to 0.

(p) The sequence $u_n = (2^{n+1} + 5)/(2^n - 7)$.

The denominator is $2^n - 7$. At $n = 1$ it is $2 - 7 = -5$, and at $n = 2$ it is $4 - 7 = -3$. For $n \geq 3$, $2^n \geq 8$ and therefore $2^n - 7 \geq 1 > 0$. These cases show that every term is defined. For example the first three terms are

$$u_1 = \frac{4+5}{2-7} = -\frac{9}{5}, \qquad u_2 = \frac{8+5}{4-7} = -\frac{13}{3}, \qquad u_3 = \frac{16+5}{8-7} = 21.$$

They do not by themselves determine the limiting behavior.

Subtract the candidate 2 exactly:

$$\begin{aligned} u_n - 2 &= \frac{2^{n+1} + 5 - 2(2^n - 7)}{2^n - 7} \\ &= \frac{2 \cdot 2^n + 5 - 2 \cdot 2^n + 14}{2^n - 7} \\ &= \frac{19}{2^n - 7}. \end{aligned}$$

For $n \geq 4$, $2^n \geq 16 \geq 14$, so $7 \leq 2^n/2$. It follows that

$$2^n - 7 \geq 2^n - \frac{2^n}{2} = \frac{2^n}{2} > 0.$$

Using this bound and Fact 2 yields

$$\begin{aligned} |u_n - 2| &= \frac{19}{2^n - 7} \\ &\leq \frac{19}{2^n/2} = \frac{38}{2^n} \leq \frac{38}{n+1} \leq \frac{38}{n}. \end{aligned}$$

Fact 1 with $C = 38$ and starting index 4 proves convergence. It also explains why the negative initial terms and the relatively large value $u_3 = 21$ do not change the tail conclusion.

Conclusion. Part (p) converges to 2.

(q) The sequence $a_n = 3^n/n!$.

Both 3^n and $n!$ are positive for every $n \in \mathbb{N}$, so $a_n > 0$ and every ratio below is defined. Compare successive terms using $(n+1)! = (n+1)n!$:

$$\begin{aligned} \frac{a_{n+1}}{a_n} &= \frac{3^{n+1}}{(n+1)!} \cdot \frac{n!}{3^n} \\ &= \frac{3 \cdot 3^n \cdot n!}{(n+1)n!\, 3^n} \\ &= \frac{3}{n+1}. \end{aligned}$$

For $n \geq 5$, $n + 1 \geq 6$, giving

$$0 < \frac{3}{n+1} \leq \frac{3}{6} = \frac{1}{2}, \qquad a_{n+1} \leq \frac{1}{2} a_n.$$

We will turn this ratio estimate into an explicit reciprocal bound; no theorem about ratios of sequences is being invoked.

The value at the starting index is

$$C = a_5 = \frac{3^5}{5!} = \frac{243}{120} = \frac{81}{40} > 0.$$

We claim, for every integer $m \geq 0$, that

$$a_{5+m} \leq C\,2^{-m}.$$

At $m = 0$, this is equality $a_5 = C$. If the estimate holds at m, the ratio bound applies at $n = 5+m \geq 5$, and gives

$$a_{5+(m+1)} = a_{6+m} \leq \frac{1}{2}a_{5+m} \leq \frac{1}{2}C\,2^{-m} = C\,2^{-(m+1)}.$$

This is the next estimate, so induction proves the claim. Putting $m = n - 5$ now yields, for every $n \geq 5$,

$$0 < a_n \leq C\,2^{-(n-5)}.$$

Fact 2, applied to $m = n - 5 \geq 0$, supplies

$$2^{n-5} \geq (n-5) + 1 = n - 4 > 0.$$

Taking reciprocals and multiplying by $C > 0$ therefore gives

$$|a_n - 0| = a_n \leq C\,2^{-(n-5)} \leq \frac{C}{n-4} = \frac{81/40}{n-4}.$$

For any $\varepsilon > 0$, choose an integer $N > \max\{5, 4 + C/\varepsilon\}$. If $n > N$, then $n - 4 > C/\varepsilon$, so the displayed error is less than ε. This is the shifted estimate of Fact 1, proving the limit directly.

Conclusion. Part (q) converges to 0.

(r) The sequence $u_n = (1 + 1/n)^2$.

Expand by multiplying its two factors, not by assuming a rule about limits of squares:

$$\begin{aligned} u_n &= \left(1 + \frac{1}{n}\right)\left(1 + \frac{1}{n}\right) \\ &= 1 + \frac{1}{n} + \frac{1}{n} + \frac{1}{n^2} \\ &= 1 + \frac{2}{n} + \frac{1}{n^2}. \end{aligned}$$

All terms after the leading 1 are positive. Since $n^2 \geq n > 0$, we have $1/n^2 \leq 1/n$, giving

$$|u_n - 1| = \frac{2}{n} + \frac{1}{n^2} \leq \frac{2}{n} + \frac{1}{n} = \frac{3}{n}.$$

Fact 1 with $C = 3$ proves convergence.

Conclusion. Part (r) converges to 1.

(s) The sequence $u_n = (4n^2 + 3)/(3n^2 - 2)$.

For all $n \geq 1$, $n^2 \geq 1$, so $3n^2 - 2 \geq 3 - 2 = 1 > 0$. Thus every term is defined and the denominator has a fixed positive sign. Subtracting $4/3$ with a common denominator gives

$$\begin{aligned} u_n - \frac{4}{3} &= \frac{3(4n^2+3) - 4(3n^2-2)}{3(3n^2-2)} \\ &= \frac{12n^2 + 9 - 12n^2 + 8}{3(3n^2-2)} \\ &= \frac{17}{3(3n^2-2)}. \end{aligned}$$

The estimate $n^2 \geq 1$ also yields

$$(3n^2 - 2) - n^2 = 2n^2 - 2 = 2(n^2 - 1) \geq 0,$$

so $3n^2 - 2 \geq n^2 > 0$. Therefore

$$\left| u_n - \frac{4}{3} \right| = \frac{17}{3(3n^2-2)} \leq \frac{17}{3n^2} \leq \frac{17}{3n}.$$

Fact 1 with $C = 17/3$ verifies the limit from the definition.

Conclusion. Part (s) converges to $4/3$.

(t) The sequence $u_n = (6n + 4)/(9n^2 + 7)$.

For $n \geq 1$, both $6n + 4$ and $9n^2 + 7$ are positive, so the absolute value of u_n is u_n itself. We have

$$9n^2 + 7 \geq 9n^2 > 0, \qquad 6n + 4 \leq 6n + 4n = 10n,$$

where $4 \leq 4n$ uses $n \geq 1$. Taking reciprocals in the denominator comparison and then applying the numerator comparison,

$$|u_n - 0| = \frac{6n+4}{9n^2+7} \leq \frac{6n+4}{9n^2} \leq \frac{10n}{9n^2} = \frac{10}{9n}.$$

Fact 1 with $C = 10/9$ proves convergence to the proposed value 0.

Conclusion. Part (t) converges to 0.

Complete answer list

In this list, "does not converge" means that there is no finite real limit. Every entry is justified in its corresponding part above. The growth cases (g) and (k), the persistent oscillations (e), (h), (l), and (n), and the shrinking oscillations (i) and (o) have been distinguished by direct estimates and the definition.

Part	Answer	Part	Answer
(a)	Converges to 1	(k)	Does not converge
(b)	Converges to 1	(l)	Does not converge
(c)	Converges to 0	(m)	Converges to 0
(d)	Converges to 1	(n)	Does not converge
(e)	Does not converge	(o)	Converges to 0
(f)	Converges to 1	(p)	Converges to 2
(g)	Does not converge	(q)	Converges to 0
(h)	Does not converge	(r)	Converges to 1
(i)	Converges to 0	(s)	Converges to 4/3
(j)	Converges to 7/2	(t)	Converges to 0

7.4 Exercise 7.4: Rational and irrational terms can have different kinds of limits

What we must show. Give (a) a sequence of irrational numbers converging to a rational number, and (b) a sequence of rational numbers converging to an irrational number.

Plan. For part (a), divide $\sqrt{2}$ by larger positive integers. For part (b), use the lower decimal truncations of $\sqrt{2}$. We verify both the nature of every term and convergence.

1. Recall why $\sqrt{2}$ is irrational

The irrationality of $\sqrt{2}$ was established in Section 2. Here is the short integer argument as well. If $\sqrt{2} = p/q$ in lowest terms, with integers p, q and $q > 0$, then $p^2 = 2q^2$, so p^2 is even. The square of an odd integer $2j + 1$ is $4j(j + 1) + 1$, which is odd. Thus p must be even; write $p = 2j$. Substitution gives

$$4j^2 = 2q^2, \qquad q^2 = 2j^2.$$

Now q^2 is even, so the same odd-square argument makes q even. The numerator and denominator would have the common factor 2, contrary to lowest terms. Hence $\sqrt{2} \notin \mathbb{Q}$.

2. Part (a): Irrational terms with rational limit

Define, for every $n \in \mathbb{N}$,

$$x_n = \frac{\sqrt{2}}{n}.$$

Since $n \geq 1$, division is legitimate. If any term x_n were rational, multiplication by the rational integer n would make $nx_n = \sqrt{2}$ rational. This contradicts Step 1. Thus every term x_n is irrational.

To prove convergence, let $\varepsilon > 0$ and choose an integer $N > \sqrt{2}/\varepsilon$ by the Archimedean property. For $n > N$,

$$|x_n - 0| = \frac{\sqrt{2}}{n} < \varepsilon.$$

Therefore $x_n \to 0$. The limit is rational because $0 = 0/1$.

3. Part (b): Define each decimal truncation using a finite maximum

For a real $t \geq 0$, the notation $\lfloor t \rfloor$ denotes the integer m satisfying

$$m \leq t < m + 1.$$

To justify existence, choose a positive integer $M > t$ using the Archimedean property. The finite set

$$F = \{j \in \mathbb{Z} : 0 \le j \le M \text{ and } j \le t\}$$

is nonempty because it contains 0. Let $m = \max F$, which exists for a nonempty finite set of integers. Then $m \le t$. If $m + 1 \le t$, the inequalities $0 \le m + 1 \le t < M$ would give $m + 1 \in F$, contradicting the maximality of m. Hence $t < m + 1$, as required.

This integer is unique. If two integers $m < j$ satisfied their respective inequalities, then $m + 1 \le j \le t$ would contradict $t < m + 1$. Interchanging them handles the other possible order. Thus the floor notation is well-defined here.

Apply this construction to $t = 10^n\sqrt{2}$ and put

$$m_n = \left\lfloor 10^n\sqrt{2} \right\rfloor, \qquad r_n = \frac{m_n}{10^n} \quad (n \in \mathbb{N}).$$

Every m_n is an integer and 10^n is a positive integer. Therefore every r_n is rational by the definition of $\mathbb{Q}$. The defining floor inequalities give

$$m_n \le 10^n\sqrt{2} < m_n + 1.$$

Dividing by $10^n > 0$ preserves order and yields

$$r_n \le \sqrt{2} < r_n + \frac{1}{10^n}.$$

After subtracting r_n, this becomes the precise truncation error estimate

$$0 \le \sqrt{2} - r_n < \frac{1}{10^n}.$$

In fact the left inequality is strict, because a rational r_n cannot equal the irrational number $\sqrt{2}$.

4. Prove convergence of the rational truncations

We may compare the error with $1/n$. The inequality $10^n \ge n$ holds for all positive integers n: the base case is $10 \ge 1$; if it holds for n, then

$$10^{n+1} = 10 \cdot 10^n \ge 10n \ge n + 1,$$

because $10n - (n + 1) = 9n - 1 \ge 8 > 0$ for $n \ge 1$. Induction proves the inequality. Taking reciprocals of positive quantities gives $1/10^n \le 1/n$.

Let $\varepsilon > 0$ and choose an integer $N > 1/\varepsilon$. For $n > N$, we obtain

$$\left|r_n - \sqrt{2}\right| = \sqrt{2} - r_n < \frac{1}{10^n} \le \frac{1}{n} < \varepsilon.$$

Thus $r_n \to \sqrt{2}$, and its limit is irrational by Step 1.

Conclusion. For (a), take $x_n = \sqrt{2}/n \to 0$, with every term irrational and the limit rational. For (b), take $r_n = \lfloor 10^n\sqrt{2} \rfloor / 10^n \to \sqrt{2}$, with every term rational and the limit irrational.

7.5 Exercise 7.5: Three differences of large square roots

What we must show. Find the limits of

$$\text{(a) } \sqrt{n^2+1}-n, \qquad \text{(b) } \sqrt{n^2+n}-n, \qquad \text{(c) } \sqrt{4n^2+n}-2n.$$

Plan. Multiply each difference by its conjugate divided by that same conjugate. The difference-of-squares identity removes the subtraction in the numerator. We then bound the error from the proposed limit directly, so no later limit laws are needed.

1. Part (a): Expand the product with the conjugate

Let $a_n = \sqrt{n^2+1}-n$. For $n \geq 1$, the square root is nonnegative and $n > 0$, so the conjugate $\sqrt{n^2+1}+n$ is positive. Dividing by it is valid. Using $(u-v)(u+v) = u^2 - v^2$ gives

$$\begin{aligned} a_n &= \frac{(\sqrt{n^2+1}-n)(\sqrt{n^2+1}+n)}{\sqrt{n^2+1}+n} \\ &= \frac{(\sqrt{n^2+1})^2 - n^2}{\sqrt{n^2+1}+n} \\ &= \frac{n^2+1-n^2}{\sqrt{n^2+1}+n} \\ &= \frac{1}{\sqrt{n^2+1}+n}. \end{aligned}$$

Since $n^2+1 > n^2$ and $n \geq 0$, the nonnegative square root $\sqrt{n^2+1}$ is greater than n. Indeed, the contrary inequality would give $n^2+1 \leq n^2$ upon squaring nonnegative quantities. Therefore

$$0 < a_n = \frac{1}{\sqrt{n^2+1}+n} \leq \frac{1}{2n}.$$

Given $\varepsilon > 0$, choose an integer $N > 1/(2\varepsilon)$. For $n > N$, this estimate gives

$$|a_n - 0| = a_n \leq \frac{1}{2n} < \varepsilon.$$

Thus the limit in part (a) is 0.

2. Part (b): Rationalize and divide by the positive index

Write $b_n = \sqrt{n^2+n}-n$. Its conjugate is positive for $n \geq 1$, and the same multiplication gives

$$\begin{aligned} b_n &= \frac{(\sqrt{n^2+n}-n)(\sqrt{n^2+n}+n)}{\sqrt{n^2+n}+n} \\ &= \frac{(n^2+n)-n^2}{\sqrt{n^2+n}+n} \\ &= \frac{n}{\sqrt{n^2+n}+n}. \end{aligned}$$

The radicand factors as $n^2+n = n^2(1+1/n)$. Because $n > 0$,

$$\sqrt{n^2+n} = n\sqrt{1+\frac{1}{n}}.$$

To check this identity, the right-hand side is nonnegative, and its square is $n^2(1+1/n) = n^2+n$; uniqueness of the nonnegative square root then gives the equality. Define $q_n = \sqrt{1+1/n}$. Substitution and cancellation of $n \neq 0$ yield

$$b_n = \frac{n}{nq_n + n} = \frac{n}{n(q_n+1)} = \frac{1}{q_n+1}.$$

Here $q_n > 1$, since $q_n \geq 0$ and $q_n^2 = 1 + 1/n > 1$; a nonnegative $q_n \leq 1$ would have square at most 1. Hence $q_n + 1 > 2$, so $0 < b_n < 1/2$.

3. Part (b): Compute the exact error from $1/2$

All the denominators below are positive. Since $b_n < 1/2$,

$$\begin{aligned}
\left| b_n - \frac{1}{2} \right| &= \frac{1}{2} - \frac{1}{q_n+1} \\
&= \frac{q_n + 1 - 2}{2(q_n+1)} \\
&= \frac{q_n - 1}{2(q_n+1)} \\
&= \frac{(q_n-1)(q_n+1)}{2(q_n+1)^2} \\
&= \frac{q_n^2 - 1}{2(q_n+1)^2} \\
&= \frac{(1+1/n) - 1}{2(q_n+1)^2} \\
&= \frac{1}{2n(q_n+1)^2} \leq \frac{1}{8n}.
\end{aligned}$$

The last inequality uses $(q_n+1)^2 \geq 2^2 = 4$ and $n > 0$. For any $\varepsilon > 0$, choose an integer $N > 1/(8\varepsilon)$. Then $n > N$ implies $|b_n - 1/2| \leq 1/(8n) < \varepsilon$. Therefore the limit in part (b) is $1/2$.

4. Part (c): Rationalize and factor the square root

Put $c_n = \sqrt{4n^2+n} - 2n$. Since $n \geq 1$, its conjugate $\sqrt{4n^2+n} + 2n$ is positive. Consequently

$$\begin{aligned}
c_n &= \frac{(\sqrt{4n^2+n} - 2n)(\sqrt{4n^2+n} + 2n)}{\sqrt{4n^2+n} + 2n} \\
&= \frac{(4n^2+n) - (2n)^2}{\sqrt{4n^2+n} + 2n} \\
&= \frac{4n^2 + n - 4n^2}{\sqrt{4n^2+n} + 2n} \\
&= \frac{n}{\sqrt{4n^2+n} + 2n}.
\end{aligned}$$

Now $4n^2 + n = n^2(4 + 1/n)$. The positive-index square-root identity gives

$$\sqrt{4n^2+n} = n\sqrt{4 + \frac{1}{n}},$$

because both sides are nonnegative and the square of the right-hand side is $n^2(4 + 1/n) = 4n^2 + n$. Let $p_n = \sqrt{4 + 1/n}$. Canceling the nonzero factor n gives

$$c_n = \frac{n}{np_n + 2n} = \frac{n}{n(p_n + 2)} = \frac{1}{p_n + 2}.$$

Since $p_n \geq 0$ and $p_n^2 = 4 + 1/n > 4$, we have $p_n > 2$. Thus $p_n + 2 > 4$ and $0 < c_n < 1/4$.

5. Part (c): Compute the exact error from $1/4$

Using $c_n < 1/4$ and then multiplying numerator and denominator by the positive number $p_n + 2$, we obtain

$$\begin{aligned}
\left| c_n - \frac{1}{4} \right| &= \frac{1}{4} - \frac{1}{p_n + 2} \\
&= \frac{p_n + 2 - 4}{4(p_n + 2)} \\
&= \frac{p_n - 2}{4(p_n + 2)} \\
&= \frac{(p_n - 2)(p_n + 2)}{4(p_n + 2)^2} \\
&= \frac{p_n^2 - 4}{4(p_n + 2)^2} \\
&= \frac{(4 + 1/n) - 4}{4(p_n + 2)^2} \\
&= \frac{1}{4n(p_n + 2)^2} \leq \frac{1}{64n}.
\end{aligned}$$

Here $(p_n + 2)^2 \geq 4^2 = 16$, so the denominator is at least $4n \cdot 16 = 64n$. For any $\varepsilon > 0$, choose an integer $N > 1/(64\varepsilon)$. Then $n > N$ gives $|c_n - 1/4| \leq 1/(64n) < \varepsilon$. This proves that the limit in part (c) is $1/4$.

Conclusion. The limits are (a) 0, (b) 1/2, and (c) 1/4. The displayed conjugate identities and error estimates justify all three values directly.

8 A discussion about proofs

Every limit proof in this section starts with an arbitrary $\varepsilon > 0$, chooses an appropriate threshold, and verifies the required inequality for every integer index beyond that threshold. An exploratory rearrangement helps discover N; the formal argument then runs forward from $n > N$. Divergence proofs instead show that the definition cannot hold for any proposed real limit. No limit law from Section 9 is assumed.

A proof template for the first exercises

Definition 7.1 says that $x_n \to L$ means

for every real $\varepsilon > 0$ there exists a number N such that every integer $n > N$ satisfies $|x_n - L| < \varepsilon$.

The order of these choices matters: the reader supplies an arbitrary positive ε; we choose N in terms of that ε; then we verify the estimate for *every* sequence index $n > N$. Our choice of N cannot depend on which later index is tested.

Ross permits a real threshold N. If an integer threshold is preferred, the Archimedean property supplies a natural number larger than our chosen N, and the same proof applies to all indices above that larger threshold. The first finitely many terms need not obey the error estimate: the definition requires it only after the threshold.

All the limit assertions below are proved from this definition. Algebraic simplifications, sign rules, and absolute-value identities are used as rules about individual real numbers, without invoking the limit theorems of Section 9.

8.1 Exercise 8.1: Four limits proved directly from the definition

What we must show. Prove each of the four stated limits with a complete ε–N argument.

Part (a): the alternating reciprocal sequence.

We must prove

$$\lim_{n\to\infty} \frac{(-1)^n}{n} = 0.$$

For an integer n, $(-1)^n$ is either 1 or -1, according to the parity of n. Its absolute value is therefore always 1. Since $n \in \mathbb{N}$ implies $n > 0$, the exact error is

$$\begin{aligned}\left|\frac{(-1)^n}{n} - 0\right| &= \frac{|(-1)^n|}{n}, \\ &= \frac{1}{n}.\end{aligned}$$

We need this error to be strictly less than ε.

Let $\varepsilon > 0$ be arbitrary and choose

$$N = \max\left\{1, \frac{1}{\varepsilon}\right\}.$$

Take any integer $n > N$. Then $n > 1/\varepsilon$. Multiplication by the positive number ε gives $\varepsilon n > 1$; division by the positive number n gives $\varepsilon > 1/n$. Consequently

$$\left|\frac{(-1)^n}{n} - 0\right| = \frac{1}{n} < \varepsilon.$$

The estimate holds for every integer $n > N$, as the definition requires.

Conclusion. For every $\varepsilon > 0$ the displayed threshold works, so $\boxed{(-1)^n/n \longrightarrow 0}$. Alternation does not interfere with the estimate because both possible signs have absolute value 1.

Part (b), Step 1: compute the error and the needed threshold.

We must prove $1/n^{1/3} \to 0$, where $n^{1/3}$ is the positive real cube root of n. Since $n^{1/3} > 0$,

$$\left|\frac{1}{n^{1/3}} - 0\right| = \frac{1}{n^{1/3}}.$$

To make this less than ε, it suffices to have $n^{1/3} > 1/\varepsilon$, or $n > 1/\varepsilon^3$. These implications concern ordinary positive real numbers. For example, strict comparison of their cubes follows from

$$u^3 - v^3 = (u - v)(u^2 + uv + v^2) \qquad (u, v > 0).$$

If $u > v > 0$, both factors on the right are positive, so $u^3 > v^3$. Conversely, if $u^3 > v^3$ but $u \le v$, cubing the nonnegative numbers would give $u^3 \le v^3$, a contradiction. Thus comparing positive cubes is equivalent to comparing their positive cube roots.

Part (b), Step 2: give the formal proof.

Let $\varepsilon > 0$ be arbitrary and choose

$$N = \max\left\{1, \frac{1}{\varepsilon^3}\right\}.$$

For an arbitrary integer $n > N$, we have

$$n > \frac{1}{\varepsilon^3} = \left(\frac{1}{\varepsilon}\right)^3.$$

Taking positive cube roots gives $n^{1/3} > 1/\varepsilon$. Multiplying this inequality by $\varepsilon > 0$ and then dividing by $n^{1/3} > 0$ gives

$$\varepsilon n^{1/3} > 1 \quad \Longrightarrow \quad \frac{1}{n^{1/3}} < \varepsilon.$$

Hence the exact error satisfies $\left|1/n^{1/3} - 0\right| < \varepsilon$ for every integer $n > N$.

Conclusion. Definition 7.1 therefore gives $\boxed{1/n^{1/3} \longrightarrow 0}$. The threshold depends on ε^{-3} because the denominator is a cube root.

Part (c), Step 1: simplify the exact difference from the proposed limit.

The sequence in the printed exercise is

$$x_n = \frac{2n - 1}{3n + 2}, \qquad \text{and the proposed limit is } \frac{2}{3}.$$

For every $n \in \mathbb{N}$, $3n+2 > 0$, so the denominator is nonzero. Putting the two fractions over the common denominator $3(3n+2)$ gives

$$\begin{aligned} x_n - \frac{2}{3} &= \frac{3(2n-1) - 2(3n+2)}{3(3n+2)}, \\ &= \frac{6n - 3 - 6n - 4}{9n+6}, \\ &= \frac{-7}{9n+6}. \end{aligned}$$

Because $9n+6 > 0$, taking absolute values yields

$$\left| x_n - \frac{2}{3} \right| = \frac{7}{9n+6}.$$

Also $9n+6 \geq 9n > 0$, so increasing the positive denominator decreases the quotient:

$$\frac{7}{9n+6} \leq \frac{7}{9n}.$$

Part (c), Step 2: choose a threshold and check all later indices.

Let $\varepsilon > 0$ and set

$$N = \max\left\{1, \frac{7}{9\varepsilon}\right\}.$$

If an integer $n > N$, then $n > 7/(9\varepsilon)$. Multiplication by $9\varepsilon > 0$ gives $9\varepsilon n > 7$; division by $9n > 0$ gives $\varepsilon > 7/(9n)$. Thus

$$\left| x_n - \frac{2}{3} \right| = \frac{7}{9n+6} \leq \frac{7}{9n} < \varepsilon.$$

Conclusion. The estimate holds for every integer $n > N$, proving

$$\boxed{\lim_{n\to\infty} \frac{2n-1}{3n+2} = \frac{2}{3}}.$$

Part (d), Step 1: isolate a positive tail of the denominator.

We must prove

$$x_n = \frac{n+6}{n^2-6} \longrightarrow 0.$$

The denominator is nonzero for every positive integer n: if $n \leq 2$, then $n^2 \leq 4 < 6$; if $n \geq 3$, then $n^2 \geq 9 > 6$. For the limit estimate, we will use only indices $n \geq 4$. For such n, we have $n^2/2 \geq 8 \geq 6$, and therefore

$$n^2 - 6 \geq n^2 - \frac{n^2}{2} = \frac{n^2}{2} > 0.$$

This is a lower bound on the denominator, with its sign recorded.

Part (d), Step 2: bound the numerator and then the entire error.

For $n \geq 1$, the inequality $6 \leq 6n$ gives

$$n + 6 \leq n + 6n = 7n.$$

If $n \geq 4$, both the numerator $n+6$ and denominator n^2-6 are positive. Thus

$$\begin{aligned} |x_n - 0| &= \frac{n+6}{n^2-6}, \\ &\leq \frac{7n}{n^2-6} && \text{by the upper bound on the numerator,} \\ &\leq \frac{7n}{n^2/2} && \text{by the positive lower bound on the denominator,} \\ &= \frac{14n}{n^2} = \frac{14}{n}. \end{aligned}$$

The cancellation in the last line is valid because $n > 0$.

Part (d), Step 3: give the formal threshold.

Let $\varepsilon > 0$ and choose

$$N = \max\left\{4, \frac{14}{\varepsilon}\right\}.$$

An arbitrary integer $n > N$ satisfies $n > 4$, so every preceding tail estimate applies. It also satisfies $n > 14/\varepsilon$; multiplying by $\varepsilon > 0$ and dividing by $n > 0$ gives $14/n < \varepsilon$. Hence

$$\left|\frac{n+6}{n^2-6} - 0\right| \leq \frac{14}{n} < \varepsilon \qquad \text{for every integer } n > N.$$

Conclusion. The threshold supplies the estimate required by Definition 7.1, so

$$\boxed{\frac{n+6}{n^2-6} \longrightarrow 0}.$$

We did not require the first few terms to have a positive denominator.

8.2 Exercise 8.2: Determine five limits and justify each one formally

What we must show. Find the limits of the five stated sequences and prove every answer directly from Definition 7.1. The candidate for a quotient of linear expressions can be suggested by its leading coefficients, but the error calculations below are what establish the limits.

Part (a), Step 1: estimate the difference from zero.

For $a_n = n/(n^2+1)$, the numerator and denominator are positive for every $n \in \mathbb{N}$. Since $n^2 + 1 \geq n^2 > 0$,

$$\begin{aligned} |a_n - 0| &= \frac{n}{n^2+1}, \\ &\leq \frac{n}{n^2}, \\ &= \frac{1}{n}. \end{aligned}$$

We will prove that 0 is the limit.

Part (a), Step 2: supply the quantified proof.

Let $\varepsilon > 0$ and set

$$N = \max\left\{1, \frac{1}{\varepsilon}\right\}.$$

For every integer $n > N$, we have $n > 1/\varepsilon$. Because ε and n are positive, this implies $1/n < \varepsilon$. The error bound from Step 1 then gives

$$|a_n - 0| \leq \frac{1}{n} < \varepsilon.$$

Conclusion. The limit is $\boxed{\lim_{n\to\infty} \frac{n}{n^2+1} = 0}$, as verified for every positive error tolerance.

Part (b), Step 1: compute the exact difference from $7/3$.

Let $b_n = (7n-19)/(3n+7)$. Since $n \geq 1$, its denominator satisfies $3n+7 \geq 10 > 0$. Compute

$$\begin{aligned} b_n - \frac{7}{3} &= \frac{3(7n-19) - 7(3n+7)}{3(3n+7)}, \\ &= \frac{21n - 57 - 21n - 49}{9n+21}, \\ &= \frac{-106}{9n+21}. \end{aligned}$$

Thus

$$\left|b_n - \frac{7}{3}\right| = \frac{106}{9n+21} \leq \frac{106}{9n},$$

because $9n + 21 \geq 9n > 0$.

Part (b), Step 2: choose and verify a threshold.

Given any $\varepsilon > 0$, choose

$$N = \max\left\{1, \frac{106}{9\varepsilon}\right\}.$$

Take any integer $n > N$. Then $n > 106/(9\varepsilon)$, so $9\varepsilon n > 106$ and hence $106/(9n) < \varepsilon$. It follows that

$$\left|b_n - \frac{7}{3}\right| = \frac{106}{9n+21} \leq \frac{106}{9n} < \varepsilon.$$

Conclusion. Since this works for every $\varepsilon > 0$,

$$\boxed{\lim_{n\to\infty} \frac{7n-19}{3n+7} = \frac{7}{3}}.$$

Part (c), Step 1: compute the exact difference from $4/7$.

Let $c_n = (4n+3)/(7n-5)$. Its denominator is positive for all natural indices, since $7n-5 \geq 7-5 = 2 > 0$. Cross-multiplication gives

$$\begin{aligned} c_n - \frac{4}{7} &= \frac{7(4n+3) - 4(7n-5)}{7(7n-5)}, \\ &= \frac{28n + 21 - 28n + 20}{49n-35}, \end{aligned}$$

$$= \frac{41}{49n - 35}.$$

The error is already positive, so its absolute value has the same expression.

Part (c), Step 2: bound the denominator below by a multiple of n.

For $n \geq 1$,

$$(7n - 5) - 2n = 5n - 5 = 5(n - 1) \geq 0.$$

Consequently $7n - 5 \geq 2n > 0$, and multiplying by $7 > 0$ gives $49n - 35 \geq 14n > 0$. Hence

$$\left| c_n - \frac{4}{7} \right| = \frac{41}{49n - 35} \leq \frac{41}{14n}.$$

Part (c), Step 3: prove the definition with an explicit threshold.

Let $\varepsilon > 0$ and put

$$N = \max\left\{1, \frac{41}{14\varepsilon}\right\}.$$

For an arbitrary integer $n > N$, the inequality $n > 41/(14\varepsilon)$ gives $14\varepsilon n > 41$ and thus $41/(14n) < \varepsilon$. Therefore

$$\left| c_n - \frac{4}{7} \right| \leq \frac{41}{14n} < \varepsilon.$$

The estimates used here hold for every such index, not merely for a selected collection of large indices.

Conclusion. The limit is

$$\boxed{\lim_{n\to\infty} \frac{4n + 3}{7n - 5} = \frac{4}{7}}.$$

Part (d), Step 1: simplify the difference from $2/5$.

For $d_n = (2n + 4)/(5n + 2)$, the denominator satisfies $5n + 2 \geq 7 > 0$. The exact calculation is

$$\begin{aligned} d_n - \frac{2}{5} &= \frac{5(2n + 4) - 2(5n + 2)}{5(5n + 2)}, \\ &= \frac{10n + 20 - 10n - 4}{25n + 10}, \\ &= \frac{16}{25n + 10}. \end{aligned}$$

Thus

$$\left| d_n - \frac{2}{5} \right| = \frac{16}{25n + 10} \leq \frac{16}{25n},$$

where the comparison uses $25n + 10 \geq 25n > 0$.

Part (d), Step 2: choose a threshold and verify the required error.

For an arbitrary $\varepsilon > 0$, set

$$N = \max\left\{1, \frac{16}{25\varepsilon}\right\}.$$

If an integer $n > N$, then $n > 16/(25\varepsilon)$. Multiplication by $25\varepsilon > 0$ gives $25\varepsilon n > 16$; division by $25n > 0$ gives $16/(25n) < \varepsilon$. Hence

$$\left| d_n - \frac{2}{5} \right| \leq \frac{16}{25n} < \varepsilon.$$

Conclusion. Definition 7.1 gives

$$\boxed{\lim_{n\to\infty} \frac{2n+4}{5n+2} = \frac{2}{5}}.$$

Part (e), Step 1: bound the error using the elementary sine bound.

The sequence is $s_n = (\sin n)/n$. For every real x, the elementary trigonometric inequality $-1 \leq \sin x \leq 1$ gives $|\sin x| \leq 1$. Applying it at $x = n$ and using $n > 0$ yields

$$\begin{aligned} |s_n - 0| &= \left|\frac{\sin n}{n}\right|, \\ &= \frac{|\sin n|}{n}, \\ &\leq \frac{1}{n}. \end{aligned}$$

The bound holds for every natural n regardless of the sign of $\sin n$.

Part (e), Step 2: carry out the formal proof.

Let $\varepsilon > 0$ and choose

$$N = \max\left\{1, \frac{1}{\varepsilon}\right\}.$$

For every integer $n > N$, multiplication of $n > 1/\varepsilon$ by $\varepsilon > 0$ and division by $n > 0$ give $1/n < \varepsilon$. Consequently

$$|s_n - 0| \leq \frac{1}{n} < \varepsilon.$$

Conclusion. Thus $\boxed{(\sin n)/n \longrightarrow 0}$. The proof uses a uniform bound on sine, not a claim that the sequence $(\sin n)$ has a limit.

8.3 Exercise 8.3: Taking square roots when the limit is zero

What we must show. Suppose $s_n \geq 0$ for every $n \in \mathbb{N}$ and $s_n \to 0$. Prove that $\sqrt{s_n} \to 0$ directly from Definition 7.1.

Plan. To make the nonnegative square root smaller than ε, it suffices to make the original number smaller than ε^2. The input error tolerance for the given convergence will therefore be $\delta = \varepsilon^2$.

Step 1: choose the input tolerance and use the hypothesis.

Let $\varepsilon > 0$ be arbitrary. Then

$$\delta = \varepsilon^2 > 0.$$

The assertion $s_n \to 0$ means that the definition applies to every positive tolerance, including this particular δ. Hence there is an N such that every integer $n > N$ satisfies

$$|s_n - 0| < \delta = \varepsilon^2.$$

Because $s_n \geq 0$, its absolute value is itself, and the last estimate is exactly

$$0 \leq s_n < \varepsilon^2.$$

Step 2: justify the strict comparison of square roots.

Fix any integer $n > N$ and put $u = \sqrt{s_n}$. The nonnegative square root exists because $s_n \geq 0$, and it satisfies $u \geq 0$ and $u^2 = s_n$. If $u \geq \varepsilon$, then

$$u^2 - \varepsilon^2 = (u - \varepsilon)(u + \varepsilon) \geq 0,$$

since $u - \varepsilon \geq 0$ and $u + \varepsilon > 0$. This would imply $s_n = u^2 \geq \varepsilon^2$, contradicting the estimate from Step 1. Thus $u < \varepsilon$.

Step 3: verify the target absolute-value estimate.

Since $u \geq 0$,

$$|\sqrt{s_n} - 0| = \sqrt{s_n} = u < \varepsilon \qquad \text{for every integer } n > N.$$

The threshold N was supplied by the given convergence for the tolerance ε^2; it does not depend on the later index n.

Conclusion. For every target tolerance $\varepsilon > 0$ we obtained the required threshold, proving $\boxed{\sqrt{s_n} \longrightarrow 0}$. No division by $\sqrt{s_n}$ was used, so the argument includes terms equal to zero.

8.4 Exercise 8.4: A sequence tending to zero times a bounded sequence

What we must show. Assume (t_n) is bounded, so there is a real number M with $|t_n| \leq M$ for every $n \in \mathbb{N}$, and assume $s_n \to 0$. Prove $s_n t_n \to 0$ directly from the definition.

Step 1: choose a positive bound that can be used as a divisor.

The given bound satisfies $M \geq |t_1| \geq 0$, so $M \geq 0$. It need not be strictly positive. Define instead

$$B = \max\{1, M\}.$$

Then $B \geq 1 > 0$, and for every n,

$$|t_n| \leq M \leq B.$$

The constant B is fixed for the whole sequence; it does not depend on the index n or on our later choice of error tolerance.

Step 2: ask the sequence tending to zero for a sufficiently small error.

Let $\varepsilon > 0$ be arbitrary and put

$$\delta = \frac{\varepsilon}{B} > 0.$$

Since $s_n \to 0$, there is an N such that, for every integer $n > N$,

$$|s_n - 0| < \delta = \frac{\varepsilon}{B}.$$

Step 3: verify the product estimate term by term.

Take an arbitrary integer $n > N$. Multiplication of $|t_n| \leq B$ by the nonnegative number $|s_n|$ gives $|s_n|\,|t_n| \leq B\,|s_n|$. Multiplication of $|s_n| < \varepsilon/B$ by the strictly positive number B preserves the strict inequality. Therefore

$$|s_n t_n - 0| = |s_n|\,|t_n|\,,$$

$$\begin{aligned} &\leq B\,|s_n|\,, \\ &< B\frac{\varepsilon}{B}, \\ &= \varepsilon. \end{aligned}$$

This bound holds for every integer $n > N$.

Step 4: check the possible zero-bound case.

If the original M was zero, then $0 \leq |t_n| \leq 0$ forces $t_n = 0$ for every n. The products are then all zero. Our proof already covers this case because it used $B = \max\{1, 0\} = 1$ rather than dividing by $M = 0$.

Conclusion. The error estimate establishes $\boxed{s_n t_n \longrightarrow 0}$ by Definition 7.1. The sequence (t_n) need only be bounded; its own convergence was not assumed or used.

8.5 Exercise 8.5: The squeeze lemma and an absolute-value comparison

What we must show. Prove (a) that $a_n \leq s_n \leq b_n$ for all n, together with $a_n \to s$ and $b_n \to s$, implies $s_n \to s$; and (b) that $|s_n| \leq t_n$ for all n, together with $t_n \to 0$, implies $s_n \to 0$.

Part (a), Step 1: obtain two thresholds for the same tolerance.

Let $\varepsilon > 0$ be arbitrary. Because $a_n \to s$, there is an N_1 such that

$$n > N_1 \quad \Longrightarrow \quad |a_n - s| < \varepsilon.$$

Because $b_n \to s$, there is an N_2 such that

$$n > N_2 \quad \Longrightarrow \quad |b_n - s| < \varepsilon.$$

Each implication is required for every integer index exceeding the corresponding threshold.

Part (a), Step 2: make both estimates valid at once.

Choose

$$N = \max\{N_1, N_2\}.$$

For an arbitrary integer $n > N$, we have both $n > N_1$ and $n > N_2$. Thus both error estimates apply at this same index. The elementary absolute-value equivalence from Exercise 3.7(a) gives

$$-\varepsilon < a_n - s < \varepsilon, \qquad -\varepsilon < b_n - s < \varepsilon.$$

Part (a), Step 3: insert the middle term between these errors.

The order assumption holds for every n, so in particular for our chosen $n > N$ we have $a_n \leq s_n \leq b_n$. Add $-s$ throughout:

$$a_n - s \leq s_n - s \leq b_n - s.$$

Combine the strict lower bound from a_n with the strict upper bound from b_n:

$$-\varepsilon < a_n - s \leq s_n - s \leq b_n - s < \varepsilon.$$

The middle inequalities are permitted to be weak. The outer strict inequalities still give $-\varepsilon < s_n - s < \varepsilon$ by transitivity. Applying Exercise 3.7(a) once more gives $|s_n - s| < \varepsilon$.

Part (a), Step 4: record the quantifiers of the conclusion.

The number $\varepsilon > 0$ was arbitrary. We chose a single N for that tolerance and verified $|s_n - s| < \varepsilon$ for every integer $n > N$. This is exactly the definition of convergence to s.

Conclusion. The squeeze lemma is

$$\boxed{a_n \le s_n \le b_n, \quad a_n \to s, \quad b_n \to s \quad \Longrightarrow \quad s_n \to s}.$$

The two enclosing sequences may approach s from either side; only their common limit and the displayed order comparisons were used.

Part (b), Step 1: extract the sign information contained in the hypothesis.

For each n, $|s_n| \ge 0$. The assumed inequality therefore gives

$$0 \le |s_n| \le t_n.$$

In particular, $t_n \ge 0$ for every index. This is a consequence of the hypothesis, not an extra assumption.

Part (b), Step 2: apply the definition to the given sequence.

Let $\varepsilon > 0$ be arbitrary. Since $t_n \to 0$, there is an N such that every integer $n > N$ satisfies

$$|t_n - 0| < \varepsilon.$$

As $t_n \ge 0$, this is the same as $t_n < \varepsilon$.

Part (b), Step 3: finish the required error estimate.

For every integer $n > N$, combine Step 1 with Step 2:

$$0 \le |s_n - 0| = |s_n| \le t_n < \varepsilon.$$

Thus $|s_n - 0| < \varepsilon$ for every such n.

Conclusion. For every $\varepsilon > 0$ the preceding threshold works, so $\boxed{s_n \longrightarrow 0}$. This direct proof records why the sign of t_n causes no difficulty.

8.6 Exercise 8.6: Absolute values at zero and an alternating counterexample

What we must show. Prove (a) $s_n \to 0$ if and only if $|s_n| \to 0$, and (b) the sequence $s_n = (-1)^n$ has convergent absolute values but is itself divergent.

Throughout these exercises, convergence is used in the precise form of Definition 7.1: $x_n \to L$ means that for every $\varepsilon > 0$ there is a real number N such that

$$n \in \mathbb{N}, \quad n > N \quad \Longrightarrow \quad |x_n - L| < \varepsilon.$$

A threshold N need not be an integer. Whenever an index above a specified threshold is needed, the Archimedean property supplies a positive integer above it.

Part (a), Step 1: compare the two errors exactly.

For every $n \in \mathbb{N}$, the number $|s_n|$ is nonnegative. Therefore

$$||s_n| - 0| = ||s_n|| = |s_n| = |s_n - 0|\,.$$

The errors in the two proposed zero limits are the same real number, not merely bounded by each other.

Part (a), Step 2: prove the forward implication.

Assume $s_n \to 0$, and let $\varepsilon > 0$ be arbitrary. The definition supplies N such that

$$n > N \quad \Longrightarrow \quad |s_n - 0| < \varepsilon.$$

For each such index, the identity in Step 1 gives

$$||s_n| - 0| = |s_n - 0| < \varepsilon.$$

Thus the same threshold N verifies convergence of $|s_n|$ to 0.

Part (a), Step 3: prove the reverse implication.

Assume $|s_n| \to 0$, and let $\varepsilon > 0$ be arbitrary. Choose N from this convergence so that

$$n > N \quad \Longrightarrow \quad ||s_n| - 0| < \varepsilon.$$

Using Step 1 again yields

$$|s_n - 0| = ||s_n| - 0| < \varepsilon \qquad (n > N).$$

This is the definition of $s_n \to 0$.

Conclusion. Both implications have been proved with the same error and the same threshold:

$$\boxed{s_n \to 0 \quad \Longleftrightarrow \quad |s_n| \to 0.}$$

Part (b), Step 1: prove convergence of the absolute values.

If $s_n = (-1)^n$, then s_n equals either 1 or -1. Hence $|s_n| = 1$ for every n. For any $\varepsilon > 0$, choose $N = 0$. Every positive integer $n > N$ satisfies

$$||s_n| - 1| = |1 - 1| = 0 < \varepsilon.$$

Therefore $|s_n| \to 1$.

Part (b), Step 2: rule out every possible limit of the original sequence.

Suppose, for a contradiction, that $s_n \to L$ for some real L. Apply the definition with the fixed positive tolerance $\varepsilon = 1$. There is a real N such that

$$n > N \quad \Longrightarrow \quad |s_n - L| < 1.$$

Choose an integer $k > \max\{N, 1\}$. Then both $2k$ and $2k + 1$ are positive integers larger than N. Their terms are

$$s_{2k} = 1, \qquad s_{2k+1} = -1.$$

The triangle inequality gives

$$\begin{aligned} 2 &= |1 - (-1)| \\ &= |(1 - L) + (L - (-1))| \end{aligned}$$

$$\begin{aligned}
&\leq |1 - L| + |L - (-1)| \\
&= |s_{2k} - L| + |s_{2k+1} - L| \\
&< 1 + 1 = 2.
\end{aligned}$$

The equality of the absolute values in the fourth line uses $|-x| = |x|$. The resulting strict inequality $2 < 2$ is impossible.

Conclusion. The absolute-value sequence converges to 1, but $((-1)^n)$ has no real limit. Thus existence of a limit for $|s_n|$ alone does not guarantee existence of a limit for s_n.

8.7 Exercise 8.7: Three divergent sequences

What we must show. Show that none of the following sequences converges:

$$\cos\left(\frac{n\pi}{3}\right), \qquad (-1)^n n, \qquad \sin\left(\frac{n\pi}{3}\right).$$

Plan. We will use only the definition of convergence and the triangle inequality. If a sequence converged to L, then for any chosen $\varepsilon > 0$, all sufficiently late terms would be within ε of L. Any two such terms would satisfy

$$|s_p - s_q| = |(s_p - L) + (L - s_q)| \leq |s_p - L| + |s_q - L| < 2\varepsilon.$$

For each proposed sequence, we will produce two arbitrarily late terms whose distance contradicts this inequality. This displayed observation has just been proved directly, so no theorem about subsequences or bounded convergent sequences is needed.

Part (a), Step 1: select a tolerance under an assumed limit.

Suppose

$$s_n = \cos\left(\frac{n\pi}{3}\right) \to L.$$

With $\varepsilon = 1$, choose N such that every positive integer $n > N$ satisfies $|s_n - L| < 1$.

Part (a), Step 2: find late terms equal to 1 and -1.

Choose an integer $k > \max\{N, 1\}$, and put

$$p = 6k, \qquad q = 6k + 3.$$

Both indices exceed N. The period 2π of cosine gives

$$s_p = \cos(2k\pi) = 1, \qquad s_q = \cos(2k\pi + \pi) = -1.$$

Their distance is therefore $|s_p - s_q| = 2$. The two-late-terms estimate gives

$$2 = |s_p - s_q| \leq |s_p - L| + |s_q - L| < 1 + 1 = 2,$$

a contradiction.

Conclusion. The sequence $(\cos(n\pi/3))$ does not converge.

Part (b), Step 1: use the definition with a fixed tolerance.

Suppose $s_n = (-1)^n n \to L$. Choose N from the definition with $\varepsilon = 1$.

Part (b), Step 2: compute the distance between late even and odd terms.

Choose an integer $k > \max\{N, 1\}$. Then $p = 2k$ and $q = 2k + 1$ are larger than N and

$$s_p = 2k, \qquad s_q = -(2k + 1).$$

Their difference is

$$s_p - s_q = 2k + (2k + 1) = 4k + 1 > 2.$$

Since this difference is positive, its absolute value is also $4k + 1$. But convergence would give

$$4k + 1 = |s_p - s_q| \leq |s_p - L| + |s_q - L| < 1 + 1 = 2.$$

This contradicts $4k + 1 > 2$.

Conclusion. The sequence $((-1)^n n)$ does not converge. The proof uses two explicit terms in every sufficiently late tail, rather than assuming a boundedness theorem from a later section.

Part (c), Step 1: choose a tolerance suited to the sine values.

Suppose

$$s_n = \sin\left(\frac{n\pi}{3}\right) \to L.$$

Choose the positive tolerance $\varepsilon = \sqrt{3}/2$. The definition gives N such that $n > N$ implies $|s_n - L| < \sqrt{3}/2$.

Part (c), Step 2: use two positions in the six-term period.

Choose an integer $k > \max\{N, 1\}$ and let

$$p = 6k + 1, \qquad q = 6k + 4.$$

Both are larger than N. By the period 2π of sine,

$$s_p = \sin\left(2k\pi + \frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}, \qquad s_q = \sin\left(2k\pi + \frac{4\pi}{3}\right) = -\frac{\sqrt{3}}{2}.$$

In particular, $|s_p - s_q| = \sqrt{3}$. The convergence estimates would imply

$$\sqrt{3} = |s_p - s_q| \leq |s_p - L| + |s_q - L| < \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = \sqrt{3}.$$

This is impossible.

Conclusion. The sequence $(\sin(n\pi/3))$ does not converge. Every sequence in the exercise has now been ruled out by the definition of convergence itself.

8.8 Exercise 8.8: Three radical limits with explicit error bounds

What we must show. Prove the three stated limits:

$$\sqrt{n^2 + 1} - n \longrightarrow 0,$$

$$\sqrt{n^2+n}-n \longrightarrow \frac{1}{2},$$
$$\sqrt{4n^2+n}-2n \longrightarrow \frac{1}{4}.$$

The index n is always a positive integer. In particular, division by n is legitimate, and $\sqrt{n^2} = n$ rather than $-n$.

Plan. Multiplication by a conjugate removes the subtraction of large radicals. We will then bound each *error from its claimed limit* by a positive multiple of $1/n$. The estimates below are ordinary algebraic identities and inequalities, not applications of limit laws.

Part (a), Step 1: rationalize the expression and check its sign.

Let $u_n = \sqrt{n^2+1}-n$. Because $n \geq 1$, both n and $\sqrt{n^2+1}$ are positive. Moreover, $\sqrt{n^2+1} > n$: otherwise squaring a supposed inequality $\sqrt{n^2+1} \leq n$ between nonnegative numbers would give $n^2+1 \leq n^2$, a contradiction. Thus $u_n > 0$. Its conjugate $\sqrt{n^2+1}+n$ is positive, so

$$\begin{aligned} u_n &= \frac{(\sqrt{n^2+1}-n)(\sqrt{n^2+1}+n)}{\sqrt{n^2+1}+n} \\ &= \frac{(n^2+1)-n^2}{\sqrt{n^2+1}+n} \\ &= \frac{1}{\sqrt{n^2+1}+n}. \end{aligned}$$

Part (a), Step 2: obtain an error estimate valid for every index.

The denominator satisfies $\sqrt{n^2+1}+n \geq 2n > 0$. Taking reciprocals of positive quantities reverses the comparison, so

$$|u_n - 0| = u_n \leq \frac{1}{2n}.$$

Part (a), Step 3: choose and verify a threshold.

Let $\varepsilon > 0$ be arbitrary, and choose

$$N = \frac{1}{2\varepsilon}.$$

For every positive integer $n > N$, multiplying by $2\varepsilon > 0$ gives $2\varepsilon n > 1$. Division by $2n > 0$ gives $1/(2n) < \varepsilon$. Hence

$$n > N \quad \Longrightarrow \quad |u_n - 0| \leq \frac{1}{2n} < \varepsilon.$$

This is precisely $u_n \to 0$.

Conclusion. $\boxed{\lim_{n\to\infty}(\sqrt{n^2+1}-n) = 0}$. An explicit error bound is $1/(2n)$.

Part (b), Step 1: rationalize and factor out the positive index.

Let $v_n = \sqrt{n^2+n}-n$. Multiplying by its positive conjugate gives

$$v_n = \frac{(n^2+n)-n^2}{\sqrt{n^2+n}+n} = \frac{n}{\sqrt{n^2+n}+n}.$$

Set

$$r_n = \sqrt{1 + \frac{1}{n}}.$$

The quantity nr_n is positive, and its square is

$$(nr_n)^2 = n^2\left(1 + \frac{1}{n}\right) = n^2 + n.$$

By uniqueness of the nonnegative square root, $\sqrt{n^2+n} = nr_n$. Substitution gives the exact identity

$$v_n = \frac{n}{nr_n + n} = \frac{1}{r_n + 1}.$$

The number r_n is greater than 1, because $r_n^2 = 1 + 1/n > 1$. Thus $r_n + 1 > 2$, and $0 < v_n < 1/2$.

Part (b), Step 2: rationalize the error from $1/2$.

The sign information in Step 1 gives

$$\begin{aligned}\left|v_n - \tfrac{1}{2}\right| &= \frac{1}{2} - \frac{1}{r_n + 1} \\ &= \frac{(r_n + 1) - 2}{2(r_n + 1)} \\ &= \frac{r_n - 1}{2(r_n + 1)}.\end{aligned}$$

A second conjugate calculation yields

$$r_n - 1 = \frac{(r_n - 1)(r_n + 1)}{r_n + 1} = \frac{r_n^2 - 1}{r_n + 1} = \frac{1/n}{r_n + 1}.$$

Therefore

$$\left|v_n - \tfrac{1}{2}\right| = \frac{1}{2n(r_n + 1)^2}.$$

Since $r_n \geq 1$, we have $(r_n + 1)^2 \geq 4$, so

$$\left|v_n - \tfrac{1}{2}\right| \leq \frac{1}{8n}.$$

Part (b), Step 3: verify the definition of convergence.

Let $\varepsilon > 0$ be arbitrary and choose $N = 1/(8\varepsilon)$. For every positive integer $n > N$, multiplication and division by positive quantities give $1/(8n) < \varepsilon$. Consequently

$$\left|v_n - \tfrac{1}{2}\right| \leq \frac{1}{8n} < \varepsilon \qquad (n > N).$$

The definition proves the desired limit.

Conclusion. $\boxed{\lim_{n\to\infty}(\sqrt{n^2+n} - n) = \tfrac{1}{2}}$. An explicit error bound is $1/(8n)$.

Part (c), Step 1: rationalize and identify a simpler denominator.

Let $w_n = \sqrt{4n^2+n} - 2n$. Its conjugate is positive, and multiplication gives

$$w_n = \frac{(4n^2+n) - 4n^2}{\sqrt{4n^2+n} + 2n} = \frac{n}{\sqrt{4n^2+n} + 2n}.$$

Define $R_n = \sqrt{4 + 1/n}$. Since $n > 0$, the quantity nR_n is the nonnegative square root of $4n^2 + n$:

$$(nR_n)^2 = n^2\left(4 + \frac{1}{n}\right) = 4n^2 + n.$$

Hence

$$w_n = \frac{n}{nR_n + 2n} = \frac{1}{R_n + 2}.$$

Because $R_n^2 = 4 + 1/n > 4$ and $R_n \geq 0$, we have $R_n > 2$. Thus $R_n + 2 > 4$ and $0 < w_n < 1/4$.

Part (c), Step 2: compute and bound the error from $1/4$.

Using the sign of the error, put the fractions over one denominator:

$$\begin{aligned}\left|w_n - \tfrac{1}{4}\right| &= \frac{1}{4} - \frac{1}{R_n + 2} \\ &= \frac{(R_n + 2) - 4}{4(R_n + 2)} \\ &= \frac{R_n - 2}{4(R_n + 2)}.\end{aligned}$$

The second rationalization gives

$$R_n - 2 = \frac{R_n^2 - 4}{R_n + 2} = \frac{1/n}{R_n + 2}.$$

It follows that

$$\left|w_n - \tfrac{1}{4}\right| = \frac{1}{4n(R_n + 2)^2} \leq \frac{1}{64n},$$

because $R_n \geq 2$ implies $(R_n + 2)^2 \geq 16$.

Part (c), Step 3: give the final tolerance-dependent threshold.

Let $\varepsilon > 0$ be arbitrary and choose $N = 1/(64\varepsilon)$. Then $n > N$ implies $1/(64n) < \varepsilon$, so

$$\left|w_n - \tfrac{1}{4}\right| \leq \frac{1}{64n} < \varepsilon \qquad (n \in \mathbb{N},\ n > N).$$

This verifies convergence directly.

Conclusion. $\boxed{\lim_{n\to\infty}(\sqrt{4n^2+n} - 2n) = \tfrac{1}{4}}$. All three limits were proved by explicit inequalities, without using algebraic limit theorems from Section 9.

8.9 Exercise 8.9: Eventual bounds pass to a convergent limit

What we must show. Let $s_n \to s$. Prove:

(a) If $s_n \geq a$ for all but finitely many indices, then $s \geq a$.

(b) If $s_n \le b$ for all but finitely many indices, then $s \le b$.

(c) If $s_n \in [a, b]$ for all but finitely many indices, then $s \in [a, b]$.

A preliminary detail: replace finitely many exceptions by one threshold.

If a property fails only on a finite set $E \subseteq \mathbb{N}$ of indices, define $K = 0$ if E is empty and $K = \max E$ otherwise. Every positive integer $n > K$ is outside E, so every such index satisfies the property. This explains each use of an eventual-bound threshold below.

Part (a), Step 1: suppose the limit violates the eventual lower bound.

Choose K so that $n > K$ implies $s_n \ge a$. Suppose, for a contradiction, that $s < a$. Then the gap $a - s$ is positive. Set

$$\varepsilon = \frac{a - s}{2} > 0.$$

Convergence supplies N_0 such that $n > N_0$ implies $|s_n - s| < \varepsilon$.

Part (a), Step 2: combine the two tail conditions.

Choose a positive integer $n > \max\{K, N_0\}$. The condition $n > K$ gives $s_n \ge a$. The condition $n > N_0$ gives

$$s_n - s < \varepsilon, \qquad s_n < s + \varepsilon = \frac{s + a}{2}.$$

But

$$a - \frac{s + a}{2} = \frac{a - s}{2} > 0.$$

Thus $s_n < (s + a)/2 < a$, contradicting $s_n \ge a$.

Conclusion. The inequality $s < a$ is impossible. Therefore $\boxed{s \ge a}$. Equality $s = a$ is permitted; for example, the constant sequence $s_n = a$ has limit a and satisfies every lower bound in the hypothesis.

Part (b), Step 1: suppose the limit violates the eventual upper bound.

Choose K so that $n > K$ implies $s_n \le b$. Suppose $s > b$, and choose the positive tolerance

$$\varepsilon = \frac{s - b}{2}.$$

Convergence gives N_0 such that $n > N_0$ implies $|s_n - s| < \varepsilon$.

Part (b), Step 2: choose one index after both thresholds.

Take a positive integer $n > \max\{K, N_0\}$. Then $s_n \le b$, whereas the convergence inequality gives

$$-\varepsilon < s_n - s, \qquad s_n > s - \varepsilon = \frac{s + b}{2}.$$

The midpoint exceeds b because

$$\frac{s + b}{2} - b = \frac{s - b}{2} > 0.$$

Hence $s_n > b$, contradicting $s_n \le b$.

Conclusion. The inequality $s > b$ is impossible. Therefore $\boxed{s \le b}$. The endpoint $s = b$ is allowed.

Part (c), Step 1: unpack membership of a closed interval.

For $a \le b$,

$$x \in [a, b] \quad \Longleftrightarrow \quad a \le x \le b.$$

If all but finitely many s_n belong to this interval, then all but finitely many satisfy $s_n \ge a$, and all but finitely many satisfy $s_n \le b$. The finite-exception threshold may be chosen from the indices where interval membership fails, so both inequalities hold beyond the same threshold.

Part (c), Step 2: apply both proved statements to the same limit.

Part (a) gives $s \ge a$, and part (b) gives $s \le b$. Together these say $a \le s \le b$, which is exactly $s \in [a, b]$. If $a = b$, the two inequalities force $s = a = b$, so the singleton case is included as well.

Conclusion. For a convergent sequence,

$$\boxed{s_n \in [a, b] \text{ for all but finitely many } n \quad \Longrightarrow \quad \lim s_n \in [a, b].}$$

The finitely many exceptional early terms do not affect any of these conclusions because an index can be chosen after every exception and after the convergence threshold.

8.10 Exercise 8.10: A strict bound for the limit becomes an eventual strict bound

What we must show. If $s_n \to s$ and $s > a$, prove that there is a real N such that

$$n \in \mathbb{N}, \quad n > N \quad \Longrightarrow \quad s_n > a.$$

Plan. The positive gap $s - a$ allows us to place a whole ε-neighborhood of the limit above a.

Step 1: choose a tolerance smaller than the gap.

Because $s > a$, the real number $s - a$ is positive. Define

$$\varepsilon = \frac{s - a}{2} > 0.$$

The tolerance is fixed from the limit and the bound; it does not depend on the index n.

Step 2: obtain a threshold from convergence.

The definition of $s_n \to s$ gives a real number N such that

$$n \in \mathbb{N}, \quad n > N \quad \Longrightarrow \quad |s_n - s| < \varepsilon.$$

For any index in this tail, the absolute-value inequality implies

$$-\varepsilon < s_n - s < \varepsilon.$$

The left inequality gives $s_n > s - \varepsilon$.

Step 3: check that the lower edge is above the desired bound.

Substitution and simplification give

$$s - \varepsilon = s - \frac{s - a}{2} = \frac{2s - s + a}{2} = \frac{s + a}{2}.$$

Also

$$\frac{s+a}{2} - a = \frac{s-a}{2} > 0,$$

so $(s+a)/2 > a$. Combining these strict comparisons yields

$$s_n > s - \varepsilon = \frac{s+a}{2} > a \qquad (n \in \mathbb{N},\ n > N).$$

Conclusion. The threshold supplied for $\varepsilon = (s-a)/2$ satisfies the required implication:

$$\boxed{n > N \quad \Longrightarrow \quad s_n > a.}$$

The strict hypothesis $s > a$ matters: if $s = a$ and $s_n = a$ for all n, the sequence converges to a but none of its terms is greater than a.

9 Limit theorems for sequences

This section proves and applies rules for limits. Each application checks the hypotheses: the sequences must have the required finite limits, and a quotient needs a nonzero limiting denominator.

Definition 9.8 extends the phrase *a limit exists* to include divergence to $+\infty$ or $-\infty$. These are not finite real limits. We use the extended convention where the exercise permits it, while checking separately whether a finite-limit theorem applies. A recurrence's fixed-point equation gives a necessary condition for a finite limit; it does not prove convergence.

9.1 Exercise 9.1: Three quotients evaluated by the finite limit laws

What we must show. Use Theorems 9.2–9.7, with their hypotheses checked, to prove

$$\text{(a)} \quad \lim_{n\to\infty} \frac{n+1}{n} = 1,$$
$$\text{(b)} \quad \lim_{n\to\infty} \frac{3n+7}{6n-5} = \frac{1}{2},$$
$$\text{(c)} \quad \lim_{n\to\infty} \frac{17n^5+73n^4-18n^2+3}{23n^5+13n^3} = \frac{17}{23}.$$

Plan. Divide each polynomial expression by the appropriate positive power of n. The resulting expressions involve constants and reciprocal powers, whose limits are known. For each use of the quotient theorem, verify that the denominator terms are nonzero and that their finite limit is nonzero.

1. Record the finite limit rules being used

A constant sequence with value $c \in \mathbb{R}$ converges to c, because $|c-c| = 0 < \varepsilon$ for every $\varepsilon > 0$. Theorem 9.7(a) gives

$$\frac{1}{n^p} \longrightarrow 0 \qquad (p > 0).$$

The scalar rule is Theorem 9.2, the sum rule is Theorem 9.3, and the product rule is Theorem 9.4. Subtraction follows by combining the scalar rule with scalar -1 and the sum rule. The quotient rule, Theorem 9.6, requires a convergent numerator, a convergent denominator, a nonzero denominator at every index, and a nonzero denominator limit. All limits used in these rules are finite real numbers.

2. Part (a): Separate the constant term

Since $n \in \mathbb{N}$, the denominator n is positive. The identity

$$\frac{n+1}{n} = \frac{n}{n} + \frac{1}{n} = 1 + \frac{1}{n}$$

therefore holds for every index. The constant sequence 1 converges to 1, and Theorem 9.7(a), with $p = 1$, gives $1/n \to 0$. Applying Theorem 9.3 yields

$$\frac{n+1}{n} = 1 + \frac{1}{n} \longrightarrow 1 + 0 = 1.$$

Here the sum rule applies after rewriting; there is no need to treat the unbounded sequence n as a finite convergent sequence.

3. Part (b): Normalize numerator and denominator

For every $n \geq 1$, the original denominator satisfies $6n - 5 \geq 1 > 0$. Dividing numerator and denominator by $n > 0$ gives

$$\frac{3n+7}{6n-5} = \frac{(3n+7)/n}{(6n-5)/n} = \frac{3+7/n}{6-5/n}.$$

Let $u_n = 3+7/n$ and $v_n = 6-5/n$. By Theorem 9.7(a), $1/n \to 0$. The scalar rule gives $7/n \to 7 \cdot 0 = 0$ and $-5/n \to -5 \cdot 0 = 0$. Adding the constant sequences by Theorem 9.3 gives

$$u_n \longrightarrow 3 + 0 = 3, \qquad v_n \longrightarrow 6 + 0 = 6.$$

For each n, we also have

$$v_n = \frac{6n-5}{n} > 0.$$

Thus every denominator term is nonzero, and its limit 6 is nonzero. All hypotheses of Theorem 9.6 have now been verified, so

$$\frac{3n+7}{6n-5} = \frac{u_n}{v_n} \longrightarrow \frac{3}{6} = \frac{1}{2}.$$

4. Part (c): Divide by the largest power in the denominator

The original denominator is positive for every index:

$$23n^5 + 13n^3 = n^3(23n^2 + 13) > 0,$$

because $n \geq 1$. Divide both numerator and denominator by $n^5 > 0$. Each term then simplifies as follows:

$$\frac{n^4}{n^5} = \frac{1}{n}, \qquad \frac{n^2}{n^5} = \frac{1}{n^3}, \qquad \frac{n^3}{n^5} = \frac{1}{n^2}.$$

Consequently

$$\frac{17n^5 + 73n^4 - 18n^2 + 3}{23n^5 + 13n^3} = \frac{17 + 73/n - 18/n^3 + 3/n^5}{23 + 13/n^2}.$$

Write the normalized numerator and denominator as U_n and V_n. Theorem 9.7(a), applied with $p = 1, 2, 3, 5$, gives

$$\frac{1}{n} \to 0, \qquad \frac{1}{n^2} \to 0, \qquad \frac{1}{n^3} \to 0, \qquad \frac{1}{n^5} \to 0.$$

Multiplication by the indicated constants, using Theorem 9.2, gives

$$\frac{73}{n} \to 0, \qquad -\frac{18}{n^3} \to 0, \qquad \frac{3}{n^5} \to 0, \qquad \frac{13}{n^2} \to 0.$$

Repeated applications of the sum rule, together with the constant limits, therefore yield

$$U_n \longrightarrow 17 + 0 + 0 + 0 = 17, \qquad V_n \longrightarrow 23 + 0 = 23.$$

Moreover, $V_n = 23 + 13/n^2 > 23 > 0$ for every n, and its limit 23 is nonzero. The quotient rule applies and gives

$$\frac{U_n}{V_n} \longrightarrow \frac{17}{23}.$$

Conclusion. The three limits are (a) 1, (b) 1/2, and (c) 17/23. Every use of the quotient theorem has a denominator that is nonzero at every index and has a nonzero finite limit.

9.2 Exercise 9.2: Limits obtained from two given convergent sequences

What we must show. Suppose $x_n \to 3$, $y_n \to 7$, and $y_n \neq 0$ for every n. Find

$$\text{(a)} \quad \lim_{n\to\infty}(x_n + y_n), \qquad \text{(b)} \quad \lim_{n\to\infty} \frac{3y_n - x_n}{y_n^2}.$$

Plan. Use the sum rule in part (a). In part (b), build the numerator from scalar multiples and a sum, build the denominator with the product rule, and then check the quotient rule.

1. Part (a): Add the two limits

The hypotheses give two finite convergent sequences. Theorem 9.3 therefore applies directly:

$$x_n + y_n \longrightarrow 3 + 7 = 10.$$

2. Part (b): Evaluate the numerator

Theorem 9.2 gives

$$3y_n \longrightarrow 3 \cdot 7 = 21, \qquad -x_n \longrightarrow (-1) \cdot 3 = -3.$$

Theorem 9.3 then gives

$$3y_n - x_n = 3y_n + (-x_n) \longrightarrow 21 + (-3) = 18.$$

The numerator here is $3y_n - x_n$; only the denominator contains a square.

3. Part (b): Evaluate the denominator and take the quotient

Since $y_n \to 7$, Theorem 9.4 applied to the two factors y_n gives

$$y_n^2 = y_n\, y_n \longrightarrow 7 \cdot 7 = 49.$$

The assumption $y_n \neq 0$ makes $y_n^2 > 0$ at every index. Thus every denominator is nonzero, and its limit 49 is also nonzero. Theorem 9.6 now gives

$$\frac{3y_n - x_n}{y_n^2} \longrightarrow \frac{18}{49}.$$

Conclusion. The limits are (a) 10 and (b) 18/49.

9.3 Exercise 9.3: A polynomial expression in convergent sequences

What we must show. Assume $a_n \to a \in \mathbb{R}$ and $b_n \to b \in \mathbb{R}$. For

$$s_n = \frac{a_n^3 + 4a_n}{b_n^2 + 1},$$

prove carefully that

$$s_n \longrightarrow \frac{a^3 + 4a}{b^2 + 1}.$$

Plan. Form the powers one multiplication at a time. Then use the scalar and sum rules. The added 1 makes the denominator and its limit positive even when $b_n = 0$ or $b = 0$.

1. Construct the square and cube of the numerator sequence

The two factors in $a_n\,a_n$ both converge to a. The product rule, Theorem 9.4, gives

$$a_n^2 = a_n\,a_n \longrightarrow a \cdot a = a^2.$$

We may now apply the product rule again, this time to a_n^2 and a_n, since both have finite limits:

$$a_n^3 = (a_n^2)a_n \longrightarrow a^2 \cdot a = a^3.$$

Separately, Theorem 9.2 gives $4a_n \to 4a$. The sum rule therefore yields

$$a_n^3 + 4a_n \longrightarrow a^3 + 4a.$$

2. Construct the denominator limit

Apply Theorem 9.4 to $b_n\,b_n$ to obtain

$$b_n^2 \longrightarrow b \cdot b = b^2.$$

The constant sequence 1 converges to 1, so Theorem 9.3 gives

$$b_n^2 + 1 \longrightarrow b^2 + 1.$$

For every real b_n, its square is nonnegative, and thus $b_n^2 + 1 \geq 1 > 0$. The denominator terms are consequently nonzero. Likewise $b^2 + 1 \geq 1 > 0$, so their limit is nonzero.

3. Apply the quotient theorem with all hypotheses established

Step 1 proves convergence of the numerator, Step 2 proves convergence of the denominator, and the positivity checks verify both nonzero requirements. Theorem 9.6 therefore gives the desired limit.

Conclusion. $\lim\limits_{n\to\infty} s_n = \dfrac{a^3 + 4a}{b^2 + 1}$ for all finite real limits a and b, including $b = 0$.

9.4 Exercise 9.4: A nested-square-root recursion

What we must show. For $s_1 = 1$ and $s_{n+1} = \sqrt{s_n + 1}$, list the first four terms and, assuming convergence, prove that the limit is $(1 + \sqrt{5})/2$.

Plan. First check that every radical is defined and every term is at least 1. Under the stated convergence assumption, square the recursion and apply the product and sum limit laws. The lower bound will select the appropriate quadratic root.

1. Verify that the recursion stays defined and positive

The initial value is $s_1 = 1 \geq 1$. If $s_n \geq 1$, then $s_n + 1 \geq 2 > 0$, so its nonnegative real square root exists. Moreover, $\sqrt{s_n + 1} \geq 1$: a nonnegative number less than 1 would have square less than 1, whereas this square is at least 2. Thus $s_{n+1} \geq 1$. Induction shows that all terms are defined and satisfy $s_n \geq 1$.

2. Part (a): Substitute successively four times

The successive terms are

$$\begin{aligned} s_1 &= 1, \\ s_2 &= \sqrt{s_1 + 1} = \sqrt{1+1} = \sqrt{2}, \\ s_3 &= \sqrt{s_2 + 1} = \sqrt{\sqrt{2}+1} = \sqrt{1+\sqrt{2}}, \\ s_4 &= \sqrt{s_3 + 1} = \sqrt{\sqrt{1+\sqrt{2}}+1} = \sqrt{1+\sqrt{1+\sqrt{2}}}. \end{aligned}$$

3. A shift of a convergent sequence has the same limit

We will use this elementary observation for this and the next two recursions. If $u_n \to u \in \mathbb{R}$, then $u_{n+1} \to u$. Indeed, given $\varepsilon > 0$, choose N so that $m > N$ implies $|u_m - u| < \varepsilon$. If $n > N$, then also $n + 1 > N$, so $|u_{n+1} - u| < \varepsilon$. This is exactly the limit definition for the shifted sequence.

4. Part (b): Record the restriction on a finite limit

Assume, as requested, that $s_n \to L \in \mathbb{R}$. Since $s_n \geq 1$ for every n, we have $L \geq 1$. To see the bound directly, if $L < 1$, set $\varepsilon = (1 - L)/2 > 0$. Eventually $|s_n - L| < \varepsilon$, giving

$$s_n < L + \varepsilon = \frac{L+1}{2} < 1,$$

contrary to the bound on the terms. This is also the eventual-bound principle of Exercise 8.9(a).

5. Pass to limits in a squared identity

The defining radical equality implies

$$s_{n+1}^2 = s_n + 1$$

for every n. By Step 3, $s_{n+1} \to L$. Theorem 9.4 applied to $s_{n+1}\, s_{n+1}$ gives $s_{n+1}^2 \to L^2$. Theorem 9.3 gives $s_n + 1 \to L + 1$. These two expressions are equal term by term, so uniqueness of finite limits, proved at the end of Section 7, gives

$$L^2 = L + 1.$$

Only the product and sum laws were used; a square-root limit rule is not needed.

6. Solve the quadratic equation and select its permitted root

Rearrange and complete the square:

$$L^2 - L - 1 = 0, \qquad 4L^2 - 4L = 4, \qquad (2L-1)^2 = 5.$$

Consequently $2L - 1 = \sqrt{5}$ or $2L - 1 = -\sqrt{5}$, which gives

$$L = \frac{1+\sqrt{5}}{2} \quad \text{or} \quad L = \frac{1-\sqrt{5}}{2}.$$

We have $\sqrt{5} > 1$, since the opposite nonnegative inequality would imply $5 \leq 1$ after squaring. Thus $(1 - \sqrt{5})/2 < 0$, contradicting $L \geq 1$. The only remaining root is $(1 + \sqrt{5})/2$.

Conclusion. The first four terms are 1, $\sqrt{2}$, $\sqrt{1+\sqrt{2}}$, and $\sqrt{1+\sqrt{1+\sqrt{2}}}$. Under the given convergence assumption, $\lim s_n = (1 + \sqrt{5})/2$.

9.5 Exercise 9.5: A positive rational recursion

What we must show. Let $t_1 = 1$ and

$$t_{n+1} = \frac{t_n^2 + 2}{2t_n} \qquad (n \geq 1).$$

Assuming that (t_n) converges to a finite real number, find its limit.

Plan. Prove positivity so that every term is defined, and strengthen it to the uniform lower bound $t_n \geq 1$. The lower bound makes the assumed limit nonzero before the quotient theorem is applied.

1. Check that the recursion is defined at every index

We start with $t_1 = 1 > 0$. If $t_n > 0$, then its denominator $2t_n$ is positive and its numerator $t_n^2 + 2$ is positive. The quotient is therefore a well-defined positive real number. Induction proves $t_n > 0$ for all n, and in particular no denominator in the recursion vanishes.

2. Obtain a positive uniform lower bound

For every n, use the positive denominator to compute

$$\begin{aligned} t_{n+1} - 1 &= \frac{t_n^2 + 2}{2t_n} - \frac{2t_n}{2t_n} \\ &= \frac{t_n^2 - 2t_n + 2}{2t_n} \\ &= \frac{(t_n - 1)^2 + 1}{2t_n} > 0. \end{aligned}$$

The last numerator is positive because $(t_n - 1)^2 \geq 0$. Thus every term after t_1 is greater than 1, and $t_1 = 1$. It follows that $t_n \geq 1$ for every n.

3. The convergence assumption gives a nonzero limit

Suppose $t_n \to L \in \mathbb{R}$. The bound $t_n \geq 1$ passes to the limit by Exercise 8.9(a), so $L \geq 1 > 0$. Equivalently, if $L < 1$, the positive tolerance $(1 - L)/2$ would force eventual terms to be less than $L + (1 - L)/2 = (L + 1)/2 < 1$, a contradiction. In particular, $2L \neq 0$.

4. Apply the finite limit laws to the recursion

The product rule, Theorem 9.4, gives

$$t_n^2 = t_n\, t_n \longrightarrow L^2.$$

The constant sequence 2 converges to 2, and hence the sum rule gives $t_n^2 + 2 \to L^2 + 2$. The scalar rule gives $2t_n \to 2L$. The terms $2t_n$ are nonzero by Step 1, and their limit $2L$ is nonzero by Step 3. Thus Theorem 9.6 applies:

$$\frac{t_n^2 + 2}{2t_n} \longrightarrow \frac{L^2 + 2}{2L}.$$

The shift observation in Exercise 9.4 also gives $t_{n+1} \to L$. The recursion identifies these two sequences, so uniqueness of finite limits gives

$$L = \frac{L^2 + 2}{2L}.$$

5. Solve and select the positive root

Since $2L \neq 0$, multiplying the last equality by $2L$ is valid:

$$2L^2 = L^2 + 2, \qquad L^2 = 2.$$

Thus $L = \sqrt{2}$ or $L = -\sqrt{2}$. The root $-\sqrt{2}$ is negative and violates $L \geq 1$. Consequently the assumed finite limit is $\sqrt{2}$.

Conclusion. Under the convergence assumption, $\lim_{n\to\infty} t_n = \sqrt{2}$. The nonzero denominator limit was established before the quotient theorem was used.

9.6 Exercise 9.6: A necessary finite-limit equation and an actual infinite limit

What we must show. For $x_1 = 1$ and $x_{n+1} = 3x_n^2$, address all three questions: (a) determine the possibilities for a finite real limit; (b) decide whether the actual sequence has a limit; (c) explain why the conclusions are consistent.

Plan. A hypothetical finite limit must satisfy the recursion's fixed-point equation. Separately, use an induction estimate for the actual terms. Distinguish finite convergence from the extended infinite-limit convention in Definition 9.8.

1. Part (a): Assume a finite limit before applying the product rule

Suppose $x_n \to a \in \mathbb{R}$. The shift observation proved in Exercise 9.4 gives $x_{n+1} \to a$. Theorem 9.4 gives $x_n^2 \to a^2$, and Theorem 9.2 then gives $3x_n^2 \to 3a^2$. Since $x_{n+1} = 3x_n^2$ for every n, uniqueness of finite limits implies

$$a = 3a^2.$$

Subtracting a and factoring gives

$$0 = 3a^2 - a = a(3a - 1).$$

The zero-product property in $\mathbb{R}$ says that one of the two factors must be zero. Thus $a = 0$ or $3a - 1 = 0$; the second equation gives $a = 1/3$ after division by $3 \neq 0$.

This proves only a necessary condition: *if the sequence has a finite real limit, that limit must be* 0 *or* $1/3$. The computation does not prove that such a finite limit exists.

2. Part (b): Bound the actual terms below by their indices

We prove $x_n \geq n$ for every $n \in \mathbb{N}$. For $n = 1$, the equality $x_1 = 1$ supplies the base case. Suppose $x_n \geq n$ for some $n \geq 1$. In particular $x_n \geq 1$, so $x_n^2 - x_n = x_n(x_n - 1) \geq 0$. It follows that

$$x_{n+1} = 3x_n^2 \geq 3x_n \geq 3n \geq n + 1.$$

The final inequality holds because $3n - (n + 1) = 2n - 1 \geq 1 > 0$ for $n \geq 1$. This completes the induction.

Now let $M > 0$ and take the threshold $N = M$. For every integer $n > N$, the bound gives

$$x_n \geq n > M.$$

This is exactly Definition 9.8 for $x_n \to +\infty$. Thus the actual sequence has the extended limit $+\infty$.

It cannot have a finite real limit. For a direct check, if $x_n \to a \in \mathbb{R}$, then eventually $|x_n - a| < 1$, which implies $x_n < a + 1$. Choose an integer n larger than both that eventual threshold and $a + 1$, using the Archimedean property. The bound $x_n \geq n$ would instead give $x_n > a + 1$, a contradiction. This is also consistent with Theorem 9.1, which says every finitely convergent sequence is bounded.

3. Part (c): Explain the two meanings of the limit statement

Part (a) began with the hypothesis of a *finite real* limit a. Its use of the finite product and scalar limit laws is valid under that hypothesis. Part (b) shows that the hypothesis fails for this initial value: the terms grow without bound. Consequently the necessary alternatives in part (a) are not attained by this sequence.

Under the convention introduced after Definition 9.8, saying that "the limit exists" also allows divergence to $+\infty$ or $-\infty$. In that stated convention the answer to part (b) is yes, with limit $+\infty$; the sequence does not converge to a finite real number. Theorems 9.2–9.6 were proved for finite real limits, so their product calculation in part (a) cannot be applied with $a = +\infty$. The extended infinite-limit conclusion therefore supplies no contradiction to the finite necessary condition.

Conclusion. (a) Any hypothetical finite real limit is 0 or 1/3. (b) The actual sequence satisfies $\lim x_n = +\infty$ in the sense of Definition 9.8, and has no finite real limit. (c) A necessary condition for finite convergence does not assert its existence, and finite product laws cannot be applied at an infinite limit.

9.7 Exercise 9.7: Finish the estimate for the root of the index

What we must show. Complete the argument that

$$s_n = n^{1/n} - 1 \longrightarrow 0,$$

and hence finish the proof of Theorem 9.7(c), $n^{1/n} \to 1$.

Plan. Use the nonnegative binomial terms in $(1 + s_n)^n$ to obtain an explicit bound on s_n. Then choose a threshold directly in terms of the prescribed positive tolerance.

1. Verify the sign and the defining power identity

Let $u_n = n^{1/n}$ denote the positive real nth root of n. Then $u_n^n = n \geq 1$. If $u_n < 1$, the positivity $0 < u_n < 1$ would give $u_n^n < 1$, a contradiction. Hence $u_n \geq 1$ and

$$s_n = u_n - 1 \geq 0, \qquad (1 + s_n)^n = u_n^n = n.$$

For $n = 1$, this gives $s_1 = 1 - 1 = 0$.

2. Derive the binomial bound for $n \geq 2$

For an integer $n \geq 2$, the binomial theorem gives

$$n = (1 + s_n)^n = 1 + ns_n + \frac{n(n-1)}{2} s_n^2 + \sum_{j=3}^{n} \binom{n}{j} s_n^j.$$

When $n = 2$, the final sum is empty and equals 0. All terms of that sum are nonnegative, since $s_n \geq 0$ and all the binomial coefficients are nonnegative. Also $ns_n \geq 0$, and the constant term 1 is strictly positive. Therefore

$$n > \frac{n(n-1)}{2} s_n^2.$$

Dividing by the positive number n gives $1 > (n-1)s_n^2/2$. Multiplying by $2/(n-1) > 0$ yields

$$s_n^2 < \frac{2}{n-1}.$$

Both s_n and the square root of the right-hand side are nonnegative, so the order of their squares gives

$$0 \le s_n < \sqrt{\frac{2}{n-1}} \qquad (n \ge 2).$$

3. Specify the threshold and verify the limit definition

Let $\varepsilon > 0$ be arbitrary and set

$$N = \max\left\{2,\ 1 + \frac{2}{\varepsilon^2}\right\}.$$

A real threshold is permitted by the definition of convergence; an integer larger than it may also be chosen by the Archimedean property. If $n > N$, then $n \ge 2$ and

$$n - 1 > \frac{2}{\varepsilon^2} > 0.$$

Taking reciprocals of positive quantities and multiplying by 2 gives $2/(n-1) < \varepsilon^2$. As $\varepsilon > 0$, this implies

$$\sqrt{\frac{2}{n-1}} < \varepsilon.$$

Using the bound from Step 2, we conclude

$$|s_n - 0| = s_n < \sqrt{\frac{2}{n-1}} < \varepsilon \qquad (n > N).$$

This is the required proof that $s_n \to 0$.

4. Return to the root sequence

The constant sequence 1 converges to 1, and $s_n \to 0$ was proved without assuming a limit for $n^{1/n}$. The sum rule, Theorem 9.3, therefore gives

$$n^{1/n} = 1 + s_n \longrightarrow 1 + 0 = 1.$$

Conclusion. The explicit estimate $0 \le n^{1/n} - 1 < \sqrt{2/(n-1)}$ for $n \ge 2$, together with $N = \max\{2, 1 + 2/\varepsilon^2\}$, proves $n^{1/n} - 1 \to 0$ and hence $n^{1/n} \to 1$.

9.8 Exercise 9.8: Finite and infinite possibilities for five limits

What we must show. Determine each of the five requested limits, using the convention introduced in Definition 9.8: an existing limit may be a real number, $+\infty$, or $-\infty$. The answer "NOT EXIST" therefore means that none of these three possibilities works.

Plan. Use the defining threshold tests. A sequence has limit $+\infty$ if, for every $M > 0$, all sufficiently late terms exceed M. It has limit $-\infty$ if, for every $M < 0$, all sufficiently late terms are below M.

Neither definition describes finite convergence, and the finite-limit arithmetic laws cannot simply be applied to expressions involving infinity.

(a) The sequence n^3.

Let $M > 0$ be arbitrary. Set $N = \max\{1, M\}$. If $n > N$, then $n \geq 1$ and

$$n^3 = n \cdot n^2 \geq n > M.$$

Thus every prescribed positive threshold is exceeded by all sufficiently late terms, exactly as Definition 9.8 requires.

Conclusion. In part (a), $\lim_{n\to\infty} n^3 = +\infty$.

(b) The sequence $-n^3$.

Let $M < 0$. The number $-M$ is positive. Choose $N = \max\{1, -M\}$. If $n > N$, the estimate used in (a) gives $n^3 \geq n > -M$. Multiplication by -1 reverses the inequality, giving $-n^3 < M$. This is the negative-infinity definition.

Conclusion. In part (b), $\lim_{n\to\infty}(-n^3) = -\infty$.

(c) The sequence $(-n)^n$.

At an even index $n = 2m$ the value is $(2m)^{2m} > 0$; at an odd index $n = 2m+1$ it is $-(2m+1)^{2m+1} < 0$. There are even and odd indices beyond every threshold: choose a positive integer $m > N$ to make both $2m$ and $2m + 1$ exceed a prescribed N.

The limit cannot be $+\infty$, because the test with $M = 1$ would require all sufficiently late terms to be greater than 1, whereas arbitrarily late odd terms are negative. It cannot be $-\infty$, because the test with $M = -1$ would require all sufficiently late terms to be less than -1, whereas arbitrarily late even terms are positive.

It also cannot be a finite real number L. If it converged to L, the definition with $\varepsilon = 1$ would give a threshold N_1 such that

$$n > N_1 \quad \Longrightarrow \quad |(-n)^n - L| < 1.$$

Choose an even positive integer $n > N_1$ with $n > \max\{1, L + 1\}$. Such an index exists by taking $n = 2m$ with m sufficiently large. Since $n \geq 1$, $n^n \geq n$, so at this even index

$$(-n)^n = n^n \geq n > L + 1.$$

Consequently $|(-n)^n - L| = n^n - L > 1$, contradicting the required error bound. All three possible types of limit have now been excluded.

Conclusion. In part (c), the limit does NOT EXIST, even in the extended sense of Definition 9.8.

(d) The sequence $(1.01)^n$.

Write $1.01 = 1 + 1/100$. The binomial theorem, or Bernoulli's inequality, gives

$$\left(1 + \frac{1}{100}\right)^n \geq 1 + \frac{n}{100}.$$

For completeness, the inequality $(1 + x)^n \geq 1 + nx$ for $x \geq 0$ follows by induction: the case $n = 1$ is equality, and from the inequality at n one obtains

$$(1 + x)^{n+1} \geq (1 + nx)(1 + x) = 1 + (n + 1)x + nx^2 \geq 1 + (n + 1)x.$$

Now let $M > 0$ and set $N = 100M$. If $n > N$, then $n/100 > M$, and hence

$$(1.01)^n \geq 1 + \frac{n}{100} > M.$$

This proves divergence to positive infinity by the definition.

Conclusion. In part (d), $\lim_{n\to\infty} (1.01)^n = +\infty$.

(e) The sequence n^n.

For $n \geq 1$, the product of n factors all equal to n satisfies $n^n \geq n$. Let $M > 0$ and set $N = \max\{1, M\}$. If $n > N$, then $n^n \geq n > M$. The definition therefore applies.

Conclusion. In part (e), $\lim_{n\to\infty} n^n = +\infty$.

9.9 Exercise 9.9: Order preserved by existing extended limits

What we must show. Assume there is a threshold N_0 such that $s_n \leq t_n$ whenever $n > N_0$. Prove the three assertions, including every possible finite or infinite case in part (c).

Plan. Combine the threshold at which the comparison becomes valid with the threshold supplied by the relevant limit. In the finite case, two limits in the wrong order would force the late terms to have the wrong order as well.

(a) From $s_n \to +\infty$ to $t_n \to +\infty$.

Fix $M > 0$. Since $s_n \to +\infty$, some N_1 satisfies $s_n > M$ for all $n > N_1$. Set $N = \max\{N_0, N_1\}$. For every $n > N$, both the comparison and this threshold test hold, so

$$t_n \geq s_n > M.$$

This holds for the arbitrary $M > 0$, proving $t_n \to +\infty$.

(b) From $t_n \to -\infty$ to $s_n \to -\infty$.

Fix $M < 0$. Since $t_n \to -\infty$, some N_1 satisfies $t_n < M$ for all $n > N_1$. With $N = \max\{N_0, N_1\}$, every $n > N$ gives

$$s_n \leq t_n < M.$$

This is the definition of $s_n \to -\infty$.

(c) Account for the infinite limit cases.

Write $S = \lim s_n$ and $T = \lim t_n$, which exist by the assumption in this part. Their possible values lie in $\mathbb{R} \cup \{-\infty, +\infty\}$, with the usual extended order. The finite and infinite alternatives cannot overlap. Indeed, convergence to a real c puts the late terms between $c - 1$ and $c + 1$. A simultaneous limit $+\infty$ would put them above $\max\{1, c + 2\}$, contradicting the upper restriction; a simultaneous limit $-\infty$ would put them below $\min\{-1, c - 2\}$, contradicting the lower restriction. Limits of both signs of infinity are incompatible as well, since they would eventually require a term to be both greater than 1 and less than -1. The following cases exhaust all possibilities involving infinity:

1. If $S = -\infty$, then $S \leq T$ for every possible T.

2. If $T = +\infty$, then $S \leq T$ for every possible S.

3. If $S = +\infty$, part (a) forces $T = +\infty$, so $S = T$ and in particular $S \le T$.

4. If $T = -\infty$, part (b) forces $S = -\infty$, so again $S = T$.

After these cases, both S and T must be finite real numbers.

(c) Prove the finite case by contradiction.

Suppose, contrary to the desired inequality, that $S > T$. Set $\delta = (S-T)/3 > 0$. By finite convergence there are thresholds N_1, N_2 such that

$$n > N_1 \Longrightarrow |s_n - S| < \delta, \qquad n > N_2 \Longrightarrow |t_n - T| < \delta.$$

Thus $s_n > S - \delta$ and $t_n < T + \delta$ at the corresponding late indices. But

$$(S-\delta) - (T+\delta) = (S-T) - 2\delta = (S-T) - \frac{2(S-T)}{3} = \frac{S-T}{3} > 0.$$

For $n > \max\{N_0, N_1, N_2\}$ this gives

$$s_n > S - \delta > T + \delta > t_n,$$

contradicting $s_n \le t_n$. Hence $S \le T$ in the finite case as well.

Conclusion. The eventual order implies both infinite-limit comparisons in (a) and (b), and $\lim s_n \le \lim t_n$ whenever both limits exist in the extended sense.

9.10 Exercise 9.10: Scaling and reversing an infinite limit

What we must show. Prove the effects of multiplying a sequence with limit $+\infty$ by a positive constant or a negative constant, and prove both directions of the sign-reversal equivalence.

Plan. Choose a threshold for s_n by dividing the target threshold by the multiplying constant. A negative constant reverses the inequality, so the signs must be recorded.

(a) Multiplication by $k > 0$.

Assume $s_n \to +\infty$ and fix $k > 0$. To prove $ks_n \to +\infty$, let $M > 0$. Then $M/k > 0$, so the assumed limit supplies N with

$$n > N \quad \Longrightarrow \quad s_n > \frac{M}{k}.$$

Multiplying by $k > 0$ preserves the direction and gives $ks_n > M$. This verifies the target definition.

In fact this implication is reversible. If $ks_n \to +\infty$ and $M > 0$, use the positive threshold kM for ks_n. Eventually $ks_n > kM$, and division by $k > 0$ gives $s_n > M$. Thus positive multiplication preserves the $+\infty$ limit in both directions.

(b) Prove $s_n \to +\infty \Longrightarrow -s_n \to -\infty$.

Let $M < 0$ be an arbitrary threshold for $-s_n$. Because $-M > 0$, the assumption $s_n \to +\infty$ supplies N such that $n > N$ implies $s_n > -M$. Multiplication by -1 reverses the inequality, giving $-s_n < M$. Thus $-s_n \to -\infty$.

(b) Prove the reverse implication.

Assume $-s_n \to -\infty$ and let $M > 0$. The number $-M$ is negative, so some N satisfies $n > N \implies -s_n < -M$. Multiplication by -1 reverses this inequality and gives $s_n > M$. Consequently $s_n \to +\infty$, completing the equivalence.

(c) Multiplication by $k < 0$.

Assume $s_n \to +\infty$ and $k < 0$. Let $M < 0$ be an arbitrary target threshold for ks_n. The quotient M/k is positive, since both numerator and denominator are negative. Hence the assumed limit supplies N such that $n > N$ implies $s_n > M/k$. Multiplying by $k < 0$ reverses the comparison:

$$n > N \quad \implies \quad ks_n < M.$$

This proves $ks_n \to -\infty$.

The corresponding reverse implication also holds. If $ks_n \to -\infty$, fix $M > 0$ and use the negative threshold kM. Eventually $ks_n < kM$, and division by $k < 0$ reverses the inequality, yielding $s_n > M$. Therefore $s_n \to +\infty$.

Conclusion. Positive scaling preserves a positive-infinity limit; negative scaling changes it to negative infinity. In particular, $s_n \to +\infty$ if and only if $-s_n \to -\infty$, with each direction justified by its own threshold test.

9.11 Exercise 9.11: Adding a sequence that is bounded below

What we must show. Prove the sum has limit $+\infty$ in each of the three stated situations. Distinguish a lower bound valid for all terms from a lower bound valid only after an initial segment.

Plan. Once $t_n \geq m$ holds in the tail, it suffices to make $s_n > M - m$. If this number is not positive, replace it by a larger positive threshold before invoking Definition 9.8.

(a) Use the stated global lower bound.

Assume $s_n \to +\infty$ and $\inf\{t_n : n \in \mathbb{N}\} > -\infty$. The set is nonempty and consists of real numbers; its infimum is at most the finite member t_1. Thus this hypothesis provides a real lower bound m, for example the real infimum itself, such that $t_n \geq m$ for every n.

Let $M > 0$ be arbitrary, and define

$$B = \max\{1, M - m\} > 0.$$

Since $s_n \to +\infty$, there is N such that $n > N$ implies $s_n > B$. For those indices,

$$s_n + t_n \geq s_n + m > B + m \geq (M - m) + m = M.$$

The first inequality uses the global lower bound; the last uses $B \geq M - m$. This proves the positive-infinity limit.

(b) Derive an eventual lower bound in every allowed limit case.

Assume $s_n \to +\infty$ and $T = \lim t_n > -\infty$. The existing limit T is either a real number or $+\infty$.

- If $T \in \mathbb{R}$, apply finite convergence with $\varepsilon = 1$. For some N_1, every $n > N_1$ has $|t_n - T| < 1$, implying $t_n > T - 1$. Thus we may take $m = T - 1$ as an eventual lower bound.

- If $T = +\infty$, use the positive threshold 1 in Definition 9.8. For some N_1, every $n > N_1$ has $t_n > 1$. Thus we may take $m = 1$.

In either case there are a real m and a threshold N_1 such that $t_n \geq m$ whenever $n > N_1$. This is a tail bound; no assertion about the earlier terms is needed.

Now fix $M > 0$. Set $B = \max\{1, M - m\} > 0$, and use $s_n \to +\infty$ to choose N_2 such that $s_n > B$ for $n > N_2$. For $n > \max\{N_1, N_2\}$, both estimates hold and

$$s_n + t_n \geq s_n + m > B + m \geq M.$$

This proves the desired limit using the eventual bound directly, without applying a global hypothesis to an unexamined prefix.

(c) A bounded sequence is bounded below.

If (t_n) is bounded, there is a real $C \geq 0$ with $|t_n| \leq C$ for every n. Therefore $t_n \geq -C$ for every n, a global lower bound, and part (a) applies. One can also see the threshold explicitly: for $M > 0$, choose N such that $s_n > M + C$ whenever $n > N$; this is a valid positive threshold because $M + C > 0$. Then

$$s_n + t_n \geq s_n - C > M + C - C = M.$$

Conclusion. In (a), (b), and (c), the sum $s_n + t_n$ tends to $+\infty$. An eventual lower bound on the added sequence already suffices, as the proof of (b) shows.

9.12 Exercise 9.12: The ratio test for the size of a sequence

What we must show. Suppose $s_n \neq 0$ for every n, and the limit

$$L = \lim_{n\to\infty} r_n, \qquad r_n = \left|\frac{s_{n+1}}{s_n}\right|,$$

exists. Prove $s_n \to 0$ if $L < 1$, and $|s_n| \to +\infty$ if $L > 1$, including the case $L = +\infty$.

Plan. A ratio eventually less than a fixed $q < 1$ gives a geometric bound. For ratios tending to a value larger than 1, apply that first result to the reciprocal magnitudes.

First locate the possible values of L.

Every r_n is positive, since its numerator and denominator are nonzero in absolute value. Therefore its existing limit can only be a nonnegative real number or $+\infty$. For example, if a finite limit were $L < 0$, convergence with $\varepsilon = -L/2 > 0$ would eventually give $r_n < L + (-L/2) = L/2 < 0$, a contradiction. A limit $-\infty$ would eventually give $r_n < -1$ by its definition, also a contradiction. This verifies the claimed restriction on L.

(a) Choose a single ratio bound valid at every sufficiently late index.

Suppose $L < 1$. Then $0 \leq L < 1$, so

$$q = \frac{L+1}{2} \quad \text{satisfies} \quad 0 < q < 1 \quad \text{and} \quad L < q.$$

By convergence of r_n to the finite number L, some threshold N_0 gives $|r_n - L| < q - L$ for all $n > N_0$. In particular $r_n < q$ at all those indices. Choose an integer $N \in \mathbb{N}$ with $N > N_0$. Then every integer $n \geq N$ has $n > N_0$, so

$$|s_{n+1}| = r_n \, |s_n| < q \, |s_n| \qquad (n \geq N).$$

The integer choice makes the starting index and the validity of every subsequent ratio inequality explicit.

(a) Prove the geometric estimate by induction.

For $n = N + 1$, the preceding inequality gives

$$|s_{N+1}| < q \, |s_N| = q^{(N+1)-N} \, |s_N| \, .$$

Assume for some integer $n > N$ that $|s_n| < q^{n-N} \, |s_N|$. Since the ratio estimate also applies at this n, multiplication by $q > 0$ gives

$$|s_{n+1}| < q \, |s_n| < q \, q^{n-N} \, |s_N| = q^{(n+1)-N} \, |s_N| \, .$$

Induction proves

$$|s_n| < q^{n-N} \, |s_N| \qquad (n > N).$$

Since $0 < q < 1$, Theorem 9.7(b) gives $q^m \to 0$ as $m \to \infty$. To spell out the use of this fact, let $\varepsilon > 0$. The number $C = |s_N|$ is positive, because $s_N \neq 0$. Choose a threshold K such that $m > K$ implies $q^m < \varepsilon / C$. If $n > N + \max\{K, 0\}$, then $m = n - N > K$ and $n > N$, so

$$|s_n - 0| < q^{n-N} C < \frac{\varepsilon}{C} C = \varepsilon.$$

This is precisely finite convergence to 0.

(b) Form the reciprocal sequence and compute its ratio.

Suppose $L > 1$, and define

$$t_n = \frac{1}{|s_n|} > 0.$$

It is defined and nonzero for every n. Its absolute ratio is

$$\begin{aligned} \left| \frac{t_{n+1}}{t_n} \right| &= \frac{1/\,|s_{n+1}|}{1/\,|s_n|} \\ &= \frac{|s_n|}{|s_{n+1}|} = \frac{1}{r_n}. \end{aligned}$$

There are two distinct cases for the ratio limit:

- If $1 < L < +\infty$, the finite reciprocal law applies to $r_n \to L$, since every $r_n \neq 0$ and $L \neq 0$. It gives $1/r_n \to 1/L$, with $0 < 1/L < 1$.
- If $L = +\infty$, every r_n is positive, so Theorem 9.10 gives $1/r_n \to 0$. This conclusion uses the infinite-limit reciprocal theorem, not the finite quotient law with an infinite denominator.

In either case, the absolute ratio for t_n has a finite limit strictly less than 1. Part (a) therefore proves $t_n \to 0$. Finally apply Theorem 9.10 to the positive sequence $|s_n|$: its reciprocal is $t_n \to 0$, so

$$|s_n| \longrightarrow +\infty.$$

Conclusion. If $L < 1$, the sequence itself converges to 0. If $L > 1$, including $L = +\infty$, its magnitudes tend to $+\infty$. No conclusion for the boundary value $L = 1$ is asserted by either part.

9.13 Exercise 9.13: All real bases in a geometric sequence

What we must show. Prove the complete classification of a^n as the fixed real base a varies, including $a = 0$, $a = 1$, $a = -1$, and both intervals of negative bases.

Plan. Use Exercise 9.12 where the terms are nonzero. For a negative base with magnitude at least 1, check that no finite limit and neither infinite limit can accommodate the alternating signs.

Case 1: $a = 0$, and then $0 < |a| < 1$.

If $a = 0$, every term a^n is 0 because $n \geq 1$. Thus the sequence converges to 0 directly. If $0 < |a| < 1$, every term is nonzero and

$$\left|\frac{a^{n+1}}{a^n}\right| = |a| < 1.$$

This constant ratio has limit $|a|$, so Exercise 9.12(a) proves $a^n \to 0$. The two cases together cover all $|a| < 1$, including negative a of magnitude less than 1.

Case 2: $a = 1$.

Every term equals $1^n = 1$. Thus for every $\varepsilon > 0$, $|a^n - 1| = 0 < \varepsilon$ at every index, giving limit 1.

Case 3: $a > 1$.

The terms a^n are positive and nonzero. Their absolute ratio is the constant $a > 1$. Exercise 9.12(b) gives $|a^n| \to +\infty$. Since $a^n > 0$, $|a^n| = a^n$, so the sequence itself tends to $+\infty$.

Case 4: $a = -1$.

The terms are 1 at even indices and -1 at odd indices. If there were a finite limit L, convergence with $\varepsilon = 1/2$ would put every sufficiently late term within $1/2$ of L. Choose a late even index p and a late odd index q. The triangle inequality would give

$$2 = |1 - (-1)| \leq |a^p - L| + |L - a^q| < \frac{1}{2} + \frac{1}{2} = 1,$$

a contradiction. A limit $+\infty$ fails the threshold $M = 1$ because late odd terms equal -1. A limit $-\infty$ fails the threshold $M = -1$ because late even terms equal 1. Thus no extended limit exists.

Case 5: $a < -1$.

Set $b = |a| = -a > 1$. By Case 3, $b^n \to +\infty$. The terms satisfy

$$a^n = (-1)^n b^n, \qquad |a^n| = b^n.$$

Suppose first that a^n had a finite limit L. Convergence with $\varepsilon = 1$ would give, eventually,

$$|a^n| \leq |a^n - L| + |L| < 1 + |L|\,.$$

But $b^n \to +\infty$ means that eventually $b^n > 1 + |L|$, contradicting $|a^n| = b^n$. Thus there is no finite limit.

There is also no limit $+\infty$, since for every threshold index there is a later odd index with $a^n < 0$, defeating the positive threshold $M = 1$. There is no limit $-\infty$, since there is always a later even index with $a^n > 0$, defeating the negative threshold $M = -1$. The required even and odd indices can be taken as $2m$ and $2m + 1$, where m is any positive integer larger than the proposed threshold index.

Conclusion. For every real a,

$$\lim_{n\to\infty} a^n = \begin{cases} 0, & |a| < 1, \\ 1, & a = 1, \\ +\infty, & a > 1, \\ \text{does not exist}, & a \le -1. \end{cases}$$

The last line excludes finite limits and both infinite limits.

9.14 Exercise 9.14: Geometric growth compared with a real power

What we must show. Fix $p > 0$ and determine the limit of a^n/n^p for every real a, using Exercise 9.12 where its hypotheses apply. In particular, include both boundary bases $a = 1$ and $a = -1$.

Plan. The absolute ratio is $|a|\ (n/(n+1))^p$. Prove its second factor tends to 1, then distinguish ratios below 1, above 1, and the boundary cases where the ratio test does not decide the answer.

1. State the real-power facts and justify the needed limit.

As in Theorem 9.7(a), positive bases with real exponents use the usual real-power algebra and order: products and quotients obey the exponent rules; and $r^v \le r^u$ whenever $0 < r < 1$ and $0 < u < v$. These are the elementary real-power properties already implicit in the statement of Theorem 9.7(a). We do not assume a continuity theorem for real powers from a later section.

Set $r_n = n/(n+1)$, so $0 < r_n < 1$. Choose an integer $m > p$, possible by the Archimedean property. Then

$$r_n^m \le r_n^p \le 1, \qquad 0 \le 1 - r_n^p \le 1 - r_n^m.$$

For the integer exponent m, finite multiplication gives the factorization

$$1 - r_n^m = (1 - r_n)(1 + r_n + \cdots + r_n^{m-1}).$$

Every one of the m terms in the parenthesized sum is at most 1, and $1 - r_n = 1/(n+1)$. Therefore

$$0 \le 1 - r_n^p \le 1 - r_n^m \le \frac{m}{n+1}.$$

The final expression tends to 0: for $\varepsilon > 0$, $n > m/\varepsilon$ makes $m/(n+1) < \varepsilon$. This direct error estimate proves

$$\left(\frac{n}{n+1}\right)^p \longrightarrow 1$$

for the possibly nonintegral exponent $p > 0$.

2. Compute the ratio when $a \neq 0$.

Let $u_n = a^n/n^p$. Since $n > 0$, its denominator is positive. If $a \neq 0$, all u_n are nonzero, and the real-power quotient rule yields

$$\left|\frac{u_{n+1}}{u_n}\right| = \left|\frac{a^{n+1}}{(n+1)^p} \cdot \frac{n^p}{a^n}\right| = |a|\ \frac{n^p}{(n+1)^p}$$

$$= |a| \left(\frac{n}{n+1} \right)^p.$$

Step 1 and the finite scalar-multiple limit theorem show that this ratio has limit $|a|$.

3. Treat $a = 0$ and the cases $0 < |a| < 1$.

For $a = 0$, every numerator is 0 and every denominator is positive, so $u_n = 0$ and the limit is 0. For $0 < |a| < 1$, Step 2 gives an absolute ratio limit less than 1, so Exercise 9.12(a) applies and gives $u_n \to 0$.

4. Treat the two boundary cases $|a| = 1$.

If $a = 1$ or $a = -1$, then $|a^n| = 1$, and hence

$$|u_n - 0| = \frac{1}{n^p}.$$

Theorem 9.7(a), whose hypothesis is precisely $p > 0$, says $1/n^p \to 0$. Thus $u_n \to 0$ in both boundary cases. The ratio limit here is 1; the conclusion comes from the direct error bound rather than from an inapplicable strict case of Exercise 9.12.

5. Treat $a > 1$.

The absolute ratio limit is $a > 1$, so Exercise 9.12(b) gives $|u_n| \to +\infty$. Since $a^n > 0$ and $n^p > 0$, the terms themselves are positive: $u_n = |u_n|$. Consequently $u_n \to +\infty$.

6. Treat $a < -1$ and exclude all three types of limit.

Exercise 9.12(b) gives $|u_n| \to +\infty$, since the absolute ratio limit is $|a| > 1$. If u_n converged to a finite L, then eventually

$$|u_n| \leq |u_n - L| + |L| < 1 + |L|$$

by the error bound $|u_n - L| < 1$. This contradicts $|u_n| \to +\infty$. At even indices $u_n > 0$, while at odd indices $u_n < 0$. Every tail has an odd index, so the positive-infinity test with $M = 1$ fails. Every tail also has an even index, so the negative-infinity test with $M = -1$ fails. Thus no extended limit exists in this last case.

Conclusion. For $p > 0$ and real a,

$$\lim_{n\to\infty} \frac{a^n}{n^p} = \begin{cases} 0, & |a| \leq 1, \\ +\infty, & a > 1, \\ \text{does not exist}, & a < -1. \end{cases}$$

In particular, the base $a = -1$ now gives limit 0, unlike the unscaled sequence in Exercise 9.13.

9.15 Exercise 9.15: Every fixed geometric sequence is dominated by a factorial

What we must show. Prove $a^n/n! \to 0$ for every $a \in \mathbb{R}$, including zero and negative values.

Plan. After the zero-base case is separated, the absolute ratio is $|a|/(n+1)$, which tends to 0. This gives the strict ratio condition from Exercise 9.12(a).

1. Handle $a = 0$ without dividing by a zero term.

When $a = 0$ and $n \geq 1$, $a^n = 0$. Since $n! > 0$, every term is 0, and the limit is 0.

2. Compute the absolute ratio for $a \neq 0$.

Fix a nonzero real a and write $s_n = a^n/n!$. Then every s_n is nonzero. By the factorial identity $(n+1)! = (n+1)n!$,

$$\begin{aligned}\left|\frac{s_{n+1}}{s_n}\right| &= \left|\frac{a^{n+1}}{(n+1)!} \cdot \frac{n!}{a^n}\right| \\ &= \left|\frac{a\, n!}{(n+1)n!}\right| \\ &= \frac{|a|}{n+1}.\end{aligned}$$

For $\varepsilon > 0$, choosing $N > |a|/\varepsilon$ makes $n > N$ imply $|a|/(n+1) < \varepsilon$. Thus the ratio limit is $L = 0 < 1$. Exercise 9.12(a) now proves $s_n \to 0$. Its hypothesis is about absolute ratios, so this argument applies to negative a as well as positive a.

Conclusion. For every real a, $\lim_{n\to\infty} a^n/n! = 0$.

9.16 Exercise 9.16: Three applications of the infinite-limit results

What we must show. Establish positive-infinity limits for all three given expressions, using the proved infinite-limit theorems or Exercises 9.9–9.15 with their hypotheses checked.

Plan. In (a), isolate a factor with limit $+\infty$ and another with a positive finite limit. In (b) and (c), use Exercise 9.14 for the growing term and then control the added term from below.

(a) The expression $(n^4 + 8n)/(n^2 + 9)$.

Every denominator $n^2 + 9$ is positive. Factor the expression exactly as

$$\frac{n^4 + 8n}{n^2 + 9} = n^2\,\frac{1 + 8/n^3}{1 + 9/n^2}.$$

Indeed, multiplying the numerator and denominator of the right-hand fraction by n^2 gives

$$n^2\,\frac{n^2 + 8/n}{n^2 + 9} = \frac{n^4 + 8n}{n^2 + 9}.$$

Theorem 9.7(a) gives $1/n^3 \to 0$ and $1/n^2 \to 0$. The finite scalar and sum laws therefore give $1 + 8/n^3 \to 1$ and $1 + 9/n^2 \to 1$. All the latter denominators are positive, and the limit denominator is $1 \neq 0$, so the finite quotient theorem applies to this fraction and gives

$$v_n = \frac{1 + 8/n^3}{1 + 9/n^2} \longrightarrow 1 > 0.$$

Also $n^2 \to +\infty$: for each $M > 0$, every $n > \sqrt{M}$ has $n^2 > M$. Thus Theorem 9.9 applies to $s_n = n^2$ and $t_n = v_n$, since its first limit is $+\infty$ and its second limit is the positive number 1. It gives the desired product limit.

Conclusion. In part (a), $\lim_{n\to\infty} \dfrac{n^4 + 8n}{n^2 + 9} = +\infty$.

(b) The expression $2^n/n^2 + (-1)^n$.

Exercise 9.14 with $a = 2 > 1$ and $p = 2 > 0$ gives

$$u_n = \frac{2^n}{n^2} \longrightarrow +\infty.$$

The second sequence is bounded: $-1 \leq (-1)^n \leq 1$ for every n. Exercise 9.11(c) therefore applies to the sum. In explicit threshold form, given $M > 0$, choose N so that $n > N$ implies $u_n > M + 1$. Then

$$\frac{2^n}{n^2} + (-1)^n \geq u_n - 1 > M.$$

This verifies the limit without requiring the oscillating sequence $(-1)^n$ itself to have a limit.

Conclusion. In part (b), $\lim_{n\to\infty} \left[\frac{2^n}{n^2} + (-1)^n\right] = +\infty$.

(c) The expression $3^n/n^3 - 3^n/n!$.

Exercise 9.14 with $a = 3$, $p = 3$ and Exercise 9.15 with $a = 3$ give, respectively,

$$u_n = \frac{3^n}{n^3} \longrightarrow +\infty, \qquad v_n = \frac{3^n}{n!} \longrightarrow 0.$$

Thus $-v_n \to 0$ by the finite scalar-multiple theorem. Exercise 9.11(b) applies to $u_n + (-v_n)$, since the second limit 0 is greater than $-\infty$. To verify the threshold directly, fix $M > 0$. Choose N_1 such that $n > N_1$ implies $u_n > M + 1$. Choose N_2 such that $n > N_2$ implies $|v_n| < 1$, using its convergence to 0. For $n > \max\{N_1, N_2\}$ we have $v_n < 1$ and hence

$$u_n - v_n > u_n - 1 > M.$$

This is a growing term minus a term tending to 0; the proof does not subtract two infinite limits.

Conclusion. In part (c), $\lim_{n\to\infty} \left[\frac{3^n}{n^3} - \frac{3^n}{n!}\right] = +\infty$.

9.17 Exercise 9.17: A definition-only proof for the sequence of squares

What we must show. Use only Definition 9.8 to prove $n^2 \to +\infty$.

Plan. Given the positive height M, solve $n^2 > M$ by requiring $n > \sqrt{M}$. Both sides are positive, so squaring this comparison preserves its direction.

Formal proof with the quantifiers explicit.

Let $M > 0$ be arbitrary, and set $N = \sqrt{M}$. If $n \in \mathbb{N}$ and $n > N$, then $n > \sqrt{M} > 0$. Multiplying positive inequalities, or using the strictly increasing square function on the positive numbers, gives

$$n^2 > (\sqrt{M})^2 = M.$$

More explicitly, $n > \sqrt{M}$ gives $n^2 > n\sqrt{M} > (\sqrt{M})^2$ after multiplication first by $n > 0$ and then by $\sqrt{M} > 0$. We have therefore proved that for every $M > 0$ there is $N = \sqrt{M}$ such that every positive integer $n > N$ satisfies $n^2 > M$. This is exactly Definition 9.8.

Conclusion. $\lim_{n\to\infty} n^2 = +\infty$, with the explicit threshold $N = \sqrt{M}$ for each $M > 0$.

9.18 Exercise 9.18: Finite geometric sums and their limits

What we must show. Prove the finite-sum identity in (a), then determine the limits in (b), (c), and (d), including the base $a = 1$ where the identity's quotient is not defined.

Plan. Subtract a times the finite sum from the original sum. Every interior power cancels. After the identity is proved, use the established geometric-sequence limit for $|a| < 1$, and a direct lower comparison for $a \geq 1$.

(a) Prove the identity for every allowed base and every index.

Fix a real $a \neq 1$ and a positive integer n, and define

$$S_n = 1 + a + a^2 + \cdots + a^n = \sum_{k=0}^{n} a^k.$$

Here a^0 denotes the constant empty product 1, including when $a = 0$. Multiplication and reindexing a finite sum give

$$aS_n = \sum_{k=0}^{n} a^{k+1} = \sum_{j=1}^{n+1} a^j = a + a^2 + \cdots + a^n + a^{n+1}.$$

The reindexing sets $j = k + 1$, so the lower bound $k = 0$ becomes $j = 1$ and the upper bound $k = n$ becomes $j = n + 1$. Subtracting the two finite sums cancels exactly the terms $a, a^2, \ldots, a^n$:

$$\begin{aligned} S_n - aS_n &= (1 + a + \cdots + a^n) - (a + \cdots + a^n + a^{n+1}) \\ &= 1 - a^{n+1}. \end{aligned}$$

The left side factors as $(1 - a)S_n$. Since $a \neq 1$, $1 - a \neq 0$, and division by this nonzero number yields

$$S_n = \frac{1 - a^{n+1}}{1 - a}.$$

The base and index were arbitrary, so this proves the identity for every real $a \neq 1$ and every positive integer n. The cancellation also works for $n = 0$, if that index is included, since then the identity is $1 = (1 - a)/(1 - a)$.

(b) Pass to the limit for $|a| < 1$.

Now fix a with $|a| < 1$. Then $a \neq 1$, so (a) applies. Exercise 9.13 gives $a^n \to 0$. Since $a^{n+1} = a\,a^n$, multiplication by the fixed real number a gives $a^{n+1} \to a \cdot 0 = 0$ by the finite scalar-multiple theorem. The denominator $1 - a$ is a fixed nonzero real number. Consequently the finite sum and quotient laws applied to the identity give

$$\lim_{n\to\infty} S_n = \frac{1 - 0}{1 - a} = \frac{1}{1 - a}.$$

One can read the same conclusion directly as an error estimate, avoiding any ambiguity about division:

$$\begin{aligned} \left| S_n - \frac{1}{1 - a} \right| &= \left| \frac{1 - a^{n+1} - 1}{1 - a} \right| \\ &= \frac{|a|^{n+1}}{|1 - a|} \longrightarrow 0. \end{aligned}$$

The denominator in this expression is positive and constant; the numerator tends to 0. This includes $a = 0$, when every sum is exactly 1.

Conclusion. For part (b), if $|a| < 1$, then $\lim_{n\to\infty}(1 + a + \cdots + a^n) = 1/(1 - a)$.

(c) Substitute $a = 1/3$ and simplify the fraction.

The displayed sequence of sums is

$$1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \cdots + \frac{1}{3^n} = \sum_{k=0}^{n} \left(\frac{1}{3}\right)^k.$$

Since $0 < 1/3 < 1$, part (b) applies. Its limit is

$$\frac{1}{1 - 1/3} = \frac{1}{(3 - 1)/3} = \frac{1}{2/3} = \frac{3}{2}.$$

Conclusion. The limit in part (c) is $3/2$.

(d) Handle $a = 1$ and $a > 1$ explicitly.

If $a = 1$, the sum contains $n + 1$ terms, each equal to 1, so $S_n = n + 1$. For an arbitrary $M > 0$, $n > M$ implies $S_n = n + 1 > M$. Therefore $S_n \to +\infty$ in this case. The quotient in (a) was not used at $a = 1$, where its denominator would be zero.

If $a > 1$, every term a^k for $k = 0, 1, \ldots, n$ is at least 1: the exponent 0 gives 1, and a positive exponent gives a product of factors greater than 1. Adding the $n + 1$ inequalities gives

$$S_n = \sum_{k=0}^{n} a^k \geq \sum_{k=0}^{n} 1 = n + 1.$$

Since $n + 1 \to +\infty$, Exercise 9.9(a) proves $S_n \to +\infty$. Equivalently, the same threshold $N = M$ works directly: $n > N$ implies $S_n \geq n + 1 > M$.

Conclusion. For part (d), every base $a \geq 1$ gives $\lim_{n\to\infty}(1 + a + \cdots + a^n) = +\infty$.

10 Monotone sequences and Cauchy sequences

An increasing sequence satisfies $s_n \le s_{n+1}$; a decreasing sequence satisfies $s_n \ge s_{n+1}$. Both definitions permit equality. A sequence is bounded if all its terms lie between two fixed real numbers.

The Cauchy condition says that, for every $\varepsilon > 0$, there is one threshold N such that

$$m > N,\ n > N \quad \Longrightarrow \quad |s_m - s_n| < \varepsilon.$$

It controls every pair of late terms. Merely making successive differences small is not enough, as Exercise 10.6 explains. The solutions first justify convergence and then identify a limit.

Conventions for monotonicity and Cauchy estimates

In Definition 10.1, "increasing" means $s_n \le s_{n+1}$ and "decreasing" means $s_n \ge s_{n+1}$. Equality is permitted in both definitions. A constant sequence therefore has both properties. Repeated application of an increasing inequality gives $s_n \le s_m$ whenever $n < m$; for a decreasing sequence it gives $s_n \ge s_m$.

Definition 10.8 says that (s_n) is Cauchy if

> for every $\varepsilon > 0$ there is an N such that every pair of integers $m, n > N$ satisfies $|s_m - s_n| < \varepsilon$.

Both indices are arbitrary after the same threshold. An estimate for consecutive terms alone is not yet the estimate for an arbitrary pair.

10.1 Exercise 10.1: Classifying six sequences

What we must show. Decide whether each sequence is increasing, decreasing, and bounded, with the weak monotonicity conventions just stated.

Part (a): the sequence $s_n = 1/n$.

For every $n \in \mathbb{N}$, the denominators n and $n+1$ are positive, and

$$s_{n+1} - s_n = \frac{1}{n+1} - \frac{1}{n} = \frac{n-(n+1)}{n(n+1)} = -\frac{1}{n(n+1)} < 0.$$

Thus $s_{n+1} < s_n$ for every n, so the sequence is decreasing. It is not increasing: for example, $s_2 = 1/2 < 1 = s_1$. Because $n \ge 1$, we also have $0 < 1/n \le 1$, so 0 and 1 are a lower and an upper bound.

Conclusion. The sequence $1/n$ is **decreasing and bounded**, and is not increasing.

Part (b), Step 1: test monotonicity with exact terms.

Here $s_n = (-1)^n/n^2$. Its first three terms are

$$s_1 = -1, \qquad s_2 = \frac{1}{4}, \qquad s_3 = -\frac{1}{9}.$$

The inequality $s_1 < s_2$ rules out a decreasing sequence, which would require $s_1 \ge s_2$. The inequality $s_2 > s_3$ rules out an increasing sequence, which would require $s_2 \le s_3$. Hence it is neither.

Part (b), Step 2: give bounds valid at every index.

If n is odd, then $s_n = -1/n^2$, so $-1 \le s_n < 0$. If n is even, then $n \ge 2$ and $s_n = 1/n^2$, so $0 < s_n \le 1/4$. Both cases give

$$-1 \le s_n \le \frac{1}{4} \qquad (n \in \mathbb{N}).$$

These bounds are attained at $n = 1$ and $n = 2$, respectively.

Conclusion. The sequence $(-1)^n/n^2$ is **bounded but neither increasing nor decreasing**.

Part (c), Step 1: check the growth of $s_n = n^5$.

For every $n \ge 1$, the binomial expansion gives

$$\begin{aligned} s_{n+1} - s_n &= (n+1)^5 - n^5, \\ &= n^5 + 5n^4 + 10n^3 + 10n^2 + 5n + 1 - n^5, \\ &= 5n^4 + 10n^3 + 10n^2 + 5n + 1 > 0. \end{aligned}$$

All five terms in the final sum are positive. Thus $s_{n+1} > s_n$ for every index, giving an increasing sequence and ruling out a decreasing one.

Part (c), Step 2: prove unboundedness above.

Since $n \ge 1$, we have $n^4 \ge 1$, so $n^5 \ge n \ge 1$. The number 1 is a lower bound. To show there is no upper bound, let $U \in \mathbb{R}$ be any proposed upper bound. By the Archimedean property, choose $n \in \mathbb{N}$ with $n > \max\{1, U\}$. Then

$$s_n = n^5 \ge n > U.$$

Thus U fails to bound the sequence above. This works for every real candidate U.

Conclusion. The sequence n^5 is **increasing and unbounded above**. It is not decreasing or bounded, although it is bounded below by 1.

Part (d), Step 1: give a rise and a fall for the sine sequence.

Let $s_n = \sin(n\pi/7)$. Put $c = \sin(\pi/7)$. Since $0 < \pi/7 < \pi$, the elementary sign property of sine gives $c > 0$. The sine values at three indices are

$$s_1 = c > 0, \qquad s_7 = \sin \pi = 0, \qquad s_{15} = \sin(2\pi + \pi/7) = c > 0.$$

An increasing sequence would satisfy $s_1 \le s_7$, contradicting $s_1 > 0 = s_7$. A decreasing sequence would satisfy $s_7 \ge s_{15}$, contradicting $s_7 = 0 < s_{15}$. Comparisons at nonadjacent indices are legitimate because monotonicity implies the corresponding comparison at any pair of ordered indices.

Part (d), Step 2: give global bounds.

The elementary trigonometric inequality $-1 \le \sin x \le 1$ is valid for every real x. Applying it at $x = n\pi/7$ gives $-1 \le s_n \le 1$ for every n.

Conclusion. The sequence $\sin(n\pi/7)$ is **bounded but neither increasing nor decreasing**.

Part (e), Step 1: rule out both monotonicity directions.

For $s_n = (-2)^n$, the first three terms are

$$s_1 = -2, \qquad s_2 = 4, \qquad s_3 = -8.$$

The rise from -2 to 4 rules out a decreasing sequence. The fall from 4 to -8 rules out an increasing sequence.

Part (e), Step 2: record an elementary power estimate.

We will use

$$2^j \geq j+1 \qquad (j \in \mathbb{N}).$$

For $j = 1$, this is $2 \geq 2$. If it holds for $j \geq 1$, then

$$2^{j+1} = 2 \cdot 2^j \geq 2(j+1) = 2j+2 \geq j+2,$$

since $(2j+2) - (j+2) = j \geq 0$. Induction proves the estimate.

Part (e), Step 3: show that both signs have arbitrarily large magnitude.

Let $R > 0$ be arbitrary. Choose $k \in \mathbb{N}$ with $k > R$ by the Archimedean property. At an even index,

$$s_{2k} = 2^{2k} \geq 2k+1 > R.$$

At the following odd index,

$$s_{2k+1} = -2^{2k+1} \leq -(2k+2) < -R.$$

Thus there are terms above R and below $-R$ for every positive R. In particular, no real number is an upper bound and no real number is a lower bound.

Conclusion. The sequence $(-2)^n$ is **neither increasing nor decreasing, and is unbounded both above and below**.

Part (f), Step 1: compare consecutive positive terms.

Here $s_n = n/3^n > 0$. Since we are dividing positive quantities, we can compute the consecutive-term ratio:

$$\begin{aligned} \frac{s_{n+1}}{s_n} &= \frac{n+1}{3^{n+1}} \frac{3^n}{n}, \\ &= \frac{n+1}{3n}. \end{aligned}$$

For $n \geq 1$, the denominator is positive and $3n-(n+1) = 2n-1 \geq 1 > 0$. Hence $0 < (n+1)/(3n) < 1$. Multiplying this strict upper inequality by $s_n > 0$ gives $s_{n+1} < s_n$.

Part (f), Step 2: bound the whole sequence.

Every term is positive. Repeated application of the decreasing inequality gives $s_n \leq s_1 = 1/3$. Therefore

$$0 < s_n \leq \frac{1}{3} \qquad (n \in \mathbb{N}).$$

The strict decrease also rules out an increasing sequence.

Conclusion. The sequence $n/3^n$ is **decreasing and bounded**, and is not increasing.

Summary of all six classifications.

Sequence	Increasing?	Decreasing?	Bounded?
$1/n$	No	Yes	Yes
$(-1)^n/n^2$	No	No	Yes
n^5	Yes	No	No
$\sin(n\pi/7)$	No	No	Yes
$(-2)^n$	No	No	No
$n/3^n$	No	Yes	Yes

10.2 Exercise 10.2: Convergence of a bounded decreasing sequence

What we must show. Prove the decreasing case of Theorem 10.2: every bounded decreasing real sequence converges.

Step 1: define the candidate limit and justify its existence.

Let (s_n) be bounded and decreasing, and let

$$S = \{s_n : n \in \mathbb{N}\}.$$

The set is nonempty because $s_1 \in S$, and it is bounded because the sequence is bounded. In particular, it is bounded below. The lower-bound form of completeness gives a real number

$$L = \inf S.$$

Being a lower bound means $L \leq s_n$ for every n.

Step 2: obtain one term close to the infimum.

Let $\varepsilon > 0$. The number $L + \varepsilon$ is strictly larger than the greatest lower bound L, so it cannot itself be a lower bound of S. Therefore there is an element of S below $L + \varepsilon$. By the definition of S, that element is s_N for some index $N \in \mathbb{N}$. Thus

$$L \leq s_N < L + \varepsilon.$$

Step 3: use decrease to control every subsequent term.

Take any integer $n > N$. Since the sequence is decreasing, $s_n \leq s_N$. Since L is a lower bound, $L \leq s_n$. Combining these facts with Step 2 gives

$$L \leq s_n \leq s_N < L + \varepsilon.$$

Subtract L throughout to get $0 \leq s_n - L < \varepsilon$. The difference $s_n - L$ is nonnegative, so

$$|s_n - L| = s_n - L < \varepsilon \qquad \text{for every integer } n > N.$$

Conclusion. For every $\varepsilon > 0$ we obtained the threshold required by Definition 7.1. Hence $\boxed{s_n \longrightarrow \inf\{s_n : n \in \mathbb{N}\}}$. The proof uses the greatest-lower-bound property, without assuming the infimum is one of the terms.

10.3 Exercise 10.3: A finite bound for decimal approximations

What we must show. Let K be a nonnegative integer and let each digit d_j belong to $\{0, 1, \ldots, 9\}$. For

$$s_n = K + \sum_{j=1}^{n} \frac{d_j}{10^j},$$

prove $s_n < K + 1$ for every $n \in \mathbb{N}$.

Step 1: bound each digit contribution.

For every $j \geq 1$, $0 \leq d_j \leq 9$ and $10^j > 0$. Division by the positive denominator gives

$$0 \leq \frac{d_j}{10^j} \leq \frac{9}{10^j}.$$

Add these finitely many inequalities and then add K:

$$K \leq s_n \leq K + \sum_{j=1}^{n} \frac{9}{10^j}.$$

Step 2: evaluate the finite upper-bound sum by cancellation.

For an individual index j,

$$\frac{9}{10^j} = \frac{10-1}{10^j} = \frac{1}{10^{j-1}} - \frac{1}{10^j}.$$

Consequently

$$\begin{aligned} \sum_{j=1}^{n} \frac{9}{10^j} &= \left(1 - \frac{1}{10}\right) + \left(\frac{1}{10} - \frac{1}{10^2}\right) + \cdots + \left(\frac{1}{10^{n-1}} - \frac{1}{10^n}\right), \\ &= 1 - \frac{1}{10^n}. \end{aligned}$$

Every intermediate negative term cancels the identical positive term in the next parentheses. This is a finite algebraic identity; no limit or infinite sum has been used to derive it.

Step 3: preserve the strict final inequality.

Substitute the identity from Step 2 into Step 1:

$$K \leq s_n \leq K + 1 - \frac{1}{10^n}.$$

Since $10^n > 0$, its reciprocal is strictly positive. Subtracting that positive number makes $K+1-1/10^n$ strictly less than $K + 1$. Thus

$$s_n \leq K + 1 - \frac{1}{10^n} < K + 1.$$

Conclusion. Every finite decimal approximation satisfies $\boxed{s_n < K + 1}$. Moreover $s_{n+1} - s_n = d_{n+1}/10^{n+1} \geq 0$, so the approximations are increasing and bounded. Their convergence follows from Theorem 10.2 *after* the finite bound has been established.

10.4 Exercise 10.4: Why the monotone and Cauchy theorems require real completeness

What we must show. Explain why Theorems 10.2 and 10.11 would be false if the permitted terms and limits were restricted to $\mathbb{Q}$.

Plan. We construct a sequence of rational decimal truncations to $\sqrt{2}$. It is increasing, bounded by rational numbers, and Cauchy, but its only real limit is the irrational number $\sqrt{2}$. The construction itself uses finite sets and rational comparisons, rather than assuming that $\mathbb{Q}$ has the completeness property being tested.

Step 1: construct each truncation using integers only.

For each $n \in \mathbb{N}$, consider the finite set

$$E_n = \{k \in \mathbb{Z} : 0 \le k \le 2 \cdot 10^n,\ k^2 < 2 \cdot 10^{2n}\}.$$

It is nonempty because $0 \in E_n$. A finite nonempty set of integers has a largest member, so define

$$k_n = \max E_n, \qquad q_n = \frac{k_n}{10^n}.$$

The denominator is a positive integer, and the numerator is an integer. Hence every q_n is rational. The operations and comparisons used to define E_n involve only integers. In particular, this construction does not presuppose a supremum in $\mathbb{Q}$ for an infinite set.

Step 2: prove rational lower and upper bounds.

The integer 10^n belongs to E_n, since it is between 0 and $2 \cdot 10^n$ and its square 10^{2n} is strictly less than $2 \cdot 10^{2n}$. Thus $k_n \ge 10^n$. The integer $2 \cdot 10^n$ does not belong to E_n, since its square is $4 \cdot 10^{2n} > 2 \cdot 10^{2n}$. Therefore $k_n < 2 \cdot 10^n$. Dividing by $10^n > 0$ gives

$$1 \le q_n < 2.$$

Both bounds are rational, so the sequence is bounded even when all bounding numbers are required to lie in $\mathbb{Q}$.

Step 3: prove that the sequence is increasing.

Since $k_n^2 < 2 \cdot 10^{2n}$, multiplication by 100 gives

$$(10k_n)^2 < 2 \cdot 10^{2(n+1)}.$$

Also $0 \le 10k_n \le 2 \cdot 10^{n+1}$. Thus $10k_n \in E_{n+1}$. The maximality of k_{n+1} implies $10k_n \le k_{n+1}$. Dividing by $10^{n+1} > 0$ yields

$$q_n = \frac{10k_n}{10^{n+1}} \le \frac{k_{n+1}}{10^{n+1}} = q_{n+1}.$$

Equality may occur at some indices, which is permitted by the definition of increasing.

Step 4: give a rational upper approximation to the same square.

The integer $k_n + 1$ is still at most $2 \cdot 10^n$, because $k_n < 2 \cdot 10^n$ and both numbers are integers. If $(k_n + 1)^2 < 2 \cdot 10^{2n}$, then $k_n + 1$ would belong to E_n, contradicting maximality. Therefore

$$k_n^2 < 2 \cdot 10^{2n} \le (k_n + 1)^2.$$

Divide by $10^{2n} > 0$. The result is

$$q_n^2 < 2 \leq \left(q_n + \frac{1}{10^n}\right)^2.$$

Every number in this display is rational, so the comparisons remain meaningful entirely within $\mathbb{Q}$.

Step 5: verify the Cauchy condition through rational estimates.

For integers $m > n$, Step 3 gives $q_n \leq q_m$. The positive numbers q_m and $q_n + 10^{-n}$ satisfy

$$q_m^2 < 2 \leq (q_n + 10^{-n})^2.$$

Hence $q_m < q_n + 10^{-n}$. Indeed, the contrary inequality $q_m \geq q_n + 10^{-n} \geq 0$ would imply the contrary comparison between their squares. We conclude

$$0 \leq q_m - q_n < 10^{-n} \qquad (m > n).$$

We may use $10^j \geq j$ for every $j \in \mathbb{N}$: the case $j = 1$ is $10 \geq 1$; if $10^j \geq j$, then $10^{j+1} \geq 10j \geq j + 1$ because $j \geq 1$. This proves the estimate by induction, and taking reciprocals gives $10^{-j} \leq 1/j$.

Now let $\varepsilon > 0$ and choose $N = \max\{1, 1/\varepsilon\}$. Take any two integer indices $m, n > N$. If $m = n$, then $|q_m - q_n| = 0 < \varepsilon$. If $m > n$, the preceding estimate gives

$$|q_m - q_n| = q_m - q_n < 10^{-n} \leq \frac{1}{n} < \varepsilon.$$

If $n > m$, interchange their roles and use the symmetry of absolute value to obtain $|q_m - q_n| < 10^{-m} \leq 1/m < \varepsilon$. Thus the Cauchy estimate holds for every pair of indices above N. In particular, it holds for every positive rational tolerance, as a Cauchy condition formulated within $\mathbb{Q}$ requires.

Step 6: identify the only possible real limit.

In the ambient real numbers, let $\alpha = \sqrt{2}$. It is positive, $\alpha^2 = 2$, and it is irrational by Section 2. Step 4 and comparison of squares of positive numbers give

$$q_n < \alpha \leq q_n + 10^{-n}.$$

The upper equality is impossible because $q_n + 10^{-n}$ is rational and α is irrational. Therefore

$$0 < \alpha - q_n < 10^{-n} \leq \frac{1}{n}.$$

For any real $\varepsilon > 0$, the choice $N = \max\{1, 1/\varepsilon\}$ makes the right side strictly smaller than ε whenever $n > N$. Hence Definition 7.1 gives $q_n \to \alpha$ as a sequence in $\mathbb{R}$.

Step 7: rule out every rational limit.

A limit in $\mathbb{Q}$ would have to be some $r \in \mathbb{Q}$. Restricting the definition to positive rational tolerances makes no difference: given any real tolerance $\varepsilon > 0$, the Archimedean property gives a positive rational $1/j < \varepsilon$, and the estimate for that smaller rational tolerance implies the estimate for ε. Thus a limit in $\mathbb{Q}$ would also be the same limit in $\mathbb{R}$.

Because r is rational and α is irrational, they are distinct. Put $d = |\alpha - r| > 0$. If q_n converged to both numbers, there would be thresholds N_1, N_2 such that, for $n > \max\{N_1, N_2\}$,

$$|q_n - \alpha| < \frac{d}{3}, \qquad |q_n - r| < \frac{d}{3}.$$

The triangle inequality would then give

$$d = |\alpha - r| \leq |\alpha - q_n| + |q_n - r| < \frac{2d}{3} < d,$$

a contradiction. Thus the rational sequence has no rational limit.

Conclusion. The same rational sequence is bounded and increasing, so it contradicts the version of Theorem 10.2 with limits restricted to $\mathbb{Q}$. It is also Cauchy, so it contradicts the corresponding version of Theorem 10.11. Both failures arise because the required limit $\sqrt{2}$ is missing from $\mathbb{Q}$, not because the sequence is unbounded or fails its Cauchy estimates.

10.5 Exercise 10.5: An unbounded decreasing sequence tends to negative infinity

What we must show. Prove Theorem 10.4(ii): if a real sequence (s_n) is decreasing and unbounded, then $s_n \to -\infty$.

Step 1: determine which direction must be unbounded.

Since the sequence is decreasing, every term satisfies $s_n \leq s_1$. Thus s_1 is an upper bound of its set of terms. If the sequence also had a real lower bound, it would be bounded above and below, contrary to the hypothesis that it is unbounded. Therefore the sequence is unbounded below.

Step 2: find a term below an arbitrary target.

Let $M < 0$ be any real target. Since the sequence is unbounded below, M cannot be a lower bound for all its terms. Hence some index $N \in \mathbb{N}$ satisfies

$$s_N < M.$$

Step 3: show that every later term remains below that target.

For every integer $n > N$, decrease gives $s_n \leq s_N$. Thus

$$s_n \leq s_N < M \qquad (n > N).$$

We have proved that for each $M < 0$ there exists a threshold N such that every later term is less than M.

Conclusion. This is the definition of divergence to negative infinity, so $\boxed{s_n \longrightarrow -\infty}$. The choice of s_N would in fact work for any real target M, not just negative ones.

10.6 Exercise 10.6: Small successive differences: one sufficient rate and one insufficient rate

What we must show. Prove (a) that $|s_{n+1} - s_n| < 2^{-n}$ for every natural n implies (s_n) is Cauchy and convergent; and (b) decide whether the same conclusion follows from the weaker estimate $|s_{n+1} - s_n| < 1/n$.

Part (a), Step 1: compare arbitrary terms by telescoping.

Fix integers $m > n$. Then

$$s_m - s_n = \sum_{k=n}^{m-1} (s_{k+1} - s_k).$$

To see the cancellation, expand the right side as $(s_{n+1} - s_n) + (s_{n+2} - s_{n+1}) + \cdots + (s_m - s_{m-1})$; every intermediate term occurs once with each sign. Applying the finite triangle inequality gives

$$|s_m - s_n| \leq \sum_{k=n}^{m-1} |s_{k+1} - s_k| < \sum_{k=n}^{m-1} 2^{-k}.$$

The final inequality is strict because the finite sum is nonempty and every one of its summands is strictly less than the corresponding 2^{-k}.

Part (a), Step 2: bound this finite geometric sum.

For an individual k, $2^{-k} = 2^{1-k} - 2^{-k}$. Substituting that identity into the sum makes the consecutive terms telescope:

$$\begin{aligned}\sum_{k=n}^{m-1} 2^{-k} &= (2^{1-n} - 2^{-n}) + (2^{-n} - 2^{-n-1}) + \cdots + (2^{2-m} - 2^{1-m}),\\ &= 2^{1-n} - 2^{1-m},\\ &< 2^{1-n}.\end{aligned}$$

The last inequality holds because $2^{1-m} > 0$. We have therefore shown

$$|s_m - s_n| < 2^{1-n} \qquad (m > n).$$

Part (a), Step 3: choose one threshold for all pairs.

Let $\varepsilon > 0$. By the Archimedean property choose $N \in \mathbb{N}$ such that

$$N > \max\left\{1, \frac{2}{\varepsilon}\right\}.$$

The elementary estimate $2^N \geq N + 1$ was proved in Exercise 10.1(e). It gives

$$2^{1-N} = \frac{2}{2^N} \leq \frac{2}{N+1} < \frac{2}{N} < \varepsilon.$$

Now let $m, n > N$ be arbitrary integer indices. If $m = n$, their terms have distance $0 < \varepsilon$. If $m > n$, Steps 1–2 give

$$|s_m - s_n| < 2^{1-n} \leq 2^{1-N} < \varepsilon.$$

If $n > m$, interchange the indices and use $|s_m - s_n| = |s_n - s_m|$ to get the same conclusion with 2^{1-m} in place of 2^{1-n}.

Conclusion. For every $\varepsilon > 0$ all pairs of terms after the chosen threshold have distance less than ε. Thus (s_n) is Cauchy by Definition 10.8, and Theorem 10.11 for real sequences implies that it converges to a real number.

Part (b), Step 1: construct a sequence satisfying the strict hypothesis.

Define the harmonic partial sums

$$H_n = \sum_{k=1}^{n} \frac{1}{k} = 1 + \frac{1}{2} + \cdots + \frac{1}{n}.$$

The consecutive difference is exactly

$$H_{n+1} - H_n = \frac{1}{n+1}.$$

Both sides are positive, and $n + 1 > n > 0$, so

$$|H_{n+1} - H_n| = \frac{1}{n+1} < \frac{1}{n} \qquad (n \in \mathbb{N}).$$

The required inequality is strict at every index, including $n = 1$.

Part (b), Step 2: show directly that the sequence is not Cauchy.

Fix the positive tolerance $\varepsilon_0 = 1/2$. Given any proposed threshold N, choose an integer $n > \max\{N, 1\}$ and put $m = 2n$. Both indices are greater than N. The difference of the terms is

$$H_{2n} - H_n = \sum_{k=n+1}^{2n} \frac{1}{k}.$$

There are exactly n summands. For each index in this sum, $k \leq 2n$, so $1/k \geq 1/(2n)$. Adding the n inequalities gives

$$|H_{2n} - H_n| = H_{2n} - H_n \geq n\frac{1}{2n} = \frac{1}{2} = \varepsilon_0.$$

Thus, no matter how N is chosen, some pair of indices $m, n > N$ violates the required strict Cauchy estimate.

Conclusion. The answer to part (b) is **no**. The sequence (H_n) satisfies every strict bound $|H_{n+1} - H_n| < 1/n$, but it is not Cauchy. Since every convergent sequence is Cauchy by Lemma 10.9, it does not converge.

10.7 Exercise 10.7: Approaching an unattained supremum by points of the set

What we must show. Let $S \subseteq \mathbb{R}$ be bounded and nonempty, and suppose $\sup S \notin S$. Construct a sequence (s_n) with $s_n \in S$ for every n and $s_n \to \sup S$.

Step 1: use the least-upper-bound property at each precision.

Completeness gives the real number $M = \sup S$. For each $n \in \mathbb{N}$, we have $1/n > 0$, and consequently $M - 1/n < M$. The smaller number $M - 1/n$ cannot be an upper bound of S, because M is its least upper bound. Therefore there is a point $s_n \in S$ satisfying $M - 1/n < s_n$. Since M is an upper bound, also $s_n \leq M$. Equality is ruled out by $M \notin S$. Thus, choosing such a point for each n, we obtain

$$M - \frac{1}{n} < s_n < M \qquad (n \in \mathbb{N}).$$

Step 2: translate the order estimate into an error estimate.

Subtracting s_n and rearranging the left inequality gives $M - s_n < 1/n$. The right inequality gives $M - s_n > 0$, so

$$|s_n - M| = M - s_n < \frac{1}{n}.$$

This bound holds at every index; the actual choices of the points s_n need not produce a monotone sequence.

Step 3: verify convergence with an explicit threshold.

Let $\varepsilon > 0$ and choose

$$N = \max\left\{1, \frac{1}{\varepsilon}\right\}.$$

For every integer $n > N$, $n > 1/\varepsilon$ gives $1/n < \varepsilon$, and therefore

$$|s_n - M| < \frac{1}{n} < \varepsilon.$$

Conclusion. The constructed sequence consists entirely of points of S and satisfies $\boxed{s_n \longrightarrow \sup S}$. The supremum itself is excluded from the set, so each approximation is strictly below its target.

10.8 Exercise 10.8: Averages of an increasing positive sequence

What we must show. Let (s_n) be increasing with $s_n > 0$ for every n, and let

$$\sigma_n = \frac{1}{n}\sum_{k=1}^{n} s_k.$$

Prove that the average sequence (σ_n) is increasing in the sense of Definition 10.1.

Step 1: compare each average with the last term it averages.

Fix $n \in \mathbb{N}$. Since (s_n) is increasing, every k with $1 \le k \le n$ satisfies $s_k \le s_n$. Add these n inequalities:

$$s_1 + s_2 + \cdots + s_n \le \underbrace{s_n + s_n + \cdots + s_n}_{n \text{ terms}} = ns_n.$$

Divide by the positive integer n to obtain $\sigma_n \le s_n$. Since $s_n \le s_{n+1}$, transitivity gives

$$\sigma_n \le s_n \le s_{n+1}.$$

In particular, $s_{n+1} - \sigma_n \ge 0$.

Step 2: express the next average in terms of the previous one.

The definition of σ_n gives $s_1 + \cdots + s_n = n\sigma_n$. Therefore

$$\begin{aligned}\sigma_{n+1} &= \frac{s_1 + \cdots + s_n + s_{n+1}}{n+1},\\ &= \frac{n\sigma_n + s_{n+1}}{n+1}.\end{aligned}$$

Step 3: calculate the exact difference between consecutive averages.

Put the terms over the same positive denominator $n+1$:

$$\begin{aligned}\sigma_{n+1}-\sigma_n &= \frac{n\sigma_n+s_{n+1}}{n+1}-\frac{(n+1)\sigma_n}{n+1},\\ &= \frac{n\sigma_n+s_{n+1}-(n+1)\sigma_n}{n+1},\\ &= \frac{n\sigma_n+s_{n+1}-n\sigma_n-\sigma_n}{n+1},\\ &= \frac{s_{n+1}-\sigma_n}{n+1}.\end{aligned}$$

The numerator is nonnegative by Step 1, and the denominator $n+1$ is positive. Hence $\sigma_{n+1}-\sigma_n \geq 0$, or $\sigma_n \leq \sigma_{n+1}$.

Conclusion. This comparison holds for every $n \in \mathbb{N}$, proving $\boxed{(\sigma_n) \text{ is increasing}}$. Equality can occur: for a constant positive sequence, all the averages are equal. Thus the weak meaning of "increasing" is the one required by the hypotheses and conclusion.

10.9 Exercise 10.9: A decreasing recurrence with a squared term

What we must show. For

$$s_1 = 1, \qquad s_{n+1} = \frac{n}{n+1}s_n^2 \quad (n \in \mathbb{N}),$$

compute s_2, s_3, s_4, prove that a finite limit exists, and prove that its value is 0.

Part (a): substitute the indices one at a time.

For $n = 1$, the recurrence gives

$$s_2 = \frac{1}{1+1}s_1^2 = \frac{1}{2}\cdot 1^2 = \frac{1}{2}.$$

For $n = 2$,

$$s_3 = \frac{2}{2+1}s_2^2 = \frac{2}{3}\left(\frac{1}{2}\right)^2 = \frac{2}{3}\cdot\frac{1}{4} = \frac{1}{6}.$$

For $n = 3$,

$$s_4 = \frac{3}{3+1}s_3^2 = \frac{3}{4}\left(\frac{1}{6}\right)^2 = \frac{3}{4}\cdot\frac{1}{36} = \frac{3}{144} = \frac{1}{48}.$$

Conclusion. $\boxed{s_2 = \frac{1}{2}, \quad s_3 = \frac{1}{6}, \quad s_4 = \frac{1}{48}}$.

Part (b), Step 1: prove positivity and an upper bound by induction.

We claim $0 < s_n \leq 1$ for every $n \in \mathbb{N}$.

The initial case is $s_1 = 1$, so $0 < s_1 \leq 1$ holds. For the induction step, assume $0 < s_n \leq 1$. Since $n \geq 1$, the coefficient satisfies

$$0 < \frac{n}{n+1} < 1.$$

Also $0 < s_n^2 \leq 1$. Multiplying these positive quantities gives

$$0 < s_{n+1} = \frac{n}{n+1}s_n^2 < 1.$$

This proves the claim by induction.

Part (b), Step 2: compare consecutive terms.

The inequality $s_n \le 1$, multiplied by the positive number s_n, gives $s_n^2 \le s_n$. The coefficient $n/(n+1)$ is strictly below 1, and $s_n^2 > 0$, so

$$s_{n+1} = \frac{n}{n+1} s_n^2 < s_n^2 \le s_n.$$

Hence the sequence is strictly decreasing. It is bounded below by 0 and above by 1.

Part (b), Step 3: apply monotone convergence before manipulating a limit.

Theorem 10.2 for bounded monotone sequences now applies. There is a real number L with $s_n \to L$. The inequalities $s_n > 0$ imply $L \ge 0$ by Exercise 8.9(a). Since the sequence is decreasing and $s_2 = 1/2$,

$$0 < s_n \le \frac{1}{2} \qquad (n \ge 2).$$

Exercise 8.9 also passes this eventual upper bound to the limit. Therefore

$$0 \le L \le \frac{1}{2}.$$

Conclusion. A finite limit exists, and it belongs to $[0, 1/2]$. Its existence has been proved independently of any equation for its value.

Part (c), Step 1: justify the shifted limit and the coefficient limit.

If $s_n \to L$, then $s_{n+1} \to L$ as well. Indeed, for any $\varepsilon > 0$ choose a threshold N with $m > N \Rightarrow |s_m - L| < \varepsilon$. For $n > N$, we have $n + 1 > N$, so the same estimate holds with $m = n + 1$.

Next put $c_n = n/(n+1)$. Its error from 1 is

$$|c_n - 1| = \left| \frac{n - (n+1)}{n+1} \right| = \frac{1}{n+1}.$$

For any $\varepsilon > 0$, if $n > 1/\varepsilon$, then $n + 1 > 1/\varepsilon$, so $1/(n+1) < \varepsilon$. Thus $c_n \to 1$.

Part (c), Step 2: apply the product limit law to the established limits.

Because $s_n \to L$ and $c_n \to 1$, the product laws from Section 9 give

$$\lim_{n\to\infty} c_n s_n^2 = 1 \cdot L \cdot L = L^2.$$

The recurrence says that this is the same sequence as s_{n+1}, whose limit is L. Uniqueness of a finite limit gives

$$L = L^2.$$

Rearranging and factoring gives

$$L^2 - L = 0, \qquad L(L-1) = 0.$$

The zero-product property therefore gives $L = 0$ or $L = 1$.

Part (c), Step 3: use the tail bound to remove the extraneous candidate.

Part (b) proved $L \le 1/2$, so $L = 1$ is impossible. The only remaining candidate is $L = 0$.

Conclusion. $\boxed{\lim_{n\to\infty} s_n = 0}$.

An additional direct error bound.

The same recurrence supplies a quantitative check. Since $s_n^2 \le s_n$,

$$s_{n+1} \le \frac{n}{n+1} s_n.$$

Starting with $s_1 = 1 = 1/1$, induction gives $s_n \le 1/n$: if $s_n \le 1/n$, then

$$s_{n+1} \le \frac{n}{n+1} \cdot \frac{1}{n} = \frac{1}{n+1}.$$

Consequently

$$|s_n - 0| = s_n \le \frac{1}{n} < \varepsilon \qquad \left(n > \frac{1}{\varepsilon}\right).$$

This also verifies the zero limit directly from Definition 7.1.

10.10 Exercise 10.10: A linear recurrence approaching one half

What we must show. For

$$s_1 = 1, \qquad s_{n+1} = \frac{s_n + 1}{3} \quad (n \in \mathbb{N}),$$

compute s_2, s_3, s_4, prove $s_n > 1/2$ by induction, prove that the sequence is decreasing, and identify its limit.

Part (a): calculate the first three new terms.

Using the recurrence successively gives

$$s_2 = \frac{1+1}{3} = \frac{2}{3},$$

$$s_3 = \frac{\frac{2}{3}+1}{3} = \frac{\frac{2}{3}+\frac{3}{3}}{3} = \frac{5/3}{3} = \frac{5}{9},$$

and

$$s_4 = \frac{\frac{5}{9}+1}{3} = \frac{\frac{5}{9}+\frac{9}{9}}{3} = \frac{14/9}{3} = \frac{14}{27}.$$

Conclusion. $\boxed{s_2 = \frac{2}{3}, \quad s_3 = \frac{5}{9}, \quad s_4 = \frac{14}{27}}$.

Part (b), Step 1: verify the initial case of the proposed lower bound.

The proposition for induction is P_n: $s_n > 1/2$. For $n = 1$, it says $1 > 1/2$, which is true.

Part (b), Step 2: prove the induction implication.

Assume $s_n > 1/2$. Adding 1 to both sides gives $s_n + 1 > 3/2$. Dividing by the positive number 3 preserves the strict inequality:

$$s_{n+1} = \frac{s_n + 1}{3} > \frac{3/2}{3} = \frac{1}{2}.$$

Thus P_n implies P_{n+1}. Induction proves the bound for every positive integer n.

Conclusion. $\boxed{s_n > \frac{1}{2} \text{ for every } n \in \mathbb{N}}$.

Part (c): subtract consecutive terms and determine the sign.

Use a common denominator:

$$s_{n+1} - s_n = \frac{s_n + 1}{3} - s_n = \frac{s_n + 1 - 3s_n}{3} = \frac{1 - 2s_n}{3}.$$

Part (b) gives $2s_n > 1$, so the numerator is negative. The denominator 3 is positive. Hence $s_{n+1} - s_n < 0$, which means $s_{n+1} < s_n$ for every n.

Conclusion. The sequence is strictly decreasing.

Part (d), Step 1: establish convergence from bounds and monotonicity.

The terms satisfy $1/2 < s_n \leq s_1 = 1$, and part (c) proves that the sequence is decreasing. Theorem 10.2 therefore gives a finite limit L. Exercise 8.9 implies

$$\frac{1}{2} \leq L \leq 1.$$

Part (d), Step 2: solve the limit equation after convergence is known.

The shifted sequence s_{n+1} has the same limit L, by the threshold argument given in Exercise 10.9(c). The addition and scalar-multiplication laws from Section 9 give

$$\lim_{n\to\infty} \frac{s_n + 1}{3} = \frac{L + 1}{3}.$$

The recurrence identifies these two sequences, so their limits agree:

$$L = \frac{L + 1}{3}.$$

Multiplying by 3 and collecting terms gives

$$3L = L + 1, \qquad 2L = 1, \qquad L = \frac{1}{2}.$$

Conclusion. $\boxed{\lim_{n\to\infty} s_n = \frac{1}{2}}$. Every term is strictly greater than $1/2$, but the limit can equal that lower bound.

10.11 Exercise 10.11: An infinite-product recurrence

What we must show. For

$$t_1 = 1, \qquad t_{n+1} = \left(1 - \frac{1}{4n^2}\right) t_n \quad (n \in \mathbb{N}),$$

(a) prove that a finite limit exists and (b) investigate its value. Part (b) asks what the limit appears to be. We will distinguish a numerical guess from a proof; an optional elementary-calculus argument will then verify the exact value.

Part (a), Step 1: bound the coefficient and prove positivity.

For every $n \geq 1$, we have $4n^2 \geq 4$, so

$$0 < \frac{1}{4n^2} \leq \frac{1}{4}.$$

Subtracting from 1 gives

$$\frac{3}{4} \leq 1 - \frac{1}{4n^2} < 1.$$

In particular, every recurrence coefficient is strictly positive. The initial value is $t_1 = 1 > 0$. If $t_n > 0$, the recurrence is a product of two positive numbers and gives $t_{n+1} > 0$. Induction shows $t_n > 0$ for all n.

Part (a), Step 2: prove monotonicity and apply the theorem.

Since $1 - 1/(4n^2) < 1$ and $t_n > 0$,

$$t_{n+1} = \left(1 - \frac{1}{4n^2}\right) t_n < t_n.$$

Thus (t_n) is strictly decreasing, bounded below by 0 and above by its first term 1. Theorem 10.2 gives a finite real limit L. The bound on the terms implies $0 \leq L \leq 1$ by Exercise 8.9.

Conclusion. The limit of (t_n) exists as a real number. This conclusion uses only positivity, monotonicity, and boundedness.

Part (b), Step 1: compute initial values and a numerical table.

The first terms are

$$t_2 = \left(1 - \frac{1}{4}\right) 1 = \frac{3}{4},$$

$$t_3 = \left(1 - \frac{1}{16}\right) \frac{3}{4} = \frac{15}{16} \cdot \frac{3}{4} = \frac{45}{64},$$

and

$$t_4 = \left(1 - \frac{1}{36}\right) \frac{45}{64} = \frac{35}{36} \cdot \frac{45}{64} = \frac{1575}{2304} = \frac{175}{256}.$$

Continuing the recurrence produces the following rounded values:

n	t_n
1	1.000000000000
2	0.750000000000
3	0.703125000000
4	0.683593750000
10	0.653587392299
100	0.638221323951
1000	0.636779026837

The numbers suggest a limit near 0.63662. The familiar constant $2/\pi$ has the decimal value $0.636619772368\ldots$, which suggests the guess

$$L = \frac{2}{\pi}.$$

The numerical agreement motivates this guess; finitely many numerical calculations by themselves do not prove the exact identification.

Part (b), Step 2: explain why a direct limit equation gives no value.

The coefficient $1 - 1/(4n^2)$ tends to 1. Taking limits in the recurrence, after convergence has been proved, only yields

$$L = 1 \cdot L.$$

Every real L satisfies this identity, so it does not determine the limit. Further information is needed to justify the guess $2/\pi$.

Optional verification of the guess: the elementary-calculus input.

The remaining argument uses integration by parts and the order property of definite integrals. Those are additional calculus tools; they were not needed for the existence proof in part (a). We derive the required product formula here rather than assume Wallis's product.

For an integer $m \geq 0$, define

$$I_m = \int_0^{\pi/2} \sin^m x \, dx,$$

where $\sin^0 x$ means the constant function 1. The integrands are continuous and are positive in the interior when $m \geq 1$, so all I_m are positive. Direct integration gives

$$I_0 = \frac{\pi}{2}, \qquad I_1 = 1.$$

Optional verification, Step 1: derive the integral recurrence.

For $m \geq 2$, write $I_m = \int_0^{\pi/2} \sin^{m-1} x \sin x \, dx$. In integration by parts take

$$u = \sin^{m-1} x, \quad dv = \sin x \, dx, \quad du = (m-1)\sin^{m-2} x \cos x \, dx, \quad v = -\cos x.$$

The boundary term $[-\sin^{m-1} x \cos x]_0^{\pi/2}$ is zero: $\cos(\pi/2) = 0$, and at 0 the factor $\sin^{m-1} 0$ is zero because $m - 1 \geq 1$. Consequently

$$\begin{aligned} I_m &= (m-1) \int_0^{\pi/2} \sin^{m-2} x \cos^2 x \, dx \\ &= (m-1) \int_0^{\pi/2} \sin^{m-2} x (1 - \sin^2 x) \, dx \\ &= (m-1)(I_{m-2} - I_m). \end{aligned}$$

Adding $(m-1)I_m$ to both sides and dividing by $m > 0$ gives

$$I_m = \frac{m-1}{m} I_{m-2}.$$

Optional verification, Step 2: express the even and odd integrals.

Repeated use of the recurrence, ending at I_0 or I_1, gives for every integer $N \geq 1$,

$$I_{2N} = \frac{\pi}{2} \prod_{k=1}^{N} \frac{2k-1}{2k}, \qquad I_{2N+1} = \prod_{k=1}^{N} \frac{2k}{2k+1}.$$

For example, the even recurrence successively contributes the factors $(2N-1)/(2N)$, $(2N-3)/(2N-2)$, down to $1/2$. The odd recurrence contributes $(2N)/(2N+1)$ down to $2/3$. These are exactly the displayed finite products.

Unrolling the recurrence for t_n gives

$$t_{N+1} = \prod_{k=1}^{N} \left(1 - \frac{1}{4k^2}\right) = \prod_{k=1}^{N} \frac{(2k-1)(2k+1)}{(2k)^2}.$$

The initial empty product for t_1 is 1, and each new recurrence step multiplies by the next factor, which justifies this finite-product identity. Dividing the formulas for the positive integrals gives

$$\frac{I_{2N}}{I_{2N+1}} = \frac{\pi}{2} \prod_{k=1}^{N} \frac{(2k-1)(2k+1)}{(2k)^2} = \frac{\pi}{2}\, t_{N+1}.$$

Optional verification, Step 3: bound the integral ratio.

On $[0, \pi/2]$, the quantity $\sin x$ lies in $[0, 1]$. Multiplication by a number in $[0, 1]$ cannot increase a nonnegative power, so

$$\sin^{2N+1} x \le \sin^{2N} x \le \sin^{2N-1} x.$$

Integrating preserves these inequalities:

$$I_{2N+1} \le I_{2N} \le I_{2N-1}.$$

Divide by $I_{2N+1} > 0$. The integral recurrence with $m = 2N + 1$ gives $I_{2N+1} = (2N)/(2N + 1)\, I_{2N-1}$, so

$$1 \le \frac{I_{2N}}{I_{2N+1}} \le \frac{I_{2N-1}}{I_{2N+1}} = \frac{2N+1}{2N} = 1 + \frac{1}{2N}.$$

Substituting $I_{2N}/I_{2N+1} = (\pi/2)t_{N+1}$ and multiplying by $2/\pi > 0$ yields

$$0 \le t_{N+1} - \frac{2}{\pi} \le \frac{1}{\pi N}.$$

For any $\varepsilon > 0$, a positive integer index $n > 1 + 1/(\pi\varepsilon)$ has $N = n - 1 > 1/(\pi\varepsilon)$. Therefore

$$\left| t_n - \frac{2}{\pi} \right| \le \frac{1}{\pi(n-1)} < \varepsilon.$$

This proves the conjectured limit by Definition 7.1.

Conclusion. Part (a) proves existence using Section 10 alone. For part (b), the computed terms suggest $2/\pi$. The explicitly marked additional calculus argument verifies that

$$\boxed{\lim_{n\to\infty} t_n = \frac{2}{\pi}.}$$

No product formula was taken on trust; the finite-product identity and its limiting estimate were derived.

10.12 Exercise 10.12: A product recurrence with an exact formula

What we must show. For

$$t_1 = 1, \qquad t_{n+1} = \left(1 - \frac{1}{(n+1)^2}\right) t_n \quad (n \in \mathbb{N}),$$

(a) establish existence of the limit, (b) guess its value, (c) prove $t_n = (n+1)/(2n)$ by induction, and (d) use the formula to verify the limit.

Part (a), Step 1: prove positivity and monotonicity.

For every $n \geq 1$, we have $n+1 \geq 2$, so

$$0 < \frac{1}{(n+1)^2} \leq \frac{1}{4}, \qquad \frac{3}{4} \leq 1 - \frac{1}{(n+1)^2} < 1.$$

The initial term is positive. If $t_n > 0$, multiplying it by this positive coefficient gives $t_{n+1} > 0$. Induction proves positivity for every term.

Because the coefficient is strictly less than 1 and $t_n > 0$, we also have $t_{n+1} < t_n$ for every n. Thus the sequence is strictly decreasing, bounded below by 0 and above by $t_1 = 1$.

Part (a), Step 2: apply bounded monotone convergence.

Theorem 10.2 now gives a finite limit L. Exercise 8.9 implies $0 \leq L \leq 1$.

Conclusion. The limit exists as a real number. Its value has not been assumed in establishing existence.

Part (b): calculate early terms to formulate a guess.

Successive substitutions give

$$t_2 = \left(1 - \frac{1}{4}\right) 1 = \frac{3}{4},$$

$$t_3 = \left(1 - \frac{1}{9}\right) \frac{3}{4} = \frac{8}{9} \cdot \frac{3}{4} = \frac{2}{3},$$

$$t_4 = \left(1 - \frac{1}{16}\right) \frac{2}{3} = \frac{15}{16} \cdot \frac{2}{3} = \frac{5}{8},$$

and

$$t_5 = \left(1 - \frac{1}{25}\right) \frac{5}{8} = \frac{24}{25} \cdot \frac{5}{8} = \frac{3}{5}.$$

The values

$$1, \quad \frac{3}{4}, \quad \frac{2}{3}, \quad \frac{5}{8}, \quad \frac{3}{5}$$

fit the proposed pattern $t_n = (n+1)/(2n)$. Since this expression equals $1/2 + 1/(2n)$, it suggests the limit $1/2$. At this stage, the pattern and the limit are guesses to be proved.

Part (c), Step 1: state the induction proposition and verify its base.

Let P_n be the assertion

$$t_n = \frac{n+1}{2n}.$$

For $n = 1$, the right side is $(1+1)/(2 \cdot 1) = 1$, which equals t_1. Thus P_1 holds.

Part (c), Step 2: substitute the induction hypothesis.

Assume P_n for an arbitrary positive integer n. The recurrence gives

$$t_{n+1} = \left(1 - \frac{1}{(n+1)^2}\right) \frac{n+1}{2n}$$

$$
\begin{aligned}
&= \frac{(n+1)^2 - 1}{(n+1)^2} \frac{n+1}{2n} \\
&= \frac{n^2 + 2n}{(n+1)^2} \frac{n+1}{2n} \\
&= \frac{n(n+2)}{(n+1)^2} \frac{n+1}{2n} \\
&= \frac{n+2}{2(n+1)} \\
&= \frac{(n+1)+1}{2(n+1)}.
\end{aligned}
$$

The cancellation in the penultimate line is legitimate because $n > 0$ and $n+1 > 0$, so neither cancelled factor is zero. The final expression is precisely the formula in P_{n+1}.

Conclusion. Mathematical induction proves

$$\boxed{t_n = \frac{n+1}{2n} \quad \text{for every } n \in \mathbb{N}.}$$

Part (d), Step 1: separate the constant term from the error.

The proved formula gives the identity

$$t_n = \frac{n+1}{2n} = \frac{n}{2n} + \frac{1}{2n} = \frac{1}{2} + \frac{1}{2n}.$$

Since $1/n \to 0$, the limit laws from Section 9 give $t_n \to 1/2$.

Part (d), Step 2: verify the conclusion directly with a threshold.

For any $\varepsilon > 0$, choose $N = 1/(2\varepsilon)$. Every positive integer $n > N$ satisfies $1/(2n) < \varepsilon$, and

$$\left| t_n - \frac{1}{2} \right| = \left| \frac{1}{2n} \right| = \frac{1}{2n} < \varepsilon.$$

This is the definition of convergence to $1/2$.

Conclusion. The guess in part (b) is now proved:

$$\boxed{\lim_{n \to \infty} t_n = \frac{1}{2}.}$$

The exact formula also gives the error $t_n - 1/2 = 1/(2n)$ for every positive integer index.

11 Subsequences

A subsequence has the form $(s_{n_k})_{k\in\mathbb{N}}$, where its original indices satisfy $n_1 < n_2 < \cdots$. The selection map is $\sigma(k) = n_k$. The index k counts selected terms, whereas n_k is their position in the original sequence.

Subsequential limits in this section may be real numbers or $+\infty$ or $-\infty$. A proposed list must account for every possible subsequential limit. If S denotes the set of those limits, we use

$$\limsup s_n = \sup S, \qquad \liminf s_n = \inf S,$$

as established in Theorem 11.8. Increasing and decreasing remain weak notions, so a constant subsequence is monotone.

11.1 Exercise 11.1: A constant subsequence of a two-valued sequence

What we must show. For $a_n = 3+2(-1)^n$, list the first eight terms and give a constant subsequence together with its selection function.

Plan. Separate even and odd indices. The selection function must choose positive integer indices in strictly increasing order; specifying a repeated value without its indices is not enough.

(a) Compute the first eight terms.

At an odd index, $(-1)^n = -1$, so $a_n = 3 + 2(-1) = 3 - 2 = 1$. At an even index, $(-1)^n = 1$, so $a_n = 3 + 2(1) = 3 + 2 = 5$. Substituting the first eight indices gives

n	1	2	3	4	5	6	7	8
$(-1)^n$	-1	1	-1	1	-1	1	-1	1
a_n	1	5	1	5	1	5	1	5

(b) Specify and verify a selection function.

Define $\sigma : \mathbb{N} \to \mathbb{N}$ by $\sigma(k) = 2k$. For every $k \geq 1$, $\sigma(k) \geq 2 > 0$, and

$$\sigma(k+1) - \sigma(k) = 2(k+1) - 2k = 2 > 0.$$

Thus this is a valid strictly increasing sequence of indices. The selected values are

$$a_{\sigma(k)} = a_{2k} = 3 + 2(-1)^{2k} = 3 + 2 = 5 \qquad (k \in \mathbb{N}).$$

They form the constant subsequence $(5, 5, 5, \ldots)$.

Conclusion. The first eight terms are $1, 5, 1, 5, 1, 5, 1, 5$. One constant subsequence is $a_{2k} = 5$, selected by $\sigma(k) = 2k$. Selecting $\sigma(k) = 2k - 1$ would instead give the constant value 1.

11.2 Exercise 11.2: Five questions for each of four elementary sequences

What we must show. For each of

$$a_n = (-1)^n, \qquad b_n = \frac{1}{n}, \qquad c_n = n^2, \qquad d_n = \frac{6n+4}{7n-3},$$

answer all five questions: (a) a monotone subsequence with its indices; (b) the entire set of subsequential limits; (c) the limsup and liminf; (d) finite convergence or divergence to either infinity; and (e) boundedness. Subsequential limits here include the symbols $+\infty$ and $-\infty$ as in Definition 11.6.

Plan. We first record the tools used throughout Exercises 11.2–11.4. Each sequence then receives its own answers (a)–(e). In particular, constructing some subsequential limits will always be followed by a reason that no additional ones have been missed.

Tools shared by the three catalogues

Tool 1: Selection indices tend to infinity.

For any strictly increasing positive integer indices $n_1 < n_2 < \cdots$, one has $n_k \geq k$. At $k = 1$, $n_1 \geq 1$. If $n_k \geq k$, then the integer $n_{k+1} > n_k$ satisfies $n_{k+1} \geq n_k + 1 \geq k + 1$. Induction proves the claim. Consequently $k > N$ implies $n_k > N$.

The identity selection $n_k = k$ is allowed: a subsequence may retain all the terms of the original sequence. A constant subsequence is monotone under the book's non-strict definition, since its consecutive values satisfy both $u_{k+1} \geq u_k$ and $u_{k+1} \leq u_k$.

Tool 2: Every subsequence inherits an existing finite or infinite limit.

For a finite limit, this is Theorem 11.3. Its proof uses Tool 1: once every original index $n > N$ satisfies an error bound, every selected index n_k with $k > N$ satisfies it too. The same index argument proves the infinite cases directly. If $u_n \to +\infty$ and $M > 0$, choose N with $n > N \Longrightarrow u_n > M$. For $k > N$, $n_k \geq k > N$ gives $u_{n_k} > M$, so $u_{n_k} \to +\infty$. If $u_n \to -\infty$ and $M < 0$, choose N with $n > N \Longrightarrow u_n < M$; the identical implication $n_k \geq k > N$ gives $u_{n_k} < M$ and hence the negative infinite limit.

These alternatives are mutually exclusive. A finite limit L puts all sufficiently late terms in $(L-1, L+1)$, which prevents either infinite limit. The two signs of infinity are incompatible because their threshold tests with 1 and -1 would require a late term to be both greater than 1 and less than -1. Finite limits are unique by Theorem 7.3. Thus, if the original sequence has any existing limit L, its set of subsequential limits is exactly $\{L\}$: every subsequence has that limit, and the identity selection itself supplies a subsequence.

Tool 3: A sequence with only finitely many values.

Suppose every term belongs to a finite set $F = \{f_1, \ldots, f_m\}$ and every member of F occurs infinitely often. For each particular sequence below, we will display constant selections proving that every member of F is a subsequential limit. There can be no other finite subsequential limit. Indeed, if $L \notin F$, then the finite set of positive distances $\{|L - f_i| : 1 \leq i \leq m\}$ has a positive minimum δ. Every term, including every selected term, satisfies $|u_n - L| \geq \delta$, so convergence to L fails the error test with $\varepsilon = \delta/2$.

There are no infinite subsequential limits either. With $C = \max\{|f_1|, \ldots, |f_m|\}$, every term lies between $-C$ and C. Neither the positive threshold $C + 1$ nor the negative threshold $-C - 1$ can ever be crossed in the required direction. Therefore the full subsequential-limit set is precisely F.

Tool 4: Extract limsup, liminf, and full-sequence behavior from the set.

If S is the full set of subsequential limits, Theorem 11.8(ii) gives

$$\limsup_{n\to\infty} u_n = \sup S, \qquad \liminf_{n\to\infty} u_n = \inf S.$$

In the explicit sets below these are their greatest and least elements in the extended order. Theorem 11.8(iii) says that the original sequence has a limit if and only if S has exactly one element. A singleton real element means finite convergence; a singleton $\{+\infty\}$ or $\{-\infty\}$ means divergence of the corresponding sign. If S contains two or more elements, there is no finite limit and neither infinite limit exists.

The sequence $a_n = (-1)^n$

(a) A monotone subsequence. Choose $n_k = 2k$. The indices start at 2 and increase by 2, so they are positive and strictly increasing. Then $a_{n_k} = (-1)^{2k} = 1$ at every k. This constant subsequence is monotone.

(b) All subsequential limits. At the indices $2k-1$ the values are -1, and at the indices $2k$ they are 1. Both selections have positive indices and successive differences 2. Thus constant subsequences give both -1 and 1 as limits. The original sequence takes only these two values, so Tool 3 excludes every other finite or infinite subsequential limit:

$$S_a = \{-1, 1\}.$$

(c) Limsup and liminf. The greatest element of S_a is 1 and the least is -1. Tool 4 therefore gives $\limsup a_n = 1$ and $\liminf a_n = -1$.

(d) Full-sequence behavior. Since S_a has two distinct elements, Tool 4 excludes all three possibilities: (a_n) does not converge to a real number, does not diverge to $+\infty$, and does not diverge to $-\infty$.

(e) Boundedness. Every term is either -1 or 1, so $-1 \le a_n \le 1$ for every n. Thus (a_n) is bounded.

The sequence $b_n = 1/n$

(a) A monotone subsequence. Take the identity selection $n_k = k$, whose indices start at 1 and increase by 1. It gives $b_{n_k} = 1/k$. Its consecutive differences are

$$\frac{1}{k+1} - \frac{1}{k} = \frac{k-(k+1)}{k(k+1)} = -\frac{1}{k(k+1)} < 0.$$

The denominator is positive for $k \ge 1$, so the selected sequence is strictly decreasing.

(b) All subsequential limits. The original sequence converges to 0: given $\varepsilon > 0$, $n > 1/\varepsilon$ implies $|1/n - 0| = 1/n < \varepsilon$. Tool 2 therefore proves that every subsequence has limit 0, and excludes any additional finite or infinite limit. Thus $S_b = \{0\}$.

(c) Limsup and liminf. The supremum and infimum of $\{0\}$ are both 0. Hence $\limsup b_n = \liminf b_n = 0$ by Tool 4.

(d) Full-sequence behavior. The preceding error estimate proves finite convergence to 0. It does not diverge to either sign of infinity, by the mutual exclusivity recorded in Tool 2.

(e) Boundedness. For every positive integer n, $0 < 1/n \le 1$. Thus $0 < b_n \le 1$, providing a lower and an upper bound.

The sequence $c_n = n^2$

(a) A monotone subsequence. Take $n_k = k$. Then $c_{n_k} = k^2$, and

$$(k+1)^2 - k^2 = k^2 + 2k + 1 - k^2 = 2k + 1 > 0.$$

Thus the identity subsequence is strictly increasing.

(b) All subsequential limits. The original sequence has limit $+\infty$: for $M > 0$, $n > \sqrt{M}$ implies $n^2 > M$. Tool 2 proves that every subsequence has this same infinite limit. Consequently the exact set is $S_c = \{+\infty\}$; in particular, no finite limit or negative-infinity limit has been omitted.

(c) Limsup and liminf. Both the greatest and least elements of this singleton are $+\infty$. Thus $\limsup c_n = \liminf c_n = +\infty$.

(d) Full-sequence behavior. The sequence diverges to $+\infty$. It has no finite real limit and does not diverge to $-\infty$.

(e) Boundedness. It is bounded below by 1, since $n \geq 1$. It is not bounded above: for any proposed upper bound B, choose $n > \max\{1, B\}$. Then $n^2 \geq n > B$. Thus (c_n) is not a bounded sequence.

The sequence $d_n = (6n + 4)/(7n - 3)$

For every $n \geq 1$, $7n - 3 \geq 4 > 0$, so each term is defined with a positive denominator. An exact subtraction identifies both its limit and its side of approach:

$$\begin{aligned} d_n - \frac{6}{7} &= \frac{7(6n+4) - 6(7n-3)}{7(7n-3)} \\ &= \frac{42n + 28 - 42n + 18}{7(7n-3)} \\ &= \frac{46}{7(7n-3)} > 0. \end{aligned}$$

(a) A monotone subsequence. Use $n_k = k$. Subtract consecutive values using the error formula just established:

$$\begin{aligned} d_{k+1} - d_k &= \frac{46}{7}\left(\frac{1}{7k+4} - \frac{1}{7k-3}\right) \\ &= \frac{46}{7}\,\frac{(7k-3) - (7k+4)}{(7k+4)(7k-3)} \\ &= \frac{46}{7}\,\frac{-7}{(7k+4)(7k-3)} \\ &= -\frac{46}{(7k+4)(7k-3)} < 0. \end{aligned}$$

Both denominator factors are positive for $k \geq 1$. Thus the full sequence, and hence the identity subsequence, is strictly decreasing.

(b) All subsequential limits. For $n \geq 1$, $7n - 3 \geq 4n$ because $(7n - 3) - 4n = 3(n - 1) \geq 0$. Therefore

$$\left|d_n - \frac{6}{7}\right| = \frac{46}{7(7n-3)} \leq \frac{46}{28n} = \frac{23}{14n}.$$

For $\varepsilon > 0$, any threshold $N > 23/(14\varepsilon)$ makes this error less than ε for $n > N$. Thus $d_n \to 6/7$. Tool 2 supplies exhaustiveness, so $S_d = \{6/7\}$.

(c) Limsup and liminf. Tool 4 applied to the singleton gives $\limsup d_n = \liminf d_n = 6/7$.

(d) Full-sequence behavior. The sequence converges to the finite value $6/7$. Neither sign of infinite divergence is possible.

(e) Boundedness. The positive error formula shows $d_n > 6/7$ for every n. Strict decrease shows no term is larger than the first:

$$d_1 = \frac{6+4}{7-3} = \frac{10}{4} = \frac{5}{2}.$$

Thus $6/7 < d_n \leq 5/2$ for every n, so the sequence is bounded.

Conclusion. In Exercise 11.2, (b_n) and (d_n) converge, to 0 and $6/7$ respectively; (c_n) diverges to $+\infty$; and (a_n) has no extended limit. None diverges to $-\infty$. Exactly $(a_n), (b_n), (d_n)$ are bounded. Each of the four sequences has been given a monotone selection and its complete subsequential-limit set above.

11.3 Exercise 11.3: Five questions for four bounded sequences

What we must show. Answer (a)–(e) of Exercise 11.2 separately for

$$s_n = \cos(n\pi/3), \qquad t_n = \frac{3}{4n+1}, \qquad u_n = \left(-\frac{1}{2}\right)^n, \qquad v_n = (-1)^n + \frac{1}{n}.$$

We continue to use the four shared tools, so all sets of subsequential limits are understood in the extended sense.

The sequence $s_n = \cos(n\pi/3)$

The cosine values are determined by the residue of n modulo 6. Write $n = 6q + r$, with $q \geq 0$ an integer and $r \in \{1, 2, 3, 4, 5, 6\}$. Then $n\pi/3 = 2q\pi + r\pi/3$, so the period 2π gives

r	1	2	3	4	5	6
$\cos(r\pi/3)$	$1/2$	$-1/2$	-1	$-1/2$	$1/2$	1

(a) A monotone subsequence. Select $n_k = 6k$. These positive indices start at 6 and increase by 6. The selected terms satisfy $s_{6k} = \cos(2k\pi) = 1$, so this is a constant monotone subsequence.

(b) All subsequential limits. The table gives precisely four different values. The following explicit selections show that each occurs as a constant subsequence:

indices n_k	first index	s_{n_k}
$6k-5$	1	$1/2$
$6k-4$	2	$-1/2$
$6k-3$	3	-1
$6k$	6	1

Each first index is positive and each sequence of indices increases by 6. The period table proves there are no other values; Tool 3 excludes any other finite or infinite subsequential limit. Hence

$$S_s = \{-1, -1/2, 1/2, 1\}.$$

(c) Limsup and liminf. The greatest element is 1 and the least is -1, so $\limsup s_n = 1$ and $\liminf s_n = -1$.

(d) Full-sequence behavior. The set S_s has four distinct elements, rather than one. Tool 4 shows that the original sequence does not converge to a finite value and does not diverge to either infinity.

(e) Boundedness. The table puts every value between -1 and 1, so $-1 \le s_n \le 1$ for every n and the sequence is bounded.

The sequence $t_n = 3/(4n+1)$

(a) A monotone subsequence. The denominators are positive because $4n+1 \ge 5$. Choose $n_k = k$. The difference between consecutive selected terms is

$$\begin{aligned} t_{k+1} - t_k &= \frac{3}{4k+5} - \frac{3}{4k+1} \\ &= \frac{3[(4k+1)-(4k+5)]}{(4k+5)(4k+1)} \\ &= -\frac{12}{(4k+5)(4k+1)} < 0. \end{aligned}$$

Thus the identity subsequence is strictly decreasing.

(b) All subsequential limits. For every $n \ge 1$,

$$0 < t_n = \frac{3}{4n+1} \le \frac{3}{4n}.$$

For $\varepsilon > 0$, $n > 3/(4\varepsilon)$ makes the last expression less than ε. Hence $t_n \to 0$. Tool 2 proves that all subsequences have the same limit, so $S_t = \{0\}$, with no additional extended limits.

(c) Limsup and liminf. Both extremal elements of this singleton are 0. Thus $\limsup t_n = \liminf t_n = 0$.

(d) Full-sequence behavior. The sequence converges to 0 and diverges to neither $+\infty$ nor $-\infty$.

(e) Boundedness. The inequality $4n+1 \ge 5$ gives $0 < t_n \le 3/5$. Therefore the sequence is bounded.

The sequence $u_n = (-1/2)^n$

(a) A monotone subsequence. Choose the even indices $n_k = 2k$, positive and increasing by 2. Then

$$u_{2k} = \left(-\frac{1}{2}\right)^{2k} = \left(\frac{1}{4}\right)^k > 0.$$

The next selected term is $u_{2(k+1)} = (1/4)u_{2k}$. Since $0 < 1/4 < 1$ and $u_{2k} > 0$, it is strictly smaller than u_{2k}. Thus this subsequence is strictly decreasing.

(b) All subsequential limits. The base has magnitude $1/2 < 1$. Theorem 9.7(b), or Exercise 9.13, gives $(-1/2)^n \to 0$. Equivalently its distance from 0 is $(1/2)^n \to 0$, independently of the alternating sign. Tool 2 then gives the exhaustive set $S_u = \{0\}$.

(c) Limsup and liminf. The singleton has both supremum and infimum 0, so $\limsup u_n = \liminf u_n = 0$.

(d) Full-sequence behavior. It converges to the finite value 0, and diverges to neither sign of infinity.

(e) Boundedness. For $n \geq 1$,

$$|u_n| = \left(\frac{1}{2}\right)^n = \frac{1}{2}\left(\frac{1}{2}\right)^{n-1} \leq \frac{1}{2}.$$

The final product is at most $1/2$ because its remaining $n-1$ factors are at most 1; at $n=1$ that remaining product is empty and equals 1. Hence $-1/2 \leq u_n \leq 1/2$, a bound valid at every index.

The sequence $v_n = (-1)^n + 1/n$

(a) A monotone subsequence. Choose $n_k = 2k$, again positive and increasing by 2. The selected terms are $v_{2k} = 1 + 1/(2k)$, and

$$\begin{aligned} v_{2(k+1)} - v_{2k} &= \frac{1}{2k+2} - \frac{1}{2k} \\ &= \frac{2k - (2k+2)}{(2k+2)(2k)} \\ &= -\frac{1}{2k(k+1)} < 0. \end{aligned}$$

This is a strictly decreasing subsequence.

(b) All subsequential limits. The even and odd selections respectively give

$$v_{2k} = 1 + \frac{1}{2k} \longrightarrow 1, \qquad v_{2k-1} = -1 + \frac{1}{2k-1} \longrightarrow -1.$$

Both index formulas start at positive integers and increase by 2. The reciprocal errors tend to 0: for example, $1/(2k) \leq 1/k$ and $1/(2k-1) \leq 1/k$ for $k \geq 1$. Thus -1 and 1 are both subsequential limits.

To exclude every other finite candidate, let $L \notin \{-1, 1\}$ and set

$$\delta = \min\{|L-1|, |L+1|\} > 0.$$

Consider any subsequence with indices n_k. Since $n_k \geq k$, we have $1/n_k \leq 1/k$. For $k > 2/\delta$, this is less than $\delta/2$. The reverse triangle inequality then yields

$$\begin{aligned} |v_{n_k} - L| &= |((-1)^{n_k} - L) + 1/n_k| \\ &\geq |(-1)^{n_k} - L| - \frac{1}{n_k} \\ &\geq \delta - \frac{1}{n_k} > \frac{\delta}{2}. \end{aligned}$$

This defeats the convergence test with error tolerance $\delta/2$ for every such subsequence. Finally,

$$|v_n| \leq |(-1)^n| + \frac{1}{n} = 1 + \frac{1}{n} \leq 2,$$

so no subsequence can tend to either infinity. The complete set is therefore $S_v = \{-1, 1\}$.

(c) Limsup and liminf. Tool 4 gives $\limsup v_n = 1$ and $\liminf v_n = -1$. The values of the sequence approach these numbers along the even and odd selections, respectively.

(d) Full-sequence behavior. There are two distinct subsequential limits. Thus the full sequence has no finite limit and diverges to neither infinity.

(e) Boundedness. The absolute-value estimate $|v_n| \le 2$ already proves boundedness. A sharper bound follows directly from parity: at even indices $n = 2k$, $1 < v_{2k} = 1 + 1/(2k) \le 3/2$; at odd indices $n = 2k-1$, $-1 < v_{2k-1} = -1 + 1/(2k-1) \le 0$. Combining the cases gives $-1 < v_n \le 3/2$ for every n.

Conclusion. In Exercise 11.3 all four sequences are bounded. The sequences (t_n) and (u_n) converge to 0; (s_n) and (v_n) have no extended limit. None diverges to either infinity. Their exact subsequential-limit sets are, in order, $\{-1, -1/2, 1/2, 1\}$, $\{0\}$, $\{0\}$, and $\{-1, 1\}$, with limsup and liminf the largest and smallest elements of these sets.

11.4 Exercise 11.4: Five questions for oscillation with and without growth

What we must show. Answer (a)–(e) of Exercise 11.2 separately for

$$w_n = (-2)^n, \qquad x_n = 5^{(-1)^n}, \qquad y_n = 1 + (-1)^n, \qquad z_n = n\cos(n\pi/4).$$

In the second expression, $(-1)^n$ is the *exponent* of 5; the expression is not multiplication by $(-1)^n$.

Plan. Parity resolves the first three sequences. For the last, a complete eight-residue table identifies the zero terms and the growing positive and negative terms. A separate bounded-tail argument then excludes any unlisted finite subsequential limit.

The sequence $w_n = (-2)^n$

(a) A monotone subsequence. Take $n_k = 2k$, whose positive indices increase by 2. Then $w_{2k} = 4^k > 0$. The next selected value is $4^{k+1} = 4 \cdot 4^k > 4^k$, so this subsequence is strictly increasing.

(b) All subsequential limits. The even selection gives $w_{2k} = 4^k \to +\infty$ by Exercise 9.13, since the base 4 is greater than 1. The odd selection $n_k = 2k - 1$ gives

$$w_{2k-1} = -2^{2k-1} = -\frac{1}{2}\,4^k \longrightarrow -\infty$$

by the negative scaling result of Exercise 9.10. Both selections are valid positive increasing indices.

There is no finite subsequential limit. Indeed, $|w_n| = 2^n \to +\infty$ by Exercise 9.13. Tool 2 applied to this positive magnitude sequence gives $|w_{n_k}| \to +\infty$ for every selection. If $w_{n_k} \to L \in \mathbb{R}$, the error tolerance 1 would give, eventually,

$$|w_{n_k}| \le |w_{n_k} - L| + |L| < 1 + |L|\,,$$

contradicting that magnitude limit. The only remaining extended candidates are the two already constructed, so

$$S_w = \{-\infty, +\infty\}.$$

(c) Limsup and liminf. The extremal elements of S_w give $\limsup w_n = +\infty$ and $\liminf w_n = -\infty$.

(d) Full-sequence behavior. Since S_w has two elements, the full sequence has no extended limit. In particular it does not converge to a real number and does not diverge to either single sign of infinity. Its large positive and negative terms do not constitute a limit of one sign.

(e) Boundedness. The even terms exceed every real upper bound and the odd terms go below every real lower bound, by their infinite limits just proved. Hence (w_n) is bounded neither above nor below, and is not a bounded sequence.

The sequence $x_n = 5^{(-1)^n}$

At even indices the exponent equals 1, and at odd indices it equals -1. Therefore

$$x_{2k} = 5^1 = 5, \qquad x_{2k-1} = 5^{-1} = \frac{1}{5}.$$

(a) A monotone subsequence. Choose $n_k = 2k$, positive and increasing by 2. Then $x_{n_k} = 5$ at every index, a constant monotone subsequence.

(b) All subsequential limits. The even and odd index selections are constant with values 5 and $1/5$ respectively, so both are subsequential limits. They are the only values the original sequence ever takes; Tool 3 therefore excludes all other finite and infinite subsequential limits. Thus $S_x = \{1/5, 5\}$.

(c) Limsup and liminf. Since $1/5 < 5$, Tool 4 gives $\limsup x_n = 5$ and $\liminf x_n = 1/5$.

(d) Full-sequence behavior. The two distinct subsequential limits show that (x_n) does not converge to a finite number and diverges to neither sign of infinity.

(e) Boundedness. Every term is either $1/5$ or 5, so $1/5 \le x_n \le 5$. Both bounding values are attained, and the sequence is bounded.

The sequence $y_n = 1 + (-1)^n$

(a) A monotone subsequence. Select $n_k = 2k$, positive and increasing by 2. Then $y_{2k} = 1 + 1 = 2$ at every selected index. This constant subsequence is monotone.

(b) All subsequential limits. At odd indices, $y_{2k-1} = 1 - 1 = 0$, while at even indices $y_{2k} = 2$. The two valid selections give constant subsequences with limits 0 and 2. Because these are the only values taken by any term, Tool 3 proves the exhaustive set $S_y = \{0, 2\}$, with no infinite subsequential limit.

(c) Limsup and liminf. The largest and smallest elements give $\limsup y_n = 2$ and $\liminf y_n = 0$.

(d) Full-sequence behavior. The set of subsequential limits has two distinct elements. Thus there is no finite limit and no divergence to either $+\infty$ or $-\infty$.

(e) Boundedness. The values satisfy $0 \le y_n \le 2$ at every index, so the sequence is bounded.

The sequence $z_n = n\cos(n\pi/4)$

Write $n = 8q + r$, with $q \ge 0$ an integer and $r \in \{1, 2, 3, 4, 5, 6, 7, 8\}$. Then $n\pi/4 = 2q\pi + r\pi/4$. The cosine period and its unit-circle values give the complete table

r	$\cos(r\pi/4)$	z_n when $n = 8q + r$
1	$1/\sqrt{2}$	$n/\sqrt{2}$
2	0	0
3	$-1/\sqrt{2}$	$-n/\sqrt{2}$
4	-1	$-n$
5	$-1/\sqrt{2}$	$-n/\sqrt{2}$
6	0	0
7	$1/\sqrt{2}$	$n/\sqrt{2}$
8	1	n

Thus every nonzero term has magnitude at least $n/\sqrt{2}$; the only other possibility is the exact value 0.

(a) A monotone subsequence. Take $n_k = 8k$, with first index 8 and successive difference $8 > 0$. The selected values are

$$z_{8k} = 8k\cos(2k\pi) = 8k.$$

Their consecutive differences equal $8 > 0$, so this is a strictly increasing subsequence.

(b) Construct the three candidate limits. The same even-period selection gives $z_{8k} = 8k \to +\infty$: given $M > 0$, every $k > M/8$ makes $8k > M$. For negative infinity, choose $n_k = 8k - 4$, starting at 4 and increasing by 8. Then

$$z_{8k-4} = (8k-4)\cos((2k-1)\pi) = -(8k-4) = -8k + 4.$$

For $M < 0$, every $k > (4-M)/8$ satisfies $-8k + 4 < M$, so this subsequence tends to $-\infty$.

For the finite value 0, choose $n_k = 4k - 2$, starting at 2 and increasing by 4. The angle is $(4k-2)\pi/4 = k\pi - \pi/2$, whose cosine is 0. Thus $z_{4k-2} = 0$ for every k, giving a constant subsequence with limit 0.

(b) Exclude every other finite limit. Suppose a subsequence z_{n_k} converges to a finite $L \in \mathbb{R}$. Convergence with error tolerance 1 gives a threshold K_1 such that for $k > K_1$,

$$|z_{n_k}| \le |z_{n_k} - L| + |L| < 1 + |L|\,.$$

Choose $K_2 > \sqrt{2}(1 + |L|)$. For $k > K_2$, Tool 1 gives $n_k \ge k > \sqrt{2}(1 + |L|)$, so

$$\frac{n_k}{\sqrt{2}} > 1 + |L|\,.$$

At an index with $k > \max\{K_1, K_2\}$, a nonzero selected term would have, by the period table, $|z_{n_k}| \ge n_k/\sqrt{2} > 1 + |L|$, contradicting the bounded-tail estimate. Thus every sufficiently late selected term must be exactly 0.

Such a subsequence has limit 0 directly from the definition. It cannot have a different finite limit $L \neq 0$, because its late errors from L would all equal $|L|$, failing the tolerance $\varepsilon = |L|/2$. Consequently 0 is the only possible finite subsequential limit. The only infinite possibilities, $+\infty$ and $-\infty$, have already been constructed. The complete set is

$$S_z = \{-\infty, 0, +\infty\}.$$

(c) Limsup and liminf. The greatest and least elements of this set are $+\infty$ and $-\infty$. Hence $\limsup z_n = +\infty$ and $\liminf z_n = -\infty$.

(d) Full-sequence behavior. There are three distinct subsequential limits. By Tool 4 the original sequence does not converge to a finite real number and does not diverge to either infinity. In particular, the growing magnitudes in some residue classes do not remove the infinitely many exact zero terms.

(e) Boundedness. The values $z_{8k} = 8k$ are unbounded above, and the values $z_{8k-4} = -8k + 4$ are unbounded below. Thus (z_n) is bounded neither above nor below, and is not a bounded sequence.

Conclusion. In Exercise 11.4 none of the four full sequences has a finite or infinite limit. The sequences (x_n) and (y_n) are bounded, while (w_n) and (z_n) are unbounded above and below. Their complete subsequential-limit sets are, respectively,

$$\{-\infty, +\infty\}, \qquad \{1/5, 5\}, \qquad \{0, 2\}, \qquad \{-\infty, 0, +\infty\}.$$

The limsup and liminf for each sequence are the greatest and least elements of its displayed set, as proved in the individual answers to (c).

11.5 Exercise 11.5: Subsequential limits of an enumeration of a rational interval

What we must show. Let (q_n) enumerate all rational numbers in $(0, 1]$. Find (a) its set of subsequential limits and (b) its limit superior and limit inferior.

Plan. Every term lies between 0 and 1, so a subsequential limit must lie in $[0, 1]$. Conversely, every neighborhood of a point of $[0, 1]$ contains infinitely many distinct rational values in $(0, 1]$. Removing a finite prefix of the enumeration cannot remove all those values. This will allow a construction with strictly increasing indices.

1. Explain the role of enumeration and distinct values

The enumeration hypothesis says that each rational number in $(0, 1]$ occurs somewhere in the list. It does not require a particular rational number to occur infinitely often. For example, the value 1 may occur only once. The proof below uses infinitely many *different* rational values near a proposed limit, so it works both with and without repeated terms.

Every nonempty open interval (c, d) contains infinitely many rational numbers. To prove this from density, suppose its rational numbers formed a finite set. Density ensures the set is nonempty, so it would have a largest member $r < d$. Density applied to (r, d) would produce a rational number larger than r still belonging to (c, d), contradicting the choice of r. Thus there are infinitely many distinct rational numbers in the interval.

2. Part (a): Find infinitely many eligible values near every target

Fix $t \in [0, 1]$ and $\varepsilon > 0$. Consider

$$c = \max\{0, t - \varepsilon\}, \qquad d = \min\{1, t + \varepsilon\}.$$

We have $c < d$. At $t = 0$, this follows from $c = 0 < \min\{1, \varepsilon\} = d$. At $t = 1$, it follows from $c = \max\{0, 1 - \varepsilon\} < 1 = d$. For $0 < t < 1$, both $0 < t$ and $t - \varepsilon < t$ give $c < t$, while $t < 1$ and $t < t + \varepsilon$ give $t < d$.

Every rational number $r \in (c, d)$ satisfies

$$0 < r < 1, \qquad t - \varepsilon < r < t + \varepsilon.$$

It therefore belongs to the enumerated set and satisfies $|r - t| < \varepsilon$. By Step 1, there are infinitely many distinct such rational values. A finite prefix $q_1, \ldots, q_N$ contains only finitely many values, so it cannot contain them all. Hence for every finite index bound N there is an index $n > N$ such that

$$|q_n - t| < \varepsilon.$$

This reasoning includes the endpoints $t = 0$ and $t = 1$.

3. Construct a subsequence converging to the chosen target

Put $n_0 = 0$. At stage $k \geq 1$, apply Step 2 with $\varepsilon = 1/k$ and with the finite index bound n_{k-1}. Choose an index $n_k > n_{k-1}$ satisfying

$$|q_{n_k} - t| < \frac{1}{k}.$$

For definiteness, one may take the least eligible index, since a nonempty set of positive integers has a least element. The inequalities $n_1 < n_2 < \cdots$ ensure that (q_{n_k}) is a subsequence.

Let $\eta > 0$ and choose an integer $K > 1/\eta$ by the Archimedean property. If $k > K$, then

$$|q_{n_k} - t| < \frac{1}{k} < \eta.$$

Thus $q_{n_k} \to t$. Since $t \in [0,1]$ was arbitrary, every point of that interval is a subsequential limit. This is also the infinite-neighborhood criterion of Theorem 11.2(i).

4. Exclude every other finite or infinite limit

Every subsequence still satisfies $0 < q_{n_k} \leq 1$. If it had a finite limit $t < 0$, use the positive tolerance $\eta = -t/2$. Eventually its terms would be less than $t + \eta = t/2 < 0$, a contradiction. If it had a finite limit $t > 1$, use $\eta = (t-1)/2$. Eventually its terms would be greater than $t - \eta = (t+1)/2 > 1$, again a contradiction. Hence every finite subsequential limit lies in $[0,1]$.

No subsequence can tend to $+\infty$, because its terms never exceed 1 and so cannot eventually exceed the positive threshold 2. No subsequence can tend to $-\infty$, because its terms are positive and so cannot eventually be less than -1. Combining this exclusion with Step 3 proves that the exact set of subsequential limits is $[0,1]$.

5. Part (b): Take the extreme subsequential limits

By Theorem 11.8(ii), the limit superior is the supremum of the set of subsequential limits, and the limit inferior is its infimum. The interval $[0,1]$ has maximum 1 and minimum 0, so

$$\limsup_{n\to\infty} q_n = 1, \qquad \liminf_{n\to\infty} q_n = 0.$$

Conclusion. (a) The set of subsequential limits is $[0,1]$. (b) The limit superior is 1 and the limit inferior is 0. No repeated occurrence of an individual rational value is assumed.

11.6 Exercise 11.6: A subsequence of a subsequence

What we must show. Prove that a subsequence of a subsequence of a given sequence is itself a subsequence of the original sequence.

Plan. Use Definition 11.1 in terms of strictly increasing selection functions. Compose the two selection functions and verify that the composition remains strictly increasing.

1. Write the two selections precisely

Let $s : \mathbb{N} \to \mathbb{R}$ represent the original sequence. Suppose t is a subsequence of s. There is a strictly increasing function $\sigma : \mathbb{N} \to \mathbb{N}$ with

$$t_k = s_{\sigma(k)} \qquad (k \in \mathbb{N}).$$

Now suppose u is a subsequence of t. There is a strictly increasing function $\tau : \mathbb{N} \to \mathbb{N}$ with

$$u_j = t_{\tau(j)} \qquad (j \in \mathbb{N}).$$

Here strictly increasing refers to the *indices* selected by the functions; the numerical sequence values themselves need not increase or be distinct.

2. Form and check the composite selection

Define $\rho = \sigma \circ \tau$; explicitly, $\rho(j) = \sigma(\tau(j))$. Because τ takes positive integer values and σ is defined on every positive integer, ρ is a function from $\mathbb{N}$ to $\mathbb{N}$. If $j < \ell$, strict increase of τ gives $\tau(j) < \tau(\ell)$. Strict increase of σ then gives

$$\rho(j) = \sigma(\tau(j)) < \sigma(\tau(\ell)) = \rho(\ell).$$

Thus ρ is strictly increasing.

3. Identify the selected values

Substituting the first selection into the second yields, for each j,

$$u_j = t_{\tau(j)} = s_{\sigma(\tau(j))} = s_{\rho(j)}.$$

The values u_j therefore select terms of s in the strictly increasing index order $\rho(1) < \rho(2) < \cdots$. Definition 11.1 says exactly that this makes u a subsequence of s.

Conclusion. The composite selection function $\sigma \circ \tau$ proves that every subsequence of a subsequence is a subsequence of the original sequence.

11.7 Exercise 11.7: A subsequence of an enumeration of all rational numbers

What we must show. If (r_n) enumerates $\mathbb{Q}$, construct a subsequence (r_{n_k}) tending to $+\infty$.

Plan. Every tail of the enumeration remains unbounded above. Prove that statement with explicit control of a finite prefix, then select successive terms that exceed their subsequence indices.

1. Show that arbitrarily large rational values occur after any prefix

Fix a positive integer N and a real threshold M. The finite prefix has a finite maximum

$$P = \max\{r_1, \ldots, r_N\}.$$

By the Archimedean property, choose a positive integer $p > \max\{M, P, 1\}$. Since $p \in \mathbb{Q}$, the enumeration contains p: there is an index j with $r_j = p$. The inequality $p > P$ prevents $j \leq N$, because every term of that prefix is at most P. Thus $j > N$ and $r_j > M$. This proves that the tail after any finite prefix is unbounded above. Before any terms are excluded, the same conclusion follows by choosing an integer $p > M$ in the enumeration.

2. Select terms recursively with increasing indices

Choose n_1 so that $r_{n_1} > 1$; this is possible by Step 1. Suppose $n_1 < \cdots < n_{k-1}$ have been chosen. Apply Step 1 to the prefix ending at n_{k-1} and to the real threshold

$$M_k = \max\{k, r_{n_{k-1}}\}.$$

It supplies an index $n_k > n_{k-1}$ with

$$r_{n_k} > M_k \geq k, \qquad r_{n_k} > r_{n_{k-1}}.$$

The resulting indices are strictly increasing, so they define a subsequence. The selected values are strictly increasing as well, although this latter property is not needed just to be a subsequence.

3. Check the definition of divergence to positive infinity

Let $M > 0$ be arbitrary. Choose an integer $K > M$. If $k > K$, the construction gives

$$r_{n_k} > k > K > M.$$

Thus the selected values eventually exceed every prescribed positive real threshold. By Definition 9.8, $r_{n_k} \to +\infty$.

Conclusion. Such a subsequence exists; the construction ensures both $n_1 < n_2 < \cdots$ and $r_{n_k} > k$, which proves its limit is $+\infty$.

11.8 Exercise 11.8: Reflection exchanges the limit inferior and limit superior

What we must show. For every real sequence (s_n), prove

$$\liminf_{n\to\infty} s_n = -\limsup_{n\to\infty}(-s_n)$$

using Definition 10.6 and the reflection identity of Exercise 5.4.

Plan. Reflect each tail before taking its extremum. If the original sequence is unbounded below, use the infinite-value conventions in Definition 10.6 directly. Otherwise all relevant tail bounds are real numbers, and their limits can be compared using finite convergence or the definition of an infinite limit.

1. Apply reflection to the nonempty tails

For each positive integer N, let

$$E_N = \{s_n : n > N\}, \qquad -E_N = \{-s_n : n > N\}.$$

Each set is nonempty, since it contains the corresponding term with index $N + 1$. Exercise 5.4 gives the extended equality

$$\inf E_N = -\sup(-E_N).$$

Write $u_N = \inf E_N$ and $v_N = \sup(-E_N)$. The equality says $u_N = -v_N$, or equivalently $v_N = -u_N$, with the negation conventions of Section 5.

2. Handle a sequence that is unbounded below

Suppose (s_n) is unbounded below. Every tail E_N is also unbounded below. Indeed, if some E_N had a real lower bound c, then

$$\min\{c, s_1, \ldots, s_N\}$$

would be a real lower bound for the entire sequence, a contradiction. Hence $u_N = -\infty$ for every N.

The reflected sequence $(-s_n)$ is unbounded above: for every real M, a term $s_n < -M$ yields $-s_n > M$. Thus $v_N = +\infty$ for every N. Definition 10.6 explicitly decrees, in these respective cases,

$$\liminf s_n = -\infty, \qquad \limsup(-s_n) = +\infty.$$

Consequently

$$-\limsup(-s_n) = -(+\infty) = -\infty = \liminf s_n.$$

This step uses the stated conventions, not a real-sequence limit law applied to a sequence of infinite symbols.

3. When the original sequence is bounded below, all these tail bounds are real

Suppose $c \in \mathbb{R}$ is a lower bound for all the s_n. Then every nonempty tail E_N is bounded below by c, so $u_N \in \mathbb{R}$ by Corollary 4.5. The reflected tail is bounded above by $-c$, so $v_N \in \mathbb{R}$ by completeness. The identity from Step 1 is now an identity of real numbers:

$$v_N = -u_N \qquad (N \in \mathbb{N}).$$

Moreover $E_{N+1} \subseteq E_N$, and restricting the set cannot decrease its infimum. Thus $u_N \leq u_{N+1}$. The real sequence (u_N) is increasing and is bounded below by c. The monotone-sequence results of Section 10 show that it either converges to a finite real number or diverges to $+\infty$. These are the only two cases to consider here.

4. Treat a finite limit of the real tail infima

If $u_N \to u \in \mathbb{R}$, then Theorem 9.2, applied to the real sequence (u_N) with scalar -1, gives $v_N = -u_N \to -u$. Definition 10.6 now identifies

$$\liminf s_n = u, \qquad \limsup(-s_n) = -u.$$

Negating the second real equality proves $-\limsup(-s_n) = u = \liminf s_n$.

5. Treat divergence of the real tail infima to positive infinity

If $u_N \to +\infty$, we show directly that $v_N = -u_N \to -\infty$. Let $M < 0$. The number $-M$ is positive, so the hypothesis supplies a threshold N_0 such that $N > N_0$ implies $u_N > -M$. Negating gives $v_N = -u_N < M$ for all such N. This is precisely the definition of $v_N \to -\infty$. Consequently Definition 10.6 gives

$$\liminf s_n = +\infty, \qquad \limsup(-s_n) = -\infty.$$

The convention $-(-\infty) = +\infty$ then proves the desired identity.

Conclusion. In all cases, $\liminf s_n = -\limsup(-s_n)$. The infinite cases follow from tail unboundedness and the infinite-limit definitions, with finite limit laws used only for real tail bounds having a finite limit.

11.9 Exercise 11.9: Closed intervals and an impossible set of subsequential limits

What we must show. (a) Show that $[a, b]$ is closed for real $a \leq b$. (b) Decide whether some real sequence can have exactly $(0, 1)$ as its set of subsequential limits.

Plan. Use the sequential meaning of closed stated with Theorem 11.9: the set contains every limit of a convergent sequence of its points. For part (b), Theorem 11.9 requires the set of subsequential limits itself to have this property.

1. Part (a): Verify the sequential closedness condition

Let (t_k) be a sequence with $t_k \in [a, b]$ for every k, and suppose $t_k \to t \in \mathbb{R}$. Thus

$$a \leq t_k \leq b \qquad (k \in \mathbb{N}).$$

If $t < a$, take $\varepsilon = (a - t)/2 > 0$. Eventually $|t_k - t| < \varepsilon$, so

$$t_k < t + \varepsilon = \frac{a + t}{2} < a,$$

contradicting $t_k \geq a$. If $t > b$, take $\varepsilon = (t-b)/2 > 0$. Eventually $|t_k - t| < \varepsilon$, so

$$t_k > t - \varepsilon = \frac{b+t}{2} > b,$$

contradicting $t_k \leq b$. Both alternatives are impossible. Hence $a \leq t \leq b$, so $t \in [a, b]$. This proves that $[a, b]$ is closed in the required sequential sense.

The argument includes $a = b$: in that case all terms equal a and the only possible limit is a. If extended limits are also considered, neither infinite symbol can arise from such a bounded sequence. For $+\infty$, the positive threshold $\max\{1, b+1\} > b$ cannot be exceeded; for $-\infty$, the negative threshold $\min\{-1, a-1\} < a$ cannot be crossed downwards.

2. Part (b): Apply Theorem 11.9 to a sequence of proposed limits

Suppose, for a contradiction, that a real sequence (s_n) has exactly $S = (0, 1)$ as its set of subsequential limits. Define

$$t_k = \frac{1}{k+1} \qquad (k \in \mathbb{N}).$$

Since $k + 1 \geq 2$, we have $0 < t_k \leq 1/2 < 1$. Thus $t_k \in S$ for every k. Also $t_k \to 0$: for any $\varepsilon > 0$, choose an integer $K > 1/\varepsilon$; then $k > K$ gives

$$|t_k - 0| = \frac{1}{k+1} < \frac{1}{k} < \varepsilon.$$

Theorem 11.9 applies to this sequence of finite members of S and forces its limit 0 to belong to S. But $0 \notin (0, 1)$, a contradiction.

Conclusion. (a) Every closed interval $[a, b]$, including a one-point interval, is closed in the sequential sense. (b) No sequence has $(0, 1)$ as exactly its set of subsequential limits.

11.10 Exercise 11.10: The alternating diagonal traversal in Figure 11.2

What we must show. For the sequence listed by the arrows of Figure 11.2, find (a) its exact set of subsequential limits and (b) its limit superior and limit inferior.

Plan. The figure has the same row repeated infinitely many times. Each reciprocal value therefore occurs at arbitrarily large indices. The top row also supplies a subsequence of values tending to 0. To prove completeness of the answer, isolate any other positive proposed limit from all reciprocal values.

1. Describe the array and its order precisely

Number rows and columns by $i, j \in \mathbb{N}$. The entry in position (i, j) is $1/j$, so the array is

$$\begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \cdots \\ 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \cdots \\ 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \cdots \\ 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

The traversal groups the entries by diagonals $i + j = m + 1$, with $m = 1, 2, 3, \ldots$. Diagonal m has exactly m entries. For odd m, the traversal lists columns $1, 2, \ldots, m$; for even m, it lists columns

$m, m-1, \ldots, 1$. Thus the successive diagonal blocks begin

$$\underbrace{1}_{m=1},\quad \underbrace{\tfrac{1}{2}, 1}_{m=2},\quad \underbrace{1, \tfrac{1}{2}, \tfrac{1}{3}}_{m=3},\quad \underbrace{\tfrac{1}{4}, \tfrac{1}{3}, \tfrac{1}{2}, 1}_{m=4},\quad \ldots.$$

This order agrees with the alternating directions of the arrows.

2. Check explicitly that choosing later diagonals gives later indices

Before diagonal m there are

$$T_{m-1} = 1 + 2 + \cdots + (m-1) = \frac{m(m-1)}{2}$$

listed entries, with $T_0 = 0$. For $m \geq j$, the occurrence of $1/j$ in diagonal m has index

$$I(m, j) = \begin{cases} \dfrac{m(m-1)}{2} + j, & m \text{ odd}, \\ \dfrac{m(m-1)}{2} + (m - j + 1), & m \text{ even}. \end{cases}$$

Indeed, its position within that diagonal is j when columns are in increasing order, and $m - j + 1$ when they are in decreasing order. In both cases

$$T_{m-1} + 1 \leq I(m, j) \leq T_m, \qquad T_m = \frac{m(m+1)}{2}.$$

Every index of diagonal $m+1$ is at least $T_m + 1$, which is strictly larger than every index of diagonal m. Consequently selections made from successive diagonals always have strictly increasing indices.

3. Part (a): Obtain every reciprocal as a subsequential limit

Fix $j \in \mathbb{N}$. Every diagonal $m \geq j$ contains column j once, at the matrix position $(m + 1 - j, j)$. Choose the indices

$$n_k = I(j + k - 1, j) \qquad (k \in \mathbb{N}).$$

They are strictly increasing by Step 2, and the corresponding sequence values satisfy $s_{n_k} = 1/j$ for every k. This constant subsequence converges to $1/j$. Thus every member of $\{1/j : j \in \mathbb{N}\}$ is a subsequential limit.

4. Part (a): Obtain zero from the top row

The top-row entry in column k lies in diagonal k and has value $1/k$. Select it at index

$$n_k = I(k, k) = \begin{cases} \dfrac{k(k+1)}{2}, & k \text{ odd}, \\ \dfrac{k(k-1)}{2} + 1, & k \text{ even}. \end{cases}$$

These indices are strictly increasing because they come from successive diagonals. For example, they begin $1, 2, 6, 7, 15, \ldots$. The selected values are exactly $s_{n_k} = 1/k$. Given $\varepsilon > 0$, an integer $K > 1/\varepsilon$ ensures $k > K$ implies $|1/k - 0| < \varepsilon$. Therefore $s_{n_k} \to 0$, and 0 is also a subsequential limit.

5. Part (a): Exclude every other proposed limit

Every term of the full sequence belongs to $D = \{1/j : j \in \mathbb{N}\} \subseteq (0, 1]$. Hence any finite subsequential limit is in $[0, 1]$, by the same eventual-bound argument used in Exercise 11.5. The bounds also rule out both infinite subsequential limits.

Suppose $t > 0$ and $t \notin D$. We show that a fixed positive neighborhood of t contains no term of the sequence. Choose an integer $J > \max\{2, 2/t\}$. For every $j \geq J$, we have

$$\frac{1}{j} \leq \frac{1}{J} < \frac{t}{2}, \qquad \left|\frac{1}{j} - t\right| = t - \frac{1}{j} > \frac{t}{2}.$$

For the finitely many indices $1 \leq j < J$, none of the values $1/j$ equals t. Their distances from t are strictly positive, so the finite minimum

$$d = \min_{1 \leq j < J} \left|\frac{1}{j} - t\right|$$

exists and is positive. Define $\delta = \min\{t/4, d/2\} > 0$. If $j \geq J$, then $|1/j - t| > t/2 > \delta$. If $j < J$, then $|1/j - t| \geq d > \delta$. Thus every term of the original sequence satisfies $|s_n - t| > \delta$. No subsequence can converge to t, since convergence would require its terms eventually to have distance less than δ from t.

Every possible finite limit is either 0, a member of D, or a positive number not in D. Steps 3–5 account for all these possibilities, and the infinite cases were excluded. Therefore

$$S = \{0\} \cup \left\{\frac{1}{j} : j \in \mathbb{N}\right\}$$

is exactly the set of subsequential limits.

6. Part (b): Use the extreme members of the computed set

The set S has minimum 0 and maximum 1. Theorem 11.8(ii) therefore gives

$$\limsup_{n \to \infty} s_n = \sup S = 1, \qquad \liminf_{n \to \infty} s_n = \inf S = 0.$$

Conclusion. (a) The exact set is $S = \{0\} \cup \{1/j : j \in \mathbb{N}\}$. (b) The limit superior is 1 and the limit inferior is 0.

11.11 Exercise 11.11: An increasing sequence approaching a supremum from the set

What we must show. For a nonempty bounded set $S \subseteq \mathbb{R}$, construct an increasing sequence (s_n) with every $s_n \in S$ and $s_n \to \sup S$.

Plan. If the supremum belongs to the set, use the constant sequence at that point. Otherwise, at each stage choose a point of S above both the previous value and a closer lower approximation to the supremum.

1. State the necessary nonemptiness condition and the meaning of increasing

The exercise's statement needs the implicit assumption $S \neq \varnothing$. For the empty set there is no sequence of points in S, and its supremum is not defined by the real-supremum conventions being used. We therefore assume that S is nonempty and bounded. In fact, the construction needs only that it be bounded above. Completeness gives the finite real number

$$M = \sup S.$$

As in Section 10 and in the note to this exercise, increasing means nondecreasing: $s_n \le s_{n+1}$. A constant sequence is allowed. When $M \notin S$, the construction below will give strict increase.

2. The case in which the supremum belongs to the set

If $M \in S$, define $s_n = M$ for every $n \in \mathbb{N}$. Then all terms belong to S and $s_n = s_{n+1}$, so the sequence is increasing in the stated sense. For every $\varepsilon > 0$,

$$|s_n - M| = 0 < \varepsilon$$

at every index. Hence $s_n \to M$.

3. Begin the construction when M is not a member

Suppose $M \notin S$. Since M is an upper bound, every $s \in S$ satisfies $s \le M$; the nonmembership of M strengthens this to $s < M$ for every $s \in S$. The number $M - 1 < M$ cannot be an upper bound for S, because M is its least upper bound. Therefore some $s_1 \in S$ satisfies

$$M - 1 < s_1 < M.$$

4. Continue the construction with an explicitly admissible threshold

Suppose $s_{n-1} \in S$ has been chosen, where $n \ge 2$. Both $s_{n-1} < M$ and $M - 1/n < M$ hold. Therefore

$$c_n = \max\left\{s_{n-1}, M - \frac{1}{n}\right\} < M.$$

As $c_n < M = \sup S$, the number c_n is not an upper bound for S. Choose $s_n \in S$ with $s_n > c_n$. Every point of S is less than M, so this choice satisfies

$$s_{n-1} < s_n < M, \qquad M - \frac{1}{n} < s_n < M.$$

The procedure can be repeated at every positive integer stage. It gives a sequence in S that is strictly increasing. The displayed approximation bound also holds at $n = 1$ by Step 3.

5. Verify convergence to the supremum

For every n, the construction gives

$$0 < M - s_n < \frac{1}{n}.$$

Let $\varepsilon > 0$. Choose an integer $N > 1/\varepsilon$ by the Archimedean property. If $n > N$, then

$$|s_n - M| = M - s_n < \frac{1}{n} < \varepsilon.$$

Thus $s_n \to M$ by the definition of convergence.

Conclusion. With the necessary hypothesis $S \neq \varnothing$, an increasing sequence in S converging to $\sup S$ always exists. It may be taken constant when $\sup S \in S$, and strictly increasing when $\sup S \notin S$.

12 Limit superior and limit inferior

For each integer $N \geq 0$, the tail after N is the nonempty set

$$E_N = \{s_n : n > N\}.$$

Definition 10.6 gives

$$\limsup s_n = \lim_{N\to\infty} \sup E_N, \qquad \liminf s_n = \lim_{N\to\infty} \inf E_N.$$

These limits may be infinite. When a tail bound is infinite, we check the case through the definitions and extended order before applying any theorem about sequences of real numbers.

A tail supremum need not be attained by a term. Likewise, a term may lie above the lim sup or below the lim inf at arbitrarily large indices. The solutions check the relevant bound and eventuality conditions explicitly.

Tail notation and the meaning of infinite bounds

For a real sequence (x_n) and an integer $N \geq 1$, write

$$X_N = \{x_n : n > N\}, \qquad u_N(x) = \inf X_N, \qquad v_N(x) = \sup X_N.$$

The set X_N is nonempty because it contains x_{N+1}. We use the extended-bound conventions of Section 5: $u_N(x) = -\infty$ if the tail is unbounded below, and $v_N(x) = +\infty$ if it is unbounded above. For a nonempty set of real terms, a tail infimum cannot be $+\infty$, and a tail supremum cannot be $-\infty$. By Definition 10.6,

$$\liminf_{n\to\infty} x_n = \lim_{N\to\infty} u_N(x), \qquad \limsup_{n\to\infty} x_n = \lim_{N\to\infty} v_N(x).$$

Removing more terms enlarges the infimum and decreases the supremum, since $X_{N+1} \subseteq X_N$.

We will also use a finite-prefix observation. If a tail of a real sequence has an upper bound U, then adjoining its finitely many omitted terms still gives a bounded-above set: take the maximum of U and those finitely many terms. The corresponding minimum proves the analogous statement for lower bounds. Thus a sequence that is unbounded above, or unbounded below, has the same property on every tail. Infinite symbols will be handled through order and these conventions, rather than through real-number algebra.

12.1 Exercise 12.1: Eventual comparison passes to both limiting bounds

What we must show. Suppose there is an N_0 such that $s_n \leq t_n$ for all $n > N_0$. Prove

$$\liminf s_n \leq \liminf t_n, \qquad \limsup s_n \leq \limsup t_n.$$

No boundedness assumption is imposed on the two sequences.

Step 1: make the eventual comparison valid on the tails under discussion.

Enlarge N_0, if necessary, to a positive integer. This keeps the hypothesis true, because it discards additional initial terms. Fix any integer $N \geq N_0$. For every $n > N$ we then have $s_n \leq t_n$.

For the upper tail bounds, if $v_N(t) = +\infty$, the extended-order comparison $v_N(s) \le v_N(t)$ holds immediately. Otherwise $v_N(t)$ is real, and

$$s_n \le t_n \le v_N(t) \qquad (n > N).$$

Thus $v_N(t)$ is an upper bound of the tail of (s_n), so its least upper bound satisfies $v_N(s) \le v_N(t)$.

For the lower tail bounds, if $u_N(s) = -\infty$, the comparison $u_N(s) \le u_N(t)$ is automatic in the extended order. Otherwise $u_N(s)$ is real, and

$$u_N(s) \le s_n \le t_n \qquad (n > N).$$

Thus $u_N(s)$ is a lower bound of the tail of (t_n), so it is at most that tail's greatest lower bound. We have proved

$$u_N(s) \le u_N(t), \qquad v_N(s) \le v_N(t) \qquad (N \ge N_0).$$

Step 2: pass to the lower limiting bounds, checking infinite cases.

If (s_n) is unbounded below, every tail infimum $u_N(s)$ equals $-\infty$. Therefore $\liminf s_n = -\infty \le \liminf t_n$, and there is nothing further to prove for the lower bounds in this case.

If (s_n) is bounded below, choose a real lower bound a. For $n > N_0$, the hypothesis gives $t_n \ge s_n \ge a$. The finite-prefix observation shows that (t_n) is bounded below too. Consequently both $u_N(s)$ and $u_N(t)$ are real for every N. Their limits exist in the extended sense by Definition 10.6. Apply Exercise 9.9(c), the order rule for existing limits of real-valued sequences, to $u_N(s) \le u_N(t)$:

$$\lim_{N\to\infty} u_N(s) \le \lim_{N\to\infty} u_N(t).$$

The limits may be $+\infty$, but the terms to which the exercise was applied are real. By definition this is $\liminf s_n \le \liminf t_n$.

Step 3: pass to the upper limiting bounds, checking infinite cases.

If (t_n) is unbounded above, then every $v_N(t) = +\infty$, so $\limsup s_n \le +\infty = \limsup t_n$. If (t_n) is bounded above by a real number b, then $s_n \le t_n \le b$ for $n > N_0$. The finite-prefix observation shows that (s_n) is bounded above as well. Hence $v_N(s)$ and $v_N(t)$ are real-valued sequences, with existing extended limits. Applying Exercise 9.9(c) to their eventual inequality gives

$$\lim_{N\to\infty} v_N(s) \le \lim_{N\to\infty} v_N(t),$$

which is $\limsup s_n \le \limsup t_n$.

Conclusion. Both comparisons hold:

$$\boxed{\liminf s_n \le \liminf t_n, \qquad \limsup s_n \le \limsup t_n}.$$

The proof includes infinite limiting bounds. It does not treat $+\infty$ or $-\infty$ as individual real terms in an application of a real-sequence limit rule.

12.2 Exercise 12.2: A zero limit superior of absolute values

What we must show. Prove the equivalence

$$\limsup |s_n| = 0 \quad \Longleftrightarrow \quad s_n \longrightarrow 0.$$

Forward direction, Step 1: derive the boundedness needed for finite tail suprema.

Assume $\limsup |s_n| = 0$ and put $a_n = |s_n| \geq 0$. If (a_n) were unbounded above, every tail would be unbounded above by the finite-prefix observation. Its every tail supremum would therefore be $+\infty$, giving $\limsup a_n = +\infty$, contrary to the assumed value 0. Thus (a_n) is bounded above. Because all $a_n \geq 0$, it is bounded below as well.

It follows that the numbers

$$V_N = \sup\{|s_n| : n > N\}$$

are finite and nonnegative for every positive integer N. Their nonnegativity follows, for example, by comparing V_N with the member $|s_{N+1}| \geq 0$ of the tail. The hypothesis and Definition 10.6 now say $V_N \to 0$.

Forward direction, Step 2: control every term in one sufficiently late tail.

Let $\varepsilon > 0$. From $V_N \to 0$, choose a positive integer N_* large enough that $|V_{N_*} - 0| < \varepsilon$. Since $V_{N_*} \geq 0$, this means $V_{N_*} < \varepsilon$. For every integer $n > N_*$, the definition of tail supremum gives

$$|s_n - 0| = |s_n| \leq V_{N_*} < \varepsilon.$$

This is Definition 7.1 for $s_n \to 0$. Boundedness was obtained from the hypothesis before any finite-tail estimate was used.

Reverse direction, Step 1: verify convergence of the absolute values.

Assume $s_n \to 0$. Let $\varepsilon > 0$. The definition gives a threshold N such that $n > N$ implies $|s_n - 0| < \varepsilon$. Since $|s_n| \geq 0$, the error in the sequence of absolute values is exactly

$$||s_n| - 0| = |s_n| = |s_n - 0| < \varepsilon.$$

Thus $|s_n| \to 0$ directly from the definition.

Reverse direction, Step 2: identify its limit superior.

Theorem 10.7 says that when a sequence has a limit, its limit inferior and limit superior equal that limit. Applying it to the convergent sequence $(|s_n|)$ gives $\limsup |s_n| = 0$.

Conclusion. The two implications prove $\boxed{\limsup |s_n| = 0 \iff s_n \to 0}$.

12.3 Exercise 12.3: Two synchronized sequences of period four

What we must show. Compute all seven requested combinations for the sequences whose repeated blocks are $(0, 1, 2, 1)$ and $(2, 1, 1, 0)$.

Step 1: keep the phases aligned in one residue table.

The index n has one of four residues modulo 4. Reading the entries at the same index gives

Residue of n	s_n	t_n	$s_n + t_n$	$s_n t_n$
1	0	2	2	0
2	1	1	2	1
3	2	1	3	2
0	1	0	1	0

For instance, at indices $n = 4j + 3$ the entries are $s_n = 2$ and $t_n = 1$, so the sum is 3 and the product is 2. Every row occurs at arbitrarily large indices: by taking the integer j sufficiently large, its index exceeds any prescribed threshold. For the residue 0, use indices $4, 8, 12, \ldots$.

Step 2: identify each tail's minimum and maximum.

Every tail of (s_n) contains the values $0, 1, 2$ and no others. Thus its infimum is 0 and its supremum is 2. The same is true for every tail of (t_n). The sum has the repeated block $(2, 2, 3, 1)$, so every tail of the sum contains exactly the values $1, 2, 3$; its infimum is 1 and its supremum is 3. The product has the repeated block $(0, 1, 2, 0)$, so its tail infima are 0 and its tail suprema are 2. Taking the limits of these constant sequences of tail bounds gives

$$\liminf s_n = 0, \quad \limsup s_n = 2, \qquad \liminf t_n = 0, \quad \limsup t_n = 2,$$

$$\liminf(s_n + t_n) = 1, \quad \limsup(s_n + t_n) = 3, \qquad \liminf(s_n t_n) = 0, \quad \limsup(s_n t_n) = 2.$$

Step 3: evaluate every requested expression.

(a) $\liminf s_n + \liminf t_n = 0 + 0 = \boxed{0}$,

(b) $\liminf(s_n + t_n) = \boxed{1}$,

(c) $\liminf s_n + \limsup t_n = 0 + 2 = \boxed{2}$,

(d) $\limsup(s_n + t_n) = \boxed{3}$,

(e) $\limsup s_n + \limsup t_n = 2 + 2 = \boxed{4}$,

(f) $\liminf(s_n t_n) = \boxed{0}$,

(g) $\limsup(s_n t_n) = \boxed{2}$.

Conclusion. The phase alignment matters. The largest value of (s_n) occurs when $t_n = 1$, whereas the largest value of (t_n) occurs when $s_n = 0$. Adding the separate maxima therefore gives 4, even though the paired sum has maximum 3.

12.4 Exercise 12.4: The limit superior of a sum is at most the sum of the limit superiors

What we must show. For bounded real sequences (s_n) and (t_n), prove

$$\limsup(s_n + t_n) \le \limsup s_n + \limsup t_n.$$

Step 1: establish that all relevant tail bounds are finite.

Choose real constants $A, B \ge 0$ with $|s_n| \le A$ and $|t_n| \le B$ for every n. The triangle inequality gives $|s_n + t_n| \le A + B$, so the sum sequence is bounded as well. For every positive integer N, the three tails are nonempty and bounded, and hence the numbers

$$S_N = \sup\{s_n : n > N\}, \quad T_N = \sup\{t_n : n > N\}, \quad W_N = \sup\{s_n + t_n : n > N\}$$

are real. Their limits as $N \to \infty$ are also finite, since the underlying sequences are bounded.

Step 2: prove the finite-tail inequality.

For every integer $n > N$, the upper-bound properties give $s_n \le S_N$ and $t_n \le T_N$. Adding yields

$$s_n + t_n \le S_N + T_N.$$

Thus the real number $S_N + T_N$ is an upper bound of the paired-sum tail. Its least upper bound satisfies

$$W_N \le S_N + T_N.$$

This argument does not require S_N or T_N to be attained. It also does not require a single index where both separate bounds are approached at once.

Step 3: take limits of the real tail-bound sequences.

By Definition 10.6,

$$S_N \to \limsup s_n, \qquad T_N \to \limsup t_n, \qquad W_N \to \limsup(s_n + t_n).$$

The sum law from Section 9 gives

$$S_N + T_N \longrightarrow \limsup s_n + \limsup t_n.$$

Apply Exercise 9.9(c) to the termwise comparison $W_N \le S_N + T_N$. The resulting comparison of limits is

$$\limsup(s_n + t_n) \le \limsup s_n + \limsup t_n.$$

Conclusion. This proves the desired inequality. The boundedness hypotheses ensure that every sum of limiting bounds used here is a sum of real numbers; no undefined combination of infinite symbols has been introduced.

12.5 Exercise 12.5: The corresponding lower bound for the limit inferior of a sum

What we must show. Use Exercises 11.8 and 12.4 to prove, for bounded real sequences,

$$\liminf(s_n + t_n) \ge \liminf s_n + \liminf t_n.$$

Step 1: verify that the reflected sequences meet the hypotheses.

The sequences $(-s_n)$ and $(-t_n)$ are bounded because $|-s_n| = |s_n|$ and $|-t_n| = |t_n|$. Their sum is $-s_n - t_n = -(s_n + t_n)$, which is bounded too. Exercise 12.4 therefore applies and gives

$$\limsup\big(-(s_n + t_n)\big) \le \limsup(-s_n) + \limsup(-t_n).$$

Step 2: replace each reflected upper limiting bound.

Exercise 11.8 states $\liminf x_n = -\limsup(-x_n)$. For the bounded sequences here, all these values are real. Taking their negatives also gives $\limsup(-x_n) = -\liminf x_n$. Substitute this identity in the three places in Step 1:

$$-\liminf(s_n + t_n) \le -\liminf s_n - \liminf t_n = -\big(\liminf s_n + \liminf t_n\big).$$

Step 3: reverse the inequality by negation.

Multiplying a real inequality by -1 reverses its direction. Negating both sides of the last comparison, and taking the negative twice in each expression, yields

$$\liminf(s_n + t_n) \geq \liminf s_n + \liminf t_n.$$

Conclusion. The claimed lower inequality follows. The sign reversal is the reason that reflecting the upper inequality from Exercise 12.4 produces a lower inequality here.

12.6 Exercise 12.6: Scaling the upper and lower limiting bounds

What we must show. For bounded (s_n) and $k \geq 0$, prove (a) $\limsup(ks_n) = k \limsup s_n$ and (b) $\liminf(ks_n) = k \liminf s_n$. Determine the correct formulas in (c) when $k < 0$.

Preliminary step: scaling preserves boundedness.

Choose $A \geq 0$ with $|s_n| \leq A$ for every n. For any fixed real k, the identity $|ks_n| = |k|\,|s_n| \leq |k|\,A$ shows that (ks_n) is bounded. Its tails are nonempty. Hence all tail bounds for (s_n) and (ks_n) below are real numbers. Write

$$m_N = \inf\{s_n : n > N\}, \qquad M_N = \sup\{s_n : n > N\}.$$

The limits of these two sequences are $\liminf s_n$ and $\limsup s_n$, respectively, and are finite by boundedness.

Parts (a) and (b), the case $k = 0$.

If $k = 0$, every scaled term is zero. Every scaled tail is the singleton $\{0\}$, whose infimum and supremum are zero. Therefore both limiting bounds of (ks_n) are zero. The expressions $0 \cdot \liminf s_n$ and $0 \cdot \limsup s_n$ are also zero, because their other factors are finite real numbers. Thus both requested formulas hold in this case.

Part (a), Step 1: prove the upper-tail scaling formula for $k > 0$.

Fix $k > 0$ and an integer $N \geq 1$. Since $s_n \leq M_N$ for all $n > N$, multiplication by k gives $ks_n \leq kM_N$. Thus kM_N is an upper bound of the scaled tail.

To check that it is the least upper bound, let U be any real upper bound of that tail. Then $ks_n \leq U$ for every $n > N$. Division by $k > 0$ gives $s_n \leq U/k$ for all these indices, so U/k is an upper bound of the original tail. The least-upper-bound property gives $M_N \leq U/k$. Multiplying by $k > 0$ gives $kM_N \leq U$. We have proved both defining properties of a supremum, so

$$\sup\{ks_n : n > N\} = kM_N.$$

Part (a), Step 2: take the limit of the real tail bounds.

By the constant-multiple limit law from Section 9,

$$kM_N \longrightarrow k \limsup s_n.$$

Definition 10.6 identifies the limit of the left side of the tail identity as $\limsup(ks_n)$. Hence

$$\boxed{\limsup(ks_n) = k \limsup s_n} \qquad (k > 0).$$

Together with the zero case, this proves part (a) for all $k \geq 0$.

Part (b), Step 1: prove the lower-tail scaling formula for $k > 0$.

For every $n > N$, $m_N \leq s_n$; multiplying by $k > 0$ gives $km_N \leq ks_n$. Thus km_N is a lower bound of the scaled tail. If L is any real lower bound of that tail, then $L \leq ks_n$ for every $n > N$. Divide by $k > 0$ to obtain $L/k \leq s_n$. Consequently L/k is a lower bound of the original tail. Since m_N is its greatest lower bound, $L/k \leq m_N$, and hence $L \leq km_N$. These two properties establish

$$\inf\{ks_n : n > N\} = km_N.$$

Taking limits of this equality and using the constant-multiple law gives

$$\boxed{\liminf(ks_n) = k \liminf s_n} \qquad (k > 0).$$

The zero case already proved extends it to all $k \geq 0$.

Part (b), Step 2: relate the formula to the reflection hint.

For completeness, the same lower formula follows from part (a) and Exercise 11.8. When $k > 0$,

$$\begin{aligned} \liminf(ks_n) &= -\limsup(-ks_n), \\ &= -\limsup(k(-s_n)), \\ &= -k\limsup(-s_n), \\ &= k\liminf s_n. \end{aligned}$$

The third equality uses part (a) for the bounded sequence $(-s_n)$; the first and last use reflection. All negatives here are negatives of finite real numbers.

Part (c), Step 1: a negative multiplier exchanges the tail bounds.

Now let $k < 0$ and fix N. Multiplication by k reverses order. Since $m_N \leq s_n$ for every $n > N$, we have $ks_n \leq km_N$; thus km_N is an upper bound of the scaled tail. For any other upper bound U, the inequalities $ks_n \leq U$ become $s_n \geq U/k$ after division by $k < 0$. Hence U/k is a lower bound of the original tail, so $U/k \leq m_N$. Multiplying this last inequality by $k < 0$ reverses it and gives $U \geq km_N$. Therefore

$$\sup\{ks_n : n > N\} = km_N \qquad (k < 0).$$

Similarly, $s_n \leq M_N$ gives $ks_n \geq kM_N$, so kM_N is a lower bound of the scaled tail. If L is any other lower bound, $L \leq ks_n$ gives $L/k \geq s_n$ after division by $k < 0$. Thus L/k is an upper bound of the original tail and $M_N \leq L/k$. Multiplying by $k < 0$ gives $kM_N \geq L$. It follows that

$$\inf\{ks_n : n > N\} = kM_N \qquad (k < 0).$$

Part (c), Step 2: take finite limits.

The constant-multiple law, now with a fixed negative constant, still applies to these real convergent tail-bound sequences. Taking limits of the two identities gives

$$\boxed{\limsup(ks_n) = k\liminf s_n, \qquad \liminf(ks_n) = k\limsup s_n} \qquad (k < 0).$$

Conclusion. A positive multiplier preserves the upper and lower limiting bounds separately. A zero multiplier makes both zero. A negative multiplier exchanges their roles, as the two formulas in part (c) state. All statements were proved without assuming that any tail supremum or infimum is attained.

12.7 Exercise 12.7: A positive multiplier preserves an infinite limit superior

What we must show. Suppose $\limsup s_n = +\infty$ and $k > 0$. Prove $\limsup(ks_n) = +\infty$ without assuming that (s_n) is bounded.

Step 1: prove that every tail of the original sequence is unbounded above.

Suppose some tail were bounded above by a real U. Then $s_n \le U$ for every index in that tail, so the eventual comparison result of Exercise 12.1, applied to (s_n) and the constant sequence U, would give

$$\limsup s_n \le \limsup U = U.$$

This contradicts $\limsup s_n = +\infty$. Thus no such tail bound exists: for each integer N and each real R, some index $n > N$ satisfies $s_n > R$.

Step 2: prove that every scaled tail is unbounded above.

Fix any positive integer N and any proposed real upper bound M for the scaled tail. Since $k > 0$, the quotient M/k is real. Step 1, with $R = M/k$, gives an index $n > N$ such that $s_n > M/k$. Multiplying by the positive number k preserves strict inequality:

$$ks_n > M.$$

Thus M is not an upper bound. Because M was arbitrary, the scaled tail is unbounded above. Because N was arbitrary, every scaled tail has this property.

Step 3: apply the definition of the infinite tail bound.

For every N, the extended-bound convention gives

$$\sup\{ks_n : n > N\} = +\infty.$$

Its limiting value is therefore $+\infty$, which is exactly $\limsup(ks_n)$ by Definition 10.6.

Conclusion. We obtain $\boxed{\limsup(ks_n) = +\infty}$. The proof divided and multiplied only real numbers, such as M and M/k, and used unboundedness to interpret the infinite result.

12.8 Exercise 12.8: A product bound for bounded nonnegative sequences

What we must show. Let (s_n) and (t_n) be bounded sequences with $s_n \ge 0$ and $t_n \ge 0$ for every n. Prove

$$\limsup(s_n t_n) \le (\limsup s_n)(\limsup t_n).$$

Step 1: give finite bounds for the three sequences.

Choose $A, B \ge 0$ such that $0 \le s_n \le A$ and $0 \le t_n \le B$ for every n. Because $t_n \ge 0$, multiplication of $s_n \le A$ by t_n gives $s_n t_n \le A t_n$. Because $A \ge 0$, multiplication of $t_n \le B$ by A gives $A t_n \le AB$. Thus

$$0 \le s_n t_n \le A t_n \le AB.$$

The product sequence is bounded and nonnegative too. For every N, all three tails are nonempty and bounded, and their suprema

$$S_N = \sup\{s_n : n > N\}, \quad T_N = \sup\{t_n : n > N\}, \quad P_N = \sup\{s_n t_n : n > N\}$$

are finite and nonnegative. For example, $0 \le s_{N+1} \le S_N$, which proves $S_N \ge 0$; the other two nonnegativity statements follow in the same way from their tails.

Step 2: prove the paired-product tail inequality.

For each $n > N$, the tail upper bounds give $s_n \le S_N$ and $t_n \le T_N$. Multiply the first inequality by $t_n \ge 0$:

$$s_n t_n \le S_N t_n.$$

Multiply the second by $S_N \ge 0$:

$$S_N t_n \le S_N T_N.$$

The sign checks are necessary for both comparisons. Combining them shows $s_n t_n \le S_N T_N$ for every $n > N$. Hence $S_N T_N$ is an upper bound of the product tail. Its least upper bound satisfies

$$P_N \le S_N T_N.$$

Neither separate tail supremum needs to be attained, and no common index attaining both was selected.

Step 3: compare the finite limits of the tail-bound sequences.

Definition 10.6 gives

$$S_N \to \limsup s_n, \qquad T_N \to \limsup t_n, \qquad P_N \to \limsup(s_n t_n).$$

All limits here are finite because the original sequences and the product sequence are bounded. The product limit law from Section 9 therefore yields

$$S_N T_N \longrightarrow (\limsup s_n)(\limsup t_n).$$

Applying Exercise 9.9(c) to $P_N \le S_N T_N$ gives the requested comparison of limiting values.

Conclusion. We have proved

$$\boxed{\limsup(s_n t_n) \le (\limsup s_n)(\limsup t_n)}.$$

The proof includes the case where either upper limiting bound is zero, because no division by one of those bounds was used.

12.9 Exercise 12.9: A positive lower bound and products tending upward

What we must show. Prove the product assertions in parts (a) and (b), and identify Exercise 12.7 as the special case requested in part (c).

A preliminary fact: extracting one positive lower bound.

Write $\alpha = \liminf t_n$, and suppose $\alpha > 0$. The value $\alpha = +\infty$ is allowed. We claim that there are a real number $c > 0$ and an integer N_0 such that

$$t_n \ge c \qquad (n > N_0).$$

If α is finite, choose $c = \alpha/2$; then $0 < c < \alpha$. If $\alpha = +\infty$, choose $c = 1$ instead. In either case, the tail infima $\inf\{t_n : n > N\}$ eventually exceed c, by their convergence to α. Each term in such a tail is

at least its infimum, and therefore exceeds c. Replacing this strict inequality by $\geq c$ gives the claim. This argument also covers an initially unbounded sequence: since $\liminf t_n > 0$, some tail has a finite lower bound.

(a) An ordinary limit of $+\infty$.

Assume $s_n \to +\infty$. To prove $s_n t_n \to +\infty$, fix an arbitrary real threshold $M > 0$. By the definition of this limit, there is N_1 such that

$$n > N_1 \quad \Longrightarrow \quad s_n > M/c.$$

Since $M/c > 0$, these s_n are positive. If $n > \max\{N_0, N_1\}$, we can multiply $t_n \geq c$ by the positive number s_n, obtaining

$$s_n t_n \geq c s_n > c(M/c) = M.$$

Thus every sufficiently late product exceeds M. Checking positive thresholds suffices: exceeding any positive number also exceeds all smaller, possibly negative, thresholds. Hence $s_n t_n \to +\infty$.

(b) A limit superior of $+\infty$.

Now assume only $\limsup s_n = +\infty$. Every tail of (s_n) is unbounded above. Indeed, a tail bounded above, together with the finitely many preceding real terms, would bound the whole sequence above and force its limit superior to be at most a real upper bound.

Fix a tail index N and a positive threshold M. The unbounded tail beyond $\max\{N, N_0\}$ contains an index n for which $s_n > M/c$. At that index $n > N$, and the same sign calculation as in part (a) gives $s_n t_n > M$. Thus every tail of the product sequence is unbounded above. Its tail suprema are all $+\infty$, so its limit superior is $+\infty$.

The distinction between (a) and (b) is in the quantifiers: part (a) puts every sufficiently late product above M; part (b) produces at least one such product in every tail.

(c) Recover Exercise 12.7.

If $t_n = k > 0$ for every n, then $\liminf t_n = k > 0$ and $s_n t_n = k s_n$. Part (b) therefore states exactly that

$$\limsup s_n = +\infty \quad \Longrightarrow \quad \limsup(k s_n) = +\infty \qquad (k > 0).$$

Conclusion. Both product conclusions hold; a positive constant second factor gives Exercise 12.7.

12.10 Exercise 12.10: Detecting boundedness with the limit superior of absolute values

What we must show. Prove both directions of

$$(s_n) \text{ is bounded} \quad \Longleftrightarrow \quad \limsup |s_n| < +\infty.$$

Plan. Absolute values express upper and lower bounds on s_n in one nonnegative sequence. An eventual bound becomes a bound on all terms after we include the finite initial segment.

A bounded sequence gives a finite limit superior.

If (s_n) is bounded, there is a finite $B \geq 0$ with $|s_n| \leq B$ for every n. Thus for every N,

$$0 \leq \sup\{|s_n| : n > N\} \leq B.$$

These tail suprema decrease as N increases. The monotone sequence theorem gives a limit in $[0, B]$, and by definition that limit is $\limsup |s_n|$. In particular it is less than $+\infty$.

A finite limit superior gives an eventual bound.

Conversely, suppose $L = \limsup |s_n| < +\infty$. Since all the absolute values are nonnegative, $L \geq 0$, so L is a real number. The sequence $(|s_n|)$ cannot be unbounded above: if it were, every tail would be unbounded above, since deleting finitely many real terms cannot remove unboundedness. All its tail suprema would then be $+\infty$, contradicting the finite value of L.

Equivalently, some tail has a finite supremum. Once that happens, all subsequent tail suprema are finite and converge to L. Their convergence yields an integer $N \geq 1$ with

$$|s_n| \leq \sup\{|s_j| : j > N\} < L + 1 \qquad (n > N).$$

Include the early terms.

The finite set of real numbers $\{|s_1|, \ldots, |s_N|, L+1\}$ has a finite maximum B. If $n \leq N$, the definition of B gives $|s_n| \leq B$. If $n > N$, the tail estimate gives $|s_n| < L + 1 \leq B$. Consequently $|s_n| \leq B$ for every index n.

Conclusion. The two conditions are equivalent. The limit superior bounds the tail, and a finite maximum absorbs the initial terms.

12.11 Exercise 12.11: The lower ratio–root inequality

What we must show. For a sequence with $s_n \neq 0$ at every index, prove the first inequality of Theorem 12.2:

$$\liminf \left| \frac{s_{n+1}}{s_n} \right| \;\leq\; \liminf |s_n|^{1/n}.$$

Plan. A lower bound on successive ratios becomes a geometric lower bound on the terms. Taking nth roots removes the fixed factor.

Name the two nonnegative extended limits.

Set

$$A = \liminf \left| \frac{s_{n+1}}{s_n} \right|, \qquad B = \liminf |s_n|^{1/n}.$$

Every ratio is well defined and positive because no s_n is zero. The roots are positive as well. Therefore $A, B \in [0, +\infty]$. If $A = 0$, then $A \leq B$ follows immediately from $B \geq 0$.

An arbitrary positive number below A is an eventual ratio bound.

Suppose $A > 0$, and choose any real a with $0 < a < A$. When $A = +\infty$, this means that a can be any positive real number. The definition of the limit inferior gives an integer $K \geq 1$ such that

$$\left| \frac{s_{j+1}}{s_j} \right| > a \qquad (j \geq K).$$

To see this directly, some tail infimum of these ratios exceeds a; each ratio in that tail is at least the infimum. Increase the starting index to K if needed.

Multiply the consecutive inequalities.

For $n > K$, cancellation of the nonzero terms gives

$$|s_n| = |s_K| \prod_{j=K}^{n-1} \left| \frac{s_{j+1}}{s_j} \right| \geq |s_K|\, a^{n-K}.$$

The weak inequality also holds when $n = K$, since the product then has no factors and equals 1. Define the fixed positive constant $C = |s_K|\, a^{-K}$. The estimate becomes

$$|s_n| \geq Ca^n \qquad (n \geq K).$$

The map $x \mapsto x^{1/n}$ is increasing on nonnegative reals, so

$$|s_n|^{1/n} \geq aC^{1/n} \qquad (n \geq K).$$

Pass to the limit inferior and then let a approach A.

Since $C > 0$, the elementary root limit gives $C^{1/n} \to 1$. Thus $aC^{1/n} \to a$, and eventual order (Exercise 12.1) yields $B \geq a$.

If A is finite and $B < A$, then B is finite and a number a strictly between B and A satisfies $a > 0$. This contradicts $B \geq a$. Hence $B \geq A$ in the finite case. If $A = +\infty$, the inequality $B \geq a$ holds for every positive real a; a finite B would be contradicted by taking $a > B$. Therefore $B = +\infty$.

Conclusion. In every case the lower ratio limit inferior is at most the root limit inferior. This proof does not rely on Corollary 12.3, whose proof uses the inequality being established here.

12.12 Exercise 12.12: Arithmetic means and the limits of the original sequence

What we must show. For $s_n \geq 0$, put

$$\sigma_n = \frac{s_1 + \cdots + s_n}{n}.$$

Prove the four-term inequality in part (a), deduce part (b), and give the counterexample requested in part (c).

(a), preparation: name the bounds and check nonnegativity.

Write $\alpha = \liminf s_n$ and $\beta = \limsup s_n$. Because $s_n \geq 0$, all averages are nonnegative, and $\alpha, \beta, \liminf \sigma_n, \limsup \sigma_n$ belong to $[0, +\infty]$. The inequality $\liminf \sigma_n \leq \limsup \sigma_n$ holds for every sequence by Theorem 10.7. We prove the two outside inequalities separately.

(a), upper bound: verify the estimate in the hint.

First suppose $\beta < +\infty$. The sequence (s_n) must then be bounded above: otherwise every tail would be unbounded above, and its limit superior would be $+\infty$. Set

$$U_N = \sup\{s_n : n > N\}, \qquad P_N = s_1 + \cdots + s_N.$$

These are finite, with $U_N, P_N \geq 0$. If $n > N$, each of the $n - N$ remaining terms is at most U_N, so

$$\sigma_n \leq \frac{P_N}{n} + \frac{n-N}{n} U_N \leq \frac{P_N}{n} + U_N.$$

The second inequality uses $U_N \geq 0$ and $(n-N)/n \leq 1$. If $n > M > N$, the inequality $P_N/n \leq P_N/M$ uses $P_N \geq 0$. Taking the supremum over $n > M$ gives the hint exactly:

$$\sup\{\sigma_n : n > M\} \leq \frac{P_N}{M} + U_N.$$

Hold N fixed and let $M \to \infty$. The left side tends to $\limsup \sigma_n$, while $P_N/M \to 0$, yielding $\limsup \sigma_n \leq U_N$. This holds for every N. Letting $N \to \infty$ and using $U_N \to \beta$, we get $\limsup \sigma_n \leq \beta$. All the relevant quantities are finite here, since a global bound on s_n also bounds each average.

If instead $\beta = +\infty$, the desired upper bound is simply $\limsup \sigma_n \leq +\infty$, which always holds. No finite-tail estimate involving an infinite supremum is required.

(a), lower bound: retain a uniformly positive tail.

If $\alpha = 0$, nonnegativity already gives $\liminf \sigma_n \geq 0 = \alpha$. If $\alpha > 0$, choose an arbitrary real c with $0 < c < \alpha$. For $\alpha = +\infty$, take any $c > 0$. There is N such that $s_j \geq c$ for all $j > N$, by the definition of the limit inferior. For $n > N$, discard the nonnegative initial sum and estimate the tail:

$$\sigma_n = \frac{P_N + s_{N+1} + \cdots + s_n}{n} \geq \frac{(n-N)c}{n} = c\left(1 - \frac{N}{n}\right).$$

The expression on the right tends to c, since N is fixed. Exercise 12.1 gives $\liminf \sigma_n \geq c$. When α is finite, letting c range over all positive numbers below α forces $\liminf \sigma_n \geq \alpha$: otherwise choose c between these two quantities. When $\alpha = +\infty$, the same inequality for every $c > 0$ forces $\liminf \sigma_n = +\infty$. Combining the three inequalities proves

$$\boxed{\liminf s_n \leq \liminf \sigma_n \leq \limsup \sigma_n \leq \limsup s_n.}$$

(b) Convergence passes to the averages.

Suppose $s_n \to L$. Nonnegativity means that either $L \geq 0$ is finite or $L = +\infty$; a limit of $-\infty$ is impossible. Then $\liminf s_n = \limsup s_n = L$, and part (a) gives

$$L \leq \liminf \sigma_n \leq \limsup \sigma_n \leq L.$$

Thus both middle quantities equal L. Theorem 10.7, which characterizes convergence by equality of the limit inferior and limit superior, implies $\sigma_n \to L$. If $L = +\infty$, this statement means that the averages also tend to $+\infty$.

(c) Convergent averages need not imply convergent terms.

Take $s_n = 1 + (-1)^n$. Its terms alternate $0, 2, 0, 2, \ldots$, and are nonnegative. The odd subsequence is constantly 0 and the even subsequence is constantly 2, so (s_n) has no limit.

For $n = 2k$, there are k zeros and k twos, so $\sigma_{2k} = 2k/(2k) = 1$. For $n = 2k - 1$, there are k zeros and $k - 1$ twos, so

$$\sigma_{2k-1} = \frac{2(k-1)}{2k-1} = 1 - \frac{1}{2k-1}.$$

In both cases $|\sigma_n - 1| \leq 1/n$. Given $\varepsilon > 0$, choose $N > 1/\varepsilon$; then $n > N$ implies $|\sigma_n - 1| \leq 1/n < \varepsilon$. Thus $\sigma_n \to 1$.

Conclusion. The averages satisfy the four-term bounds and inherit every limit of a nonnegative sequence. The converse fails for alternating zeros and twos.

12.13 Exercise 12.13: Limit inferior and superior as eventual bounds

What we must show. For a bounded real sequence define

$$A = \{a \in \mathbb{R} : \{n \in \mathbb{N} : s_n < a\} \text{ is finite}\},$$
$$B = \{b \in \mathbb{R} : \{n \in \mathbb{N} : s_n > b\} \text{ is finite}\}.$$

Prove $\sup A = \liminf s_n$ and $\inf B = \limsup s_n$.

Plan. A finite set of exceptional indices can be placed before a single cutoff. Thus A consists of eventual lower bounds and B of eventual upper bounds. The boundary value itself need not belong to the respective set.

Check existence of the real suprema and infima.

Choose $C \geq 0$ with $-C \leq s_n \leq C$ for every n. For $a = -C$, no term satisfies $s_n < a$, so $-C \in A$. If $a > C$, every index satisfies $s_n < a$, so $a \notin A$. Thus A is nonempty and bounded above by C. Similarly, $C \in B$, and every $b < -C$ lies outside B, since $s_n > b$ for every n. Hence B is nonempty and bounded below by $-C$. Completeness gives finite real values $\sup A$ and $\inf B$. Write

$$\alpha = \liminf s_n, \quad \beta = \limsup s_n, \quad L_N = \inf\{s_n : n > N\}, \quad U_N = \sup\{s_n : n > N\}.$$

All these numbers are finite, and $L_N \to \alpha$, $U_N \to \beta$.

Every eventual lower bound is at most α.

If $a \in A$, its finitely many exceptional indices lie at most at some index N_0; if there are none, take $N_0 = 1$. Then $s_n \geq a$ whenever $n > N_0$. For $N \geq N_0$, it follows that $L_N \geq a$, and passage to the limit gives $\alpha \geq a$. Thus α is an upper bound of A, so $\sup A \leq \alpha$.

No number below α can be an upper bound of A.

If $a < \alpha$, convergence $L_N \to \alpha$ implies $L_N > a$ for some N. For every $n > N$, we have $s_n \geq L_N > a$. Consequently the indices with $s_n < a$ are contained in $\{1, \ldots, N\}$, a finite set; hence $a \in A$. Now fix any $d < \alpha$ and take $a = (d + \alpha)/2$. Then $d < a < \alpha$, so $a \in A$ and $a > d$. Therefore d fails to be an upper bound of A. This proves that α is the least upper bound: $\sup A = \alpha$.

Repeat the two directions for eventual upper bounds.

If $b \in B$, there is N_0 such that $s_n \leq b$ whenever $n > N_0$. Then $U_N \leq b$ for all $N \geq N_0$; taking limits gives $\beta \leq b$. Thus β is a lower bound of B.

If $b > \beta$, convergence $U_N \to \beta$ gives some N with $U_N < b$. For every $n > N$, we have $s_n \leq U_N < b$, so the indices satisfying $s_n > b$ form a finite set. Thus $b \in B$. If $d > \beta$, the number $b = (\beta + d)/2$ satisfies $\beta < b < d$ and belongs to B, so d cannot be a lower bound of B. Hence β is its greatest lower bound: $\inf B = \beta$.

Why the boundary is not asserted to lie in the set.

For $s_n = -1/n$, one has $\alpha = 0$, but every $s_n < 0$, so $0 \notin A$. For $s_n = 1/n$, one has $\beta = 0$, but every $s_n > 0$, so $0 \notin B$. Suprema and infima need not be attained.

Conclusion. The least eventual upper bound is $\limsup s_n$, and the greatest eventual lower bound is $\liminf s_n$, in the precise supremum/infimum sense above.

12.14 Exercise 12.14: The growth of the geometric mean of the first integers

What we must show. Calculate the limits of $(n!)^{1/n}$ and $(n!)^{1/n}/n$.

Plan. Apply the ratio–root corollary to two different positive sequences. For the second limit we also justify the elementary sequence that defines e, without using Stirling's formula.

(a) Apply the ratio–root corollary to the factorials.

Let $v_n = n!$, which is positive. Cancelling the common product $1 \cdot 2 \cdots n$ gives

$$\frac{v_{n+1}}{v_n} = \frac{(n+1)!}{n!} = n+1 \longrightarrow +\infty.$$

Corollary 12.3 states that a limit of the absolute successive ratios is also the limit of the nth roots, including the value $+\infty$. Therefore

$$\boxed{\lim_{n\to\infty} (n!)^{1/n} = +\infty.}$$

(b), preparation: justify the sequence defining e.

Set $b_n = (1 + 1/n)^n$. The standard constant e is the limit of this sequence, as described in the text's Section 7, Example 1(e). For completeness we verify that this limit exists and is positive. By the binomial theorem,

$$b_n = \sum_{k=0}^{n} \binom{n}{k} \frac{1}{n^k} = \sum_{k=0}^{n} \frac{1}{k!} \prod_{j=0}^{k-1} \left(1 - \frac{j}{n}\right).$$

For $k = 0$ the product is empty and equals 1. For fixed $k \le n$, each displayed factor is nonnegative and does not decrease when n is replaced by $n+1$. Thus each of the old summands does not decrease. The extra summand with $k = n+1$ in the expression for b_{n+1} is positive, so $b_{n+1} > b_n$.

Every factor is at most 1. For $k \ge 2$, the $k-1$ factors $2, 3, \ldots, k$ in $k!$ are each at least 2, whence $k! \ge 2^{k-1}$. It follows that

$$b_n \le \sum_{k=0}^{n} \frac{1}{k!} \le 2 + \sum_{k=2}^{n} \frac{1}{2^{k-1}} < 3.$$

For $n = 1$ the last sum is empty, and the same bound holds. Since $b_1 = 2$, the monotone sequence theorem now gives $b_n \to e$ for a finite number $e \ge 2 > 0$.

(b), normalization: put the denominator inside the root.

Define $u_n = n!/n^n > 0$. Its positive nth root is

$$u_n^{1/n} = \frac{(n!)^{1/n}}{n},$$

because the nth power of the expression on the right is $n!/n^n$ and the positive nth root is unique. The successive ratio simplifies one factor at a time:

$$\begin{aligned} \frac{u_{n+1}}{u_n} &= \frac{(n+1)!}{(n+1)^{n+1}} \frac{n^n}{n!} \\ &= \frac{(n+1)n^n}{(n+1)^{n+1}} = \left(\frac{n}{n+1}\right)^n = \frac{1}{(1+1/n)^n} = \frac{1}{b_n}. \end{aligned}$$

Since $b_n \to e \neq 0$, the reciprocal limit law gives $u_{n+1}/u_n \to 1/e$. Corollary 12.3 therefore gives $u_n^{1/n} \to 1/e$, which is exactly

$$\boxed{\lim_{n\to\infty} \frac{(n!)^{1/n}}{n} = \frac{1}{e}.}$$

Conclusion. The unnormalized geometric mean tends to $+\infty$; after division by n, it tends to $1/e$.

13 Some topological concepts in metric spaces

Throughout this section, k is a positive integer. In $\mathbb{R}^k$, d without a subscript denotes Euclidean distance. In an arbitrary metric space (S, d), write

$$B(x, r) = \{y \in S : d(y, x) < r\} \qquad (r > 0).$$

The ambient space matters: complements, interiors, and closures are taken in the space specified in the exercise. We write $\overline{E}$ for the book's closure E^-, and E° for its interior.

We will use the following elementary ball fact. If $y \in B(x, r)$, then $\rho = r - d(y, x) > 0$. For $z \in B(y, \rho)$, the triangle inequality gives

$$d(z, x) \le d(z, y) + d(y, x) < \rho + d(y, x) = r.$$

Thus $B(y, \rho) \subseteq B(x, r)$, which proves that every open ball in every metric space is open. No assertion about the boundary of a ball in a general metric space is being assumed.

13.1 Exercise 13.1: Two complete metrics on finite-dimensional space

What we must show. For $x, y \in \mathbb{R}^k$, verify that

$$d_1(x, y) = \max_{1 \le j \le k} |x_j - y_j|, \qquad d_2(x, y) = \sum_{j=1}^{k} |x_j - y_j|$$

are metrics, and prove that each metric is complete.

Plan. Check the three metric axioms directly. For completeness, first obtain a limit of each coordinate, then combine the finitely many coordinate estimates using one cutoff.

(a), first axiom: finite nonnegative distances, with the correct zeros.

Each $|x_j - y_j|$ is a finite nonnegative real number. A finite maximum and a finite sum of these quantities are therefore finite and nonnegative. If $x = y$, every coordinate difference is zero, and both distances are zero. Conversely, if the maximum is zero, then every nonnegative coordinate difference is at most zero and so is zero. If the sum is zero, every nonnegative summand must also be zero. In either case $x_j = y_j$ for all j, hence $x = y$. Equivalently, if $x \neq y$, at least one difference is positive, and both distances are positive.

(a), second and third axioms: symmetry and triangle inequalities.

The identity $|x_j - y_j| = |y_j - x_j|$ gives symmetry for both formulas. For any $x, y, z \in \mathbb{R}^k$, the real triangle inequality gives

$$|x_j - z_j| \le |x_j - y_j| + |y_j - z_j|.$$

For d_1, the right side is at most $d_1(x, y) + d_1(y, z)$ for every j. Taking the maximum of the left sides proves $d_1(x, z) \le d_1(x, y) + d_1(y, z)$. For d_2, summing the coordinate inequalities gives

$$d_2(x, z) \le \sum_{j=1}^{k} |x_j - y_j| + \sum_{j=1}^{k} |y_j - z_j| = d_2(x, y) + d_2(y, z).$$

This checks all metric axioms.

(b), coordinate limits: start with a Cauchy sequence in either metric.

Let $(x^{(n)})$ be Cauchy for d_i, where $i = 1$ or $i = 2$. For every coordinate j,

$$\left|x_j^{(m)} - x_j^{(n)}\right| \leq d_i(x^{(m)}, x^{(n)}).$$

For d_1, this is the defining maximum bound; for d_2, it is the bound of one nonnegative summand by the whole sum. Given $\varepsilon > 0$, the Cauchy cutoff for d_i therefore also makes every coordinate difference smaller than ε. Each coordinate sequence is Cauchy in $\mathbb{R}$. Completeness of $\mathbb{R}$ gives a real limit x_j for each j. Put $x = (x_1, \ldots, x_k) \in \mathbb{R}^k$.

(b), convergence in the maximum metric.

Fix $\varepsilon > 0$. For each j, choose N_j so that $n > N_j$ implies $\left|x_j^{(n)} - x_j\right| < \varepsilon$. Take $N = \max\{N_1, \ldots, N_k\}$. If $n > N$, every one of the finitely many differences is strictly less than ε. Their maximum is therefore strictly less than ε: $d_1(x^{(n)}, x) < \varepsilon$.

(b), convergence in the sum metric.

For d_2, instead choose N_j to give coordinate errors smaller than ε/k, which is positive because $k \geq 1$. Beyond the maximum of these cutoffs,

$$d_2(x^{(n)}, x) = \sum_{j=1}^{k} \left|x_j^{(n)} - x_j\right| < k\frac{\varepsilon}{k} = \varepsilon.$$

Thus a Cauchy sequence for either metric has a limit in that same metric and in the same space $\mathbb{R}^k$.

Conclusion. Both d_1 and d_2 are complete metrics on $\mathbb{R}^k$.

13.2 Exercise 13.2: Euclidean distance and convergence of coordinates

What we must show. Prove both inequalities in Lemma 13.3(1), then prove that a sequence in $\mathbb{R}^k$ converges if and only if every coordinate sequence converges in $\mathbb{R}$.

(a) Compare one summand, the sum, and its largest summand.

For $x, y \in \mathbb{R}^k$, let $M = \max_{1 \leq i \leq k} |x_i - y_i|$. Every square is nonnegative, so for each j,

$$|x_j - y_j|^2 \leq \sum_{i=1}^{k} |x_i - y_i|^2 \leq kM^2.$$

The first inequality retains one summand and discards only nonnegative summands. The second replaces each of the k summands by the upper bound M^2. Taking nonnegative square roots, which preserves order, gives

$$\boxed{|x_j - y_j| \leq d(x, y) \leq \sqrt{k} \max_{1 \leq i \leq k} |x_i - y_i|\,.}$$

(b), forward direction: Euclidean convergence gives coordinate convergence.

Suppose $x^{(n)} \to x$ in Euclidean distance. Given $\varepsilon > 0$, there is N such that $d(x^{(n)}, x) < \varepsilon$ for $n > N$. The first inequality in (a) then gives $\left|x_j^{(n)} - x_j\right| < \varepsilon$ for every j and $n > N$. Thus each $x_j^{(n)} \to x_j$ in $\mathbb{R}$.

(b), reverse direction: select one cutoff for all the coordinates.

Suppose $x_j^{(n)} \to x_j \in \mathbb{R}$ for $j = 1, \ldots, k$, and define $x = (x_1, \ldots, x_k)$. Given $\varepsilon > 0$, choose N_j such that

$$n > N_j \quad \Longrightarrow \quad \left|x_j^{(n)} - x_j\right| < \frac{\varepsilon}{\sqrt{k}}.$$

Set $N = \max_j N_j$. For $n > N$, the maximum of all coordinate errors is smaller than $\varepsilon/\sqrt{k}$. The second inequality in (a) gives $d(x^{(n)}, x) < \varepsilon$.

Conclusion. The distance bounds are proved, and coordinate convergence is equivalent to Euclidean convergence.

13.3 Exercise 13.3: The supremum metric on bounded sequences

What we must show. Let B be the set of all bounded real sequences. Prove that

$$d(x, y) = \sup_{j \in \mathbb{N}} |x_j - y_j|$$

is a metric on B, and decide whether the analogous infinite sum defines a metric on all of B.

(a) First verify that the supremum is a finite real number.

For $x, y \in B$, choose finite nonnegative bounds C_x, C_y with $|x_j| \le C_x$ and $|y_j| \le C_y$ for all j. Then

$$0 \le |x_j - y_j| \le |x_j| + |y_j| \le C_x + C_y.$$

The set of coordinate differences is nonempty and bounded above, so its supremum exists and belongs to $[0, C_x + C_y]$.

(a) Check the zeros, symmetry, and triangle inequality.

If $x = y$, every difference is zero and so is the supremum. If $d(x, y) = 0$, then each $0 \le |x_j - y_j| \le 0$, implying $x_j = y_j$ for all j, hence $x = y$. Symmetry follows coordinate by coordinate from symmetry of absolute value. Finally, for every j,

$$|x_j - z_j| \le |x_j - y_j| + |y_j - z_j| \le d(x, y) + d(y, z).$$

The final expression is an upper bound for all coordinate differences between x and z. Their least upper bound is thus at most that expression. This is the triangle inequality for d.

(b) A sum can be infinite even when both sequences are bounded.

Take $x = (0, 0, 0, \ldots)$ and $y = (1, 1, 1, \ldots)$. Both belong to B, but

$$\sum_{j=1}^{\infty} |x_j - y_j| = \sum_{j=1}^{\infty} 1 = +\infty,$$

since its nth partial sum is n. A metric in Definition 13.1 must give a finite real distance between each pair of points.

Conclusion. The supremum formula is a metric on B. The sum formula is not a metric on all of B, because it need not be finite.

13.4 Exercise 13.4: Unions and finite intersections of open sets

What we must show. Prove that arbitrary unions and finite intersections of open subsets of a metric space (S, d) are open.

(iii) For a point of a union, use one set that contains it.

Let $U = \bigcup_{\lambda\in\Lambda} U_\lambda$, where each U_λ is open. If $x \in U$, there is a λ_0 with $x \in U_{\lambda_0}$. Since that set is open, some $r > 0$ satisfies $B(x, r) \subseteq U_{\lambda_0} \subseteq U$. Thus every point of U is interior, which says U is open. If the collection is empty, its union is the empty set. That set is open because there is no point for which the interior condition could fail.

(iv) For a point of a finite intersection, take the smallest radius.

Let $V = U_1 \cap \cdots \cap U_m$, where $m \geq 1$ and all the U_i are open. For $x \in V$, choose $r_i > 0$ with $B(x, r_i) \subseteq U_i$. The number $r = \min\{r_1, \ldots, r_m\}$ is positive because this is a finite nonempty set of positive numbers. If $d(y, x) < r$, then $d(y, x) < r_i$ for every i, so $y \in U_i$ for every i. Hence $B(x, r) \subseteq V$, proving openness. The intersection of no subsets of S is S, which is open: every ball with center in S is a subset of S.

An infinite intersection need not be open: in $\mathbb{R}$, $\bigcap_{n\geq 1}(-1/n, 1/n) = \{0\}$, and no positive-radius interval about 0 fits in $\{0\}$. This explains why the intersection statement requires finiteness.

Conclusion. Arbitrary unions and finite intersections preserve openness.

13.5 Exercise 13.5: De Morgan's law and intersections of closed sets

What we must show. Verify the stated complement identity and deduce that an arbitrary intersection of closed subsets of S is closed.

(a) Prove equality by testing membership of a point.

For any $x \in S$, the following statements are equivalent:

$$\begin{aligned} x \in \bigcap_{U\in\mathcal{U}} (S \setminus U) \quad &\iff \quad x \notin U \text{ for every } U \in \mathcal{U} \\ &\iff \quad x \notin \bigcup_{U\in\mathcal{U}} U \\ &\iff \quad x \in S \setminus \bigcup_{U\in\mathcal{U}} U. \end{aligned}$$

Since every point has the same membership on the two sides, the sets are equal. For the empty collection, the left side is S and the right side is $S \setminus \varnothing = S$, so the convention is consistent.

(b) Take complements, then apply the open-union result.

Let F_λ be closed subsets of S, and put $F = \bigcap_{\lambda\in\Lambda} F_\lambda$. A point fails to belong to this intersection exactly when it fails to belong to at least one F_λ. Thus

$$S \setminus F = \bigcup_{\lambda\in\Lambda} (S \setminus F_\lambda).$$

Each complement on the right is open by the definition of a closed set. Their union is open by Exercise 13.4, so F is closed by that same definition. If Λ is empty, $F = S$ and its complement is empty and open.

Conclusion. The complement identity holds, and every intersection of closed sets is closed.

13.6 Exercise 13.6: Four equivalent descriptions involving closure

What we must show. Prove all four assertions of Proposition 13.9 for a subset E of an arbitrary metric space (S, d).

Plan. First derive a ball characterization directly from the definition of closure as an intersection of closed supersets. Then obtain the sequence characterizations without assuming them.

Preparation: closure is a closed superset, and a ball detects its points.

The family of closed sets containing E is nonempty, because S belongs to it. Every set in that family contains E; therefore their intersection $\overline{E}$ contains E. Exercise 13.5 shows that the intersection is closed.

For any $x \in S$, we claim

$$x \in \overline{E} \quad \Longleftrightarrow \quad B(x, r) \cap E \neq \varnothing \text{ for every } r > 0. \tag{*}$$

If some ball $B(x, r)$ misses E, then its complement is a closed set containing E, since balls are open. This closed superset excludes x, so $x \notin \overline{E}$. Conversely, if $x \notin \overline{E}$, then at least one closed set C containing E excludes x; otherwise x would belong to their intersection. The open set $S \setminus C$ contains x, so it contains some $B(x, r)$, and that ball misses E. These two implications prove (*).

(a) A set is closed exactly when it equals its closure.

If E is closed, it is one of the closed supersets in the defining intersection, so $\overline{E} \subseteq E$. The reverse inclusion was already proved, giving equality. If $E = \overline{E}$, then E is closed because $\overline{E}$ was proved closed. This proves both directions.

(c) Closure points are exactly limits of sequences of points of E.

We establish (c) before (b) so that there is no circular argument. If $x \in \overline{E}$, apply (*) with radius $1/n$ to choose $x_n \in E$ with $d(x_n, x) < 1/n$. For any $\varepsilon > 0$, choose an integer $N > 1/\varepsilon$. If $n > N$, then $d(x_n, x) < 1/n < \varepsilon$. Hence $x_n \to x$.

Conversely, if $x_n \in E$ and $x_n \to x \in S$, fix $r > 0$. Convergence gives some index n with $d(x_n, x) < r$, so $x_n \in B(x, r) \cap E$. Every ball about x thus meets E, and (*) gives $x \in \overline{E}$. For $E = \varnothing$, no such sequence exists and its closure is empty, so the equivalence still holds.

(b) Closed sets are exactly those containing limits of their convergent sequences.

If E is closed and $x_n \in E$ tends to a point $x \in S$, part (c) gives $x \in \overline{E}$, and part (a) gives $\overline{E} = E$, so $x \in E$. Conversely, suppose E contains every such limit. For every $x \in \overline{E}$, part (c) supplies a sequence in E converging to x, so the assumption gives $x \in E$. Thus $\overline{E} \subseteq E$. Combined with the always-valid inclusion $E \subseteq \overline{E}$, this gives equality, and part (a) makes E closed.

(d) Boundary points see both the set and its complement.

By definition, the boundary is $\overline{E} \setminus E^{\circ}$. A point x lies outside E° exactly when no ball about x is contained in E. Indeed, the existence of such a ball would also put x in E, since its center belongs to the ball, and would make x interior. Saying no ball is contained in E is equivalent to saying every

ball meets $S \setminus E$. By (*) applied to this complement,

$$x \notin E^\circ \quad \Longleftrightarrow \quad x \in \overline{S \setminus E}.$$

Consequently

$$\overline{E} \setminus E^\circ = \overline{E} \cap \overline{S \setminus E},$$

which is precisely the assertion in (d).

Conclusion. All four claims follow from the closure definition and the ball criterion (*), with no sequence characterization assumed in advance.

13.7 Exercise 13.7: Decomposing an open real set into disjoint intervals

What we must show. Express every open $U \subseteq \mathbb{R}$ as a finite or countably infinite disjoint union of open intervals. Unbounded intervals are allowed; the empty set is the union of an empty family.

1. Group points that have no gap of U between them.

For $x, y \in U$, write $x \sim y$ when the whole closed segment with endpoints x, y lies in U. This relation is reflexive, because $[x, x] = \{x\} \subseteq U$, and symmetric because the segment is unchanged when its endpoints are interchanged. For transitivity, if $x \sim y$ and $y \sim z$, every point between x and z lies in the union of the segments joining x to y and y to z. That union is contained in U. Thus $x \sim z$, so $\sim$ is an equivalence relation. Its classes partition U into nonempty pairwise disjoint sets.

2. Show that each class is an interval and is open.

Let I be a class, and let $p < q$ belong to it. The equivalence relation gives $[p, q] \subseteq U$. For $p < t < q$, the segment $[p, t]$ is contained in $[p, q]$, so $p \sim t$ and $t \in I$. Thus every point between two points of I belongs to I.

For $z \in I$, openness of U gives $r > 0$ with $(z - r, z + r) \subseteq U$. Every t in that interval has its segment to z inside the interval, hence inside U. Therefore $t \sim z$ and $t \in I$. We have proved $(z - r, z + r) \subseteq I$, so I is itself open.

3. Identify the endpoints, including infinite endpoints.

Let $a = \inf I$ and $b = \sup I$ in the extended real sense; use $a = -\infty$ or $b = +\infty$ if the corresponding bound does not exist in $\mathbb{R}$. Since I is nonempty and open, it contains an interval around one of its points, and $a < b$. For $a < t < b$, the definitions of infimum and supremum give $p, q \in I$ with $p < t < q$; otherwise t would be a lower or upper bound contradicting $a < t < b$. The interval property puts $t \in I$. Conversely, a finite endpoint cannot lie in I: openness there would supply a point of I still smaller than a or still larger than b. Hence $I = (a, b)$, with the usual interpretations $(a, +\infty)$, $(-\infty, b)$, or $\mathbb{R}$.

4. Prove that there are at most countably many classes.

Fix an enumeration $q_1, q_2, \ldots$ of $\mathbb{Q}$. Each nonempty open interval I contains a rational by density. Assign to I the least index $m(I)$ for which $q_{m(I)} \in I$. This least index exists by the least-element property of $\mathbb{N}$. Distinct classes have distinct assigned indices, because they are disjoint and cannot contain the same rational. Thus the classes inject into $\mathbb{N}$. If there are infinitely many, list their assigned indices in increasing order and use that order to list the classes; every assigned index appears, since only finitely many positive integers precede it.

Conclusion. The equivalence classes give the required disjoint interval decomposition. The family may be empty, finite, or countably infinite, and intervals may be unbounded.

13.8 Exercise 13.8: Intervals, balls, half-spaces, and examples that are neither

What we must show. Verify the assertions in the first paragraph of Example 3 and all the assertions in Example 4. In the interval assertions assume $a < b$; in the ball assertions assume $r > 0$ and $k \geq 1$. We address the degenerate cases separately below.

(a) Open and closed intervals in the real line.

For $x \in (a, b)$, let $\rho = \min\{x - a, b - x\} > 0$. If $|y - x| < \rho$, then $a < y < b$. Thus (a, b) is open. The complement of $[a, b]$ is $(-\infty, a) \cup (b, +\infty)$. Each ray is open: for a point strictly on its allowed side, half its distance to the endpoint provides a ball staying on that side. Their union is open, so $[a, b]$ is closed.

Every $x \in (a, b)$ is interior to $[a, b]$ by the first estimate. The endpoint a is not interior: every ball about it contains $a - \rho/2 < a$ for its radius $\rho > 0$. Similarly b is not interior because $b + \rho/2$ is outside $[a, b]$. Hence $[a, b]^\circ = (a, b)$.

The closed set $[a, b]$ contains (a, b), so $\overline{(a, b)} \subseteq [a, b]$. Every ball about a meets (a, b): choose a displacement $0 < t < \min\{\rho, b - a\}$ and use $a + t$. The point $b - t$ does the same for b. The ball criterion of Exercise 13.6 therefore puts both endpoints in the closure. Interior points already belong to the set. Thus $\overline{(a, b)} = [a, b] = \overline{[a, b]}$. Subtracting the interior gives boundary $\{a, b\}$ for both intervals.

If $a = b$, the open interval is empty and has empty closure and boundary. The closed interval is the singleton $\{a\}$, which is closed, has empty interior, and has boundary $\{a\}$. This is why the boundary assertion for both intervals presumes $a < b$.

(b), balls: openness, closedness, and the possible boundary points.

For a center $c \in \mathbb{R}^k$, put

$$D = \{x : d(x, c) < r\}, \qquad K = \{x : d(x, c) \leq r\}.$$

The opening ball argument of this section shows D is open. If $x \notin K$, let $\rho = d(x, c) - r > 0$. For $d(y, x) < \rho$, the triangle inequality rearranges to

$$d(y, c) \geq d(x, c) - d(x, y) > d(x, c) - \rho = r.$$

Thus a ball about x stays outside K, so the complement of K is open and K is closed.

If $d(x, c) < r$, then x is interior to both sets, using the radius $r - d(x, c)$. If $d(x, c) > r$, the preceding complement ball shows x lies outside the closure of either set. Only points with $d(x, c) = r$ can therefore be boundary points.

(b), spheres: every point on the sphere really is a boundary point.

Let $d(x, c) = r > 0$, and let $\varepsilon > 0$. Choose $0 < t < \min\{1, \varepsilon/r\}$. Define

$$x_- = c + (1 - t)(x - c), \qquad x_+ = c + (1 + t)(x - c).$$

The Euclidean distance formula gives

$$d(x_-, c) = (1-t)r < r, \quad d(x_+, c) = (1+t)r > r, \quad d(x_-, x) = d(x_+, x) = tr < \varepsilon.$$

Thus every ball about x meets D (at x_-) and the complement of K (at x_+). It meets both either set and its complement, so Exercise 13.6(d) puts x on both boundaries. Consequently both boundaries are exactly $\{x : d(x, c) = r\}$. This radial argument uses Euclidean space; the same sphere assertion is not valid for all metrics. If $r = 0$, the open ball is empty with empty boundary, while the closed ball is the singleton $\{c\}$ with boundary $\{c\}$.

(b), strict coordinate inequalities give open half-spaces and quadrants.

In $\mathbb{R}^2$, consider $H = \{x : x_1 > 0\}$ and $Q = \{x : x_1 > 0,\ x_2 > 0\}$. For $x \in H$, choose $\rho = x_1/2 > 0$. Since $|y_1 - x_1| \le d(y, x)$, every y in $B(x, \rho)$ satisfies $y_1 > x_1 - \rho = x_1/2 > 0$. Thus H is open. For $x \in Q$, choose $\rho = \min\{x_1, x_2\}/2$. The same coordinate estimate, separately for $j = 1, 2$, gives $y_j > x_j - \rho \ge x_j/2 > 0$. Hence Q is open.

For $H_0 = \{x : x_1 \ge 0\}$, its complement is $\{x : x_1 < 0\}$, an open half-space by the same radius argument with $-x_1$ in place of x_1. Thus H_0 is closed. For $Q_0 = \{x : x_1 \ge 0,\ x_2 \ge 0\}$, the complement is $\{x : x_1 < 0\} \cup \{x : x_2 < 0\}$, a union of open half-spaces. Thus Q_0 is closed as asserted.

(b), the half-open interval is neither open nor closed.

The point $0 \in [0, 1)$ is not interior, since every ball about 0 contains negative points. Hence $[0, 1)$ is not open. The sequence $1 - 1/n$ belongs to $[0, 1)$ and tends to 1, which is outside the set. The closed-set sequence criterion of Exercise 13.6(b) shows it is not closed.

(b), the mixed quadrant is neither open nor closed.

Let $M = \{x : x_1 > 0,\ x_2 \ge 0\}$. The point $(1, 0)$ belongs to M, but every ball of radius $\rho > 0$ about it contains $(1, -\rho/2)$, outside M. Thus M is not open. The sequence $(1/n, 0)$ lies in M and tends in Euclidean distance to $(0, 0)$, outside M. It follows that M is not closed either.

Conclusion. All the interval, Euclidean ball, half-space, quadrant, and mixed examples have the stated properties, with the positive radius and nondegenerate interval conditions made explicit.

13.9 Exercise 13.9: Three closures in the real line

What we must show. Find the closures of $\{1/n : n \in \mathbb{N}\}$, $\mathbb{Q}$, and $\{q \in \mathbb{Q} : q^2 < 2\}$ in $\mathbb{R}$.

(a) Include the reciprocal terms and their one additional limit.

Let $A = \{1/n : n \in \mathbb{N}\}$. Every point of A belongs to its closure, and $1/n \to 0$ gives $0 \in \overline{A}$ by Exercise 13.6(c). We show no other point belongs. If $x < 0$, the ball of radius $-x/2$ about x is negative and misses A. If $x > 1$, the radius $(x-1)/2$ gives a ball wholly above 1, which also misses A.

It remains to exclude $0 < x \le 1$ with $x \notin A$. Choose N so large that $1/N < x/2$. All terms $1/n$ for $n \ge N$ are then below $x/2$. The finite set $\{1, 1/2, \ldots, 1/(N-1)\}$ has no member equal to x. Its finitely many positive distances from x have a positive minimum η. Here $N \ge 2$ because $1/N < x/2 \le 1/2$. A ball about x of radius $\min\{x/2, \eta\}/2$ misses the finite set and all the remaining reciprocals. The ball criterion excludes x. Therefore

$$\boxed{\overline{A} = A \cup \{0\}.}$$

(b) Density fills the whole real line.

For any $x \in \mathbb{R}$ and $r > 0$, the interval $(x - r, x + r)$ contains a rational by density of $\mathbb{Q}$. Hence every real x satisfies the closure ball criterion, giving $\boxed{\overline{\mathbb{Q}} = \mathbb{R}}$.

(c) The rational points between the two square-root endpoints.

For real q, the inequality $q^2 < 2$ is equivalent to $-\sqrt{2} < q < \sqrt{2}$. Thus the given set is $C = \mathbb{Q} \cap (-\sqrt{2}, \sqrt{2})$. If $-\sqrt{2} < x < \sqrt{2}$, then every neighborhood of x contains a smaller nonempty interval lying inside $(-\sqrt{2}, \sqrt{2})$; density supplies a rational in that interval. For $x = \sqrt{2}$, an interval $(\sqrt{2} - t, \sqrt{2})$ with $0 < t < \min\{r, 2\sqrt{2}\}$ lies inside the radius-r neighborhood and inside the allowed range, and contains a rational. The interval $(-\sqrt{2}, -\sqrt{2} + t)$ proves the same statement for $x = -\sqrt{2}$. Therefore the whole closed interval belongs to $\overline{C}$.

If $x > \sqrt{2}$, take radius $(x - \sqrt{2})/2$; if $x < -\sqrt{2}$, take radius $(-\sqrt{2} - x)/2$. The resulting ball misses the closed interval and hence misses C. No outside point belongs to the closure, so

$$\boxed{\overline{C} = [-\sqrt{2}, \sqrt{2}].}$$

Conclusion. The closures are respectively $\{0\} \cup \{1/n : n \in \mathbb{N}\}$, $\mathbb{R}$, and $[-\sqrt{2}, \sqrt{2}]$.

13.10 Exercise 13.10: Three sets with no interior points

What we must show. Prove that the reciprocal set, the rationals, and the Cantor set all have empty interior in $\mathbb{R}$.

(a) A rational-only set cannot contain a real interval.

Every reciprocal $1/n$ is rational. Every open interval contains an irrational number. For completeness, to find one in (a, b), choose a rational q in $(a - \sqrt{2}, b - \sqrt{2})$ by rational density; then $q + \sqrt{2} \in (a, b)$ is irrational, since its being rational would make $\sqrt{2} = (q + \sqrt{2}) - q$ rational. Thus every ball about any point of the reciprocal set contains a point outside that set. No point of the set is interior.

(b) Apply the same irrational-density argument to $\mathbb{Q}$.

For any $q \in \mathbb{Q}$ and $r > 0$, the preceding argument gives an irrational in $(q - r, q + r)$. That ball is not a subset of $\mathbb{Q}$. Thus $\mathbb{Q}$ also has no interior points. Notice that dense closure does not imply nonempty interior: Exercise 13.9 gave $\overline{\mathbb{Q}} = \mathbb{R}$.

(c) The intervals remaining at late Cantor stages are too short.

Use the numbering in Example 5: $F_1 = [0, 1]$, and F_n is the union of 2^{n-1} disjoint closed intervals, each of length $3^{-(n-1)}$. The Cantor set is $F = \bigcap_{n \geq 1} F_n$. Suppose, for contradiction, that $x \in F^\circ$. Then some real interval $(x - r, x + r)$, with $r > 0$, lies in F. The closed interval $J = [x - r/2, x + r/2]$ has length r and also lies in F, hence in every F_n.

Choose n so large that $3^{-(n-1)} < r$. The interval J would have to lie in one of the component intervals of F_n. Indeed, if it met two distinct components, it would also contain the points of the nonempty removed gap between those components, because it contains every point between any two of its points. That would contradict $J \subseteq F_n$. But an interval of length r cannot be contained in an interval of smaller length $3^{-(n-1)}$. This is a contradiction.

Conclusion. All three interiors are empty. The Cantor proof uses the explicit shrinking intervals of its construction.

13.11 Exercise 13.11: Compactness and subsequences in Euclidean space

What we must show. Show that $E \subseteq \mathbb{R}^k$ is compact if and only if every sequence of points of E has a subsequence converging to a point of E.

Plan. Use Heine–Borel and Bolzano–Weierstrass. For the converse, show that the stated subsequence property forces both boundedness and closedness.

Preparation: uniqueness and inheritance of metric limits.

If $x_n \to x$ and $x_n \to y$, then $d(x,y) \le d(x,x_n) + d(x_n,y) \to 0$, so $d(x,y) = 0$ and $x = y$. Every subsequence of a convergent sequence has the same limit: strictly increasing selection indices n_j satisfy $n_j \ge j$, so every eventual distance estimate for the original sequence also holds for all sufficiently late selected terms. Also, a convergent sequence in $\mathbb{R}^k$ is bounded: its tail lies within distance 1 of its limit x, and for each coordinate the triangle inequality gives $\left|x_j^{(n)}\right| \le |x_j| + 1$ on that tail. A finite maximum accommodates the omitted initial coordinates.

Forward direction: combine boundedness, subsequences, and closedness.

If E is compact, Heine–Borel (Theorem 13.12) says it is closed and bounded. Any sequence in E is then bounded. Bolzano–Weierstrass (Theorem 13.5) gives a subsequence tending to some point of $\mathbb{R}^k$. Since all its terms lie in the closed set E, Exercise 13.6(b) puts that limit in E.

Reverse direction, boundedness: unbounded points would escape every subsequence.

Assume the stated subsequence property. If E were unbounded, then for each $n \in \mathbb{N}$ we could choose $x^{(n)} \in E$ with

$$\max_{1 \le j \le k} \left|x_j^{(n)}\right| > n.$$

For any subsequence $x^{(n_m)}$, the selected indices satisfy $n_m \ge m$, so its coordinate maximum exceeds m. It is therefore unbounded: for any proposed finite bound C, take $m > C$. It cannot converge, since a convergent sequence was just shown to be bounded. This contradicts the assumed property. Thus E is bounded.

Reverse direction, closedness: recover every closure point.

Let $x \in \overline{E}$. By Exercise 13.6(c), choose a sequence $x_n \in E$ converging to x. The assumed property supplies a subsequence $x_{n_m} \to y$ for some $y \in E$. Inheritance gives $x_{n_m} \to x$ as well, and uniqueness of the limit gives $x = y \in E$. Thus $\overline{E} \subseteq E$. The reverse inclusion always holds, so Exercise 13.6(a) makes E closed. Having proved both closedness and boundedness, Heine–Borel gives compactness.

If $E = \varnothing$, it is compact because the empty subfamily is a finite subcover of every cover of it. There is no sequence of points of E, so the sequential condition is vacuously true.

Conclusion. The two conditions are equivalent for all subsets of $\mathbb{R}^k$, including the empty set.

13.12 Exercise 13.12: Closed subsets and finite unions of compact sets

What we must show. In any metric space (S, d), prove that a closed subset of a compact set is compact, and that a finite union of compact sets is compact.

Plan. Work directly with open covers, so neither result depends on the finite-dimensional Heine–Borel theorem.

(a) Add one open set to cover the rest of the larger compact set.

Suppose $E \subseteq F$, E is closed in S, and F is compact. Let $\mathcal{U}$ be any family of open subsets of S covering E. The set $V = S \setminus E$ is open, and $\mathcal{U} \cup \{V\}$ covers F: a point of F either belongs to E and is covered by $\mathcal{U}$, or lies in V. Compactness of F gives a finite subfamily covering F. Remove V if it was selected. The remaining sets still cover E, since no point of E belongs to V. They are a finite subfamily of $\mathcal{U}$, so E is compact.

If "closed subset of F" is interpreted as closed relative to F, the same conclusion holds: relative closedness means $F \setminus E = F \cap V$ for some open $V \subseteq S$. Use that V in the preceding cover argument; it still misses E and covers every point of $F \setminus E$.

(b) Take a finite subcover for each compact piece.

Let $K = K_1 \cup \cdots \cup K_m$ with each K_i compact, and let $\mathcal{U}$ be an open cover of K. It also covers each K_i. For each i, select a finite subfamily $\mathcal{U}_i \subseteq \mathcal{U}$ covering K_i. The family $\mathcal{U}_1 \cup \cdots \cup \mathcal{U}_m$ is finite, since a finite union of finite sets is finite, and it covers every piece, hence their union K. This proves compactness. When $m = 0$, the union is empty and is compact by the empty-subcover argument.

Conclusion. Both permanence properties hold in every metric space.

13.13 Exercise 13.13: A compact real set contains its extreme bounds

What we must show. For a nonempty compact $E \subseteq \mathbb{R}$, show $\sup E \in E$ and $\inf E \in E$.

1. Ensure the bounds are finite and use the approximation property.

Heine–Borel makes E closed and bounded. Since it is also nonempty, completeness of $\mathbb{R}$ supplies finite real numbers $M = \sup E$ and $m = \inf E$. For every $n \in \mathbb{N}$, $M - 1/n$ is strictly below the least upper bound, so it cannot be an upper bound. Choose $u_n \in E$ with

$$M - \frac{1}{n} < u_n \leq M.$$

Similarly, $m + 1/n$ is strictly above the greatest lower bound, so it cannot be a lower bound. Choose $v_n \in E$ with

$$m \leq v_n < m + \frac{1}{n}.$$

2. Check convergence, then use closedness.

The first inequalities give $|u_n - M| < 1/n$, so $u_n \to M$. The second give $|v_n - m| < 1/n$, so $v_n \to m$. Both sequences lie in E. Closedness and Exercise 13.6(b) therefore put both limits in E.

Conclusion. The supremum and infimum are attained: E has a maximum and a minimum, namely M and m.

13.14 Exercise 13.14: The diameter of a compact set is attained

What we must show. For nonempty compact $E \subseteq \mathbb{R}^k$, put

$$\delta = \sup\{d(x, y) : x, y \in E\}.$$

Find $x^0, y^0 \in E$ such that $d(x^0, y^0) = \delta$.

Plan. Approximate the supremum by pairs, select subsequences for both entries of the pairs, and justify passage to the limit using the triangle inequality.

1. Verify that the distance supremum is a finite real number.

Heine–Borel makes E bounded. Choose $C \geq 0$ so that every coordinate of every $x \in E$ has absolute value at most C. For $x, y \in E$, each $|x_j - y_j| \leq 2C$, so $d(x, y) \leq 2C\sqrt{k}$ by Exercise 13.2(a). The set of distances is nonempty because E is nonempty, and contains $d(x, x) = 0$. Its supremum therefore exists with $0 \leq \delta \leq 2C\sqrt{k}$.

2. Choose pairs whose distances approach the supremum.

For each $n \in \mathbb{N}$, the real number $\delta - 1/n$ is not an upper bound of the distance set. Hence choose $x_n, y_n \in E$ such that

$$\delta - \frac{1}{n} < d(x_n, y_n) \leq \delta.$$

This works even when $\delta = 0$; any pair then has distance $0 > \delta - 1/n$. The displayed error bound gives $d(x_n, y_n) \to \delta$.

3. Pass to a common subsequence for the two endpoints.

By Exercise 13.11, some subsequence x_{n_j} tends to a point $x^0 \in E$. Apply the same result to the sequence y_{n_j} in E to find further indices j_ℓ with $y_{n_{j_\ell}} \to y^0 \in E$. The matching $x_{n_{j_\ell}}$ still tends to x^0, since a subsequence inherits a limit. This further selection is also a subsequence of the original sequence of pairs because the composition of two increasing index selections is increasing.

4. Prove that distances pass to the limit.

Two uses of the triangle inequality give

$$d(x, y) \leq d(x, x^0) + d(x^0, y^0) + d(y^0, y),$$
$$d(x^0, y^0) \leq d(x^0, x) + d(x, y) + d(y, y^0).$$

Subtracting $d(x^0, y^0)$ in the first and $d(x, y)$ in the second combines into

$$\Big|d(x, y) - d(x^0, y^0)\Big| \leq d(x, x^0) + d(y, y^0).$$

At the selected pairs, the right side tends to zero. Thus $d(x_{n_{j_\ell}}, y_{n_{j_\ell}}) \to d(x^0, y^0)$. The same distances tend to δ, being a subsequence of the distance sequence in Step 2. Uniqueness of real limits gives $d(x^0, y^0) = \delta$.

Conclusion. Two points of E realize its diameter, including the zero-diameter case.

13.15 Exercise 13.15: A closed bounded set that is not compact

What we must show. In the bounded-sequence space (B, d) of Exercise 13.3, let

$$F = \left\{ x \in B : \sup_{j \in \mathbb{N}} |x_j| \leq 1 \right\}.$$

Show that F is closed and bounded, but not compact.

Plan. Verify closedness using an open complement. Then use the prescribed cover by open balls of radius 1, together with infinitely many points whose pairwise distance is 2.

(a), boundedness: use the zero sequence as the center.

The sequence $0 = (0, 0, \ldots)$ belongs to B. For $x \in F$,

$$d(x, 0) = \sup_j |x_j| \leq 1.$$

Thus F lies in the closed metric ball with center 0 and radius 1, which is exactly the boundedness condition stated in the exercise.

(a), closedness: exhibit a ball about each point outside F.

If $x \in B \setminus F$, then $\sup_j |x_j| > 1$. At least one coordinate, say j_0, satisfies $|x_{j_0}| > 1$; otherwise 1 would be an upper bound for all these absolute values and hence for their supremum. Set $\rho = (|x_{j_0}| - 1)/2 > 0$. If $d(x, y) < \rho$, the real triangle inequality gives

$$|y_{j_0}| \geq |x_{j_0}| - |x_{j_0} - y_{j_0}| > |x_{j_0}| - \rho = \frac{|x_{j_0}| + 1}{2} > 1.$$

Such a y lies outside F. Hence a ball about each point of the complement stays in the complement, proving it open and F closed in B.

(b), the open cover: give every point its radius-one ball.

For $x \in F$, define $U(x) = \{y \in B : d(y, x) < 1\}$. These sets are open in B by the general ball fact at the start of this section. They cover F, since $d(x, x) = 0 < 1$ puts each $x \in F$ in its own $U(x)$.

(b), separated points: compute the distance exactly.

For each $n \in \mathbb{N}$, let $x^{(n)}$ have coordinate -1 at index n and coordinate 1 at every other index. Every one of its coordinates has absolute value 1, so $x^{(n)} \in F$. If $m \neq n$, the two sequences differ by 2 in absolute value at coordinates m and n, and agree at all other coordinates. Therefore

$$d(x^{(m)}, x^{(n)}) = 2.$$

No ball $U(x)$ can contain two distinct such points: if it contained both, then the metric triangle inequality would give

$$2 = d(x^{(m)}, x^{(n)}) \leq d(x^{(m)}, x) + d(x, x^{(n)}) < 1 + 1 = 2,$$

an impossibility. The strict inequalities in the open-ball definition are essential here.

(b), exclude every finite subcover.

A union of q members of this cover can contain at most q of the separated points, since each member contains at most one. But F contains the infinitely many distinct points $x^{(1)}, x^{(2)}, \ldots$. For example, among its first $q+1$ separated points at least one must be missed. Thus no finite subfamily covers F.

Conclusion. F is closed and bounded in the specified metric space, but this explicit open cover has no finite subcover, so F is not compact. Heine–Borel's closed-and-bounded criterion is specific to Euclidean space and does not apply to all metric spaces.

14 Series

For a sequence (a_n), its Nth partial sum is $S_N = \sum_{n=1}^{N} a_n$. The series $\sum_{n=1}^{\infty} a_n$ converges precisely when (S_N) has a finite real limit. A limit of $+\infty$ or $-\infty$ is divergence, even though it specifies a direction of divergence.

The necessary condition $a_n \to 0$ is not sufficient for convergence. The Comparison, Ratio, and Root Tests supply sufficient conditions in the situations specified below. When terms are nonnegative, convergence and absolute convergence are the same assertion. Unless a lower index is explicitly given, the series in the catalogs begin at $n = 1$; changing finitely many initial terms does not affect convergence, as proved in Exercise 14.9. The logarithm is the natural logarithm.

14.1 Exercise 14.1: Six initial convergence tests

What we must show. Decide convergence for each of the six given series, with a reason that checks the hypotheses of the chosen test.

(a) The series with term $n^4/2^n$.

Let $a_n = n^4/2^n > 0$. Since no term is zero, the successive ratio is well defined. Canceling powers gives

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^4}{2^{n+1}} \frac{2^n}{n^4} = \frac{1}{2} \left(1 + \frac{1}{n}\right)^4 \longrightarrow \frac{1}{2}.$$

The last limit follows from $1/n \to 0$ and the finite product limit law. Its value is strictly below 1, so the Ratio Test proves absolute convergence. Since the terms are positive, this proves convergence of the requested series.

(b) The series with term $2^n/n!$.

Put $a_n = 2^n/n! > 0$. The identity $(n+1)! = (n+1)n!$ yields

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1}}{(n+1)!} \frac{n!}{2^n} = \frac{2}{n+1} \longrightarrow 0.$$

Here $0 < 1$, so the Ratio Test proves absolute convergence.

(c) The series with term $n^2/3^n$.

With $a_n = n^2/3^n > 0$, the same cancellation gives

$$\frac{a_{n+1}}{a_n} = \frac{1}{3} \left(1 + \frac{1}{n}\right)^2 \longrightarrow \frac{1}{3} < 1.$$

The Ratio Test again gives absolute convergence. The power of n in the numerator does not prevent this conclusion because the computed ratio limit is strictly less than 1.

(d) The series with term $n!/(n^4+3)$.

Let $a_n = n!/(n^4+3) > 0$. Its ratio is

$$\frac{a_{n+1}}{a_n} = (n+1) \frac{n^4+3}{(n+1)^4+3} = (n+1) \frac{1 + 3/n^4}{(1+1/n)^4 + 3/n^4}.$$

The second factor tends to 1, since its numerator and denominator both tend to 1 and the denominator is never zero. In particular, it exceeds $1/2$ eventually. Hence the displayed ratio is eventually at least $(n+1)/2$, and therefore tends to $+\infty$. The divergent part of the Ratio Test applies, since the ratio limit inferior is greater than 1.

We can also see why the term condition fails. There is N after which every ratio is at least 2. Induction then gives $a_{N+k} \geq 2^k a_N$, with $a_N > 0$, so the terms tend to $+\infty$ rather than to zero. The positive partial sums are unbounded, and the series diverges to $+\infty$.

(e) The series with term $\cos^2 n/n^2$.

The trigonometric bound $-1 \leq \cos n \leq 1$ gives $0 \leq \cos^2 n \leq 1$. Since $n^2 > 0$, division preserves order:

$$0 \leq \frac{\cos^2 n}{n^2} \leq \frac{1}{n^2}.$$

The comparison series $\sum 1/n^2$ converges by the p-series criterion with $p = 2 > 1$. The Comparison Test therefore proves convergence of the given nonnegative series.

(f) The series with term $1/\log n$, beginning at $n = 2$.

For $n \geq 2$, the logarithm is positive. The elementary logarithm bound is

$$0 < \log n = \int_1^n \frac{dt}{t} \leq \int_1^n 1\,dt = n - 1 < n.$$

The integral comparison here only justifies the inequality for one logarithm; no integral test for a series is being invoked. Taking reciprocals of positive quantities reverses the inequality, so $1/\log n \geq 1/n$. The harmonic series diverges to $+\infty$, and deleting its first term leaves divergent partial sums. Thus the Comparison Test proves divergence to $+\infty$.

Conclusion. Parts (a), (b), (c), and (e) converge absolutely. Parts (d) and (f) diverge to $+\infty$.

14.2 Exercise 14.2: Seven further convergence tests

What we must show. Classify the seven series, explaining the actual numerator and denominator in each term before applying a test.

(a) The series with term $(n-1)/n^2$.

All terms are nonnegative. For $n \geq 2$, the inequality $n - 1 \geq n/2$ follows by subtracting $n/2$ from both sides: $n/2 - 1 \geq 0$. Therefore

$$\frac{n-1}{n^2} \geq \frac{n/2}{n^2} = \frac{1}{2n} \qquad (n \geq 2).$$

The series $\sum_{n=2}^{\infty} 1/(2n)$ is one half of a tail of the harmonic series, so it diverges to $+\infty$. The Comparison Test proves divergence to $+\infty$ of the requested series. The first term is zero and has no effect on this lower bound for the later partial sums.

(b) The series with term $(-1)^n$.

Every term has absolute value 1, so the terms do not tend to zero. Corollary 14.5 therefore rules out convergence. Directly, the partial sums are $-1, 0, -1, 0, \ldots$: an odd number of terms gives -1, and

each pair $(-1)+1$ sums to zero. Thus the partial sums have two different subsequential limits and do not tend to either infinity.

(c) The series with term $3n/n^3$.

Since $n \neq 0$, cancellation gives $3n/n^3 = 3/n^2$. The p-series $\sum 1/n^2$ converges, and multiplication of all partial sums by the constant 3 preserves convergence by the constant multiple limit law. Hence this positive series converges absolutely.

(d) The series with term $n^3/3^n$.

For $a_n = n^3/3^n > 0$, compute

$$\frac{a_{n+1}}{a_n} = \frac{1}{3}\left(1+\frac{1}{n}\right)^3 \longrightarrow \frac{1}{3} < 1.$$

All ratios are defined; the Ratio Test gives absolute convergence.

(e) The series with term $n^2/n!$.

Put $a_n = n^2/n! > 0$. Cancel $n!$ and one factor of $n+1$:

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^2}{(n+1)!}\frac{n!}{n^2} = \frac{n+1}{n^2} = \frac{1}{n} + \frac{1}{n^2} \longrightarrow 0 < 1.$$

The Ratio Test proves absolute convergence.

(f) The series with term $1/n^n$.

The terms are positive and their positive nth roots satisfy

$$\left(\frac{1}{n^n}\right)^{1/n} = \frac{1}{n} \longrightarrow 0.$$

Thus the limit superior of the roots is $0 < 1$. The convergent part of the Root Test proves absolute convergence.

(g) The series with term $n^n/2^n$.

Here the positive nth root is

$$\left(\frac{n^n}{2^n}\right)^{1/n} = \frac{n}{2} \longrightarrow +\infty.$$

The root limit superior is greater than 1, so the Root Test proves divergence. More explicitly, for $n \geq 4$ one has $n/2 \geq 2$, so $(n/2)^n \geq 2^n$. These positive terms tend to $+\infty$; hence the partial sums do too.

Conclusion. Parts (c), (d), (e), and (f) converge absolutely. Parts (a) and (g) diverge to $+\infty$. Part (b) diverges with bounded, oscillating partial sums.

14.3 Exercise 14.3: Factorials, geometric comparisons, and a periodic term

What we must show. Classify each of the six series and justify convergence or failure of the necessary term condition.

(a) The series with term $1/\sqrt{n!}$.

Set $a_n = 1/\sqrt{n!} > 0$. The factorial is inside the square root. Using $(n+1)! = (n+1)n!$ and positivity,

$$\frac{a_{n+1}}{a_n} = \frac{\sqrt{n!}}{\sqrt{(n+1)!}} = \frac{1}{\sqrt{n+1}} \longrightarrow 0.$$

The Ratio Test applies and gives absolute convergence.

(b) The series with term $(2+\cos n)/3^n$.

Since $-1 \le \cos n \le 1$, adding 2 gives $1 \le 2 + \cos n \le 3$. Consequently

$$0 \le \frac{2+\cos n}{3^n} \le \frac{3}{3^n}.$$

The comparison series is geometric, with positive constant factor 3 and ratio $1/3 < 1$. It converges, so the Comparison Test proves absolute convergence of the given nonnegative series.

(c) The series with term $1/(2^n + n)$.

We have $2^n + n \ge 2^n > 0$. Reciprocation yields

$$0 < \frac{1}{2^n + n} \le \frac{1}{2^n}.$$

The upper comparison is a geometric series of ratio $1/2$, so the requested series converges absolutely.

(d) The series with term $(1/2)^n(50 + 2/n)$.

For $n \ge 1$, the inequality $0 < 2/n \le 2$ implies $0 < 50 + 2/n \le 52$. Multiply by the positive number 2^{-n}:

$$0 < \left(\frac{1}{2}\right)^n \left(50 + \frac{2}{n}\right) \le \frac{52}{2^n}.$$

The series on the right is a constant multiple of a convergent geometric series. The Comparison Test therefore gives absolute convergence. In particular, the expression in parentheses is $50 + 2/n$, with n in the denominator of its second term.

(e) The series with term $\sin(n\pi/9)$.

Consider the strictly increasing indices $n_k = 18k + 3$, $k \in \mathbb{N}$. The corresponding terms are

$$\sin\left(\frac{(18k+3)\pi}{9}\right) = \sin(2k\pi + \pi/3) = \sin(\pi/3) = \frac{\sqrt{3}}{2}.$$

This nonzero constant subsequence prevents the whole term sequence from tending to zero: every subsequence of a sequence tending to zero would also tend to zero. Thus Corollary 14.5 proves that the series diverges. A term test does not require the terms to be nonnegative or the partial sums to tend to $+\infty$.

(f) The series with term $100^n/n!$.

For $a_n = 100^n/n! > 0$, factorial cancellation gives

$$\frac{a_{n+1}}{a_n} = \frac{100}{n+1} \longrightarrow 0 < 1.$$

The Ratio Test gives absolute convergence, despite the relatively large fixed base 100 in the numerator.

Conclusion. Every series in this exercise except (e) converges absolutely. Part (e) diverges because its terms do not tend to zero.

14.4 Exercise 14.4: Oscillating denominators, telescoping roots, and factorials

What we must show. Classify the three series, keeping track of the starting index and the placement of the powers.

(a) The series with term $1/[n+(-1)^n]^2$, for $n \geq 2$.

For these indices,

$$n + (-1)^n \geq n - 1 \geq \frac{n}{2} > 0.$$

The middle inequality is equivalent to $n \geq 2$. Because both sides are positive, squaring preserves order and taking reciprocals reverses it. We obtain

$$0 < \frac{1}{[n+(-1)^n]^2} \leq \frac{4}{n^2} \qquad (n \geq 2).$$

The comparison series converges by the p-series criterion with $p = 2$ and the constant multiple limit law. Hence the given series converges absolutely. Beginning at $n = 2$ also avoids the zero denominator that would occur at $n = 1$.

(b) The series with term $\sqrt{n+1} - \sqrt{n}$.

Each term is positive. Its Nth partial sum telescopes:

$$\begin{aligned}\sum_{n=1}^{N} (\sqrt{n+1} - \sqrt{n}) &= (\sqrt{2} - 1) + (\sqrt{3} - \sqrt{2}) + \cdots + (\sqrt{N+1} - \sqrt{N}) \\ &= \sqrt{N+1} - 1.\end{aligned}$$

All intermediate square roots cancel exactly in this finite sum. For a given $M > 0$, any $N > (M+1)^2$ makes the last expression greater than M. Thus the partial sums tend to $+\infty$ and the series diverges to $+\infty$. Individual terms tending to zero would not change this conclusion about the partial sums.

(c) The series with term $n!/n^n$.

Let $a_n = n!/n^n > 0$. The ratio simplifies as follows:

$$\begin{aligned}\frac{a_{n+1}}{a_n} &= \frac{(n+1)!}{(n+1)^{n+1}} \frac{n^n}{n!} = \frac{(n+1)n^n}{(n+1)^{n+1}} \\ &= \left(\frac{n}{n+1}\right)^n = \frac{1}{(1+1/n)^n}.\end{aligned}$$

Exercise 12.14 justified $(1+1/n)^n \to e$ with finite $e \geq 2$. The reciprocal limit law therefore gives ratio limit $1/e$, which lies in $(0, 1)$. The Ratio Test proves absolute convergence.

Conclusion. Parts (a) and (c) converge absolutely. Part (b) diverges to $+\infty$.

14.5 Exercise 14.5: Linearity of series and failure of a termwise product formula

What we must show. If $\sum a_n = A$ and $\sum b_n = B$ are finite real sums, prove the two linearity formulas and examine the proposed product formula.

Preparation: use partial sums, whose limits are given.

Write $S_N = \sum_{n=1}^{N} a_n$ and $T_N = \sum_{n=1}^{N} b_n$. The hypotheses mean $S_N \to A$ and $T_N \to B$. These are finite limits, so the sum and constant multiple laws from Section 9 apply.

(a) Addition of two series.

Finite distributivity gives

$$\sum_{n=1}^{N} (a_n + b_n) = S_N + T_N.$$

Taking limits yields $A + B$. By the definition of a series sum, this proves $\sum_{n=1}^{\infty}(a_n + b_n) = A + B$.

(b) Multiplication by a fixed real constant.

For any $k \in \mathbb{R}$ and every N,

$$\sum_{n=1}^{N} ka_n = kS_N.$$

The right side tends to kA, including when $k = 0$ or $k < 0$. Hence $\sum_{n=1}^{\infty} ka_n = kA$.

(c) Test the termwise product formula on finite series.

Let $a_1 = 1$ and $a_n = 0$ for $n \geq 2$. Let $b_2 = 1$ and $b_n = 0$ for every $n \neq 2$. Each series has sum 1, so $AB = 1$. At $n = 1$ the second factor is zero; at $n = 2$ the first factor is zero; at every other index both are zero. Thus $a_n b_n = 0$ for all n, and

$$\sum_{n=1}^{\infty} a_n b_n = 0 \neq 1 = AB.$$

The conjectured identity is false even for series with only one nonzero term each. The finite product $S_N T_N$ contains all the terms $a_i b_j$ with $1 \leq i, j \leq N$, whereas the partial sum $\sum_{n=1}^{N} a_n b_n$ includes only the terms with equal indices. The product limit law applies to $S_N T_N$, not to that shorter sum.

Conclusion. The two linearity identities hold. The termwise product identity does not follow and is false in general.

14.6 Exercise 14.6: Absolutely summable terms times a bounded sequence

What we must show. Prove (a) that $\sum a_n b_n$ converges when $\sum |a_n|$ converges and (b_n) is bounded, and (b) recover Corollary 14.7.

(a), first step: put one bound on every second factor.

Boundedness supplies a finite $M > 0$ with $|b_n| \leq M$ for every n. We may insist on $M > 0$ by enlarging a given bound, even if every b_n is zero.

(a), second step: verify the Cauchy criterion for the product series.

Let $\varepsilon > 0$. Since the nonnegative series $\sum |a_n|$ converges, its Cauchy criterion gives an index N such that

$$n \geq m > N \quad \Longrightarrow \quad \sum_{j=m}^{n} |a_j| < \frac{\varepsilon}{M}.$$

For any such m, n, the finite triangle inequality and the bound on b_j imply

$$\left|\sum_{j=m}^{n} a_j b_j\right| \leq \sum_{j=m}^{n} |a_j|\,|b_j| \leq M \sum_{j=m}^{n} |a_j| < M\frac{\varepsilon}{M} = \varepsilon.$$

This is exactly the Cauchy criterion for $\sum a_n b_n$. Theorem 14.4 therefore proves convergence. The same estimate without the outer absolute value also proves the Cauchy criterion for $\sum |a_n b_n|$, so the product series actually converges absolutely.

(b) Use a constant second sequence.

Choose $b_n = 1$ for all n. This sequence is bounded and $a_n b_n = a_n$. Part (a) then says that convergence of $\sum |a_n|$ implies convergence of $\sum a_n$, precisely Corollary 14.7.

Conclusion. Every bounded multiplier preserves absolute summability; the constant multiplier 1 gives the cited corollary.

14.7 Exercise 14.7: Powers greater than one of a summable nonnegative sequence

What we must show. If $a_n \geq 0$, $\sum a_n$ converges, and $p > 1$, prove that $\sum a_n^p$ converges.

1. A convergent series has terms tending to zero.

Corollary 14.5 gives $a_n \to 0$. Hence there is N_0 such that $0 \leq a_n < 1$ for all $n > N_0$.

2. Compare the powers only where that comparison is justified.

If $a_n = 0$, then $a_n^p = 0 \leq a_n$. If $0 < a_n \leq 1$, the positive exponent $p-1$ gives $a_n^{p-1} \leq 1$. Multiplication by $a_n > 0$ yields

$$0 \leq a_n^p = a_n a_n^{p-1} \leq a_n \qquad (n > N_0).$$

The restriction to sufficiently late indices matters: a number larger than 1 has its size increased by taking a power $p > 1$.

3. Include the finite initial part.

Let $A = \sum_{n=1}^{\infty} a_n < +\infty$ and $C = \sum_{n=1}^{N_0} a_n^p$, both nonnegative and finite. For $N > N_0$,

$$\sum_{n=1}^{N} a_n^p = C + \sum_{n=N_0+1}^{N} a_n^p \leq C + \sum_{n=N_0+1}^{N} a_n \leq C + A.$$

For $N \leq N_0$, the partial sum is at most $C \leq C + A$ as well. The partial sums of $\sum a_n^p$ are increasing because their terms are nonnegative, and the displayed inequalities bound them above. The monotone sequence theorem proves a finite limit.

Conclusion. The series $\sum a_n^p$ converges for every real $p > 1$.

14.8 Exercise 14.8: A geometric-mean comparison for two nonnegative series

What we must show. If $a_n, b_n \geq 0$ and both $\sum a_n$ and $\sum b_n$ converge, prove convergence of $\sum \sqrt{a_n b_n}$.

1. Derive the needed pointwise inequality.

The square of any real number is nonnegative. Apply this to $\sqrt{a_n} - \sqrt{b_n}$, whose square expands to

$$0 \le (\sqrt{a_n} - \sqrt{b_n})^2 = a_n + b_n - 2\sqrt{a_n b_n}.$$

Rearranging and dividing by $2 > 0$ gives

$$0 \le \sqrt{a_n b_n} \le \frac{a_n + b_n}{2} \le a_n + b_n.$$

The last inequality uses $a_n + b_n \ge 0$. The square-root identity also holds when one or both of the numbers are zero.

2. Apply the Comparison Test to a convergent upper series.

By Exercise 14.5, $\sum(a_n + b_n)$ converges, with sum equal to the sum of the two original sums. Its terms are nonnegative. The displayed pointwise inequality therefore satisfies all the hypotheses of the Comparison Test, which proves convergence of $\sum \sqrt{a_n b_n}$.

Conclusion. The geometric-mean series converges. In fact its sum is at most one half of the sum of the two given series.

14.9 Exercise 14.9: Changing finitely many terms

What we must show. Suppose $a_n = b_n$ except for finitely many indices. Prove that $\sum a_n$ and $\sum b_n$ either both converge or both diverge.

1. Place all exceptional indices before a single cutoff.

If the exceptional set is nonempty, let N_0 be its maximum. If the set is empty, take $N_0 = 1$. In either case $a_n = b_n$ for every $n > N_0$. Define the finite real constant

$$D = \sum_{n=1}^{N_0} (a_n - b_n).$$

2. Compare partial sums exactly.

For each $N \ge N_0$, all the later differences are zero, so

$$\sum_{n=1}^{N} a_n - \sum_{n=1}^{N} b_n = \sum_{n=1}^{N_0} (a_n - b_n) = D.$$

Thus the two partial-sum sequences differ by a fixed constant from that point onward.

3. Prove both directions of convergence.

If $\sum a_n = A$ is finite, the last identity gives $\sum_{n=1}^{N} b_n = \sum_{n=1}^{N} a_n - D \to A - D$, so the second series converges. Conversely, if $\sum b_n = B$ is finite, then $\sum_{n=1}^{N} a_n = \sum_{n=1}^{N} b_n + D \to B + D$. Therefore convergence of either series implies convergence of the other. If one diverges and the other converged, one of these implications would be contradicted; hence divergence is shared too.

Conclusion. Finite changes preserve convergence. They can change a finite sum by the constant D, which is why equality of sums is not asserted.

14.10 Exercise 14.10: Divergence detected by roots but not by ratios

What we must show. Give a series for which the Root Test proves divergence while neither conclusion of the Ratio Test applies.

1. Define positive terms with alternating adjacent ratios.

Take

$$a_n = 2^{\,n+(-1)^n} \qquad (n \geq 1).$$

Every term is positive, so its root and ratio are defined. The nth root is $a_n^{1/n} = 2 \cdot 2^{(-1)^n/n}$. Since $-1/n \leq (-1)^n/n \leq 1/n$ and powers of 2 preserve order,

$$2 \cdot 2^{-1/n} \leq a_n^{1/n} \leq 2 \cdot 2^{1/n}.$$

Both outer expressions tend to 2, using the root limit for the fixed positive number 2 and its reciprocal. The squeeze theorem gives $a_n^{1/n} \to 2$.

2. Apply the divergent part of the Root Test.

The root limit superior is $2 > 1$, so the Root Test proves divergence of $\sum a_n$.

3. Check both possible ratio-test inequalities.

When n is even, $(-1)^n = 1$ and $(-1)^{n+1} = -1$, giving

$$\frac{a_{n+1}}{a_n} = 2^{(n+1)-1-(n+1)} = 2^{-1} = \frac{1}{2}.$$

When n is odd, the two signs are reversed, giving

$$\frac{a_{n+1}}{a_n} = 2^{(n+1)+1-(n-1)} = 2^3 = 8.$$

Each value occurs infinitely often, and there are no other ratio values. Hence the ratio limit inferior is $1/2$ and its limit superior is 8. The convergence condition, ratio limit superior less than 1, fails. The divergence condition, ratio limit inferior greater than 1, fails as well. Thus the Ratio Test is inconclusive even though the Root Test settles the question.

Conclusion. The series $\sum 2^{n+(-1)^n}$ has the required properties: root limit 2, ratio limit inferior $1/2$, and ratio limit superior 8.

14.11 Exercise 14.11: A constant successive ratio gives a geometric series

What we must show. Suppose every $a_n \neq 0$ and $a_{n+1}/a_n = r$ is independent of n. Prove that $\sum a_n$ is geometric.

1. Convert the quotient statement to a recurrence.

Since $a_n \neq 0$, multiplication by a_n gives $a_{n+1} = ra_n$ for every n. Also $r = a_2/a_1 \neq 0$ because both a_1 and a_2 are nonzero.

2. Solve the recurrence by induction.

At $n = 1$, the formula $a_n = a_1 r^{n-1}$ reads $a_1 = a_1 r^0$, which holds. Suppose $a_n = a_1 r^{n-1}$ for some $n \geq 1$. Then

$$a_{n+1} = r a_n = r(a_1 r^{n-1}) = a_1 r^n.$$

This is the same formula with n replaced by $n + 1$, so induction proves it for every positive integer.

3. Recognize the series and its possible sum.

The series is therefore

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} a_1 r^{n-1} = a_1 + a_1 r + a_1 r^2 + \cdots,$$

which is the definition of a geometric series with first term a_1 and common ratio r. It converges when $|r| < 1$ and then has sum $a_1/(1 - r)$. If $|r| \geq 1$, the nonzero first term ensures that the terms do not tend to zero, so it diverges.

Conclusion. A constant successive ratio forces the exact geometric form $a_n = a_1 r^{n-1}$.

14.12 Exercise 14.12: Selecting an absolutely summable subsequence

What we must show. From $\liminf |a_n| = 0$, construct strictly increasing indices n_k for which $\sum_{k=1}^{\infty} a_{n_k}$ converges.

1. Arbitrarily small absolute values occur after every finite prefix.

Fix a number $\eta > 0$ and an index cutoff $N \geq 0$. If no $n > N$ satisfied $|a_n| < \eta$, every term beyond N would have $|a_n| \geq \eta$. All later tail infima would then be at least η, which would imply $\liminf |a_n| \geq \eta > 0$, a contradiction. Thus such an index always exists.

2. Select a geometric error bound at each stage.

Put $n_0 = 0$. After n_{k-1} has been chosen, use Step 1 with $N = n_{k-1}$ and $\eta = 2^{-k}$ to choose

$$n_k > n_{k-1}, \qquad |a_{n_k}| < 2^{-k}.$$

For definiteness, choose the least eligible index. Existence is provided by Step 1, and the least positive integer principle makes the recursive choice well defined. Strict increase of the indices ensures that (a_{n_k}) is a subsequence of the original sequence.

3. Compare the absolute-value series to a geometric series.

The geometric series satisfies $\sum_{k=1}^{\infty} 2^{-k} = 1$. Its terms dominate $|a_{n_k}| \geq 0$, so the Comparison Test gives convergence of $\sum |a_{n_k}|$. Absolute convergence implies convergence of $\sum a_{n_k}$ by Corollary 14.7. The geometric bound is essential to this argument: merely choosing a subsequence whose terms tend to zero would not by itself prove that its series converges.

Conclusion. There is an absolutely convergent subsequence series, obtained by imposing $|a_{n_k}| < 2^{-k}$ at stage k.

14.13 Exercise 14.13: Four exact sums from geometric and telescoping identities

What we must show. Calculate the two geometric sums in part (a), prove the identities in parts (b) and (c), and deduce the sum in part (d).

(a) The ratios $2/3$ and $-2/3$, starting at $n = 1$.

For any $|r| < 1$, shifting the index of the geometric series gives

$$\sum_{n=1}^{\infty} r^n = r \sum_{j=0}^{\infty} r^j = \frac{r}{1-r}.$$

Both ratios here have absolute value $2/3 < 1$. Substitution yields

$$\sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n = \frac{2/3}{1-2/3} = 2, \qquad \sum_{n=1}^{\infty} \left(-\frac{2}{3}\right)^n = \frac{-2/3}{1+2/3} = -\frac{2}{5}.$$

The negative sign in the second first term matters because its sum begins at exponent 1, rather than exponent 0.

(b) The series with term $1/[n(n+1)]$.

The finite algebraic identity

$$\frac{1}{n} - \frac{1}{n+1} = \frac{n+1-n}{n(n+1)} = \frac{1}{n(n+1)}$$

gives, for every $N \geq 1$,

$$\begin{aligned} \sum_{n=1}^{N} \frac{1}{n(n+1)} &= (1 - \tfrac{1}{2}) + (\tfrac{1}{2} - \tfrac{1}{3}) + \cdots + (\tfrac{1}{N} - \tfrac{1}{N+1}) \\ &= 1 - \frac{1}{N+1}. \end{aligned}$$

Taking the limit $N \to \infty$ gives the finite sum 1.

(c) The series with term $(n-1)/2^{n+1}$.

Put the right side of the following identity over a common denominator to verify it:

$$\frac{n}{2^n} - \frac{n+1}{2^{n+1}} = \frac{2n-(n+1)}{2^{n+1}} = \frac{n-1}{2^{n+1}}.$$

The first term of the series, at $n = 1$, is zero. Summing the identity to N causes all intervening terms to cancel:

$$\sum_{n=1}^{N} \frac{n-1}{2^{n+1}} = \frac{1}{2} - \frac{N+1}{2^{N+1}}.$$

To justify the last term's limit explicitly, set $u_N = (N+1)/2^{N+1} > 0$. For $N \geq 1$,

$$\frac{u_{N+1}}{u_N} = \frac{N+2}{2(N+1)} \leq \frac{3}{4},$$

because the last inequality is equivalent to $N \geq 1$. Induction gives $0 < u_N \leq u_1(3/4)^{N-1}$, whose upper bound tends to zero. Hence $u_N \to 0$ by the squeeze theorem, and

$$\sum_{n=1}^{\infty} \frac{n-1}{2^{n+1}} = \frac{1}{2}.$$

(d) Deduce the sum with term $n/2^n$.

For each index,

$$\frac{n}{2^n} = 2\frac{n-1}{2^{n+1}} + \frac{1}{2^n}.$$

The first series on the right converges by part (c), and the second is geometric with sum 1. The linearity formulas of Exercise 14.5 therefore apply and give

$$\sum_{n=1}^{\infty} \frac{n}{2^n} = 2\left(\frac{1}{2}\right) + 1 = 2.$$

Conclusion. The requested sums are, in order, 2, $-2/5$, 1, $1/2$, and 2. All telescoping cancellations were made in finite partial sums before taking a limit.

14.14 Exercise 14.14: A blockwise lower comparison for the harmonic series

What we must show. Prove divergence of $\sum_{n=1}^{\infty} 1/n$ by comparing it with the displayed repeated-block sequence.

1. Express the displayed sequence by exact index ranges.

For each integer $j \geq 1$, the block

$$2^{j-1} \leq n \leq 2^j - 1$$

contains $2^j - 1 - 2^{j-1} + 1 = 2^{j-1}$ indices. On that block define $a_n = 2^{-j}$. Thus the first block is the single term $1/2$; the second contains two terms $1/4$; the third contains four terms $1/8$; and the fourth contains eight terms $1/16$. These blocks exactly reproduce the sequence in the question.

2. Every complete block contributes the same positive amount.

For the block labeled j, its sum is

$$2^{j-1} \cdot 2^{-j} = \frac{1}{2}.$$

The comparison series in the exercise begins at $n = 2$, so it omits the first block and retains all blocks with $j \geq 2$. At the end of block $J \geq 2$ its partial sum is

$$\sum_{n=2}^{2^J-1} a_n = \sum_{j=2}^{J} \frac{1}{2} = \frac{J-1}{2}.$$

These values tend to $+\infty$ as J grows. All intermediate partial sums are increasing, since $a_n > 0$. Therefore the entire comparison series, not only these block-end partial sums, diverges to $+\infty$.

3. Compare each harmonic term with its block value.

Given $n \geq 2$, there is a unique block index $j \geq 2$ with $2^{j-1} \leq n \leq 2^j - 1$. Since $n < 2^j$ and both are positive,

$$\frac{1}{n} > \frac{1}{2^j} = a_n.$$

The Comparison Test now shows that $\sum_{n=2}^{\infty} 1/n$ diverges to $+\infty$. Adding the first harmonic term, which is 1, preserves this divergence.

Conclusion. The harmonic series diverges to $+\infty$ because it dominates a positive series whose every complete block adds $1/2$.

15 Alternating series and integral tests

In this section, log means the natural logarithm. A series is *conditionally convergent* if it converges but its series of absolute values diverges. Changing or removing finitely many terms does not change convergence, by Exercise 14.9. We use this fact when a comparison or monotonicity estimate holds only after a fixed index.

The alternating-series theorem requires nonnegative magnitudes that decrease to zero. An alternating sign by itself does not guarantee convergence. An integral test requires a nonnegative decreasing function matching the terms; the precise statement and proof are given in Exercise 15.8 below. As in the textbook's Section 15, the usual elementary integral and derivative formulas are available.

15.1 Exercise 15.1: Two alternating series with very different magnitudes

What we must show. Decide whether the series in parts (a) and (b) converge, and justify each answer.

Plan. Check the magnitudes before applying the alternating-series theorem. In particular, a convergent series must have terms tending to zero, regardless of their signs.

(a) The series $\sum_{n=1}^{\infty}(-1)^n/n$.

Put $b_n = 1/n$. For every positive integer n, we have $b_n > 0$. Also $n+1 > n > 0$, so taking reciprocals reverses the inequality:

$$b_{n+1} = \frac{1}{n+1} < \frac{1}{n} = b_n.$$

Finally, $b_n \to 0$. Indeed, for $\varepsilon > 0$, choose an integer $N > 1/\varepsilon$. If $n > N$, then $|b_n - 0| = 1/n < \varepsilon$.

Thus $\sum_{n=1}^{\infty}(-1)^{n+1}b_n$ converges by the alternating-series theorem. Multiplication by -1 preserves convergence, by Exercise 14.5(b), and gives the series requested here. Its series of absolute values is $\sum_{n=1}^{\infty} 1/n$, the divergent harmonic series. Hence the requested series converges conditionally.

(b) The series $\sum_{n=1}^{\infty}(-1)^n n!/2^n$.

Let $c_n = n!/2^n > 0$. Cancellation of the common factorial and power gives

$$\frac{c_{n+1}}{c_n} = \frac{(n+1)!}{2^{n+1}}\frac{2^n}{n!} = \frac{n+1}{2}.$$

For $n \geq 3$, this ratio is at least 2. Repeated multiplication therefore gives, for $n \geq 3$,

$$c_n \geq c_3\, 2^{n-3} = \frac{3}{4}\, 2^{n-3}.$$

The right side tends to $+\infty$. In particular c_n does not tend to zero. If the signed terms $(-1)^n c_n$ tended to zero, their absolute values c_n would tend to zero too, a contradiction. The necessary term condition, Corollary 14.5, now proves divergence. Alternation does not compensate for these growing magnitudes.

Conclusion. Part (a) converges conditionally. Part (b) diverges because its terms do not tend to zero.

15.2 Exercise 15.2: Powers of periodically sampled sine values

What we must show. Determine convergence of

$$\text{(a)} \quad \sum_{n=1}^{\infty} \left[\sin\left(\frac{n\pi}{6}\right)\right]^n, \qquad \text{(b)} \quad \sum_{n=1}^{\infty} \left[\sin\left(\frac{n\pi}{7}\right)\right]^n.$$

The exponent n applies to the whole sine value in each case.

Plan. In part (a), the sampled sine values reach 1 infinitely often. In part (b), there are only finitely many sampled values, and every one has absolute value strictly less than 1.

(a) Exhibit terms that stay equal to 1.

For every integer $k \geq 0$, choose $n_k = 12k + 3$. These are strictly increasing positive indices, and

$$\frac{n_k\pi}{6} = 2k\pi + \frac{\pi}{2}.$$

Sine has period 2π and $\sin(\pi/2) = 1$, so

$$\left[\sin\left(\frac{n_k\pi}{6}\right)\right]^{n_k} = 1^{n_k} = 1.$$

The terms of a convergent series must tend to zero, and every subsequence of a sequence tending to zero must also tend to zero. This constant subsequence contradicts that requirement. Therefore the series in part (a) diverges.

(b) Bound all the absolute sine values by one number below 1.

The values $\sin(n\pi/7)$ repeat when n is increased by 14, because the angle then increases by 2π. Thus all their absolute values occur in the finite list

$$\left\{\left|\sin\left(\frac{r\pi}{7}\right)\right| : r = 1, 2, \ldots, 14\right\}.$$

None of these absolute values equals 1. Otherwise the angle $r\pi/7$ would have the form $\pi/2 + \ell\pi$ for some integer ℓ. After division by π and multiplication by 14, this would give $2r = 7(2\ell + 1)$. The left side is even and the right side is odd, which is impossible.

Let c be the maximum of that finite list. Every entry is less than 1, so its attained maximum satisfies $c < 1$ as well; also $c \geq 0$. In fact $c > 0$, since $\sin(\pi/7) > 0$. For every $n \geq 1$,

$$\left|\left[\sin\left(\frac{n\pi}{7}\right)\right]^n\right| = \left|\sin\left(\frac{n\pi}{7}\right)\right|^n \leq c^n.$$

The inequality is preserved when taking the positive integer power because both sides are nonnegative. The geometric series $\sum_{n=1}^{\infty} c^n$ converges, since $0 < c < 1$. Comparison proves that the series of absolute values converges. Hence the series in part (b) converges absolutely.

Conclusion. Part (a) diverges by the necessary term condition. Part (b) converges absolutely by comparison with a geometric series.

15.3 Exercise 15.3: The logarithmic refinement of the harmonic series

What we must show. For a real parameter p, prove

$$\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p} \text{ converges} \quad \Longleftrightarrow \quad p > 1.$$

Plan. For positive p, use the integral test with the substitution $u = \log x$. For nonpositive p, compare directly with $1/n$.

1. Check positivity and the case $p \leq 0$.

For $n \geq 2$, $\log n > 0$, so every real power $(\log n)^p$ is defined and positive. Choose an integer $N > e$. For $n \geq N$, $\log n > 1$. If $p \leq 0$, then $(\log n)^p \leq 1$, and therefore

$$\frac{1}{n(\log n)^p} \geq \frac{1}{n}.$$

The harmonic series diverges. Comparison shows that the series under consideration diverges to $+\infty$ for $p \leq 0$.

2. For $p > 0$, verify the hypotheses of the integral test.

Define

$$f_p(x) = \frac{1}{x(\log x)^p} \qquad (x \geq 2).$$

The function is positive and continuous on this interval. Differentiating the product $x^{-1}(\log x)^{-p}$ gives

$$\begin{aligned} f_p'(x) &= -x^{-2}(\log x)^{-p} - px^{-2}(\log x)^{-p-1} \\ &= -\frac{\log x + p}{x^2(\log x)^{p+1}} < 0. \end{aligned}$$

The denominator is positive, and the numerator $\log x + p$ is positive because $x \geq 2$ and $p > 0$. Thus the function decreases and matches the series terms at every integer $n \geq 2$.

3. Evaluate the integrals and distinguish the parameter cases.

For $R \geq 2$, put $u = \log x$, so $du = dx/x$. Then

$$\int_2^R \frac{dx}{x(\log x)^p} = \int_{\log 2}^{\log R} u^{-p}\, du.$$

If $p \neq 1$, the power rule gives

$$\int_2^R \frac{dx}{x(\log x)^p} = \frac{(\log R)^{1-p} - (\log 2)^{1-p}}{1-p}.$$

For $p > 1$, one has $1 - p < 0$, so $(\log R)^{1-p} \to 0$ as $R \to \infty$. The integral tends to the finite positive number $(\log 2)^{1-p}/(p-1)$. The integral test proves convergence.

For $0 < p < 1$, one has $1 - p > 0$, so $(\log R)^{1-p} \to +\infty$. The denominator $1 - p$ is positive, and the integral tends to $+\infty$. The integral test proves divergence.

For $p = 1$, integration instead gives

$$\int_2^R \frac{dx}{x \log x} = \log(\log R) - \log(\log 2) \longrightarrow +\infty.$$

The arguments of both outer logarithms are positive, even though $\log(\log 2)$ itself is negative. This case also diverges.

Conclusion. Exactly the parameters $p > 1$ give convergence. For every $p \leq 1$, the nonnegative series diverges to $+\infty$.

15.4 Exercise 15.4: Four series involving logarithms

What we must show. Determine convergence of each of the four requested series.

Plan. Compare the first two series with the harmonic series. For the other two, check the integral-test hypotheses and compute the relevant integrals.

(a) The series $\sum_{n=2}^{\infty} 1/(\sqrt{n} \log n)$.

We first justify the eventual inequality $\log x \leq \sqrt{x}$. Let $g(x) = \sqrt{x} - \log x$ for $x > 0$. Its derivative is

$$g'(x) = \frac{1}{2\sqrt{x}} - \frac{1}{x} = \frac{\sqrt{x} - 2}{2x} \geq 0 \qquad (x \geq 4).$$

Also $\log 2 = \int_1^2 dt/t < 1$, since $1/t < 1$ on $(1, 2]$. Thus $g(4) = 2 - \log 4 = 2 - 2\log 2 > 0$. As g is nondecreasing on $[4, \infty)$, we have $\log x < \sqrt{x}$ there.

For integers $n \geq 4$, both logarithms and square roots are positive, and the inequality just proved gives

$$\sqrt{n} \log n \leq n, \qquad \frac{1}{\sqrt{n} \log n} \geq \frac{1}{n}.$$

Taking reciprocals reverses the positive-denominator inequality. Comparison with the divergent harmonic series proves divergence to $+\infty$.

(b) The series $\sum_{n=2}^{\infty} (\log n)/n$.

Choose an integer $N > e$. If $n \geq N$, monotonicity of the logarithm gives $\log n > \log e = 1$. Consequently

$$\frac{\log n}{n} \geq \frac{1}{n}.$$

All terms are positive for $n \geq 2$. Comparison with the harmonic series, with the first finitely many indices ignored, proves divergence to $+\infty$.

(c) The series $\sum_{n=4}^{\infty} 1/[n(\log n)(\log \log n)]$.

For $x \geq 4$, we have $\log x > 1$. One direct check at the left endpoint is

$$\log 2 = \int_1^2 \frac{dt}{t} > \frac{1}{2}, \qquad \log 4 = 2 \log 2 > 1.$$

Therefore $\log \log x > 0$ on this whole interval. The three factors x, $\log x$, and $\log \log x$ are positive, continuous, and increasing. Their product is increasing, so its positive reciprocal

$$f(x) = \frac{1}{x(\log x)(\log \log x)}$$

is decreasing and continuous on $[4, \infty)$.

Two substitutions, first $u = \log x$ and then $v = \log u$, give

$$\begin{aligned}\int_4^R \frac{dx}{x(\log x)(\log\log x)} &= \int_{\log 4}^{\log R} \frac{du}{u \log u} \\ &= \int_{\log\log 4}^{\log\log R} \frac{dv}{v} \\ &= \log(\log(\log R)) - \log(\log(\log 4)).\end{aligned}$$

Every argument of a logarithm in these formulas is positive for $R \geq 4$: in particular $\log\log 4 > 0$. As $R \to \infty$, the first triple logarithm tends to $+\infty$. The integral test proves that this series diverges to $+\infty$.

(d) The series $\sum_{n=2}^{\infty} (\log n)/n^2$.

Let $h(x) = (\log x)/x^2$ for $x \geq 2$. This function is positive and continuous. Differentiation gives

$$h'(x) = \frac{1 - 2\log x}{x^3} < 0 \qquad (x \geq 2),$$

because $\log x \geq \log 2 > 1/2$, as established in part (c). Thus the integral test applies. For integration by parts, take $u = \log x$ and $dv = x^{-2}\,dx$. Then $du = dx/x$ and $v = -1/x$, so

$$\begin{aligned}\int_2^R \frac{\log x}{x^2}\,dx &= \left[-\frac{\log x}{x}\right]_2^R + \int_2^R \frac{dx}{x^2} \\ &= \frac{\log 2 + 1}{2} - \frac{\log R + 1}{R}.\end{aligned}$$

Part (a) gives $0 \leq (\log R)/R \leq 1/\sqrt{R}$ for $R \geq 4$. The right side tends to zero, as does $1/R$. Hence the integral tends to $(\log 2 + 1)/2$, a finite number. The integral test proves convergence of the series.

Conclusion. Parts (a), (b), and (c) diverge to $+\infty$. Part (d) converges.

15.5 Exercise 15.5: Why a direct comparison does not establish all positive powers

What we must show. Explain why the text used an integral argument to prove the convergence of $\sum 1/n^p$ for every $p > 1$.

Plan. A comparison proof needs an independently known convergent upper series. It cannot assume a nearby case of the very theorem it is trying to establish.

1. Check which direction of comparison is useful.

Knowing that $\sum 1/n^2$ converges immediately handles $p \geq 2$, since $0 \leq 1/n^p \leq 1/n^2$ for $n \geq 1$. For $1 < p < 2$, however, the inequality reverses:

$$\frac{1}{n^p} \geq \frac{1}{n^2}.$$

Being larger than the terms of a convergent nonnegative series does not establish either convergence or divergence. Comparison with $1/n$ also fails to settle the question: for $p > 1$ the terms $1/n^p$ are smaller than the terms of that divergent series.

2. Explain the missing starting exponent.

If convergence were already known at some $p_0 > 1$, comparison would prove it for every $p \geq p_0$. It would leave the interval $1 < p < p_0$ unresolved. There is no smallest real exponent greater than 1: between 1 and p_0 lies $(1 + p_0)/2$. Thus no single known case $p_0 > 1$ settles the whole range by this direct comparison. Choosing an unproved exponent q with $1 < q < p$ merely replaces the original convergence problem with another case of the same problem. The integral calculation supplies an independent argument for every $p > 1$.

3. Comparison can still work after grouping the terms.

The explanation above does not mean a comparison proof is impossible. For example, fix $p > 1$ and group the indices into blocks

$$2^j \leq n < 2^{j+1} \qquad (j = 0, 1, 2, \ldots).$$

There are exactly 2^j integers in this block, and each satisfies $n^{-p} \leq 2^{-jp}$. Hence the sum of the block is at most

$$2^j 2^{-jp} = 2^{j(1-p)}.$$

The bounding block sums form a geometric series with ratio $2^{1-p} < 1$. Every original partial sum is bounded above by the sum of this geometric series, since its terms are nonnegative and its indices occupy finitely many blocks. The original partial sums are increasing and bounded above, so they converge by Theorem 10.2. This is a valid alternative proof using an extra grouping argument.

Conclusion. The integral proof avoids relying on an unproved nearby p-series. A plain comparison with one known exponent does not cover all $p > 1$; a geometric comparison of grouped blocks does.

15.6 Exercise 15.6: What squaring the terms can change

What we must show. Supply the examples in parts (a) and (c), and justify the nonnegative-term assertion in part (b).

(a) A divergent series whose squared-term series converges.

Take $a_n = 1/n$. Then $\sum a_n$ is the divergent harmonic series, whereas

$$\sum_{n=1}^{\infty} a_n^2 = \sum_{n=1}^{\infty} \frac{1}{n^2}$$

converges by the p-series theorem, with $p = 2 > 1$. These two facts verify both required properties of the example.

(b) Squaring preserves convergence when the original terms are nonnegative.

Suppose $a_n \geq 0$ and $\sum a_n$ converges. Corollary 14.5 gives $a_n \to 0$. There is therefore an integer N with $0 \leq a_n < 1$ whenever $n > N$. Multiplication of $a_n < 1$ by the nonnegative number a_n gives

$$0 \leq a_n^2 \leq a_n \qquad (n > N).$$

The comparison test proves convergence of $\sum_{n>N} a_n^2$. Adding its finitely many omitted real terms preserves convergence, so the whole squared-term series converges. This is also Exercise 14.7 with the power $p = 2$.

(c) A convergent signed series whose squared-term series diverges.

Take

$$a_n = \frac{(-1)^{n+1}}{\sqrt{n}}.$$

Its magnitudes $b_n = 1/\sqrt{n}$ are positive and decrease, because the positive square root increases with its argument and taking reciprocals reverses order. They tend to zero: given $\varepsilon > 0$, choose $N > 1/\varepsilon^2$; then $n > N$ implies $1/\sqrt{n} < \varepsilon$. The alternating-series theorem proves that $\sum a_n$ converges. However,

$$a_n^2 = \frac{[(-1)^{n+1}]^2}{(\sqrt{n})^2} = \frac{1}{n},$$

so $\sum a_n^2$ is the divergent harmonic series.

Conclusion. Use $a_n = 1/n$ for part (a) and $a_n = (-1)^{n+1}/\sqrt{n}$ for part (c). Part (b) follows from the eventual bound $0 \le a_n^2 \le a_n$.

15.7 Exercise 15.7: A necessary decay rate for decreasing summable terms

What we must show. Prove that if (a_n) is a decreasing real sequence and $\sum a_n$ converges, then $na_n \to 0$. Then apply this conclusion to the harmonic series.

Plan. First deduce that every a_n is nonnegative. A late block of roughly $n/2$ terms is small by the Cauchy criterion, yet every term in that block is at least a_n.

(a), first: derive the sign condition from the actual hypotheses.

Since $\sum a_n$ converges, Corollary 14.5 gives $a_n \to 0$. The word decreasing means $a_{n+1} \le a_n$; strict decrease is not required. If $a_j < 0$ at any index j, then $a_n \le a_j < 0$ for all $n \ge j$. Such a tail cannot tend to zero: with $\varepsilon = -a_j/2 > 0$, its terms satisfy $|a_n| \ge -a_j > \varepsilon$. This contradicts $a_n \to 0$. Thus $a_n \ge 0$ for every index n.

(a), second: use a tail block with an explicit number of terms.

Fix $\varepsilon > 0$. By the Cauchy criterion for the convergent series, choose an integer $N \ge 1$ such that

$$q \ge m > N \quad \Longrightarrow \quad \left| \sum_{j=m}^{q} a_j \right| < \frac{\varepsilon}{2}.$$

For any integer $n > 2N$, set $k = \lfloor n/2 \rfloor$. Here $\lfloor x \rfloor$ is the greatest integer at most x. We have $k \ge N$, so the block starting at $k+1$ begins after N. It contains exactly $n - k$ terms, and $n - k \ge n/2$. For $k + 1 \le j \le n$, decreasingness gives $a_j \ge a_n$. Consequently

$$0 \le \frac{n}{2} a_n \le (n-k) a_n \le \sum_{j=k+1}^{n} a_j = \left| \sum_{j=k+1}^{n} a_j \right| < \frac{\varepsilon}{2}.$$

The equality with the absolute value uses nonnegativity of the terms. Multiplying the resulting strict upper bound by 2 gives $0 \le na_n < \varepsilon$ whenever $n > 2N$. This verifies the definition of $na_n \to 0$.

(b) Apply the necessary condition to $a_n = 1/n$.

The sequence $1/n$ is decreasing. If the harmonic series converged, part (a) would imply

$$n\left(\frac{1}{n}\right) \longrightarrow 0.$$

The expression on the left equals 1 for every n, so its limit is 1, a contradiction. Therefore the harmonic series diverges. Since its terms are nonnegative, its partial sums are increasing; having no finite limit, they tend to $+\infty$ by the monotone sequence theorem for unbounded increasing sequences.

Conclusion. Every decreasing real sequence with a convergent series satisfies $na_n \to 0$. The constant value $n(1/n) = 1$ therefore provides another proof of divergence of the harmonic series.

15.8 Exercise 15.8: A general integral test and its error bounds

What we must show. State and prove a precise integral test of the type described in Discussion 15.2.

Plan. Compare each unit-width rectangle with the integral below and above it. Nonnegativity makes both the partial sums and the integrals increasing, so a finite upper bound implies convergence.

1. State the assumptions and the conclusion.

Let $m \geq 1$ be an integer, and let $f : [m, \infty) \to [0, \infty)$ be continuous and nonincreasing. Set $a_n = f(n)$ for every integer $n \geq m$. Define the improper integral by

$$\int_m^\infty f(x)\,dx = \lim_{R\to\infty} \int_m^R f(x)\,dx,$$

where the value $+\infty$ is allowed. Then

$$\sum_{n=m}^{\infty} a_n \text{ converges} \quad \Longleftrightarrow \quad \int_m^\infty f(x)\,dx < +\infty.$$

If either side is infinite, both the integral and the series diverge to $+\infty$. If they converge and $R_N = \sum_{n=N+1}^{\infty} a_n$ for $N \geq m$, then additionally

$$\int_{N+1}^\infty f(x)\,dx \leq R_N \leq \int_N^\infty f(x)\,dx.$$

Continuity is a sufficient condition ensuring that all the finite integrals used here exist. The proof also works whenever those integrals exist, with the same nonnegativity and monotonicity assumptions.

2. Establish the unit-interval inequalities.

Fix an integer $k \geq m$. If $k \leq x \leq k+1$, nonincreasingness gives $f(k+1) \leq f(x) \leq f(k)$. Integrating these constant bounds over the interval of length 1 yields

$$f(k+1) \leq \int_k^{k+1} f(x)\,dx \leq f(k).$$

In particular, summing the right-hand inequalities for $k = m, m+1, \ldots, N$ gives

$$\int_m^{N+1} f(x)\,dx \leq \sum_{k=m}^{N} f(k).$$

For the opposite comparison, use $f(k) \le \int_{k-1}^{k} f(x)\,dx$ for $k \ge m+1$. Keep the first term $f(m)$ separate and sum to obtain

$$\sum_{k=m}^{N} f(k) \le f(m) + \int_{m}^{N} f(x)\,dx.$$

At $N = m$, the second sum has no remaining terms and the integral is zero, so this estimate still holds. Combining the two bounds, with $S_N = \sum_{k=m}^{N} f(k)$, gives

$$\boxed{\int_{m}^{N+1} f(x)\,dx \le S_N \le f(m) + \int_{m}^{N} f(x)\,dx.}$$

3. A finite integral bounds the partial sums.

Suppose the improper integral has finite value I. Since $f \ge 0$, one has $\int_m^N f \le I$. Therefore $0 \le S_N \le f(m) + I$ for every $N \ge m$. Also $S_{N+1} - S_N = f(N+1) \ge 0$, so (S_N) is increasing. The monotone sequence theorem gives a finite limit of (S_N), which is precisely convergence of the series.

4. A convergent series bounds all the finite integrals.

Conversely, suppose $S_N \to S < +\infty$. The nonnegative terms imply $S_N \le S$. If $R \ge m$ is any real number, choose an integer $N \ge m$ with $R \le N+1$. Nonnegativity and the first rectangle estimate give

$$0 \le \int_{m}^{R} f(x)\,dx \le \int_{m}^{N+1} f(x)\,dx \le S_N \le S.$$

Thus all the integrals are bounded above by S, not only those with integer endpoints. They increase with R, again because $f \ge 0$. Their supremum is finite, and their limit as $R \to \infty$ equals that supremum: given a positive tolerance, choose an endpoint whose integral is within that tolerance of the supremum; every larger endpoint gives an integral between that value and the supremum. Hence the improper integral converges.

5. Explain divergence and derive the tail estimates.

If the integral tends to $+\infty$, the lower bound $S_N \ge \int_m^{N+1} f$ forces $S_N \to +\infty$. If instead $S_N \to +\infty$, the upper rectangle bound rearranges to $\int_m^N f \ge S_N - f(m)$, which forces the integral to $+\infty$ as well. Nonnegative increasing sums and integrals have no other type of divergence.

Finally, in the convergent case, apply the same unit-interval comparisons to the terms with indices $N+1, \dots, M$, where $M > N \ge m$:

$$\int_{N+1}^{M+1} f(x)\,dx \le \sum_{k=N+1}^{M} f(k) \le \int_{N}^{M} f(x)\,dx.$$

Letting $M \to \infty$ gives the stated remainder bounds. These limits are finite since the whole integral and series converge.

The same test applies if the assumptions hold only after some integer index: start at that index and use Exercise 14.9 to restore the finite initial segment.

Conclusion. For a continuous nonnegative nonincreasing function matching the terms, the series converges exactly when its improper integral is finite. The two neighboring tail integrals bound the omitted tail of the series.

16 Decimal expansions of real numbers

A bar marks exactly the digits that repeat. Thus $0.0\overline{2} = 0.02222\ldots$, whereas $0.\overline{02} = 0.020202\ldots$. A terminating decimal is understood to continue with zeros. For a decimal $K.d_1d_2\cdots$, its value means the convergent series

$$K + \sum_{j=1}^{\infty} d_j 10^{-j}, \qquad 0 \le d_j \le 9.$$

Convergence follows by comparison with the geometric series $\sum_{j=1}^{\infty} 9 \cdot 10^{-j} = 1$. Theorem 16.3 describes the only possible ambiguity: a positive terminating decimal also has a representation ending in nines.

16.1 Exercise 16.1: Two expansions for eleven fourths

What we must show. Verify both decimal representations in part (a), and identify which one long division produces in part (b).

(a) Evaluate the expansion ending in zeros.

The expression $2.75\overline{0}$ means

$$2 + \frac{7}{10} + \frac{5}{100} = \frac{200 + 70 + 5}{100} = \frac{275}{100} = \frac{11}{4}.$$

Every remaining term is zero, so there is no additional contribution.

(a) Evaluate the expansion ending in nines.

In $2.74\overline{9}$ the first nine is in the third decimal place. The sum of the nines is

$$\sum_{j=3}^{\infty} 9 \cdot 10^{-j} = 9 \cdot 10^{-3} \sum_{k=0}^{\infty} (10^{-1})^k = \frac{9 \cdot 10^{-3}}{1 - 10^{-1}} = \frac{1}{100}.$$

Consequently

$$2.74\overline{9} = 2 + \frac{7}{10} + \frac{4}{100} + \frac{1}{100} = 2 + \frac{7}{10} + \frac{5}{100} = \frac{11}{4}.$$

The extra tail of nines supplies exactly the missing hundredth.

(b) Carry out the long division.

First $11 = 4 \cdot 2 + 3$, so the integer part is 2 and the initial remainder is 3. Multiplying the remainder by ten gives

$$30 = 4 \cdot 7 + 2, \qquad 20 = 4 \cdot 5 + 0.$$

The successive decimal digits are therefore 7 and 5, after which the remainder is zero. All subsequent divisions have $10 \cdot 0 = 4 \cdot 0 + 0$, so all subsequent digits are zero.

Conclusion. Both $2.74\overline{9}$ and $2.75\overline{0}$ equal $11/4$. The long division algorithm produces $2.75\overline{0}$.

16.2 Exercise 16.2: Trading a terminating tail for a tail of nines

What we must show. Verify the two identities in the opening paragraph of the proof of Theorem 16.3: the identity for a positive integer, and the identity for a decimal with a last nonzero digit.

First calculate a general tail of nines.

For any integer $r \geq 0$, a geometric series gives

$$\sum_{j=r+1}^{\infty} 9 \cdot 10^{-j} = \frac{9 \cdot 10^{-(r+1)}}{1 - 10^{-1}} = 10^{-r}.$$

In particular, when $r = 0$ this tail has value 1.

A positive integer has the asserted alternative representation.

Let K be a positive integer. Then $K - 1$ is a nonnegative integer, so it is an allowed integer part for a nonnegative decimal. The tail formula with $r = 0$ gives

$$(K-1).\overline{9} = (K-1) + \sum_{j=1}^{\infty} 9 \cdot 10^{-j} = (K-1) + 1 = K = K.\overline{0}.$$

These digit strings are distinct even though their values agree.

A last nonzero digit can be reduced by one.

Suppose

$$x = K.d_1 \cdots d_{r-1} d_r \overline{0}, \qquad r \geq 1, \qquad d_r \in \{1, \ldots, 9\}.$$

The condition $d_r \neq 0$ makes $d_r - 1$ an allowed digit. The value of the proposed other expansion is

$$\begin{aligned} &K + \sum_{j=1}^{r-1} d_j 10^{-j} + (d_r - 1)10^{-r} + \sum_{j=r+1}^{\infty} 9 \cdot 10^{-j} \\ &\quad = K + \sum_{j=1}^{r-1} d_j 10^{-j} + (d_r - 1)10^{-r} + 10^{-r} \\ &\quad = K + \sum_{j=1}^{r-1} d_j 10^{-j} + d_r 10^{-r} = x. \end{aligned}$$

Thus decreasing the last nonzero digit by one and replacing the following zeros by nines preserves the represented number.

Check the zero case separately.

The first identity requires $K > 0$, and the second requires a last nonzero digit. Neither condition holds for $0.\overline{0}$. Indeed, if a nonnegative decimal represents zero, its nonnegative integer part must be zero. Every digit must then be zero too: a positive digit d_j would contribute the positive amount $d_j 10^{-j}$ to a sum of nonnegative terms. Hence zero has just the expansion $0.\overline{0}$.

Conclusion. The two claimed alternative expansions follow from the geometric tail identity. Every positive terminating decimal has the indicated second expansion; zero is the exceptional case.

16.3 Exercise 16.3: Strict comparison of convergent nonnegative series

What we must show. Suppose $\sum a_n$ and $\sum b_n$ converge, all their terms are nonnegative, $a_n \le b_n$ for every n, and $a_m < b_m$ at at least one index m. Prove the inequality of their sums is strict.

Plan. Keep the positive gap at that one index when passing from finite sums to infinite sums. A strict inequality at each partial sum alone would not guarantee a strict inequality of limits.

Name the sums and the fixed positive gap.

Write

$$A = \sum_{n=1}^{\infty} a_n, \qquad B = \sum_{n=1}^{\infty} b_n, \qquad \delta = b_m - a_m > 0.$$

Both A and B are finite by hypothesis. For every $N \ge m$,

$$\sum_{n=1}^{N} b_n - \sum_{n=1}^{N} a_n = \sum_{n=1}^{N} (b_n - a_n) \ge b_m - a_m = \delta.$$

The inequality holds because all the other differences are nonnegative; deleting them cannot increase the sum.

Take the limit with the gap still present.

The difference rule for limits gives

$$B - A = \lim_{N\to\infty} \left(\sum_{n=1}^{N} b_n - \sum_{n=1}^{N} a_n \right) \ge \delta > 0.$$

Here the weak inequality is preserved by passage to finite limits, and the final strict inequality comes from the fixed number δ, which does not shrink with N.

Conclusion. $\displaystyle\sum_{n=1}^{\infty} a_n < \sum_{n=1}^{\infty} b_n.$

16.4 Exercise 16.4: Converting six repeating decimals to fractions

What we must show. Express each given repeating decimal as a fraction of integers.

Plan. Group the repeated block as an integer and use a geometric series. A block of r digits with integer value D, repeated immediately after the decimal point, represents

$$\frac{D}{10^r} + \frac{D}{10^{2r}} + \cdots = \frac{D10^{-r}}{1 - 10^{-r}} = \frac{D}{10^r - 1}.$$

A leading zero counts as a digit of the block: the block 02 has length two and integer value 2.

(a) The decimal $0.\overline{2}$.

Here the block has length one and value 2, so

$$0.\overline{2} = \frac{2}{10 - 1} = \boxed{\frac{2}{9}}.$$

(b) The decimal $0.0\overline{2}$.

Only the digit 2 repeats; the first decimal digit is the single nonrepeating zero. Moving the decimal point one place right gives $10x = 0.\overline{2} = 2/9$, where $x = 0.0\overline{2}$. Therefore

$$x = \frac{1}{10}\frac{2}{9} = \frac{2}{90} = \boxed{\frac{1}{45}}.$$

This differs from part (c), where the zero repeats as part of the block.

(c) The decimal $0.\overline{02}$.

The repeated block has length two, so

$$0.\overline{02} = \frac{2}{10^2 - 1} = \boxed{\frac{2}{99}}.$$

(d) The decimal $3.\overline{14}$.

Separate the integer part before evaluating the length-two block:

$$3.\overline{14} = 3 + \frac{14}{99} = \frac{297 + 14}{99} = \boxed{\frac{311}{99}}.$$

(e) The decimal $0.\overline{10}$.

The block is 10, of length two and integer value 10. Hence

$$0.\overline{10} = \boxed{\frac{10}{99}}.$$

(f) The decimal $0.\overline{1492}$.

This block has length four and integer value 1492, giving

$$0.\overline{1492} = \frac{1492}{10^4 - 1} = \boxed{\frac{1492}{9999}}.$$

Conclusion. The six fractions, in order, are 2/9, 1/45, 2/99, 311/99, 10/99, and 1492/9999.

16.5 Exercise 16.5: Decimal expansions of six rational numbers

What we must show. Find the decimal expansions of the six prescribed fractions. For terminating values, include both expansions supplied by Theorem 16.3.

(a) Expand $1/8$.

At each stage divide ten times the current remainder by 8:

$$10 = 8 \cdot 1 + 2, \qquad 20 = 8 \cdot 2 + 4, \qquad 40 = 8 \cdot 5 + 0.$$

The quotients give digits $1, 2, 5$. A zero remainder produces zeros forever. Reducing the last nonzero digit by one gives the other expansion, so

$$\boxed{\frac{1}{8} = 0.125\overline{0} = 0.124\overline{9}}.$$

(b) Expand $1/16$**.**

The successive divisions are

$$10 = 16 \cdot 0 + 10, \quad 100 = 16 \cdot 6 + 4,$$
$$40 = 16 \cdot 2 + 8, \qquad 80 = 16 \cdot 5 + 0.$$

The initial quotient zero matters: it is the first decimal digit. The digits are $0, 6, 2, 5$, followed by zeros. Thus

$$\boxed{\frac{1}{16} = 0.0625\overline{0} = 0.0624\overline{9}}.$$

(c) Expand $2/3$**.**

The first division is $20 = 3 \cdot 6 + 2$. The new remainder is the original remainder 2, so the same digit 6 and remainder 2 recur at every step. Therefore

$$\boxed{\frac{2}{3} = 0.\overline{6}}.$$

As a check, the geometric-series formula gives $0.\overline{6} = 6/9 = 2/3$.

(d) Expand $7/9$**.**

Here $70 = 9 \cdot 7 + 7$. The remainder 7 returns immediately, so every digit is 7:

$$\boxed{\frac{7}{9} = 0.\overline{7}}.$$

(e) Expand $6/11$**.**

The divisions $60 = 11 \cdot 5 + 5$ and $50 = 11 \cdot 4 + 6$ return to the initial remainder 6 after the two digits $5, 4$. Hence

$$\boxed{\frac{6}{11} = 0.\overline{54}}.$$

The block formula confirms $54/99 = 6/11$.

(f) Expand $22/7$**.**

First $22 = 7 \cdot 3 + 1$, giving integer part 3 and remainder 1. The next six divisions are

$$10 = 7 \cdot 1 + 3, \quad 30 = 7 \cdot 4 + 2,$$
$$20 = 7 \cdot 2 + 6, \quad 60 = 7 \cdot 8 + 4,$$
$$40 = 7 \cdot 5 + 5, \quad 50 = 7 \cdot 7 + 1.$$

The remainder has returned to 1, so the six-digit block repeats:

$$\boxed{\frac{22}{7} = 3.\overline{142857}}.$$

Account for all possible expansions.

The last four displayed expansions have neither an eventual tail of zeros nor an eventual tail of nines. Theorem 16.3 says they are unique. The first two are positive terminating decimals, so the two representations displayed in each of (a) and (b) are their only representations.

Conclusion. The expansions are the six boxed answers above, including both representations in the two terminating cases.

16.6 Exercise 16.6: The cyclic decimal blocks for the sevenths

What we must show. Find the expansions of $1/7, 2/7, \ldots, 6/7$ and explain the pattern in their repeating blocks.

Record the long division rule once for each possible nonzero remainder.

For a current remainder r, the next digit d and next remainder r' satisfy $10r = 7d + r'$ with $0 \le r' < 7$. The six possible nonzero remainders give

r	1	2	3	4	5	6
d	1	2	4	5	7	8
r'	3	6	2	5	1	4

For instance, the entry $r = 3$ comes from $30 = 7 \cdot 4 + 2$, so the next digit is 4 and the next remainder is 2.

Follow the cycle starting from remainder one.

The remainder rule traces

$$1 \longmapsto 3 \longmapsto 2 \longmapsto 6 \longmapsto 4 \longmapsto 5 \longmapsto 1.$$

The digits produced on these six steps are respectively $1, 4, 2, 8, 5, 7$. All six remainders in the cycle are distinct, so returning to the initial remainder takes exactly six steps.

Start the same cycle at each of the other remainders.

The fraction $a/7$ begins with remainder a, for $a = 1, \ldots, 6$. Starting the displayed cycle at that remainder gives

Fraction	Remainders before the successive digits	Expansion
$1/7$	$1, 3, 2, 6, 4, 5$	$0.\overline{142857}$
$2/7$	$2, 6, 4, 5, 1, 3$	$0.\overline{285714}$
$3/7$	$3, 2, 6, 4, 5, 1$	$0.\overline{428571}$
$4/7$	$4, 5, 1, 3, 2, 6$	$0.\overline{571428}$
$5/7$	$5, 1, 3, 2, 6, 4$	$0.\overline{714285}$
$6/7$	$6, 4, 5, 1, 3, 2$	$0.\overline{857142}$

Each row returns to its initial remainder after the six digits, so the same block repeats forever. These expansions are unique because none has an eventual tail of zeros or nines.

Conclusion. The six blocks are cyclic rotations of 142857. Long division uses the same cycle of six nonzero remainders, with a different starting position for each fraction.

16.7 Exercise 16.7: The decimal formed by concatenating integers

What we must show. Decide whether the number

$$c = 0.12345678910111213141516171819202122 23\cdots$$

is rational. Here its digits are obtained by writing $1, 2, 3, \ldots$ in their ordinary decimal notation and concatenating those strings.

Plan. Prove this digit string is not eventually periodic. It has arbitrarily long strings of zeros arbitrarily far out, but it also has nonzero digits arbitrarily far out. Both facts are needed: a decimal ending in zeros would have the first property alone.

Find long zero strings after any prescribed position.

For every positive integer m, the integer 10^m appears in the concatenation. Its ordinary decimal notation is the digit 1 followed by m zeros. As m increases, the starting positions of these integers go arbitrarily far out: before writing 10^m, we have already written the $10^m - 1$ preceding positive integers, each of which contributes at least one digit. Thus, after any specified digit position and for any specified length p, there is a string of at least p consecutive zeros. Choose $m \geq p$ large enough that the occurrence of 10^m starts after that position.

Find nonzero digits after any prescribed position.

The initial digit 1 in each of those occurrences of 10^m is nonzero, and these occurrences start arbitrarily far out. Consequently the decimal does not eventually consist only of zeros. It does not eventually consist only of nines either, since zero strings also occur arbitrarily far out.

Rule out an eventual repeating block.

Suppose the digits were periodic with period $p \geq 1$ after some position N. If the repeating block contained only zeros, all digits beyond N would be zero, contradicting the preceding step. Otherwise the block contains a nonzero digit. Every run of p consecutive positions beyond N passes through every position of the repeating block once, so it must contain a nonzero digit. But the first step supplies a run of p zeros entirely beyond N. This is a contradiction in both cases.

Apply the decimal characterization of rational numbers.

The given decimal is not eventually periodic. By Theorem 16.5, a rational number has a repeating decimal expansion; by Theorem 16.3 its only possible second expansion would have tails of zeros and nines. Those tails have already been excluded here. Therefore the displayed decimal cannot represent a rational number.

Conclusion. The concatenation decimal is irrational.

16.8 Exercise 16.8: Constructing a number outside a proposed list

What we must show. Given any sequence (s_n) in $(0,1)$, with selected expansions $s_n = 0.d_1^{(n)}d_2^{(n)}\cdots$, define

$$e_n = \begin{cases} 6, & d_n^{(n)} \neq 6, \\ 7, & d_n^{(n)} = 6. \end{cases}$$

Show that $0.e_1e_2\cdots$ represents a number $y \in (0,1)$ different from every s_n.

Check that the new digit string represents a number.

Every e_n is either 6 or 7, hence an allowed decimal digit. Define the partial sums $y_N = \sum_{n=1}^{N} e_n 10^{-n}$. They increase and are bounded above by the convergent geometric series $\sum_{n=1}^{\infty} 7 \cdot 10^{-n} = 7/9$. The monotone sequence theorem gives a finite limit y, which by the definition of a decimal is the number $0.e_1e_2\cdots$. Termwise comparison of the finite partial sums and passage to limits give

$$\frac{2}{3} = \sum_{n=1}^{\infty} 6 \cdot 10^{-n} \leq y \leq \sum_{n=1}^{\infty} 7 \cdot 10^{-n} = \frac{7}{9}.$$

Because $0 < 2/3 \leq 7/9 < 1$, these bounds imply $y \in (0,1)$.

Verify the diagonal difference at every index.

Fix n. If $d_n^{(n)} \neq 6$, the construction gives $e_n = 6$, so $e_n \neq d_n^{(n)}$. If $d_n^{(n)} = 6$, it gives $e_n = 7$, so again $e_n \neq d_n^{(n)}$. Thus the new expansion differs from the selected expansion of s_n at least at its nth decimal digit.

Explain why different strings cannot secretly represent the same number.

Different decimal strings can represent the same real number in the zero-tail/nine-tail case of Theorem 16.3. That issue must be checked rather than ignored. Every digit of the expansion of y is 6 or 7, so this expansion ends neither in zeros nor in nines. Theorem 16.3 therefore makes it the unique expansion of y. If $y = s_n$, the selected expansion of s_n would be a second expansion of y differing at digit n, which is impossible. Hence $y \neq s_n$ for every n.

Interpret the conclusion for a supposed enumeration.

If a sequence listed every number in $(0,1)$, the number y constructed from that sequence would belong to $(0,1)$ and yet would be absent from the list. This contradiction proves that no such listing exists.

Conclusion. The constructed y lies in $(0,1)$ and differs from every listed term. Thus $(0,1)$ is uncountable.

16.9 Exercise 16.9: Existence of Euler's constant

What we must show. For

$$H_n = \sum_{k=1}^{n} \frac{1}{k}, \qquad \gamma_n = H_n - \log n = H_n - \int_1^n \frac{dt}{t},$$

prove (a) (γ_n) decreases, (b) $0 < \gamma_n \leq 1$, and (c) its limit exists and is finite. The logarithm here is natural.

(a) Subtract two neighboring terms without discarding the logarithm.

Since $H_{n+1} = H_n + 1/(n+1)$, we have

$$\begin{aligned}\gamma_n - \gamma_{n+1} &= H_n - \log n - H_n - \frac{1}{n+1} + \log(n+1) \\ &= \int_n^{n+1} \frac{dt}{t} - \frac{1}{n+1} \\ &= \int_n^{n+1} \left(\frac{1}{t} - \frac{1}{n+1}\right) dt.\end{aligned}$$

For $n \le t < n+1$ the integrand is positive, because $t < n+1$ implies $1/t > 1/(n+1)$. It is continuous and positive on the interior of this interval, so its integral is strictly positive. For an explicit lower estimate, on $[n, n+1/2]$ the integrand is at least $1/(n+1/2) - 1/(n+1) > 0$, and that interval has positive length $1/2$. Hence $\gamma_n > \gamma_{n+1}$ for every n.

(b) Bound the integral between two rectangle sums.

For $t \in [k, k+1]$ with $k \ge 1$, reciprocation gives

$$\frac{1}{k+1} \le \frac{1}{t} \le \frac{1}{k}.$$

Integrating over an interval of length one preserves these bounds:

$$\frac{1}{k+1} \le \int_k^{k+1} \frac{dt}{t} \le \frac{1}{k}.$$

For $n \ge 2$, sum from $k = 1$ to $k = n-1$ to obtain

$$H_n - 1 \le \int_1^n \frac{dt}{t} \le H_n - \frac{1}{n}.$$

Subtract the middle expression from H_n. Reversal of the inequalities under subtraction gives

$$\frac{1}{n} \le \gamma_n \le 1.$$

The same conclusion holds for $n = 1$ directly: $\gamma_1 = 1 - \log 1 = 1$. In particular $\gamma_n > 0$ at every index.

(c) Apply monotone convergence with a real lower bound.

Part (a) makes (γ_n) decreasing, and part (b) bounds it below by zero. The monotone sequence theorem therefore gives a finite real limit

$$\gamma = \lim_{n\to\infty} \gamma_n \in [0, 1].$$

The fact that the two pieces H_n and $\log n$ both grow without bound does not obstruct convergence of their difference. The monotonicity and bounds control that difference directly.

Conclusion. The sequence is strictly decreasing, satisfies $0 < \gamma_n \le 1$, and has a finite limit called Euler's constant.

16.10 Exercise 16.10: Irrationality of the indicated powers and roots of pi

What we must show. Starting with the irrationality of π^2 established in Example 6(b), determine what follows for π, $\sqrt{\pi}$, and $\sqrt[3]{\pi}$. Also address π^4 and distinguish the additional fact needed for that last conclusion.

Recall the elementary closure property of rational powers.

If $q = a/b$ is rational, with $a, b \in \mathbb{Z}$ and $b \neq 0$, then for every positive integer m,

$$q^m = \frac{a^m}{b^m} \in \mathbb{Q},$$

since the numerator and denominator are integers and $b^m \neq 0$. Consequently, if a positive integer power of a number is irrational, the number itself must be irrational.

Apply this property to π and to its square root.

If π were rational, its square would be rational, contrary to the given result. Therefore π is irrational. If $\sqrt{\pi}$ were rational, its square would be π and would be rational, giving the same contradiction. Equivalently, its fourth power is π^2, which is already known to be irrational. Thus $\sqrt{\pi}$ is irrational too.

Apply the same argument to the cube root.

If $\sqrt[3]{\pi}$ were rational, its cube π would be rational. Since π is irrational, this is impossible. One can also apply the closure property directly to the sixth power: $(\sqrt[3]{\pi})^6 = \pi^2$. Hence $\sqrt[3]{\pi}$ is irrational.

Explain what the premise alone says about π^4.

The implication just used runs from irrationality of a power to irrationality of its base. It cannot be reversed: the square of an irrational number can be rational. For a concrete comparison, let $x = \sqrt[4]{2}$. Then $x^2 = \sqrt{2}$ is irrational, but $x^4 = 2$ is rational. Thus the statement "x^2 is irrational" does not by itself imply "x^4 is irrational." In particular, the given irrationality of π^2 alone does not settle the question about π^4 by this rational-power argument.

Determine the actual status of π^4 using the stronger fact stated in the text.

Example 6(c) additionally states the deeper theorem that π is not algebraic. This means it is not a zero of any nonzero polynomial with integer coefficients. That stated theorem does settle the question: if $\pi^4 = a/b$ were rational, where $a, b \in \mathbb{Z}$ and $b \neq 0$, then

$$b\pi^4 - a = 0.$$

The polynomial $P(X) = bX^4 - a$ has integer coefficients and is nonzero because its leading coefficient b is nonzero. The displayed equation would make π algebraic, contradicting Example 6(c). Therefore π^4 is irrational.

This final step cites the stronger theorem expressly stated in the chapter; it is not a consequence of the irrationality of π^2 alone.

Conclusion. π, $\sqrt{\pi}$, and $\sqrt[3]{\pi}$ are irrational by elementary rational-power arguments from π^2. The number π^4 is also irrational, using the stronger nonalgebraicity result stated in Example 6(c).